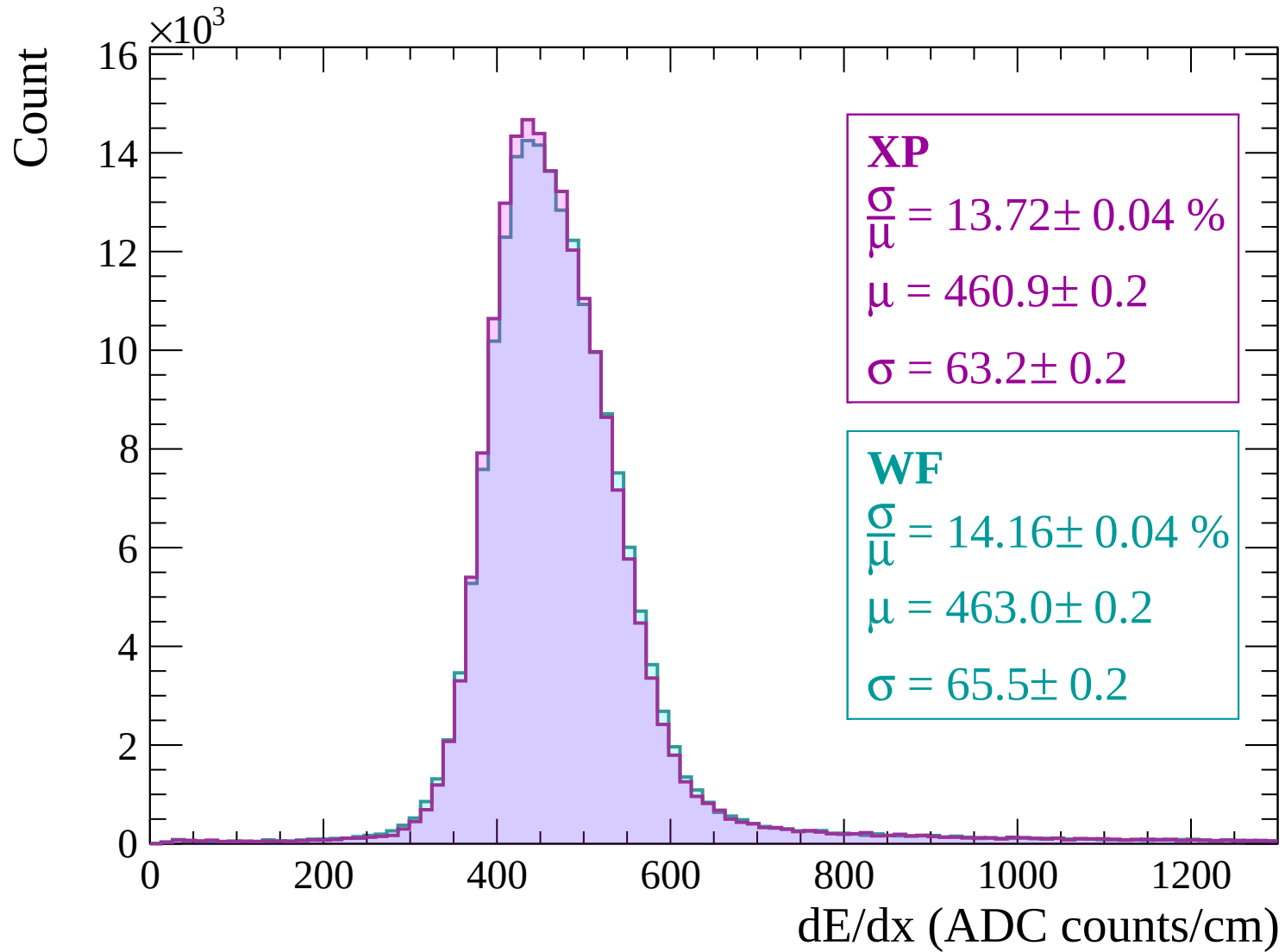
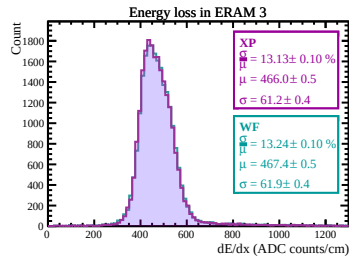
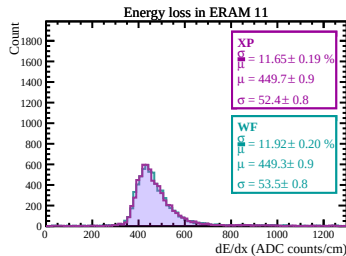
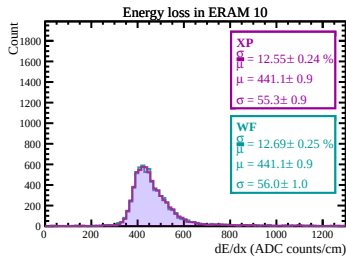
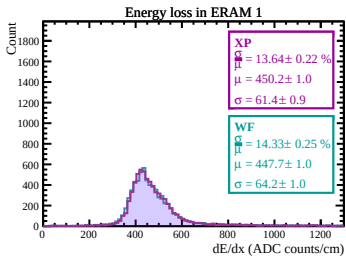
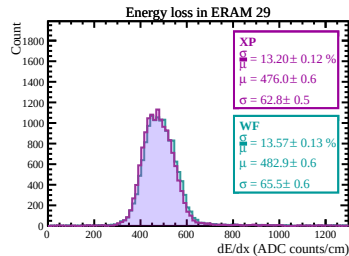
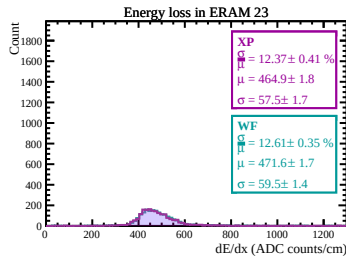
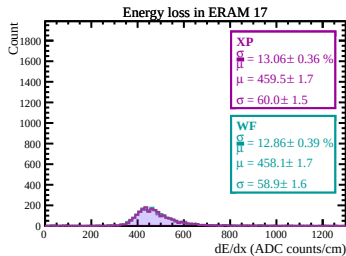
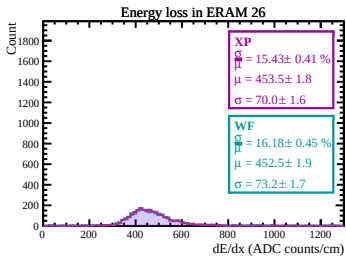
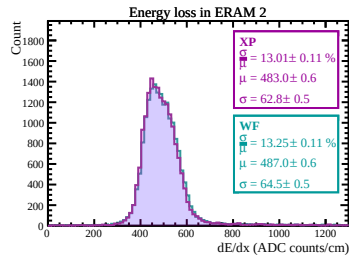
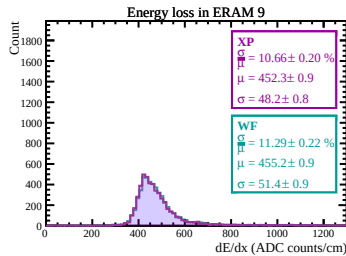
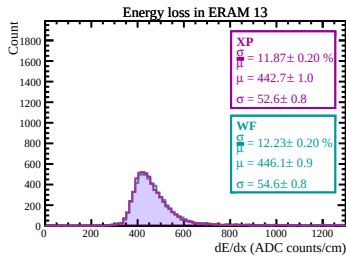
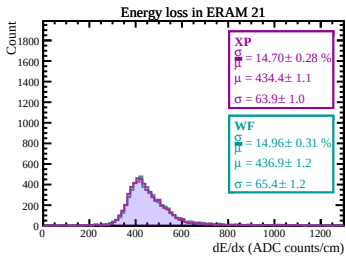
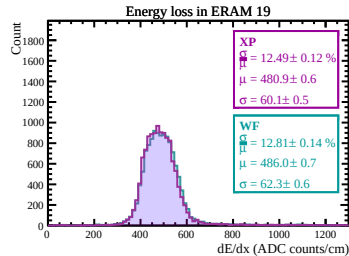
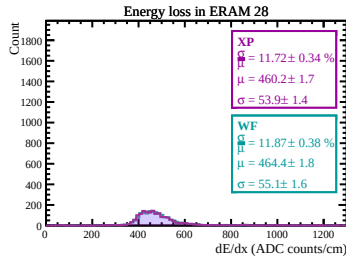
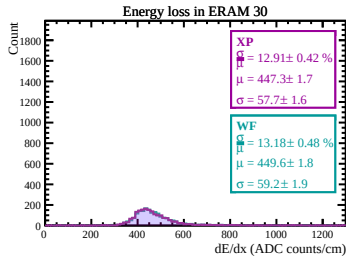
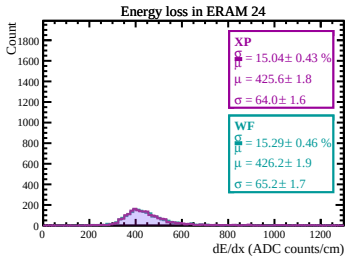
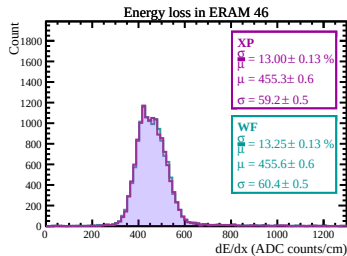
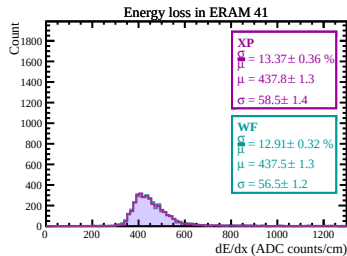
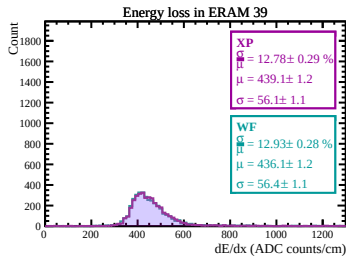
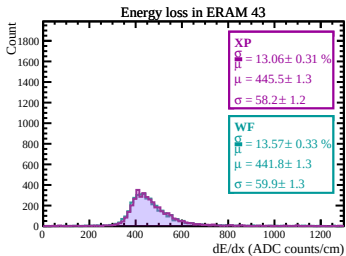
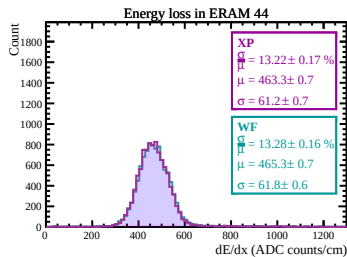
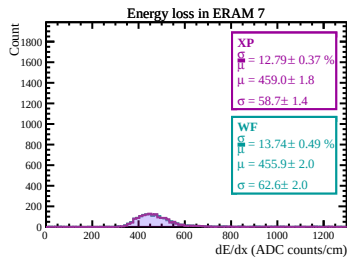
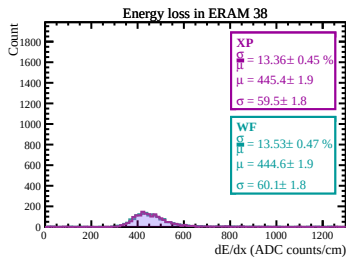
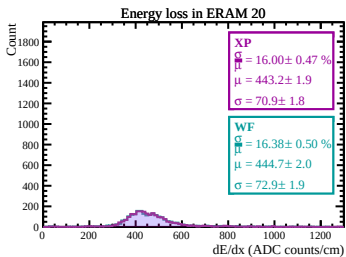
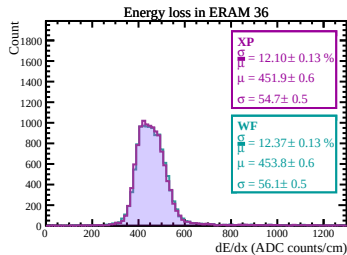
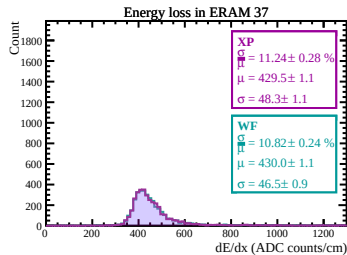
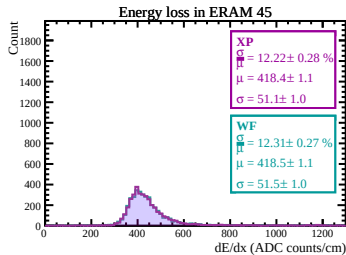
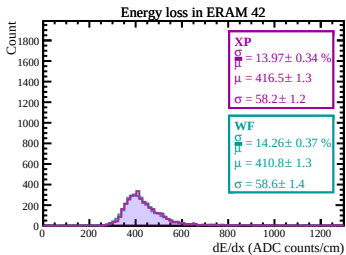
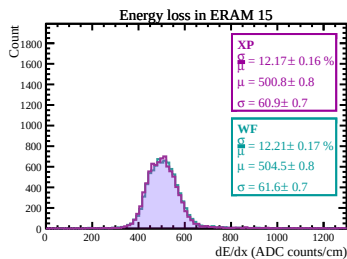
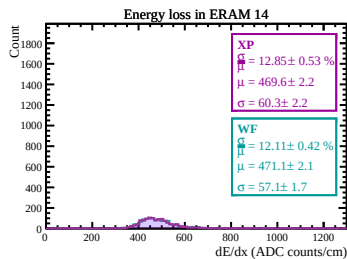
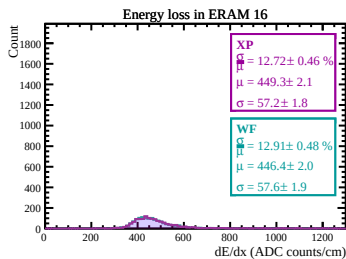
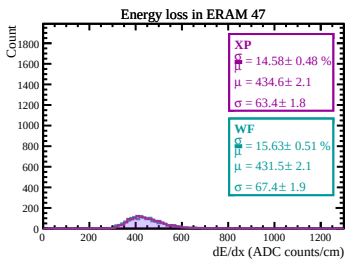
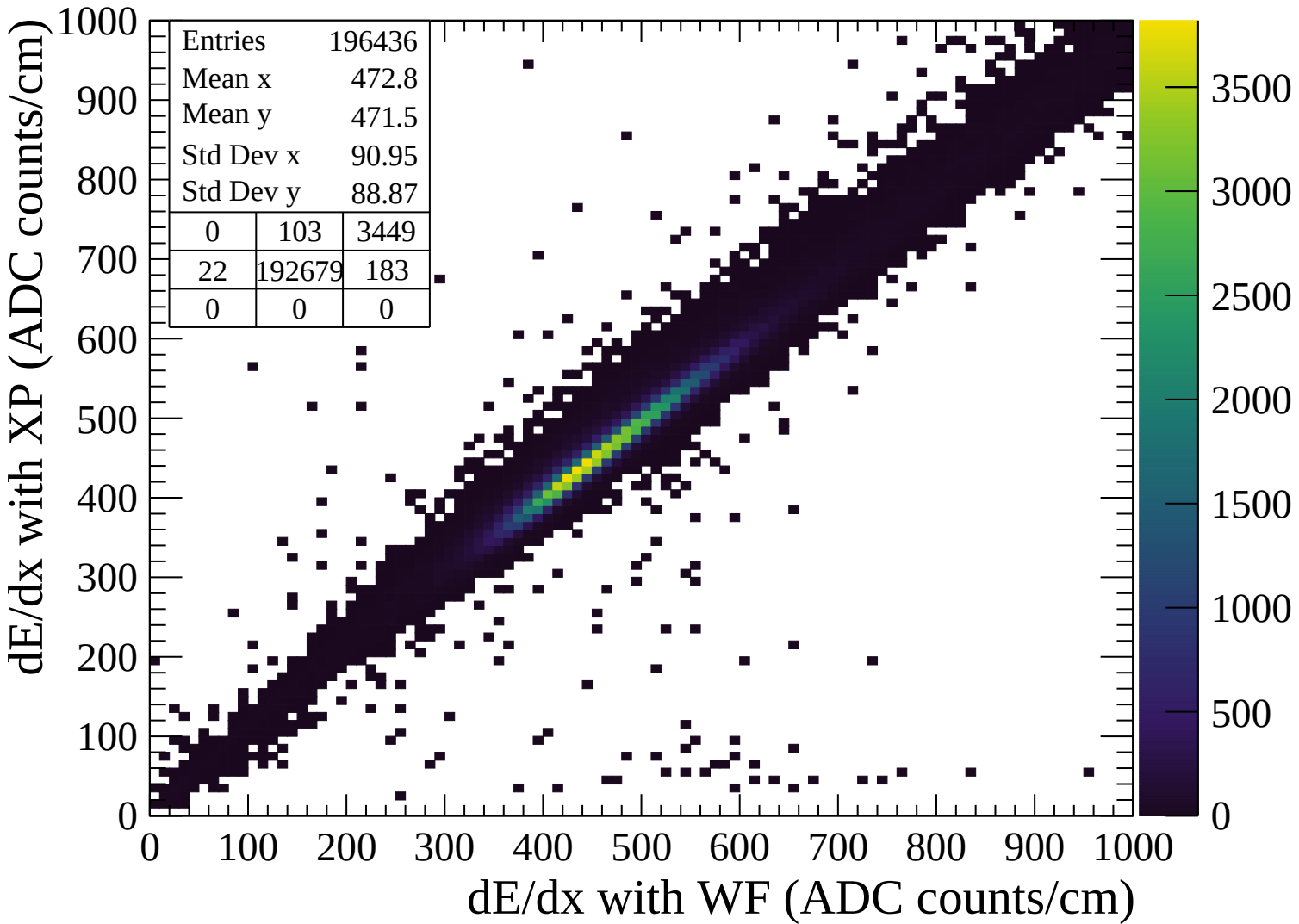
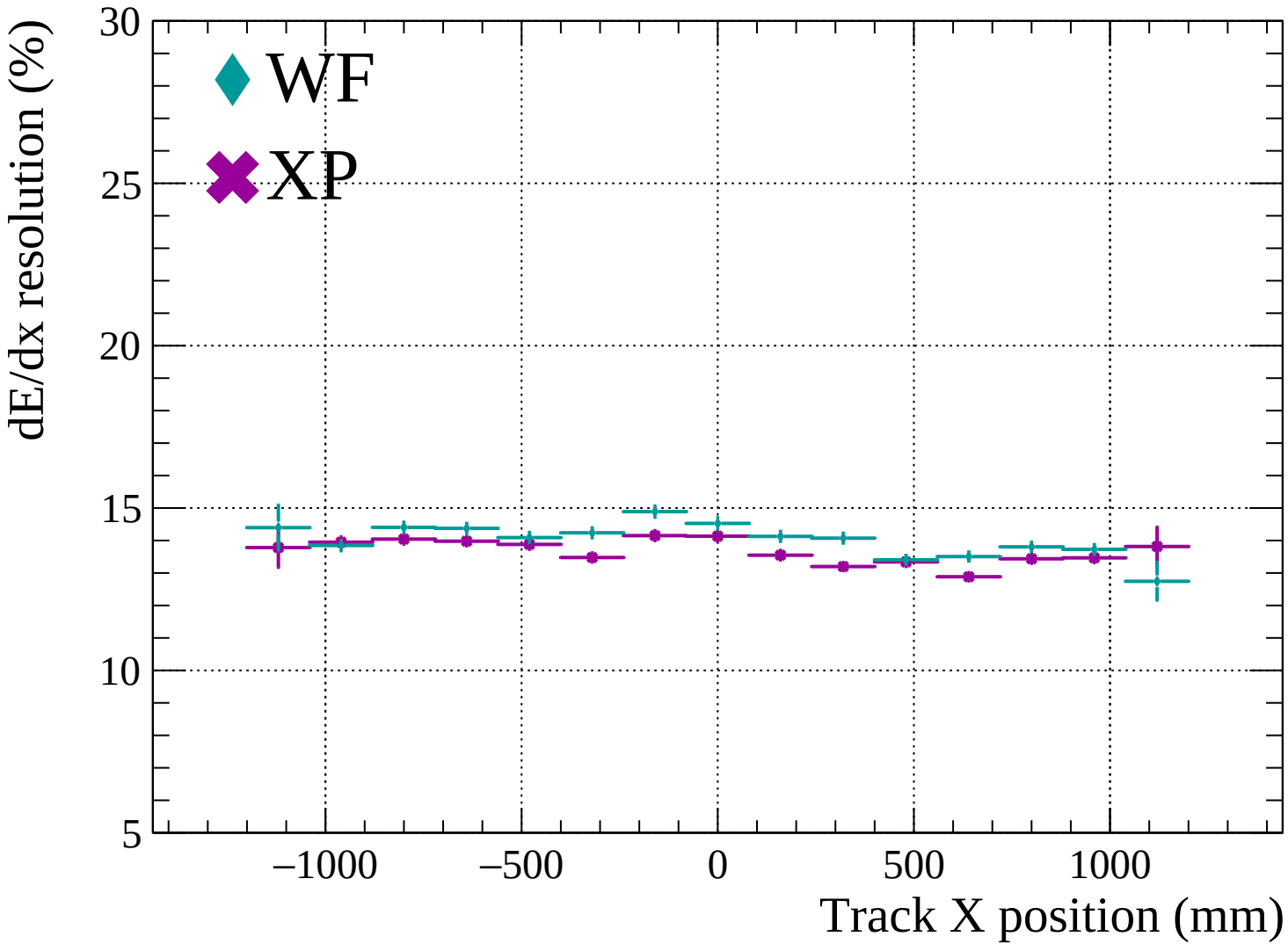


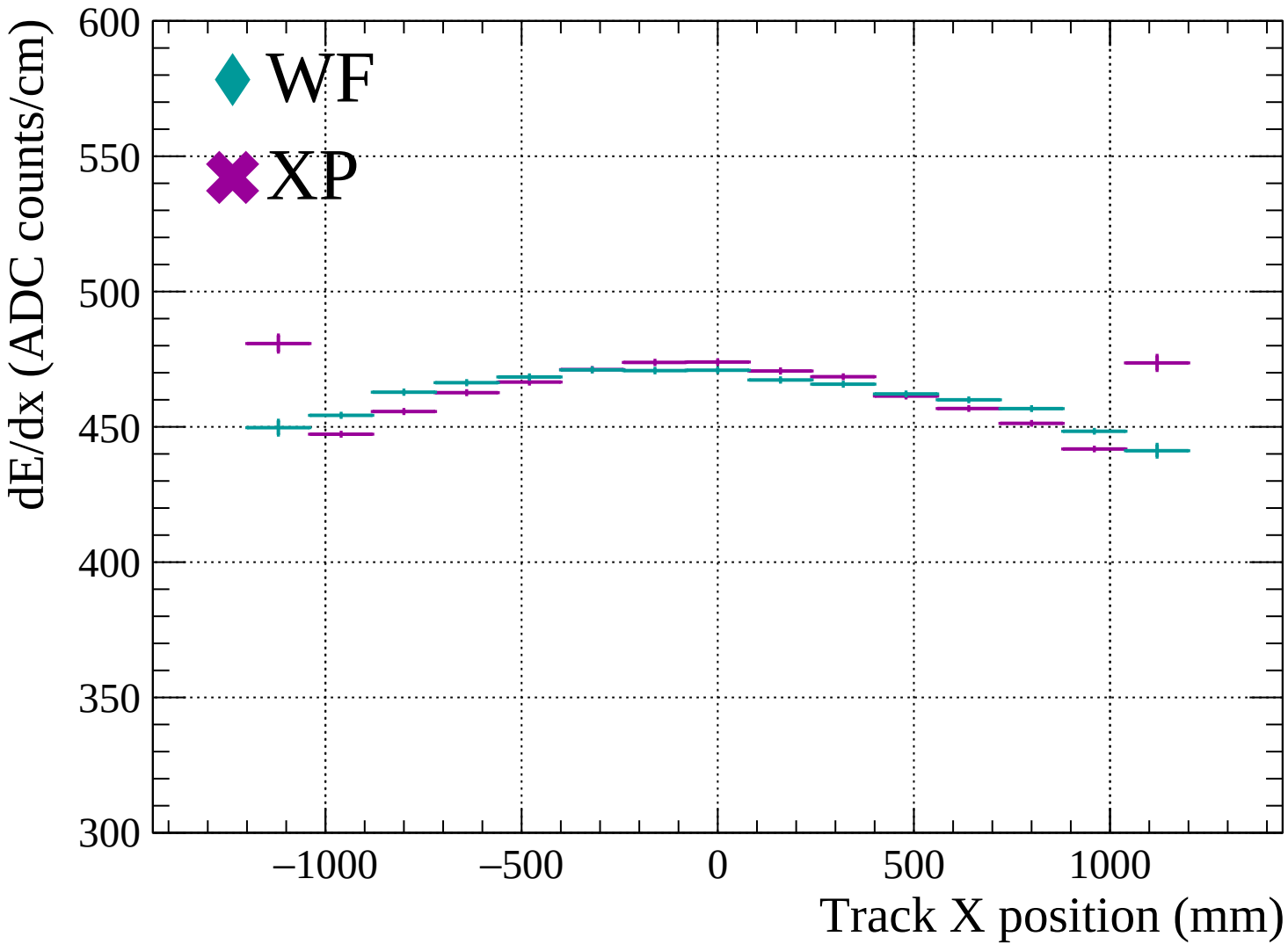


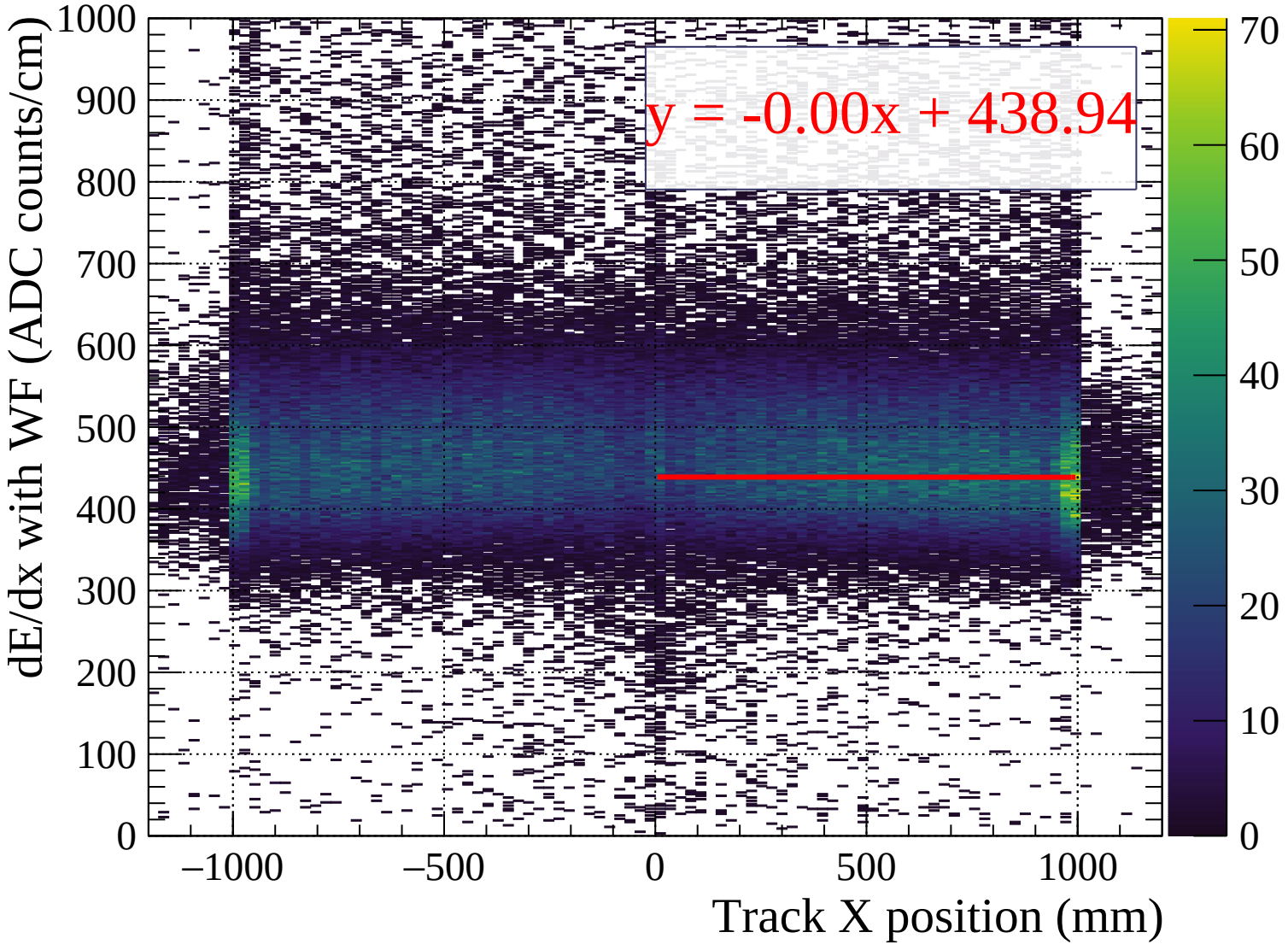


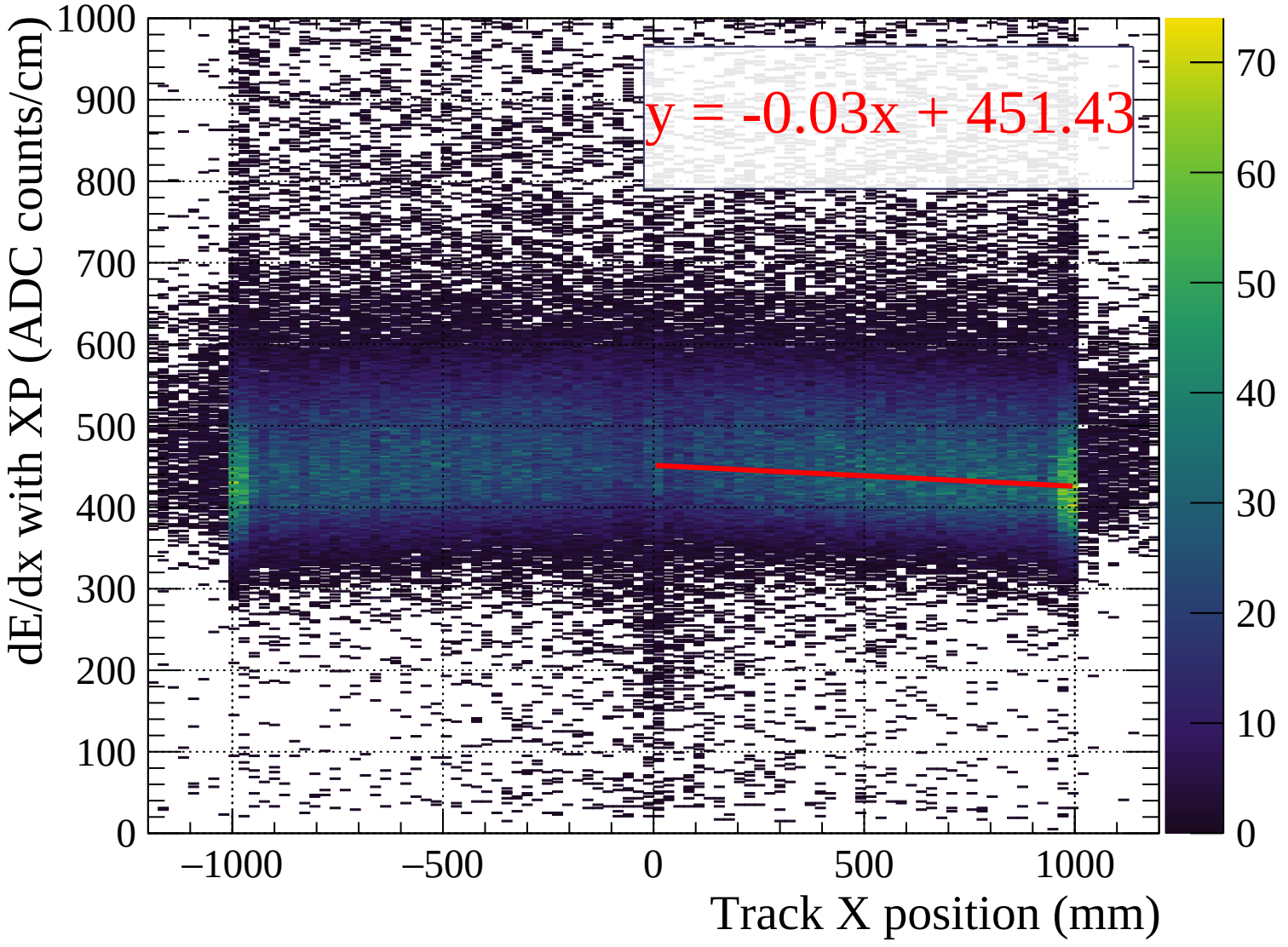




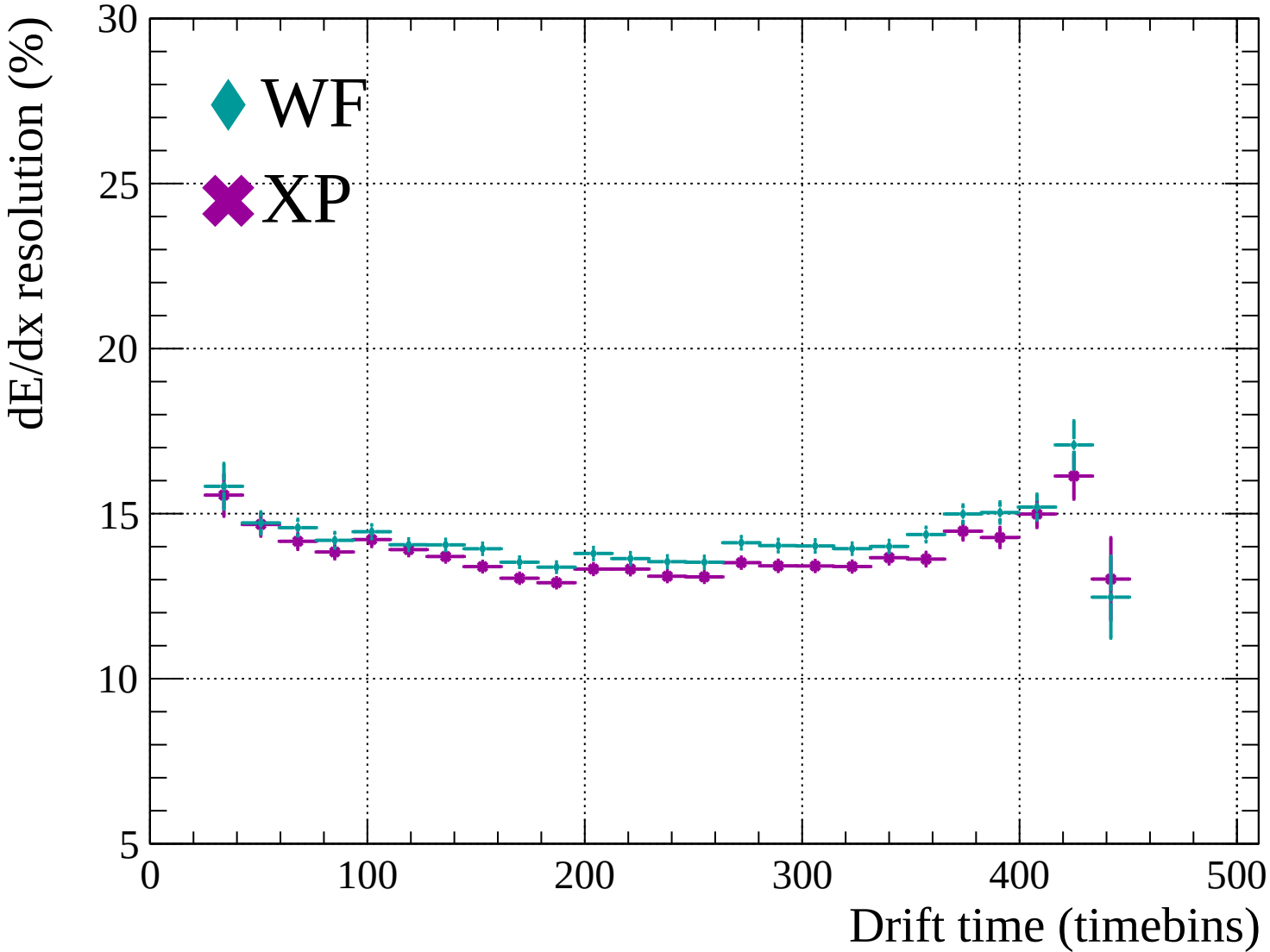


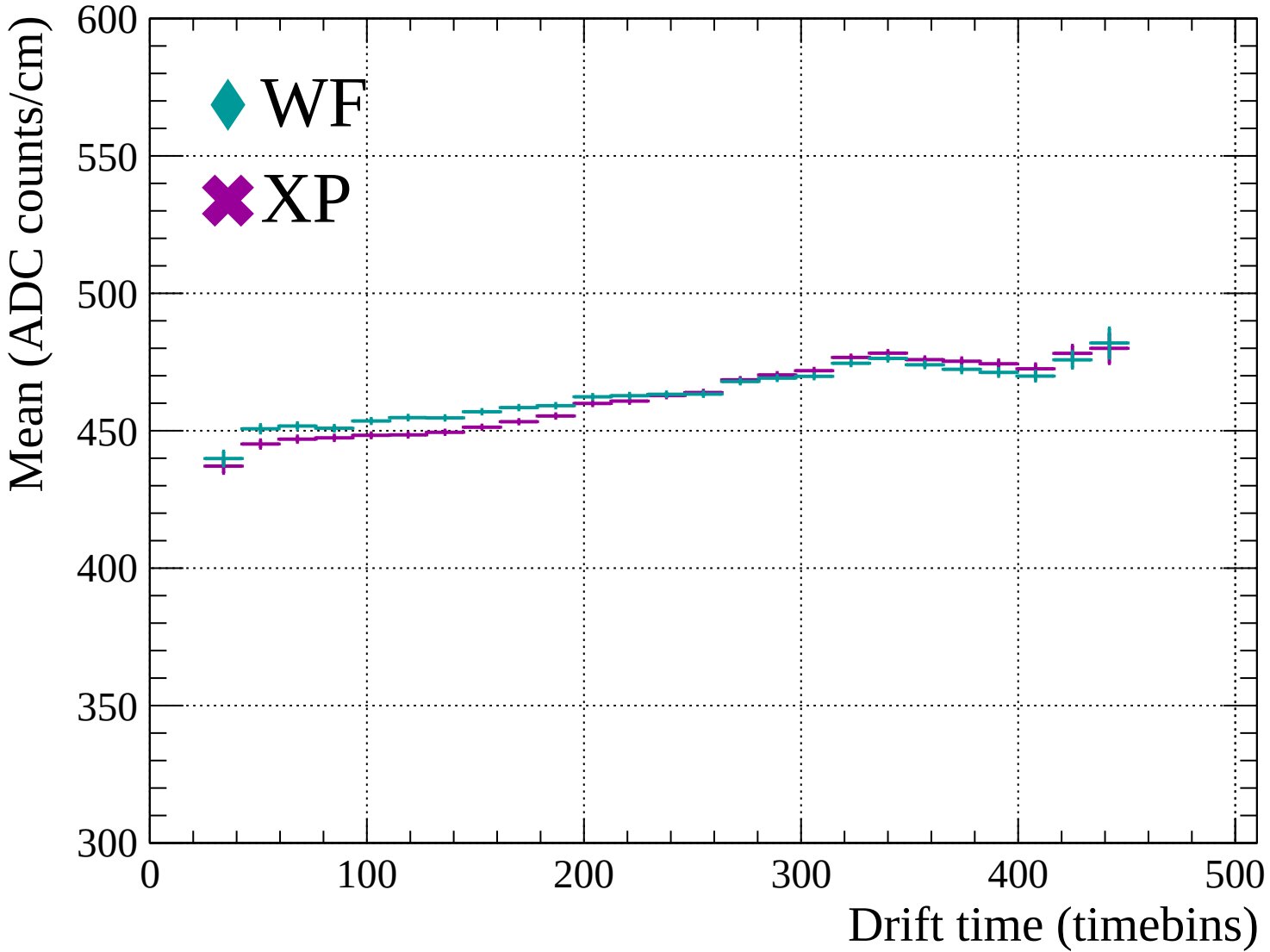


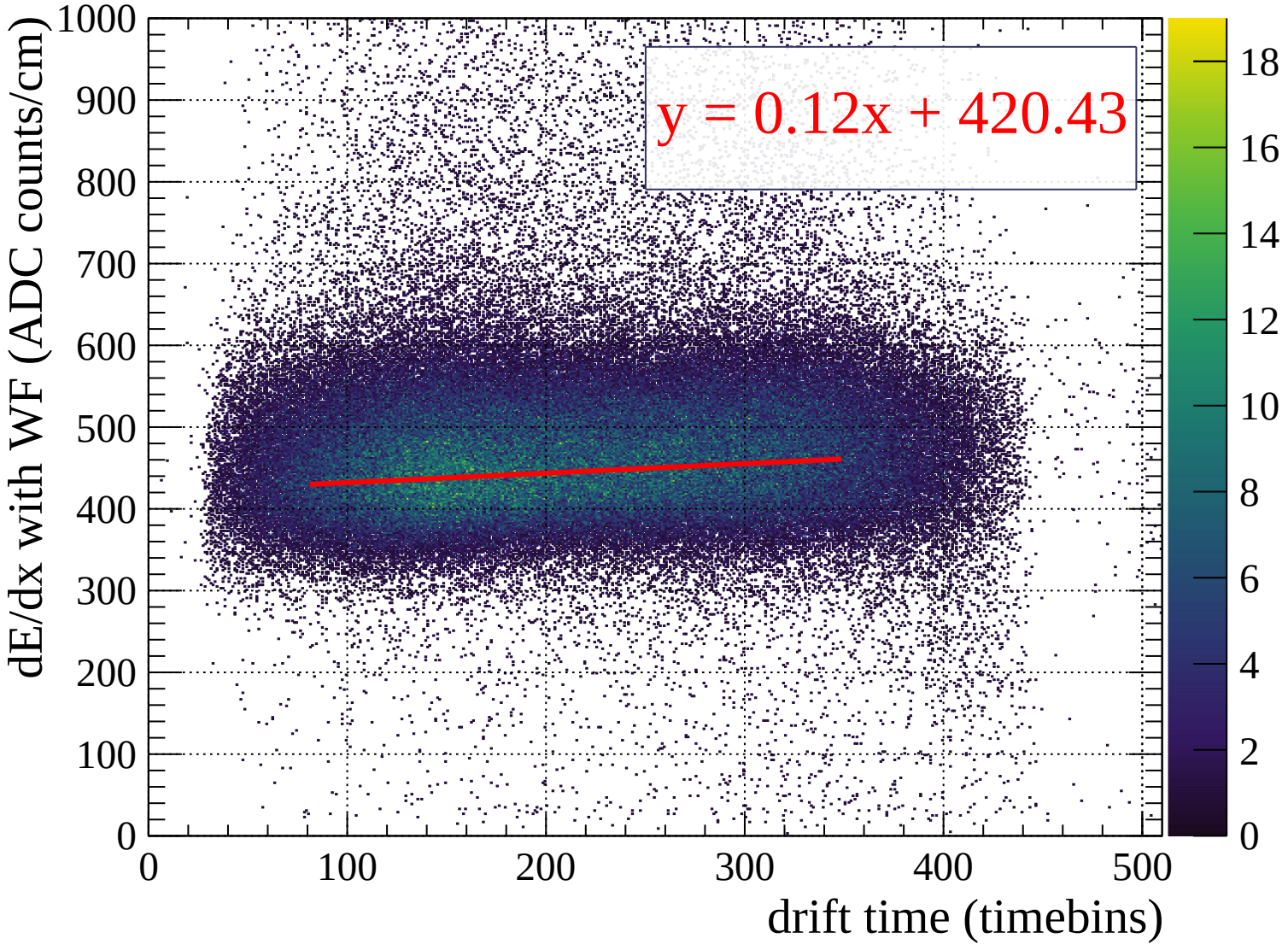


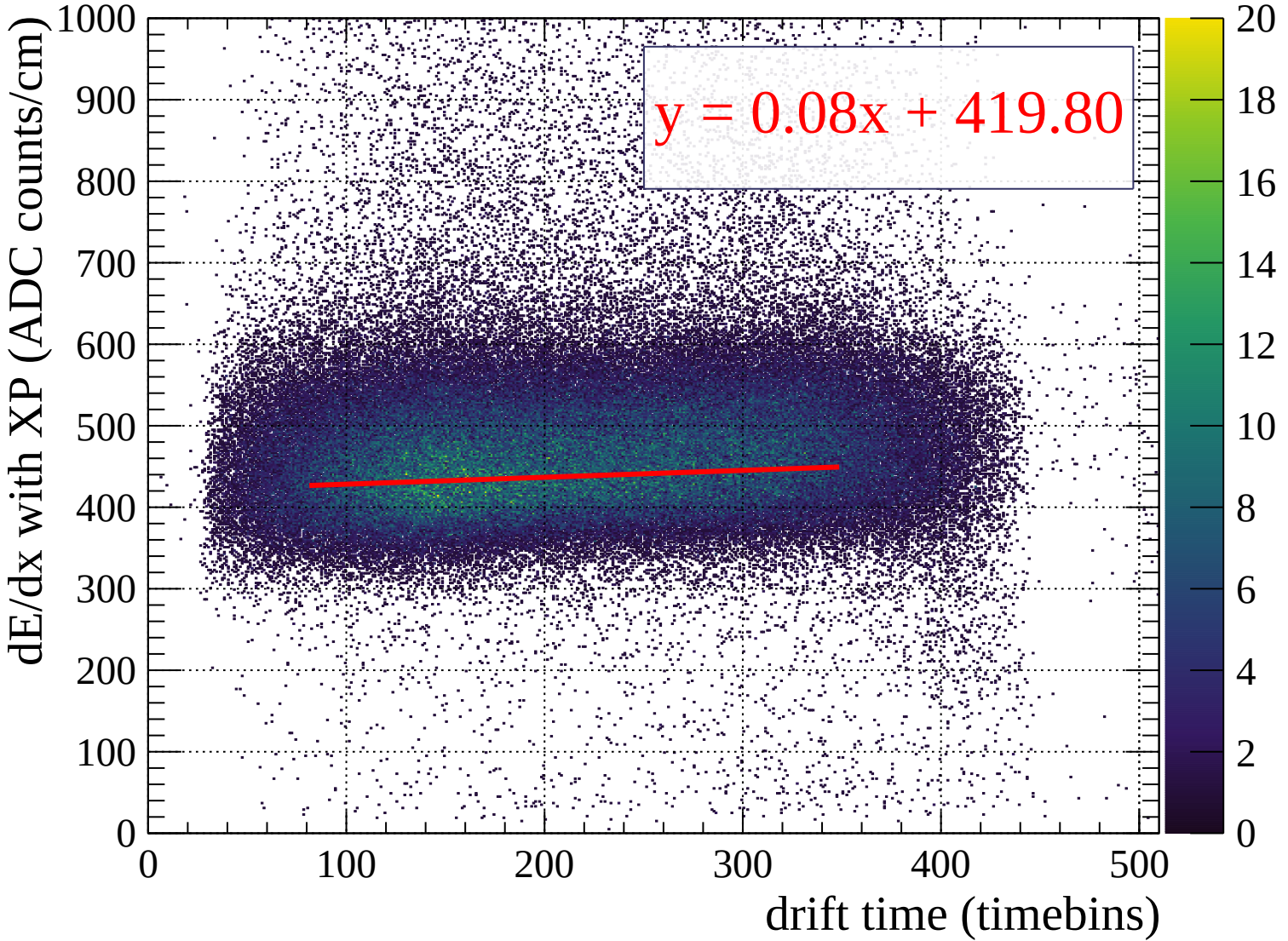


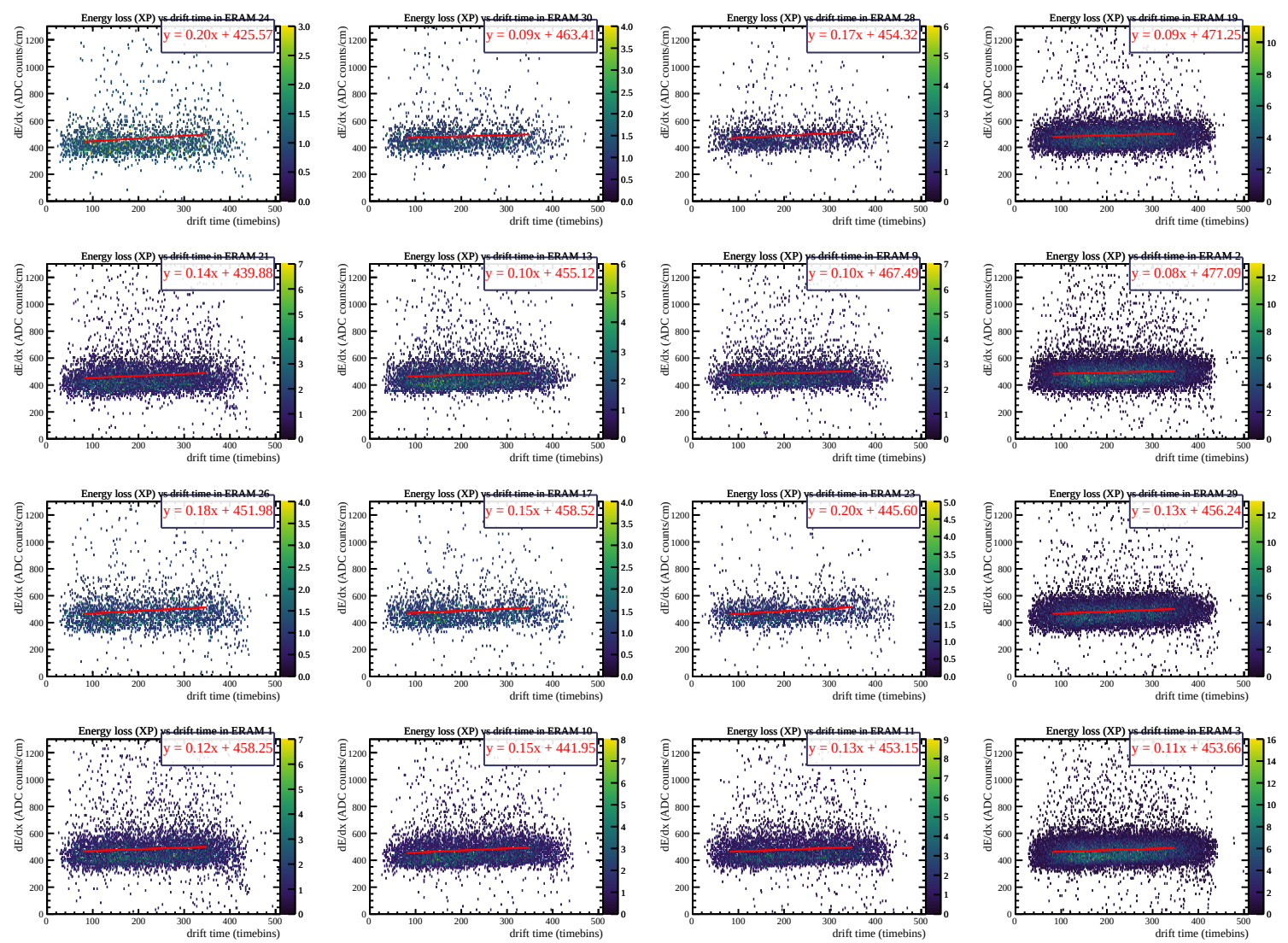


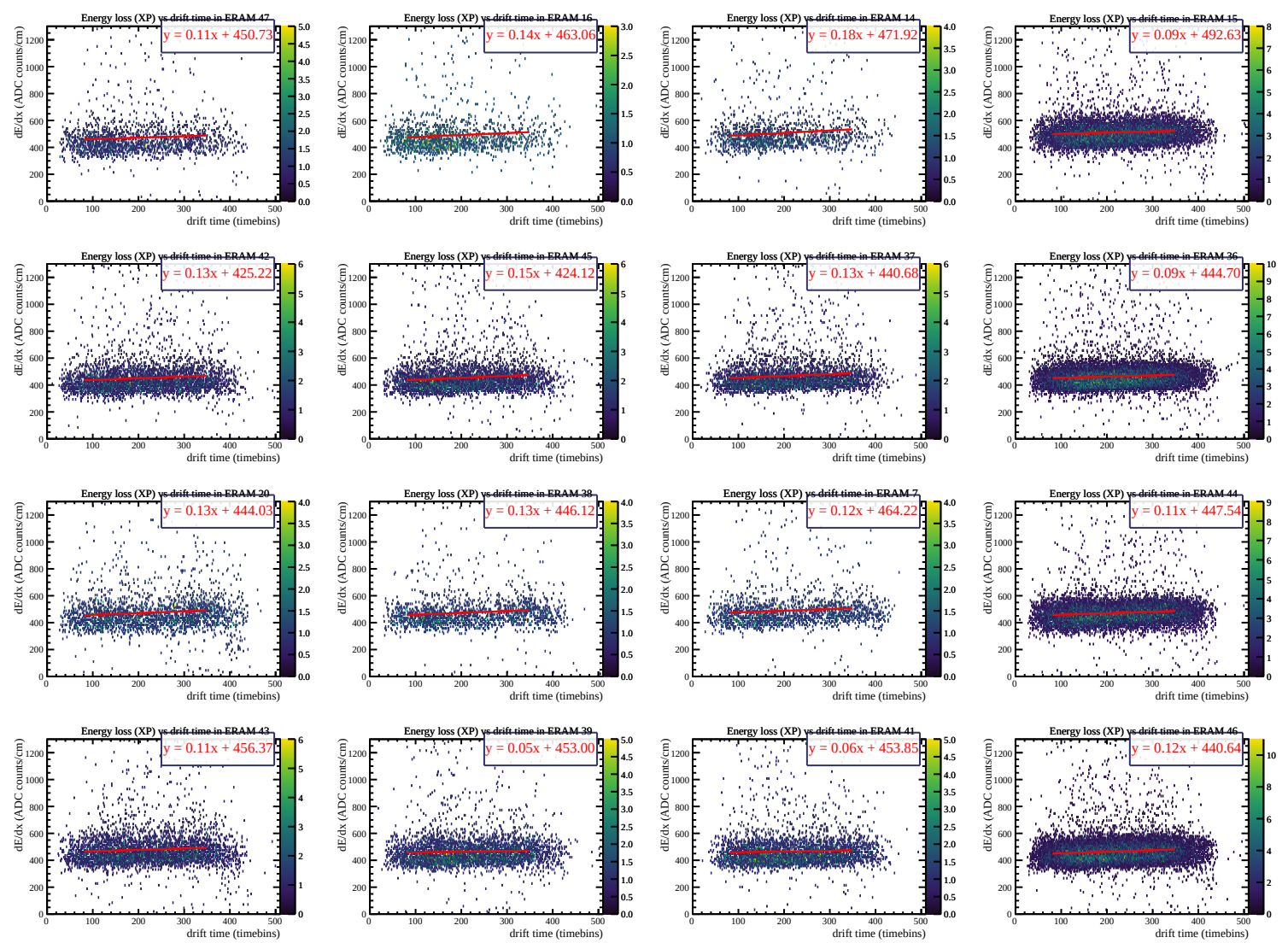


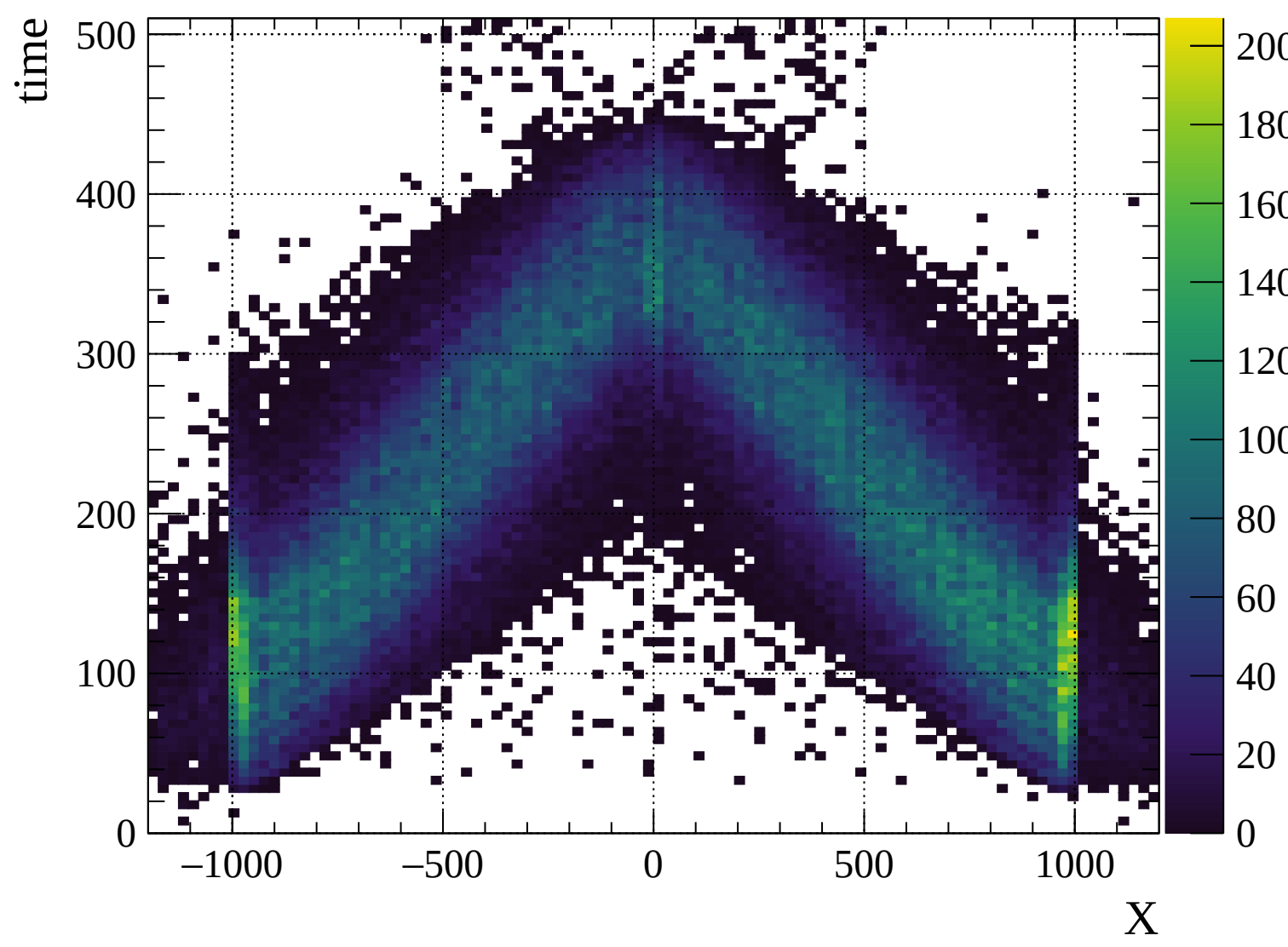




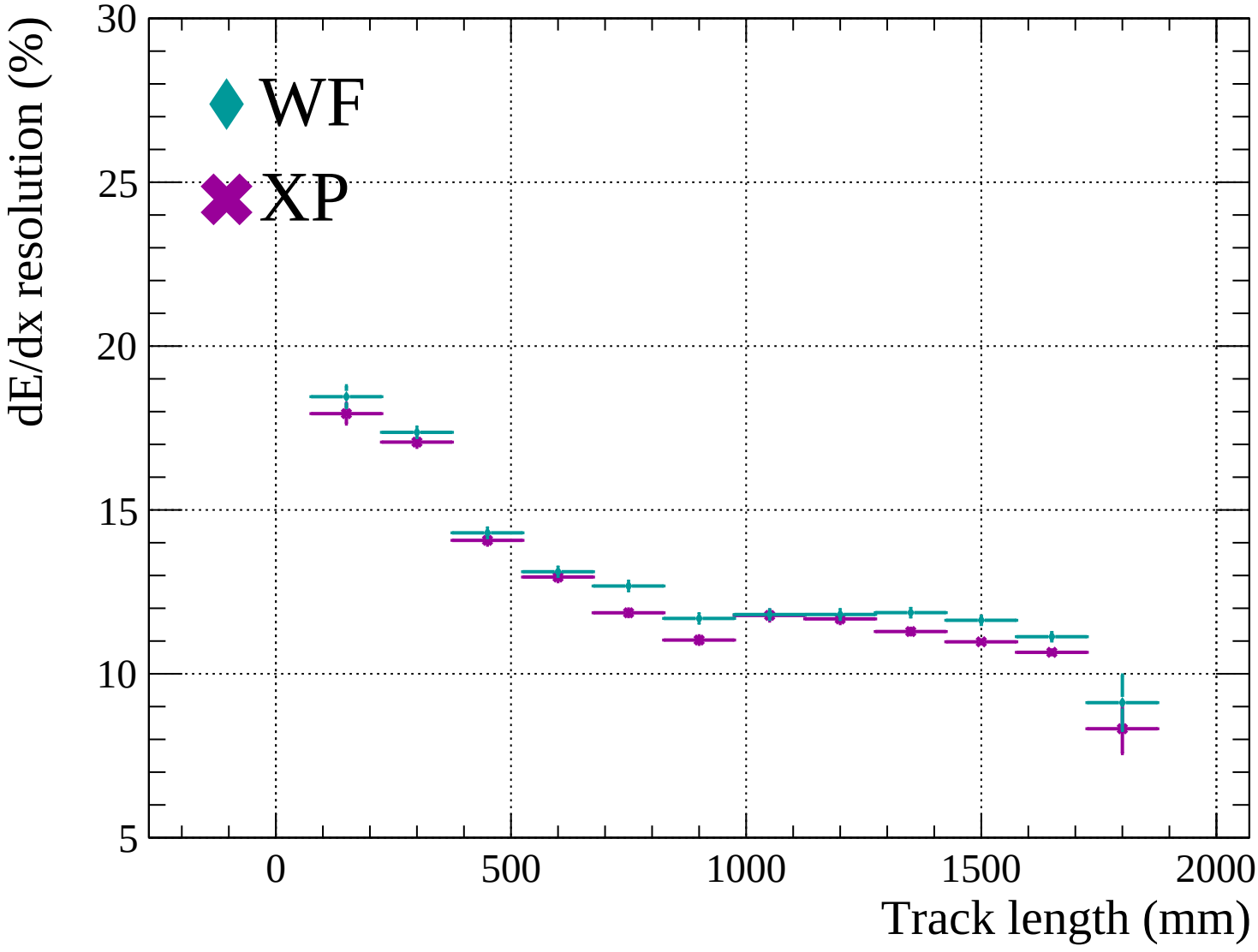




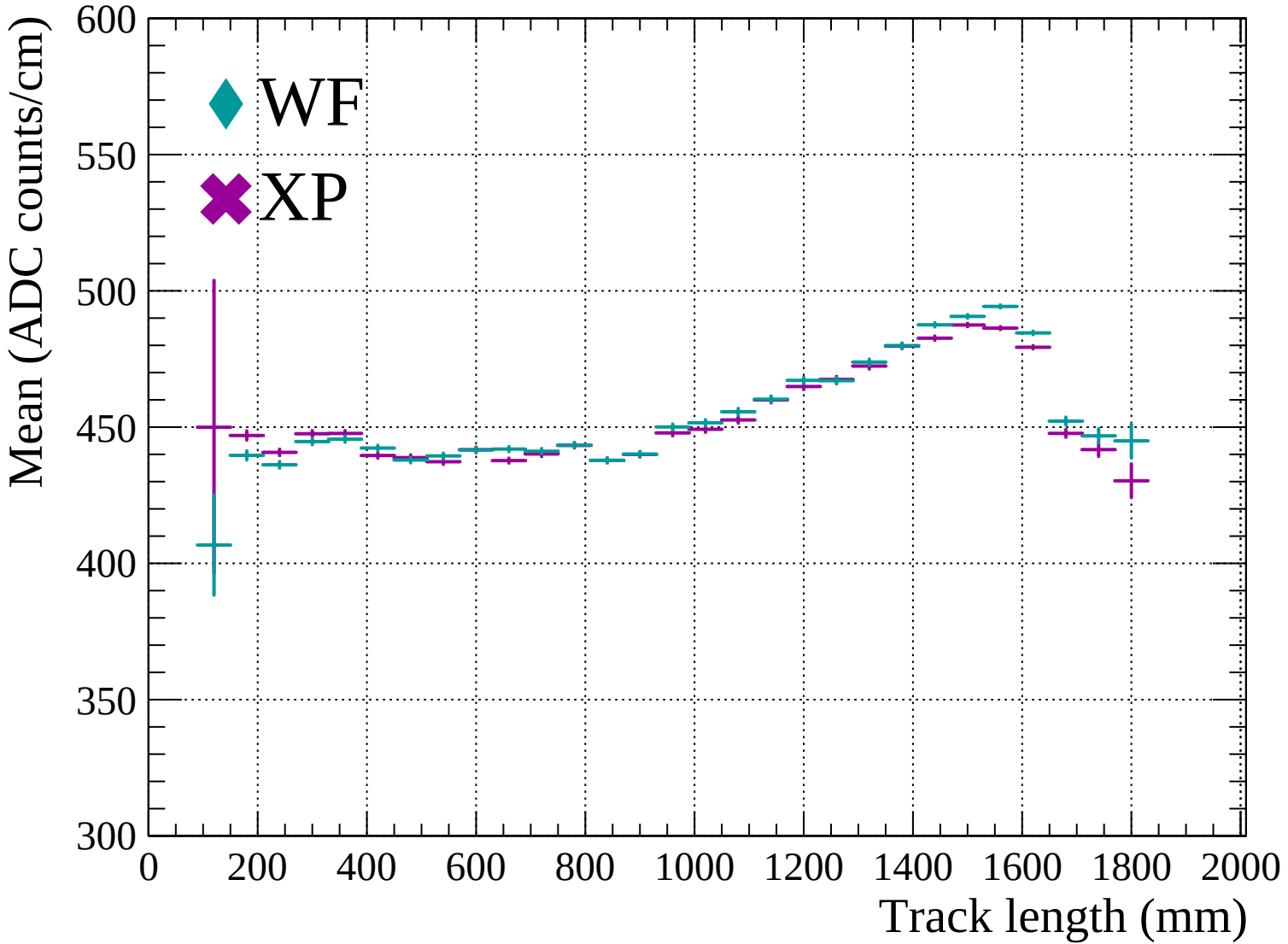


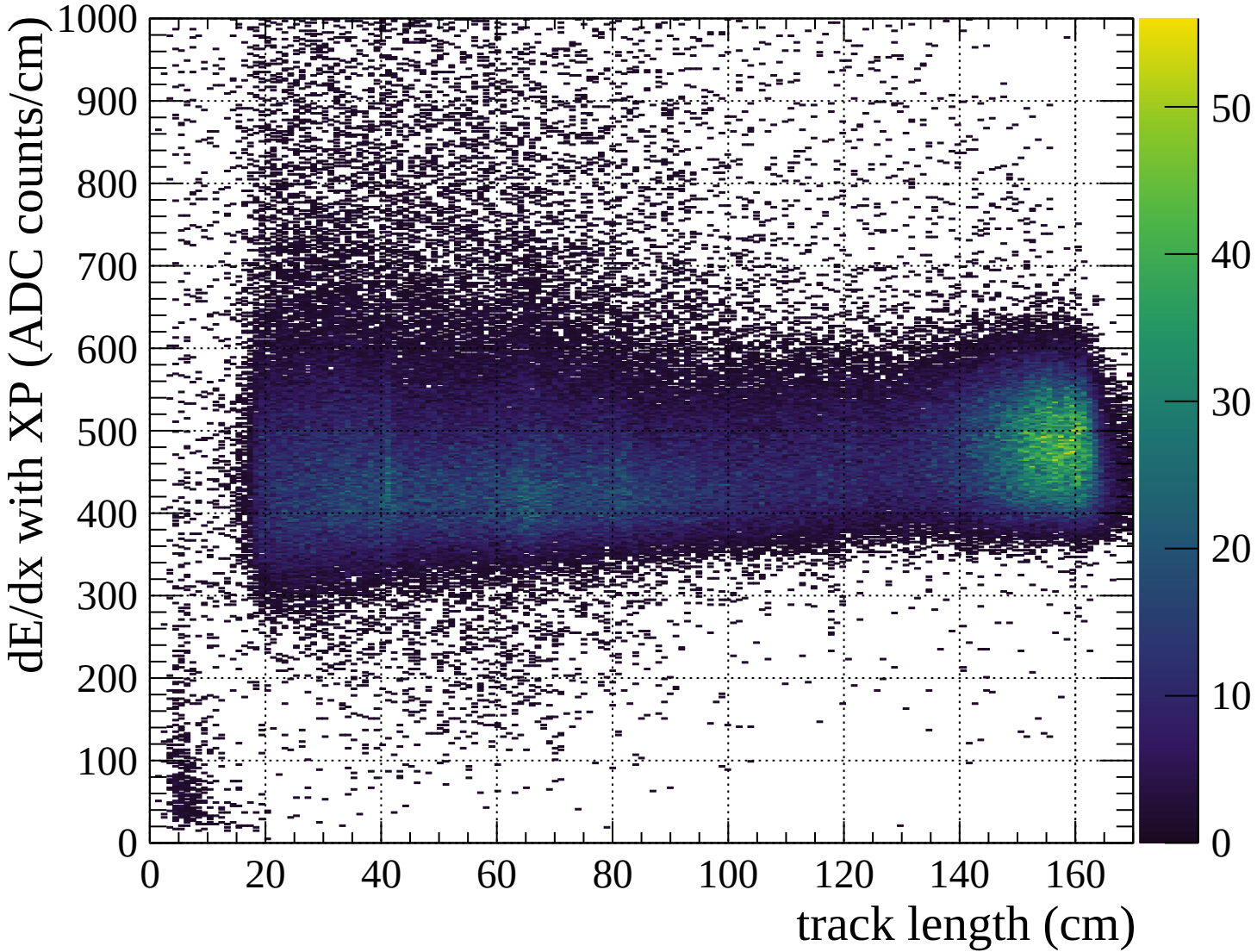


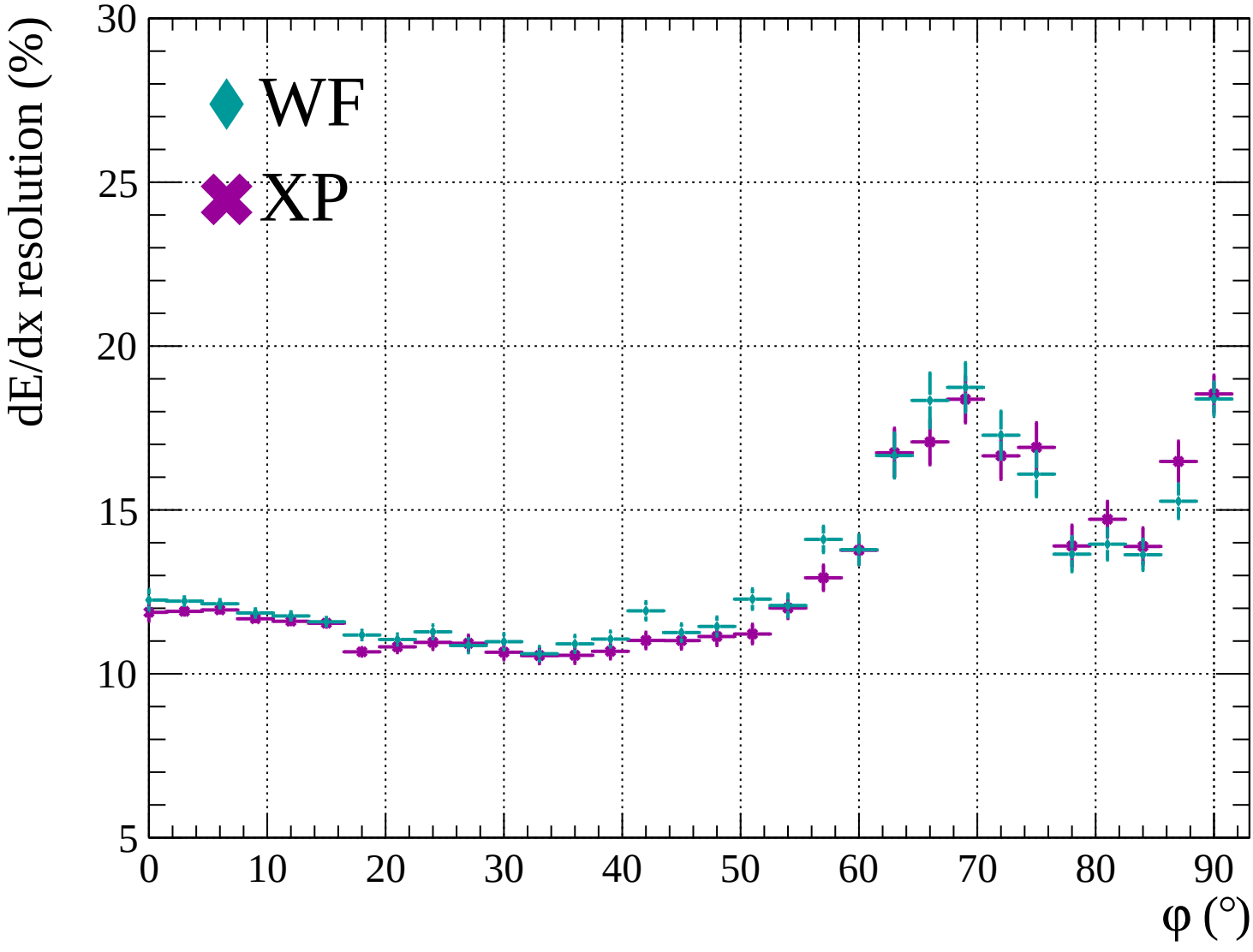


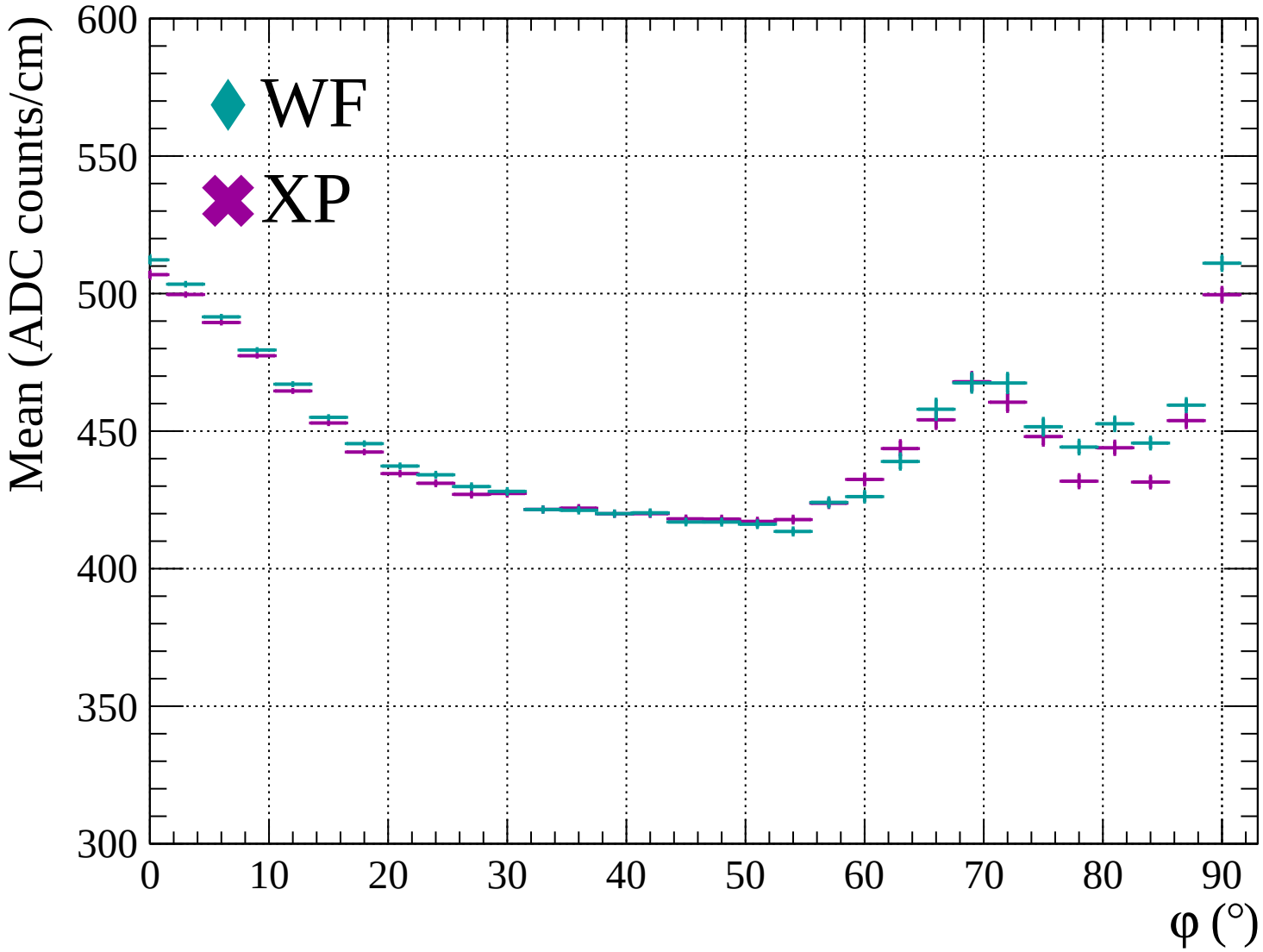


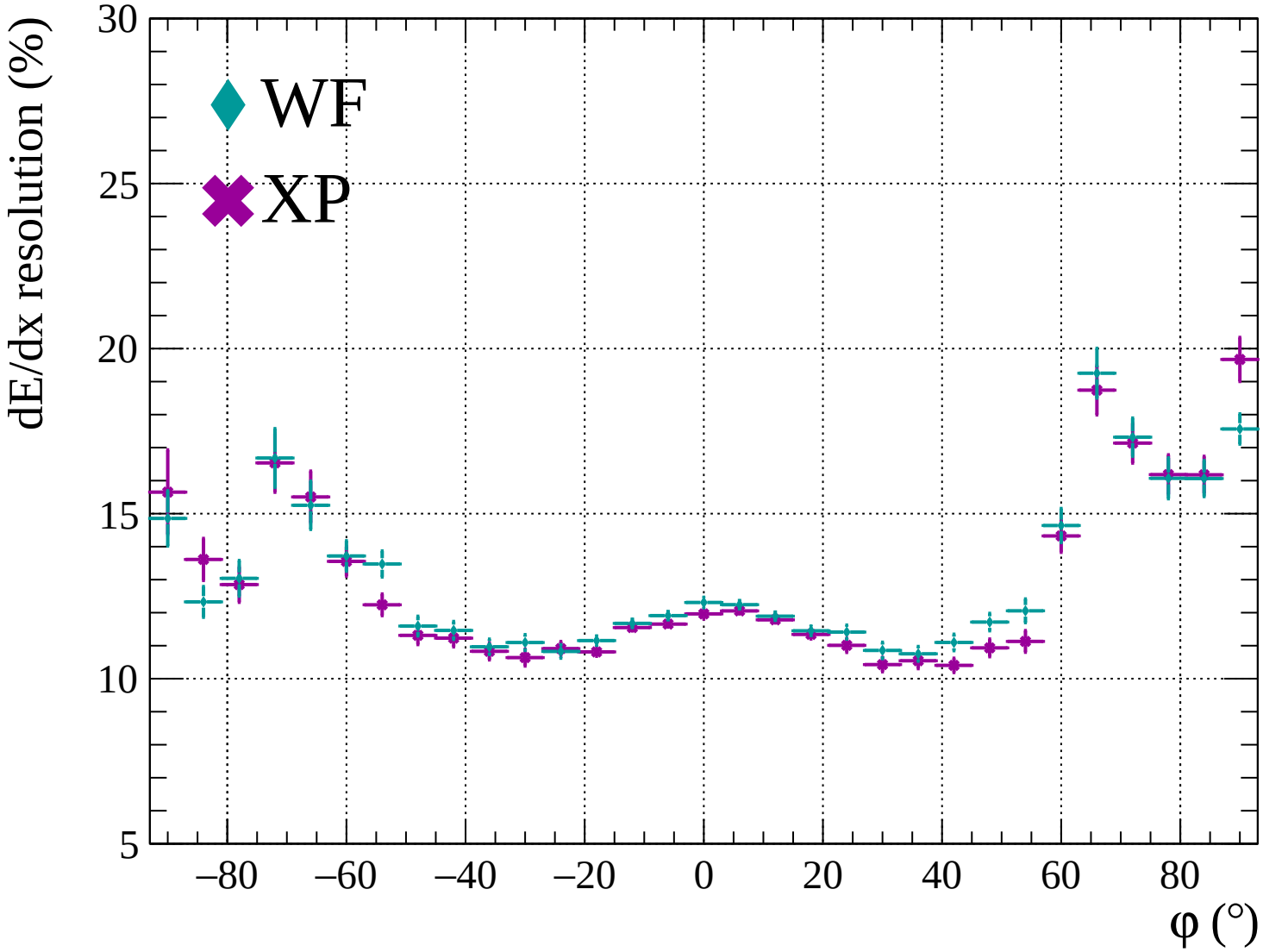


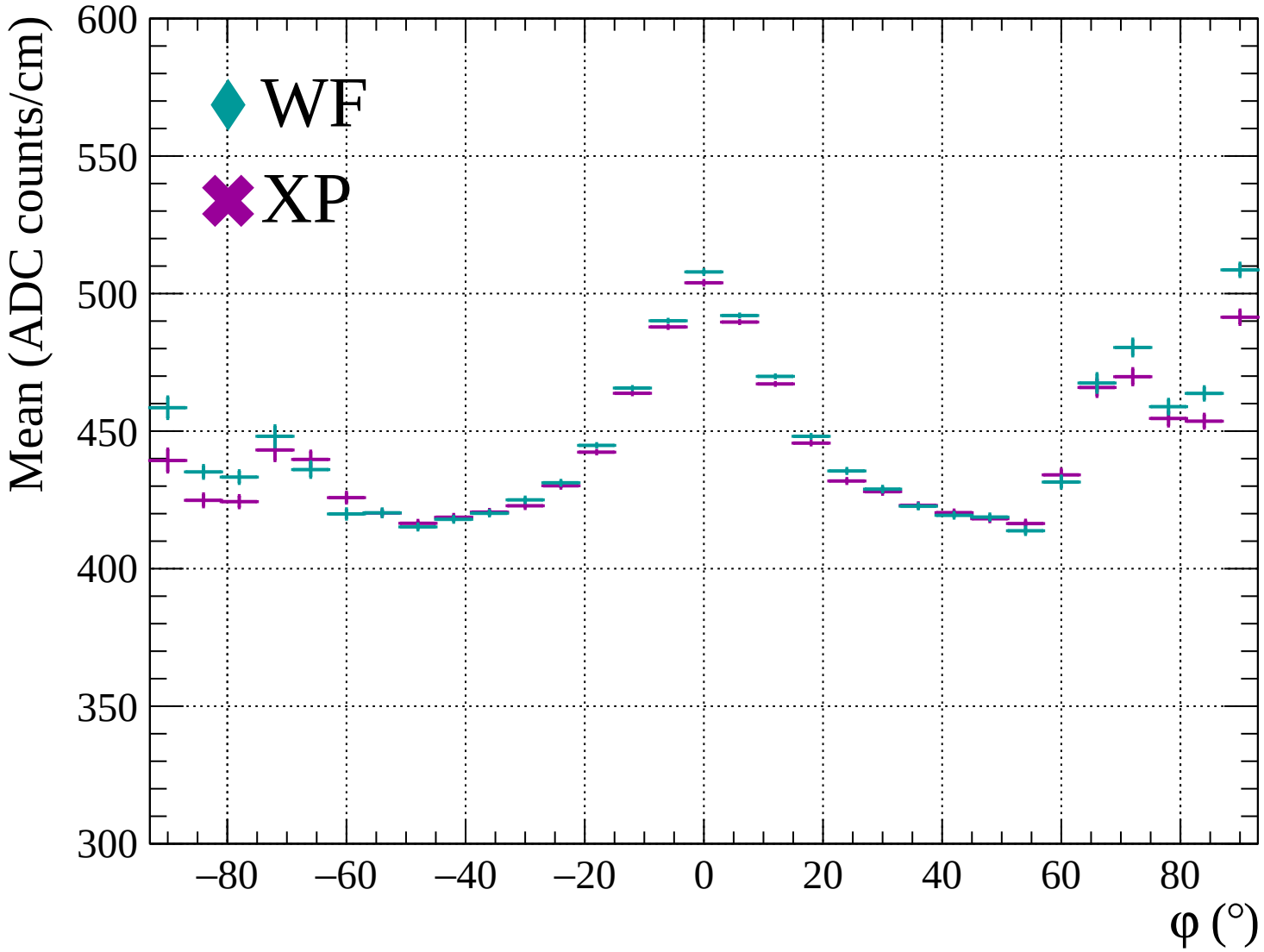


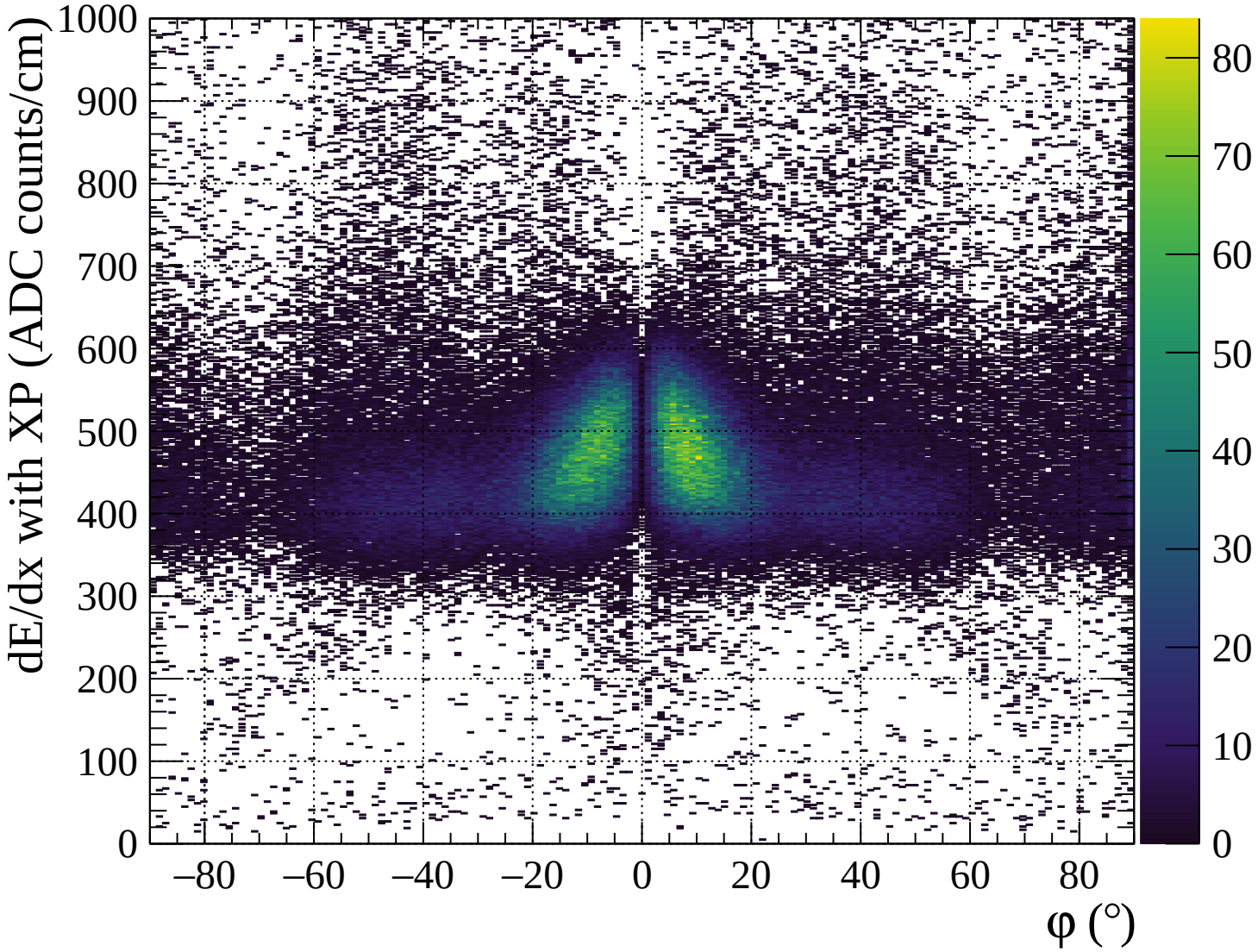


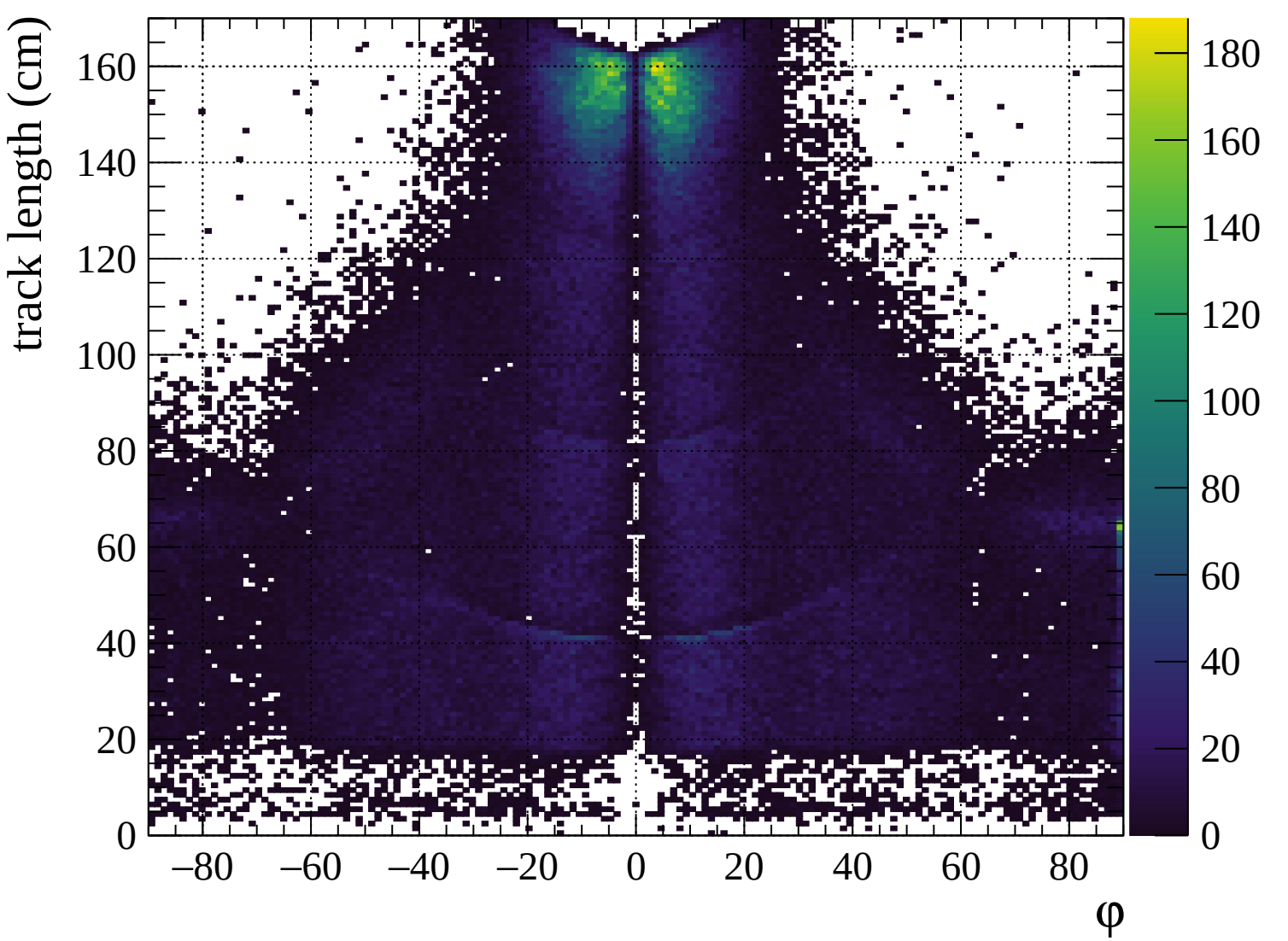




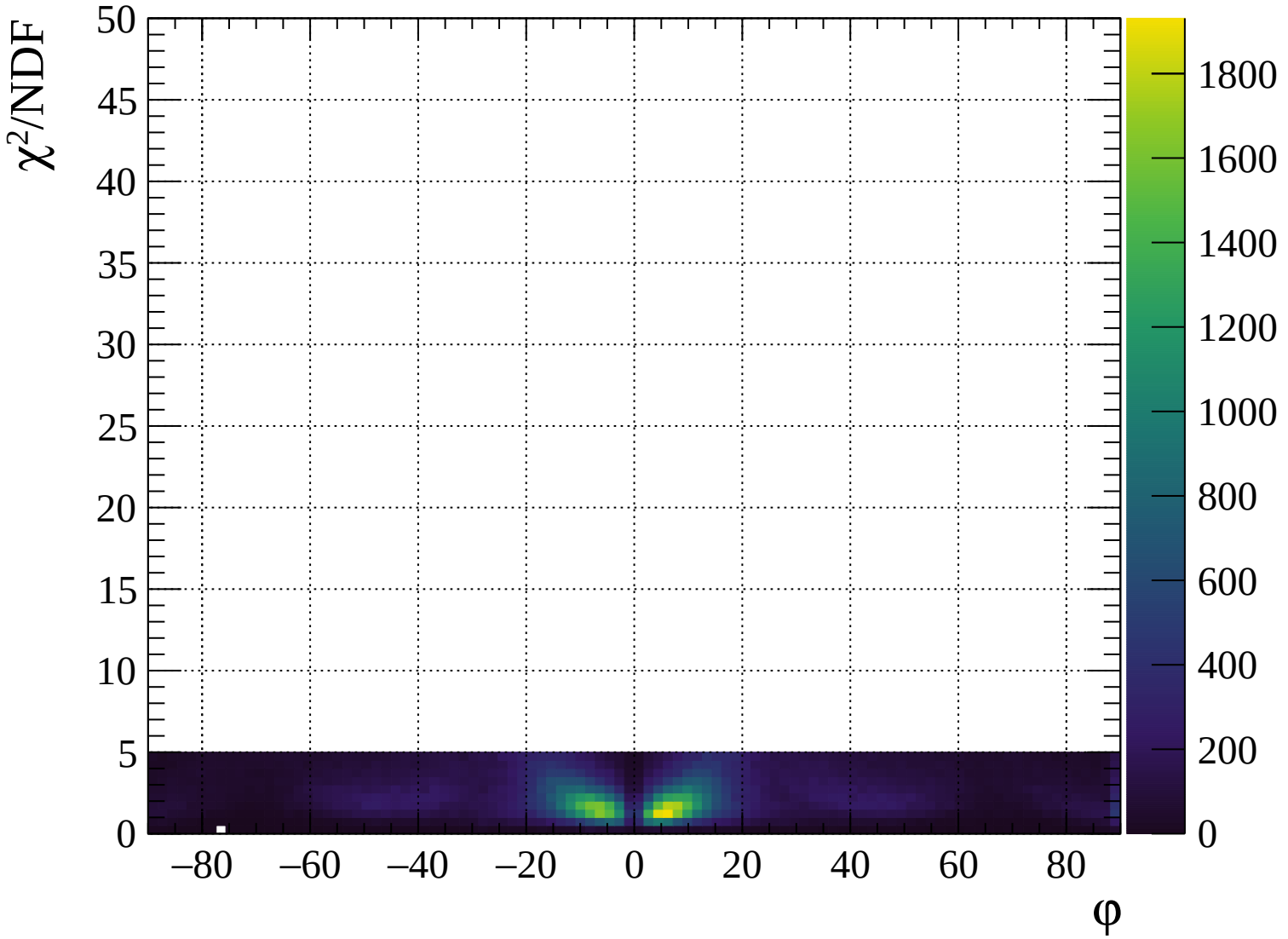


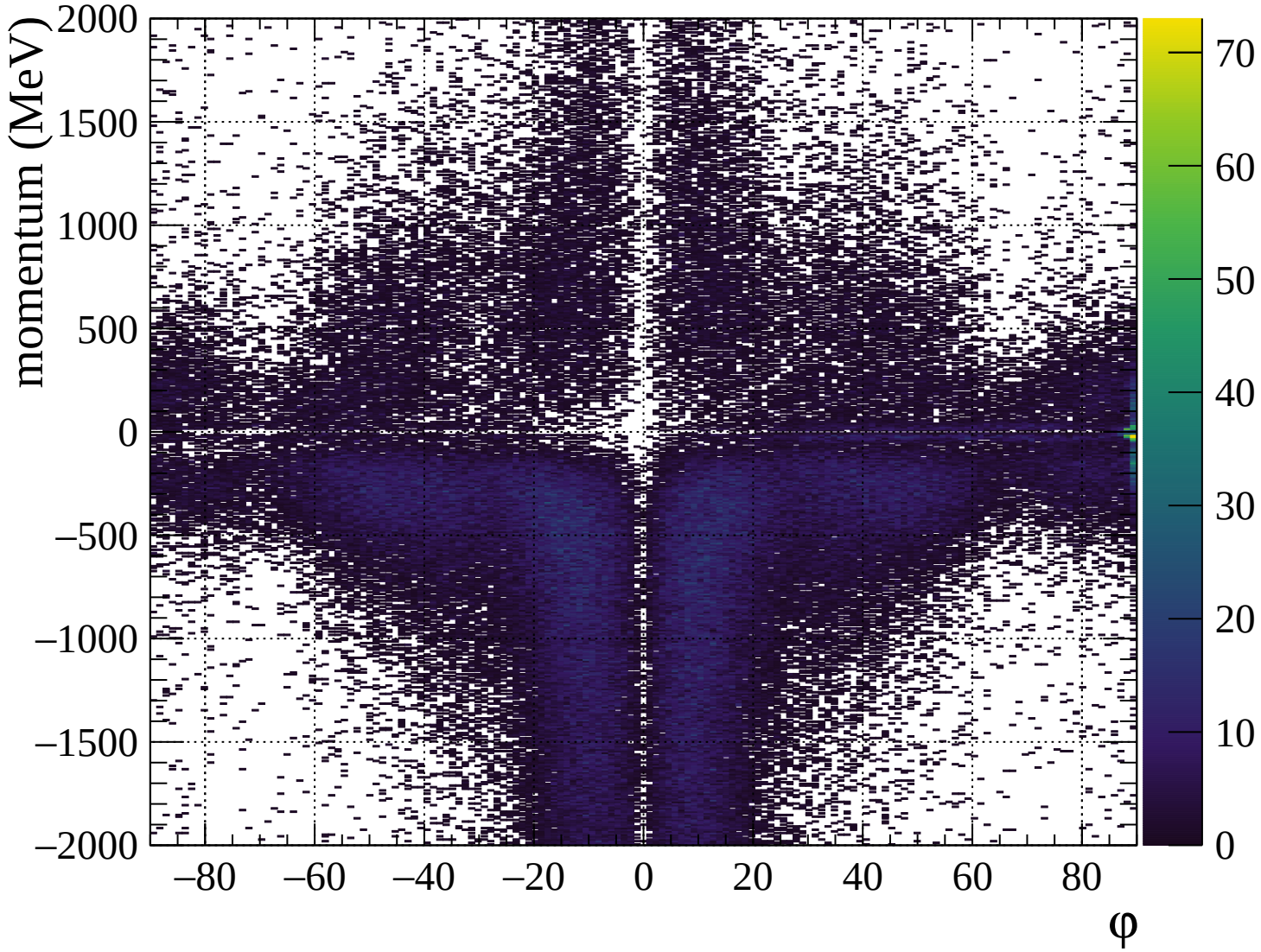


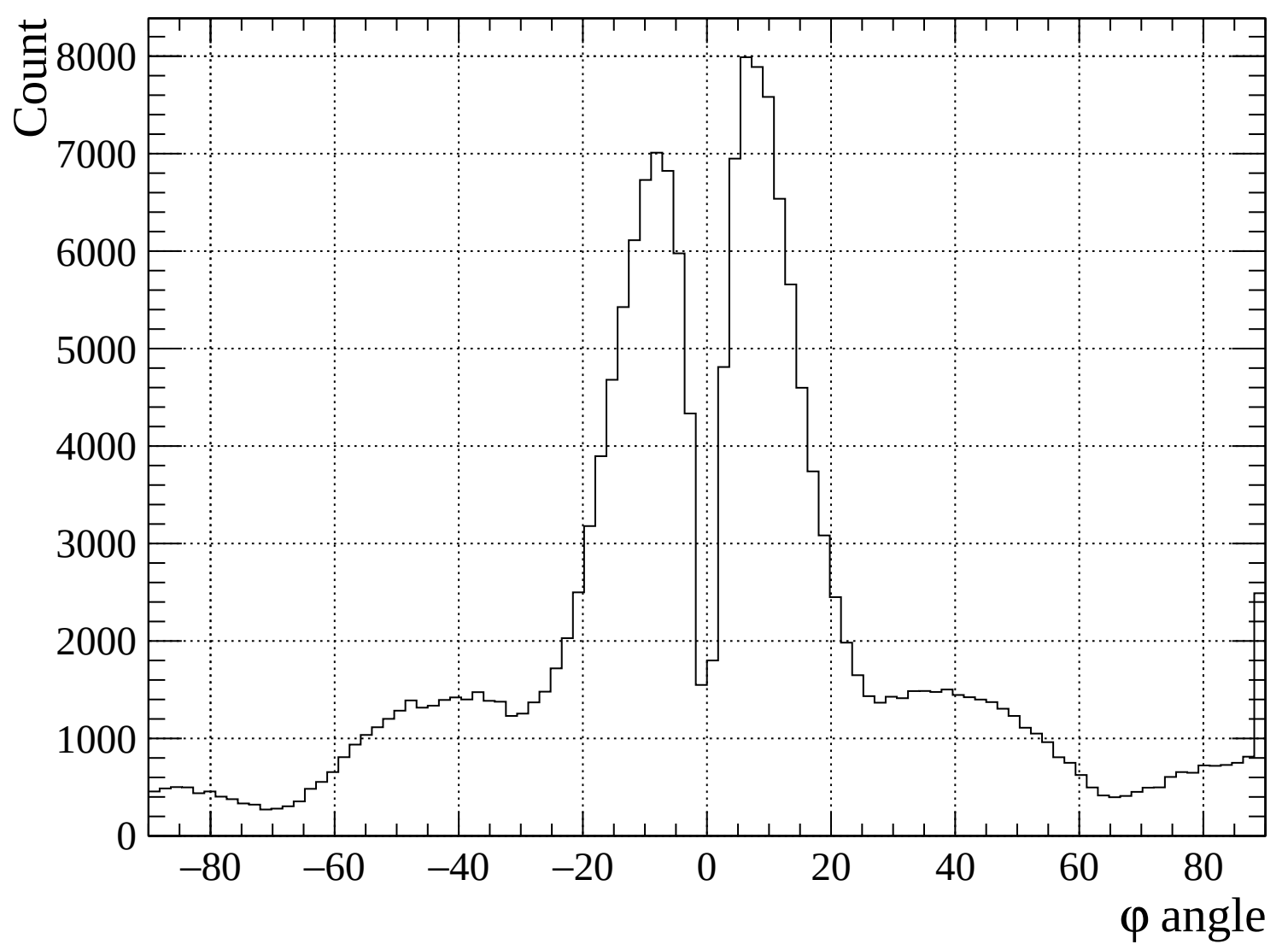


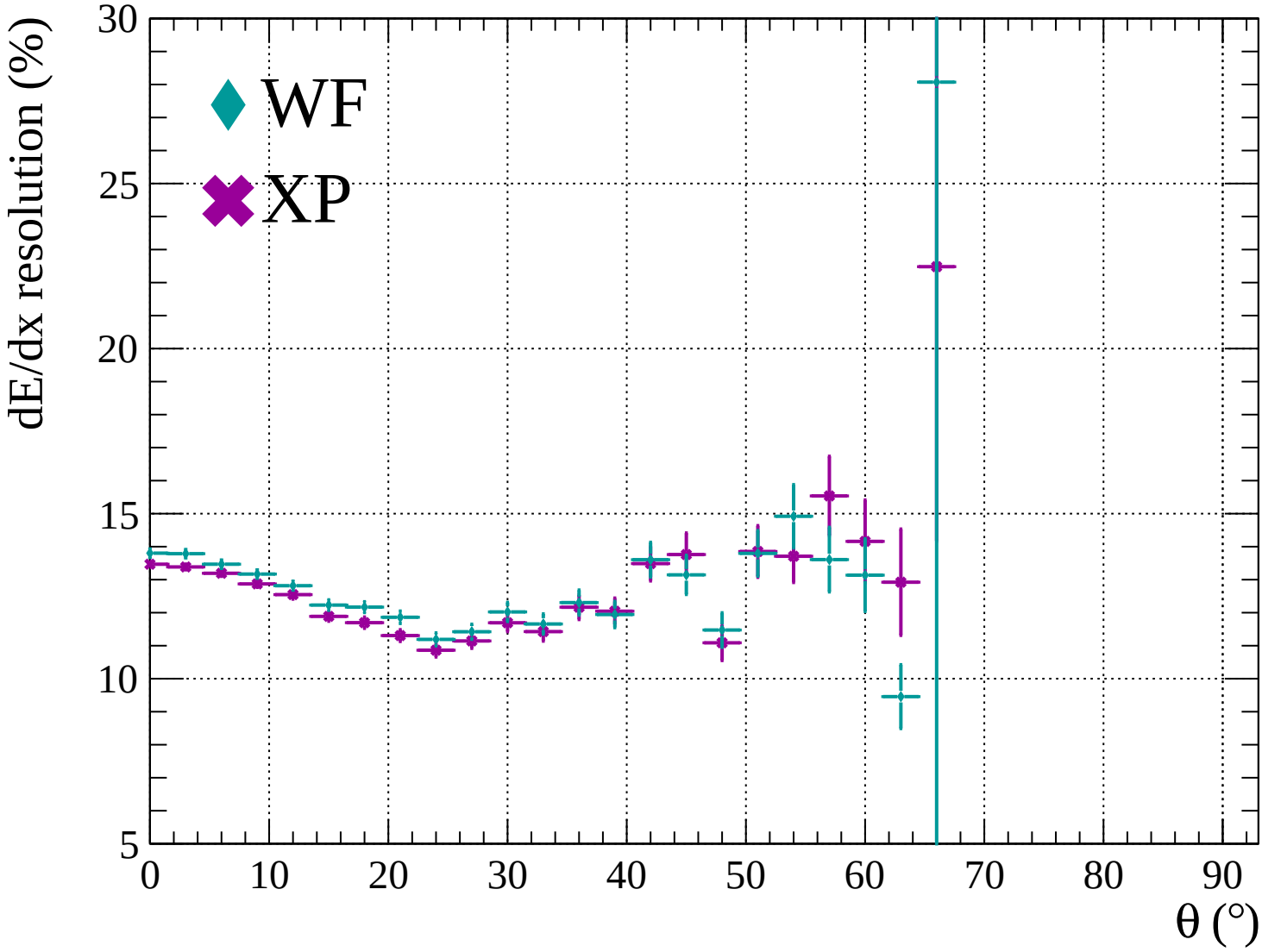


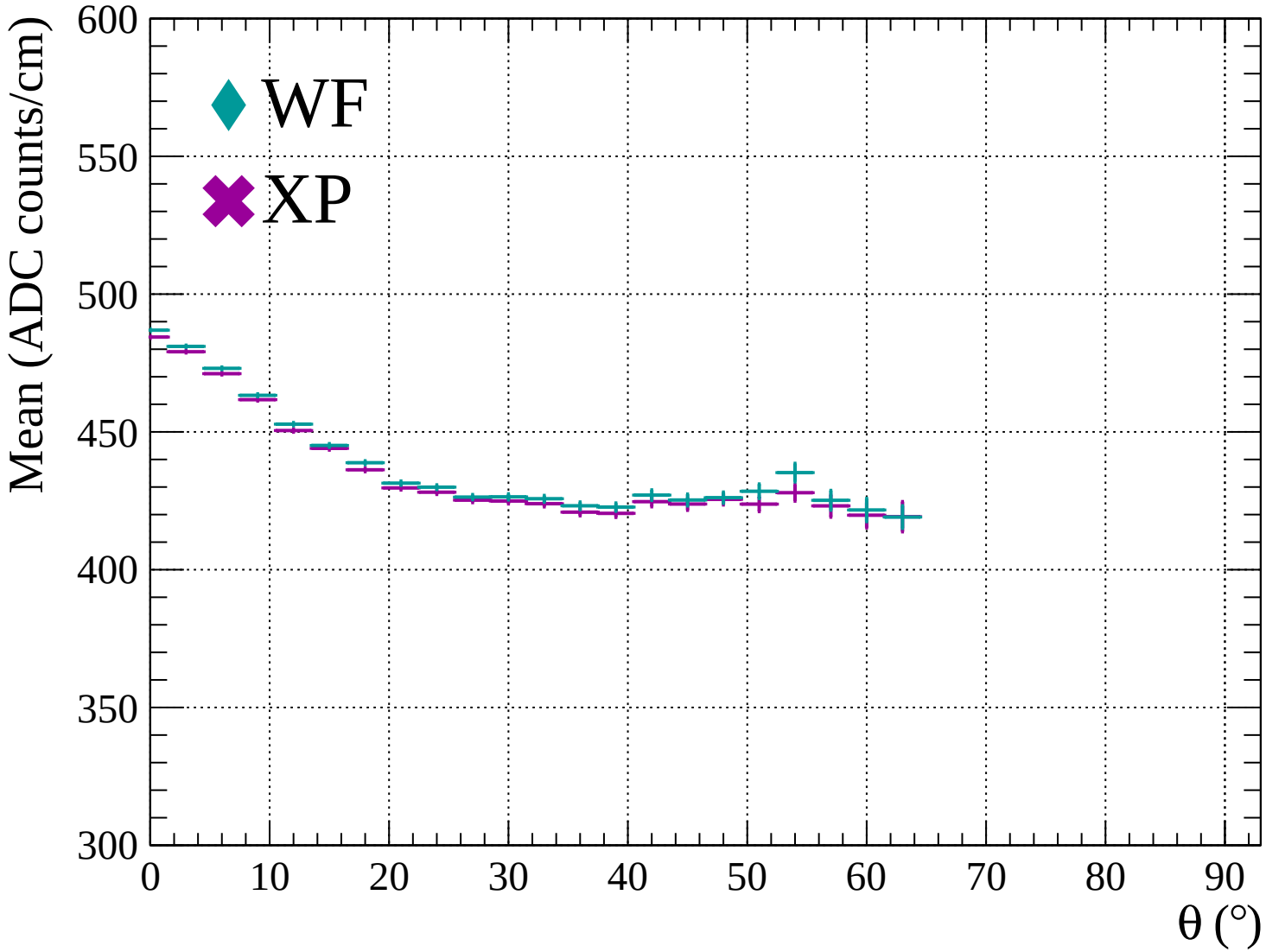


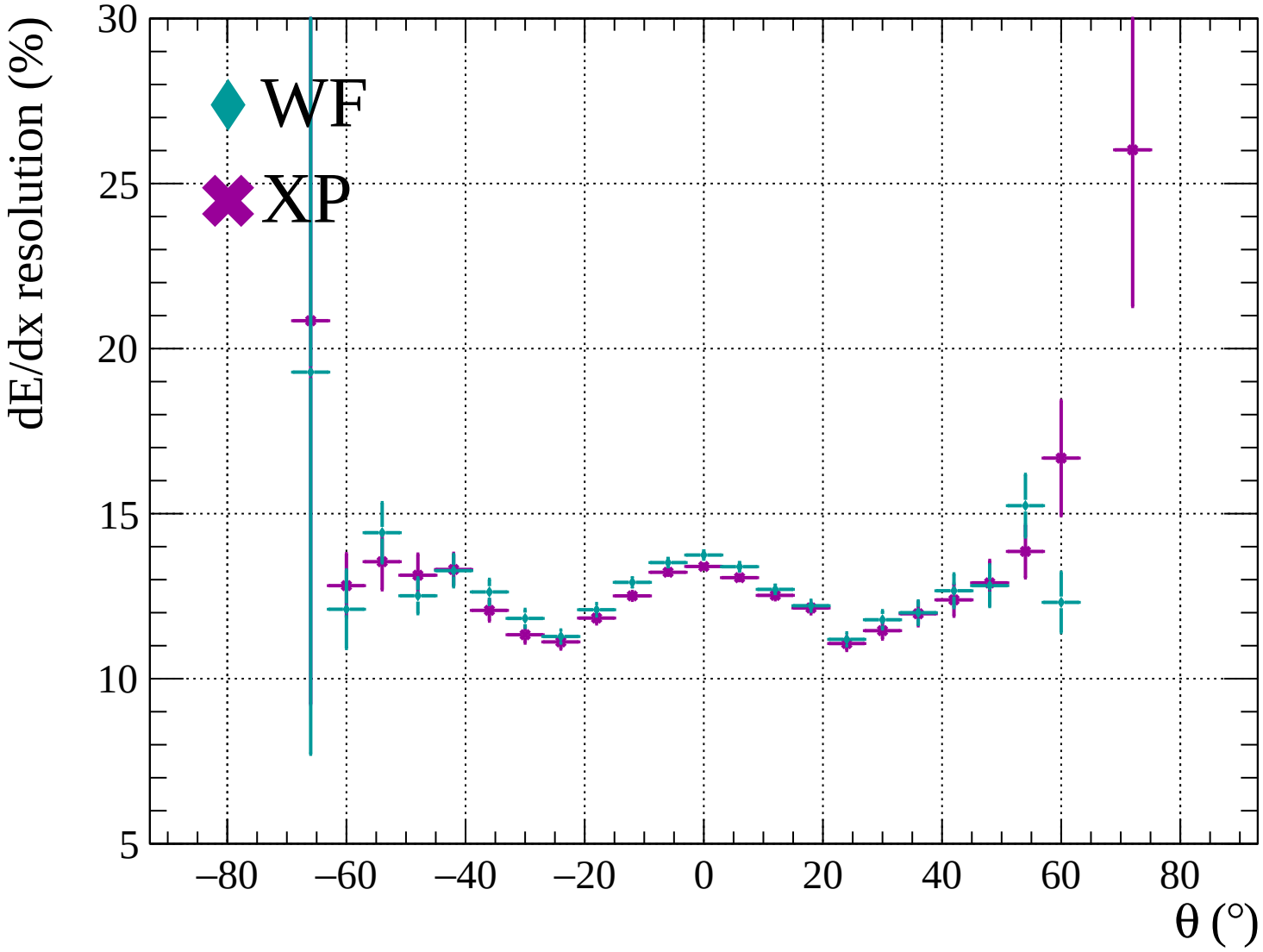


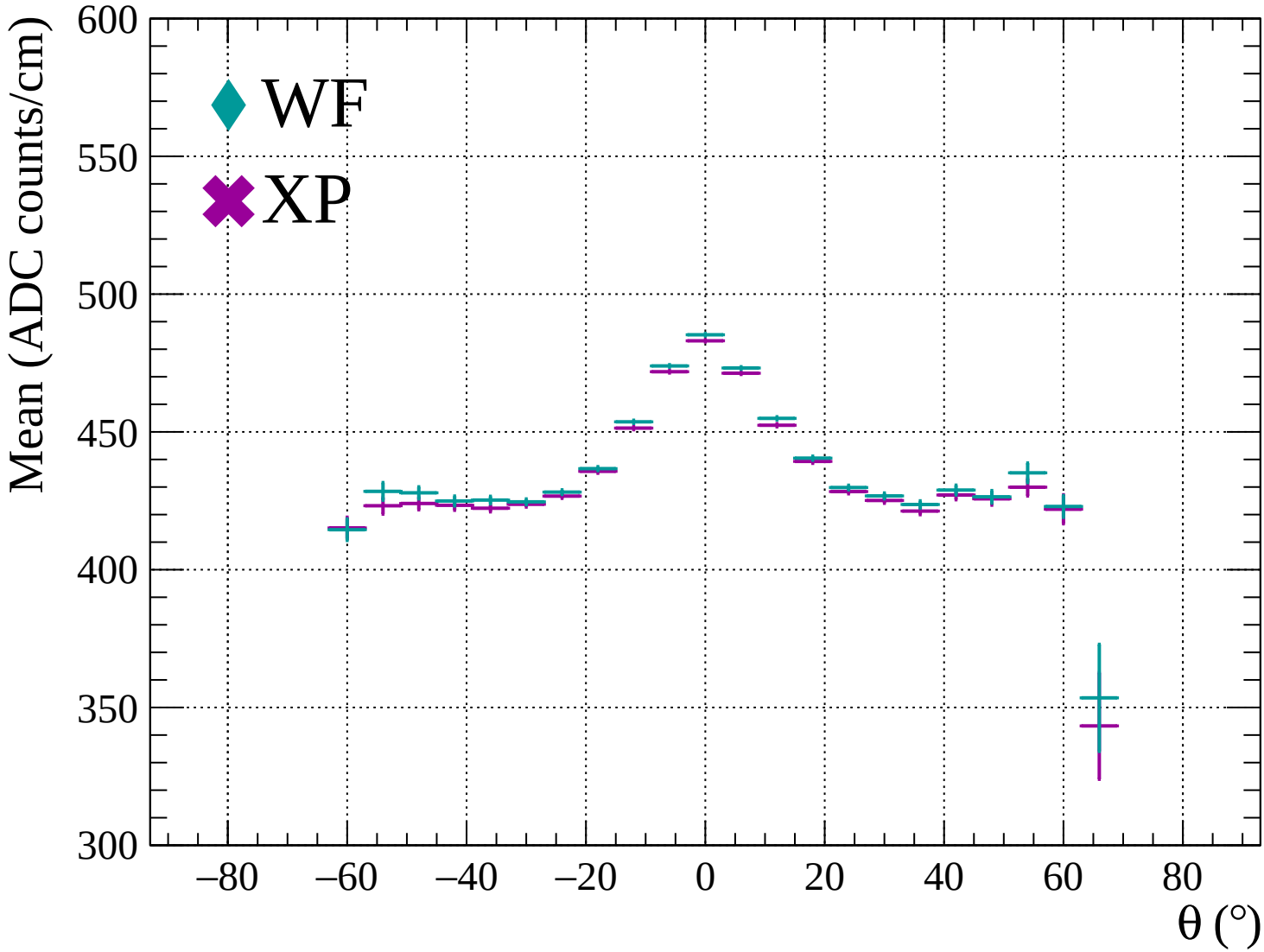


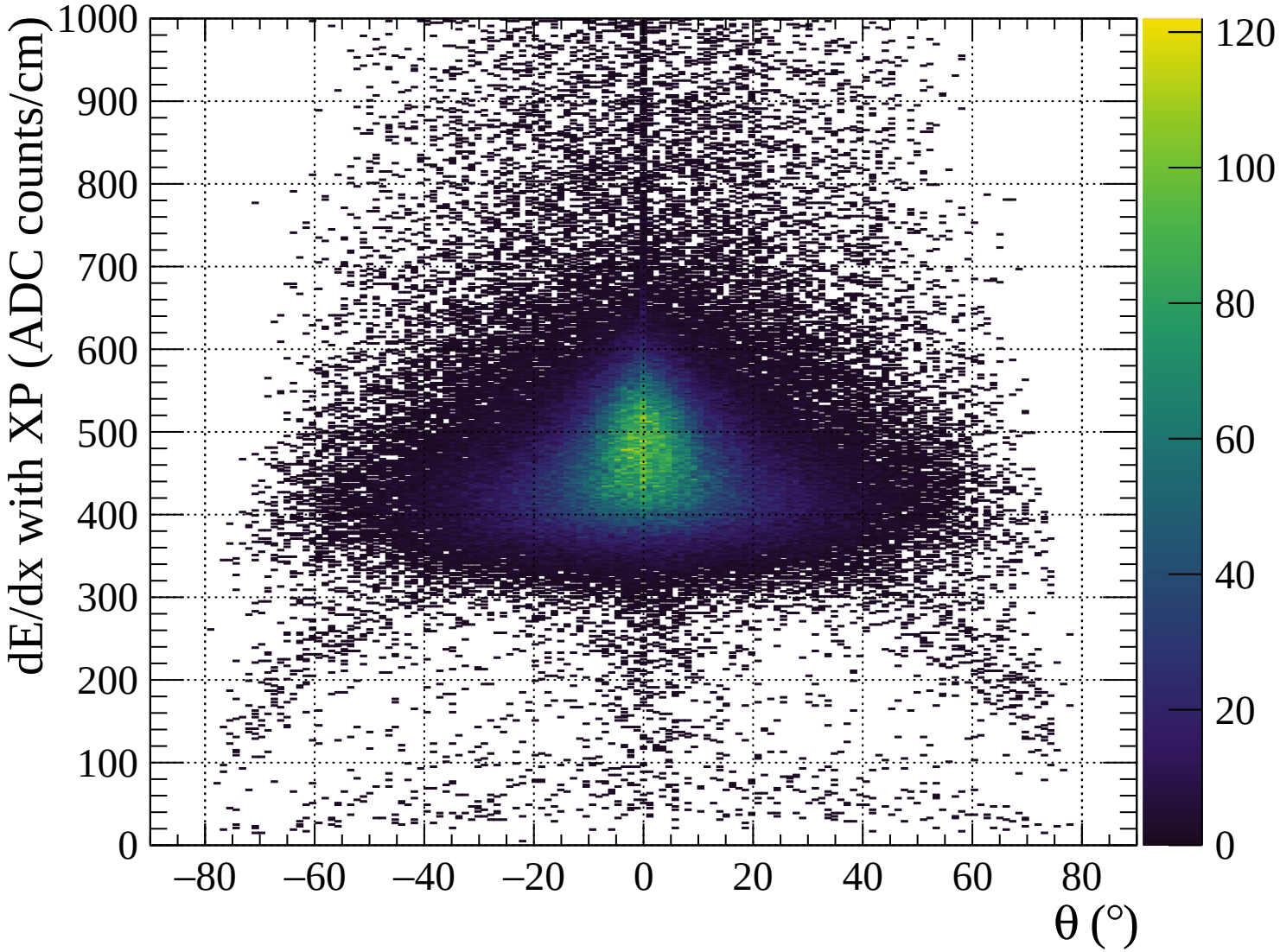




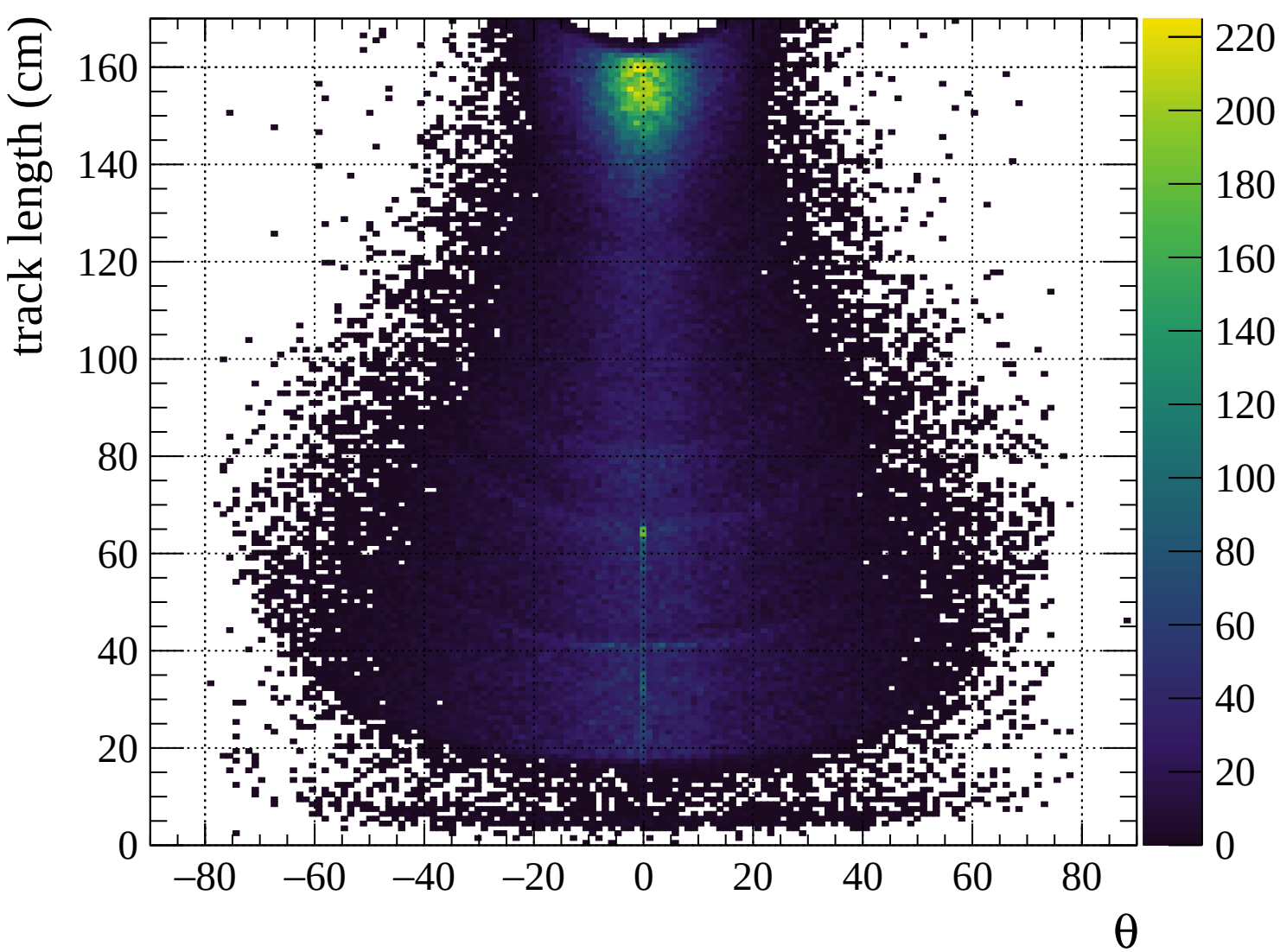


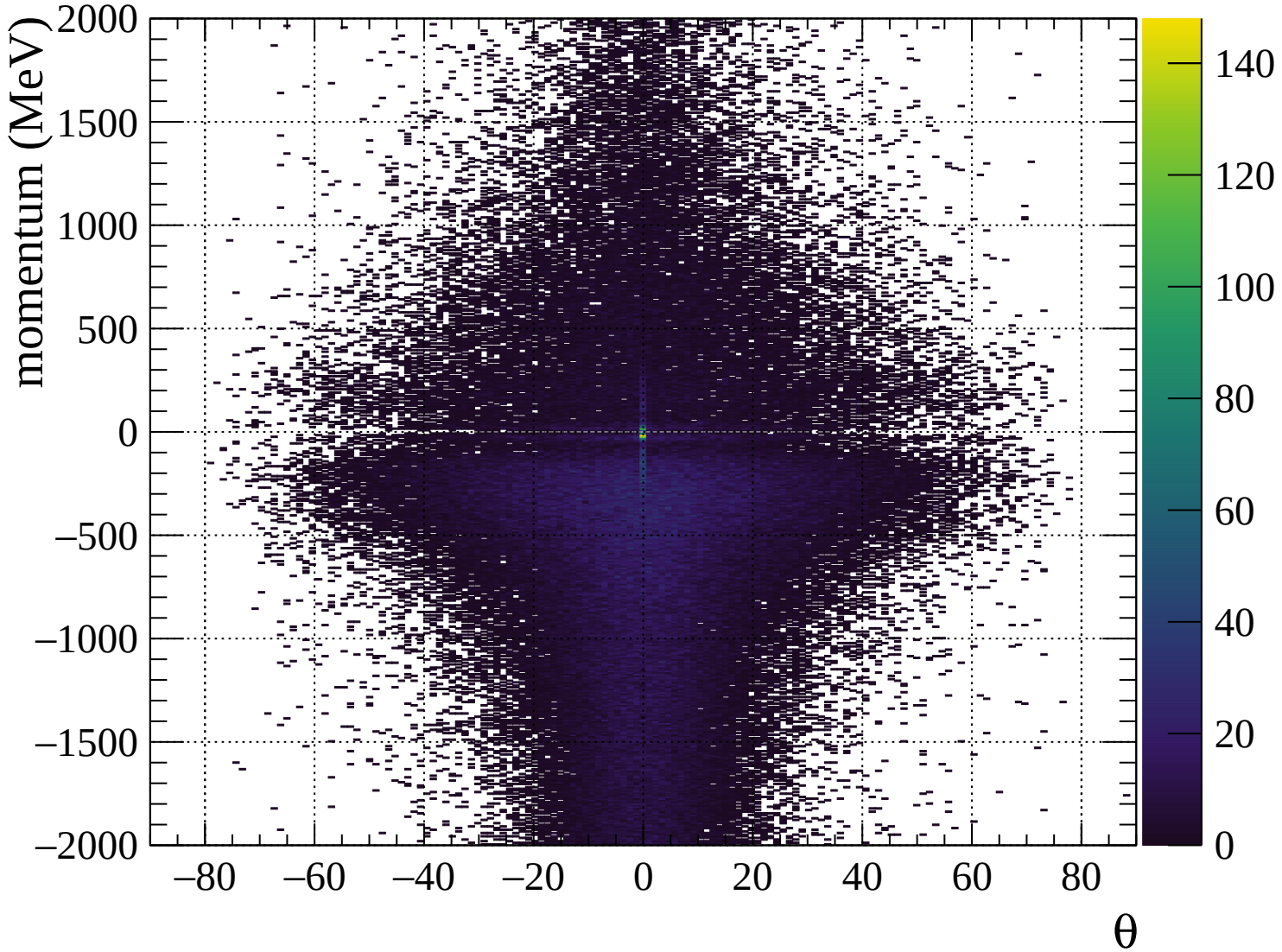


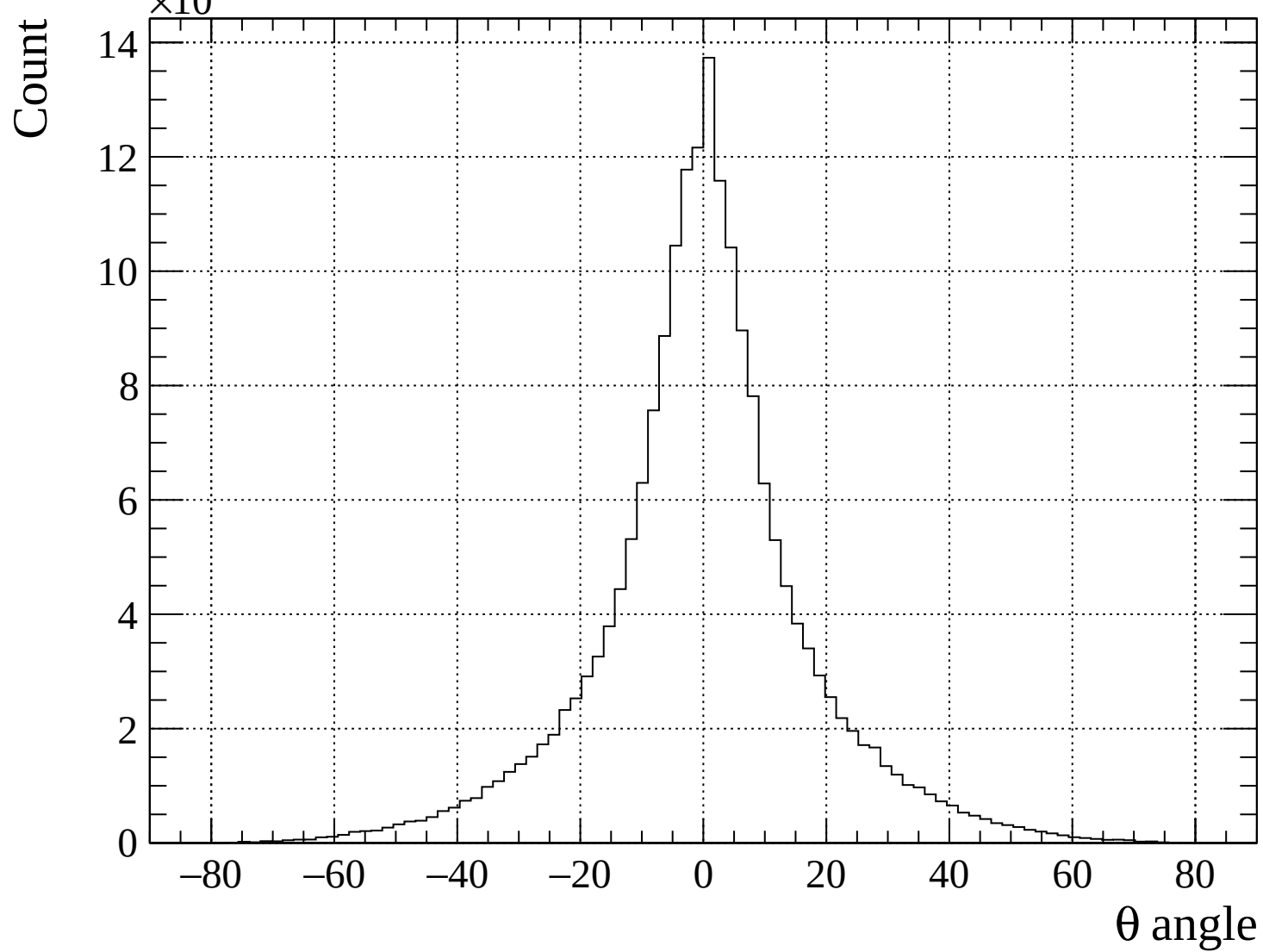


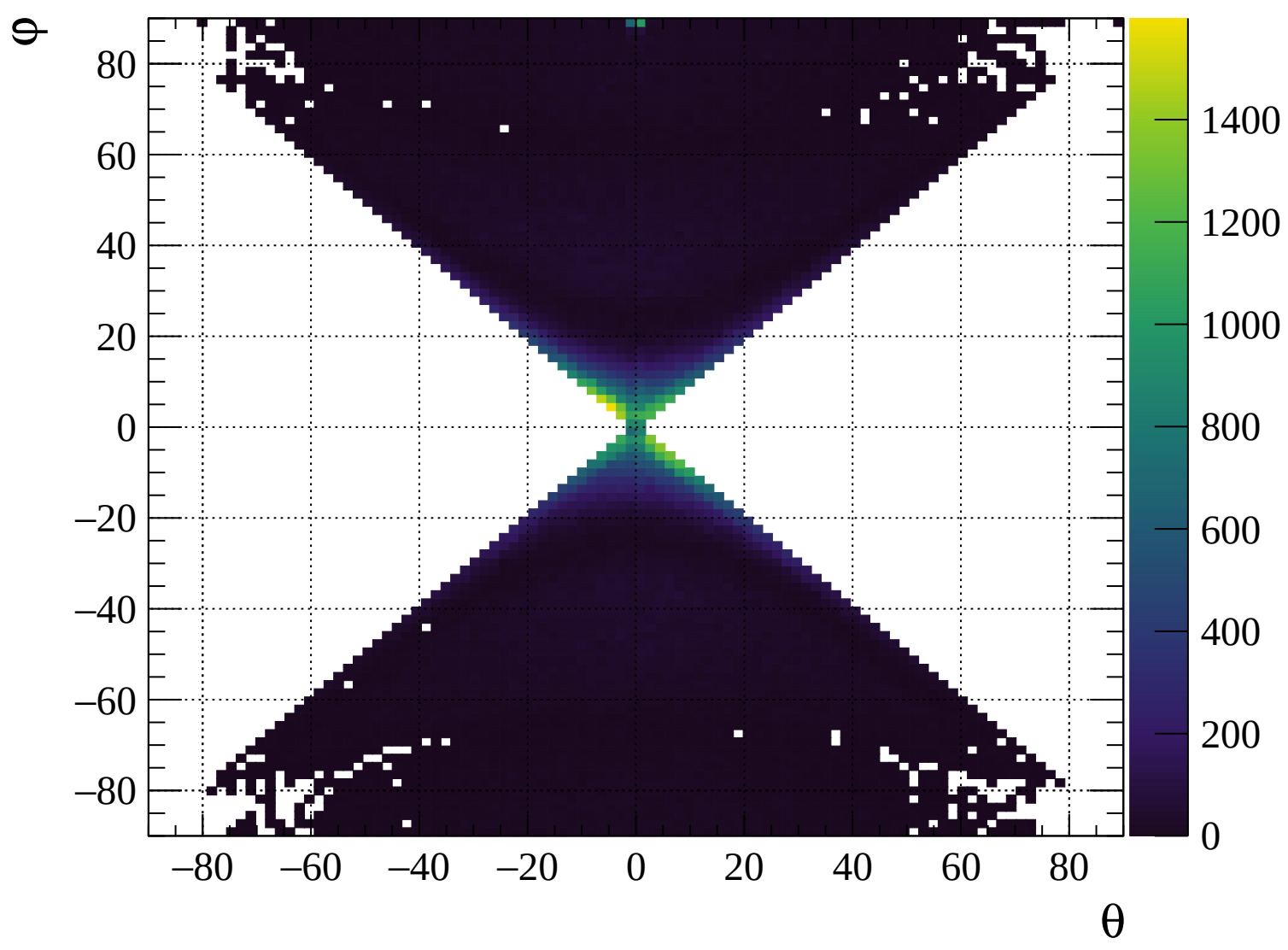


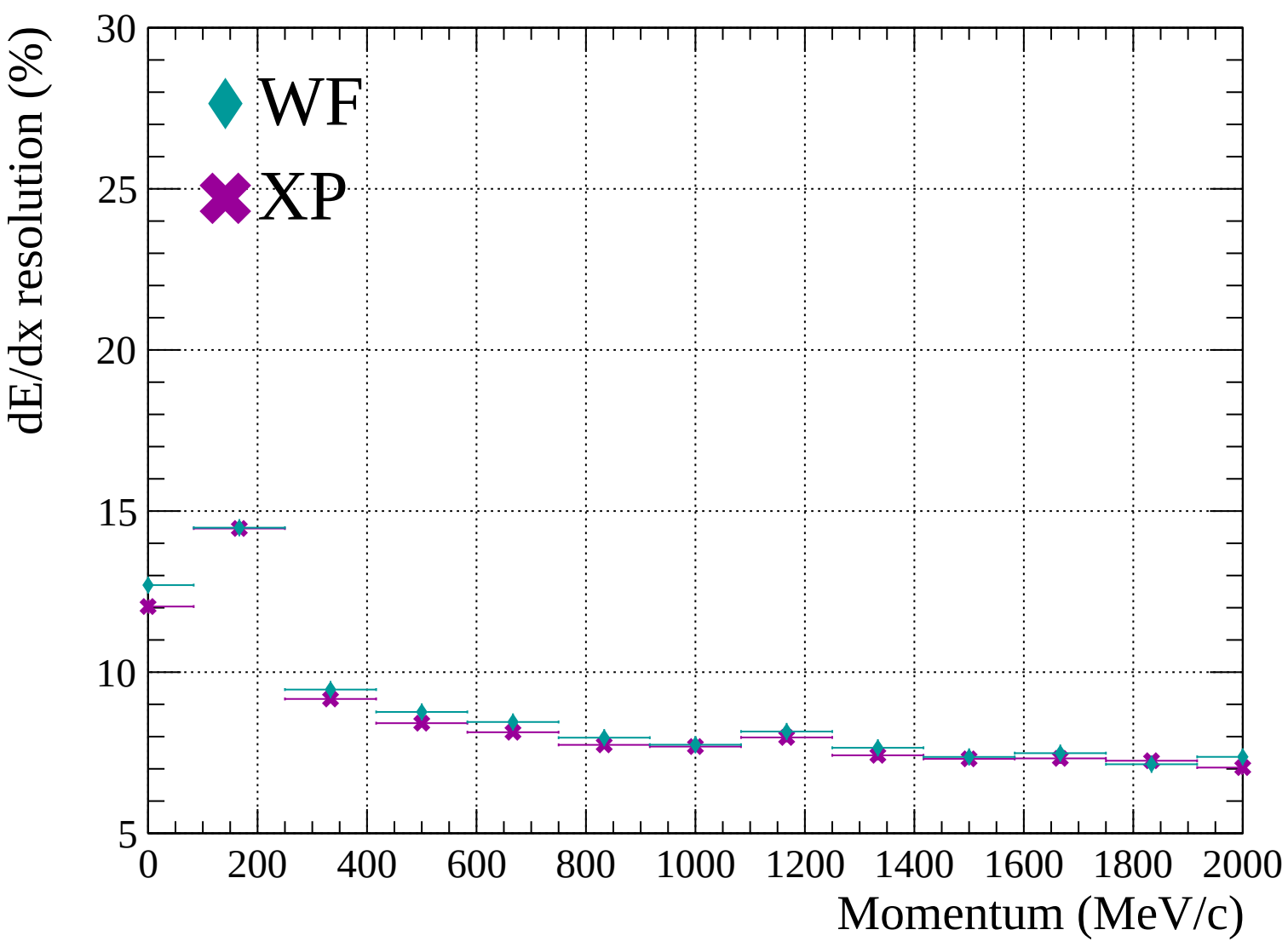


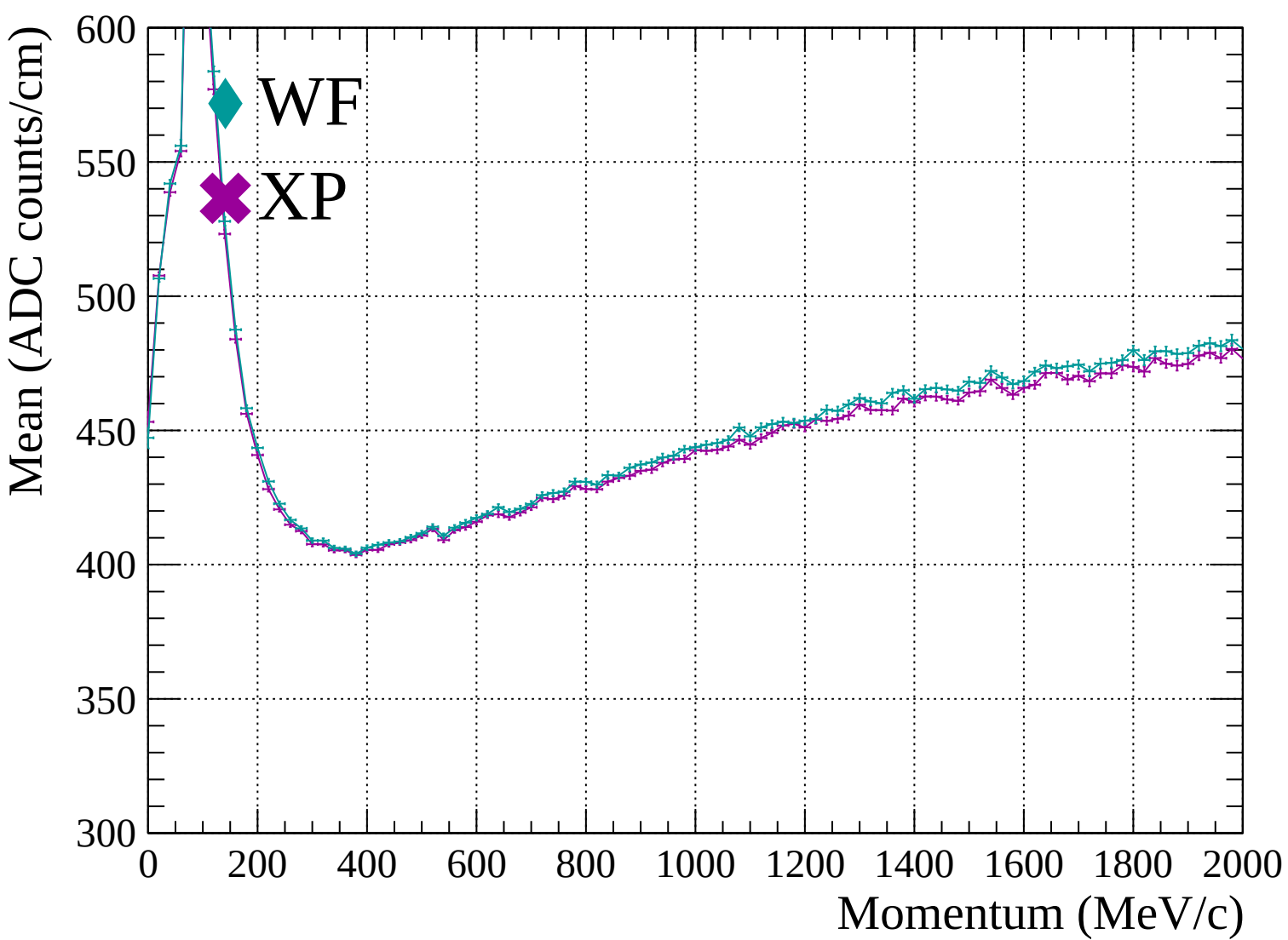


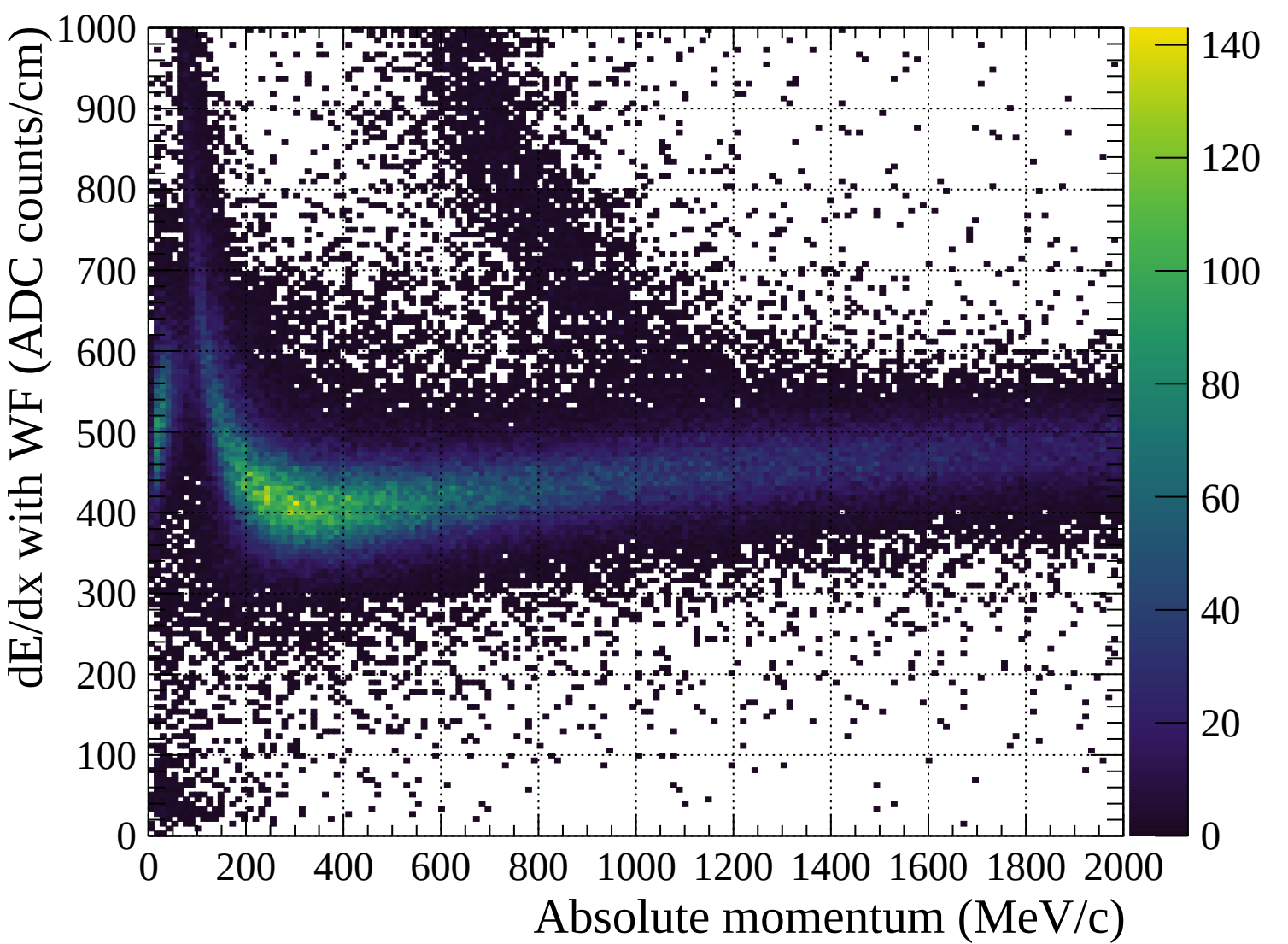


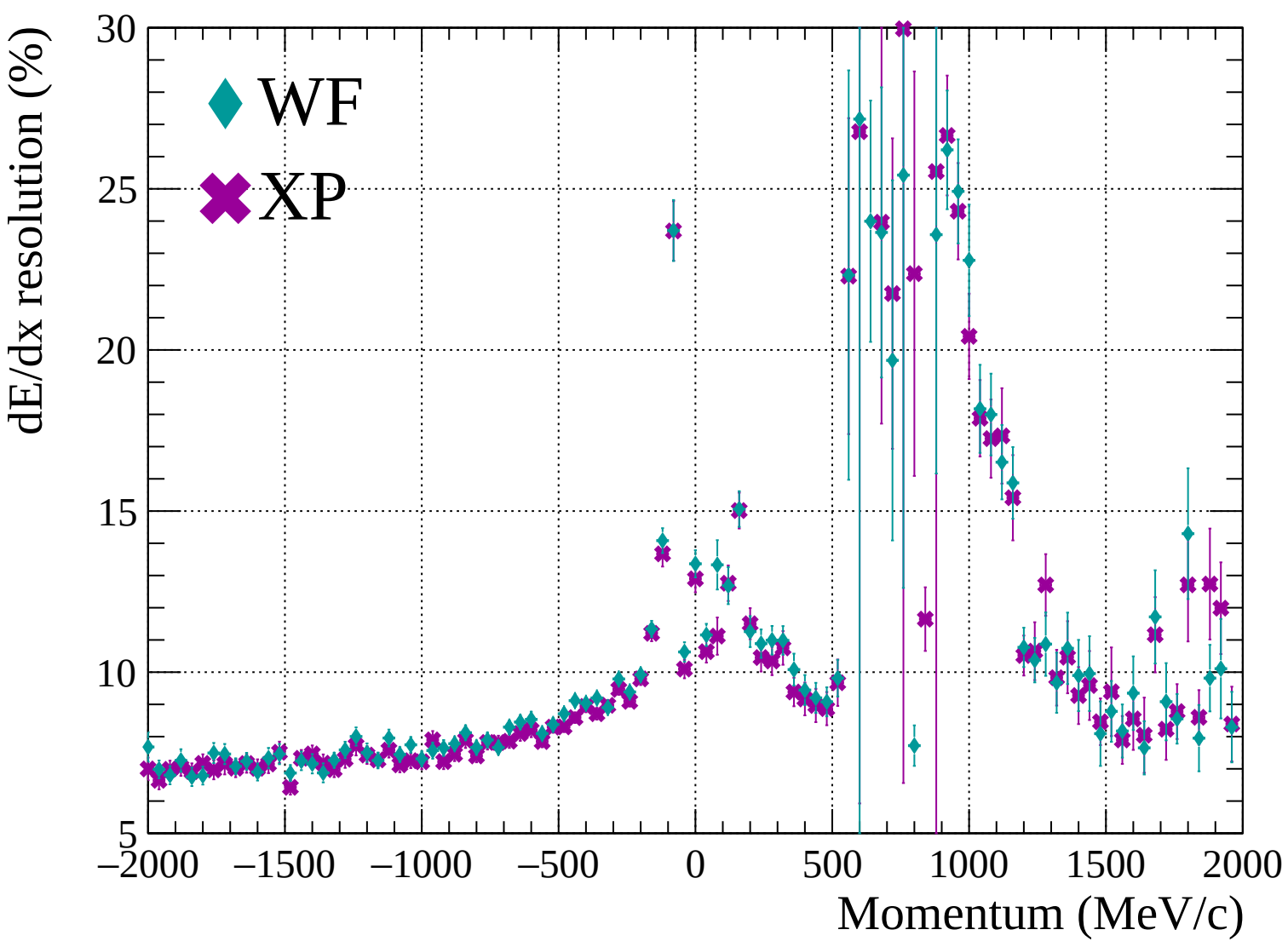




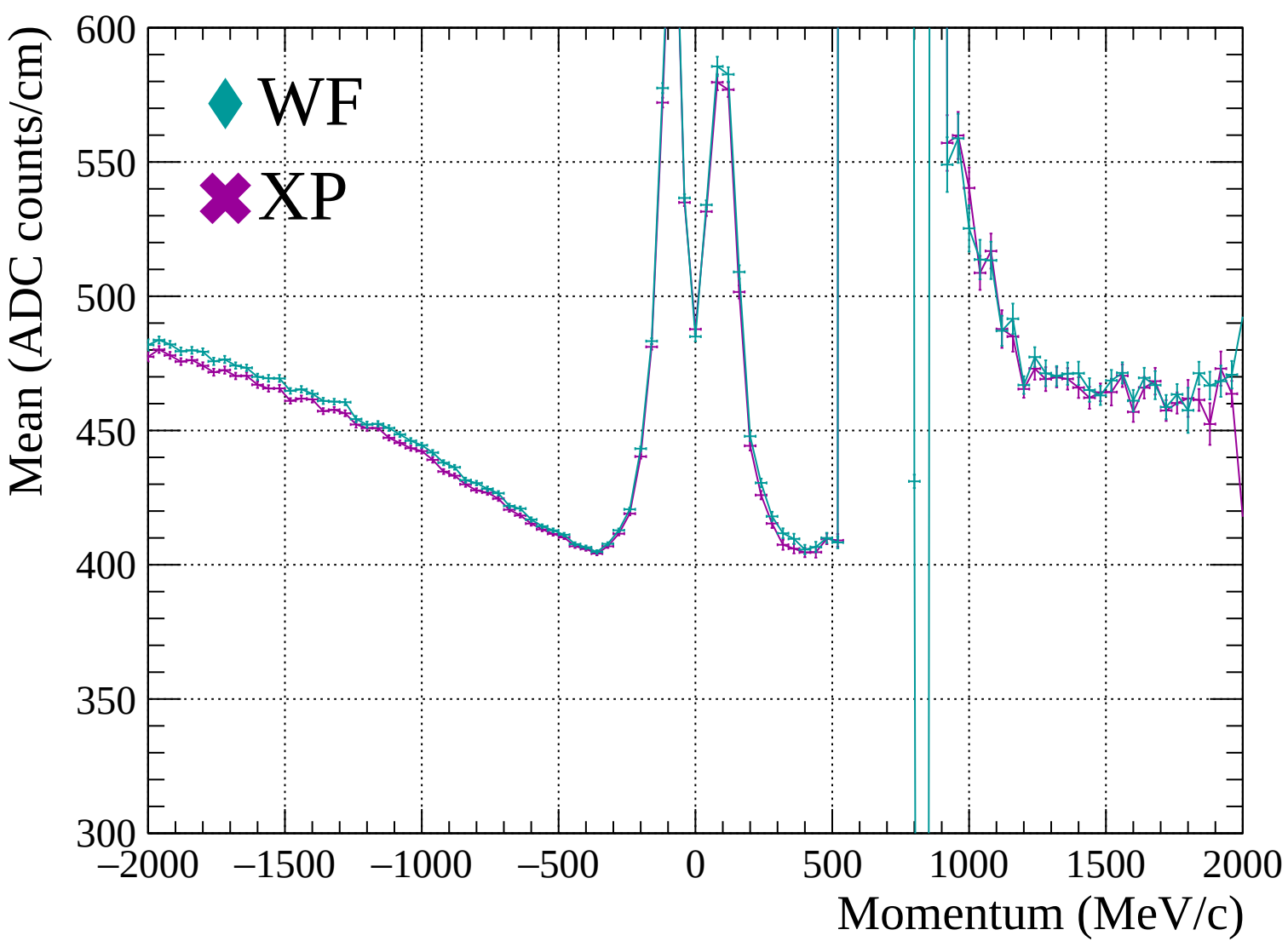


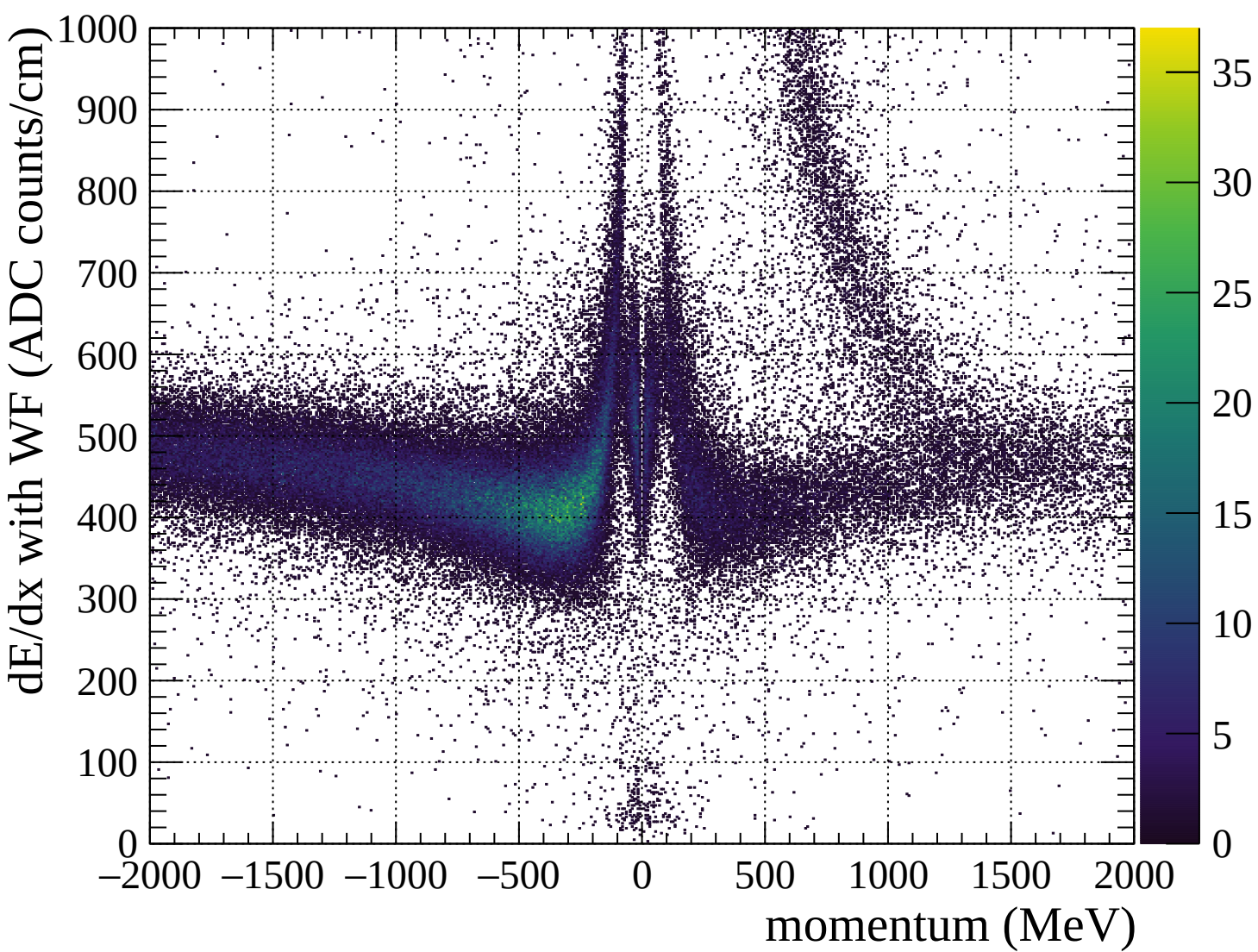


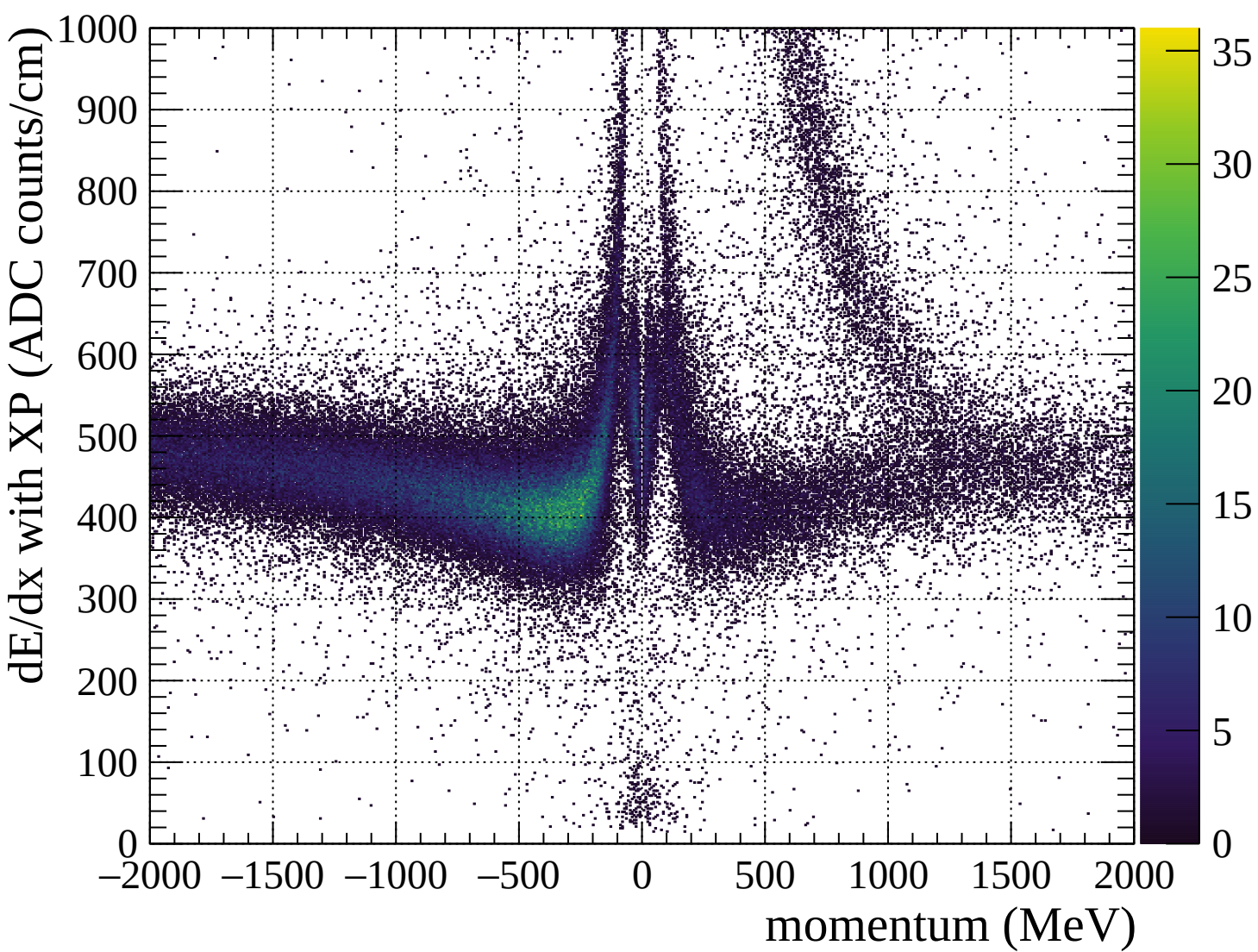


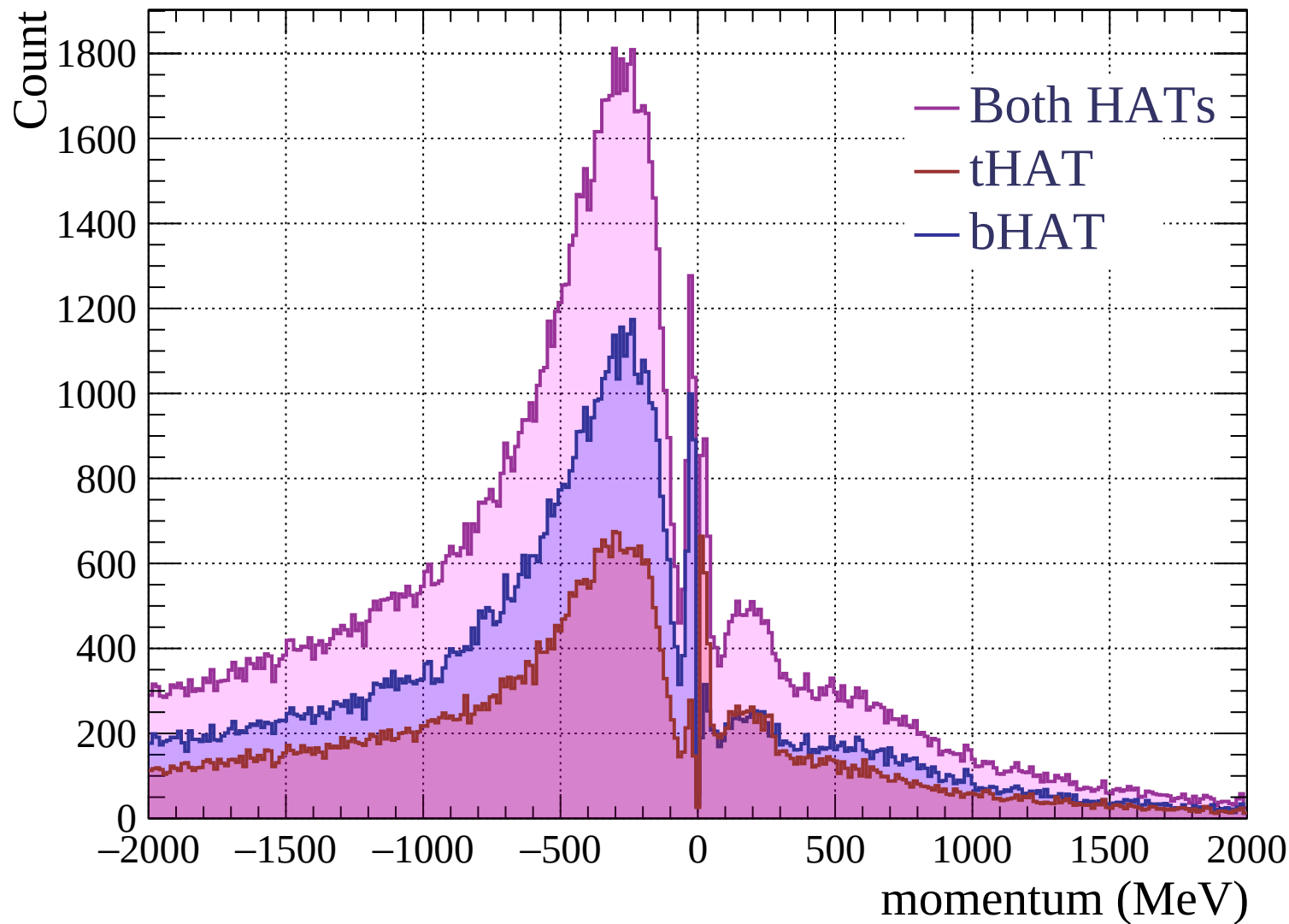


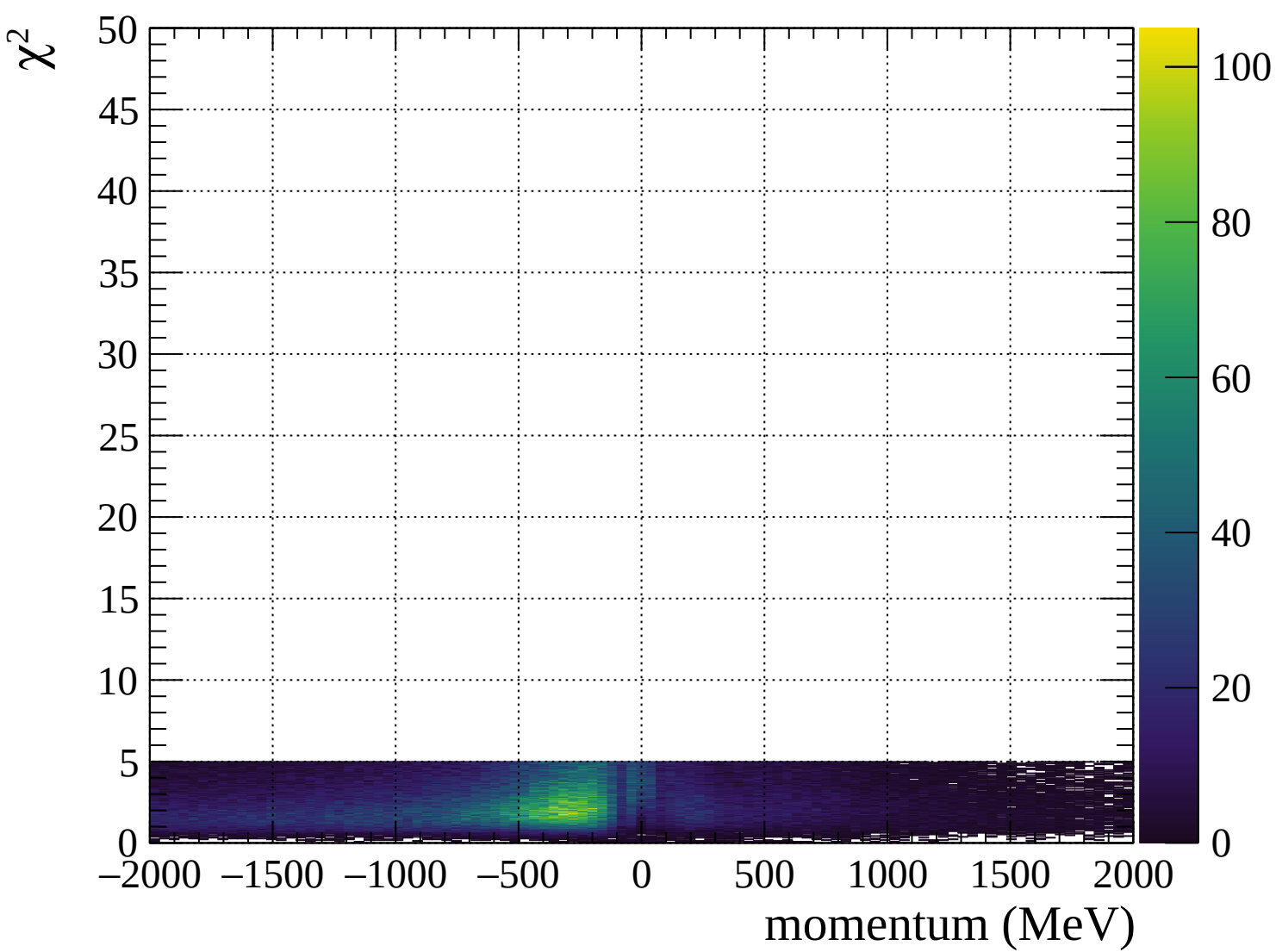




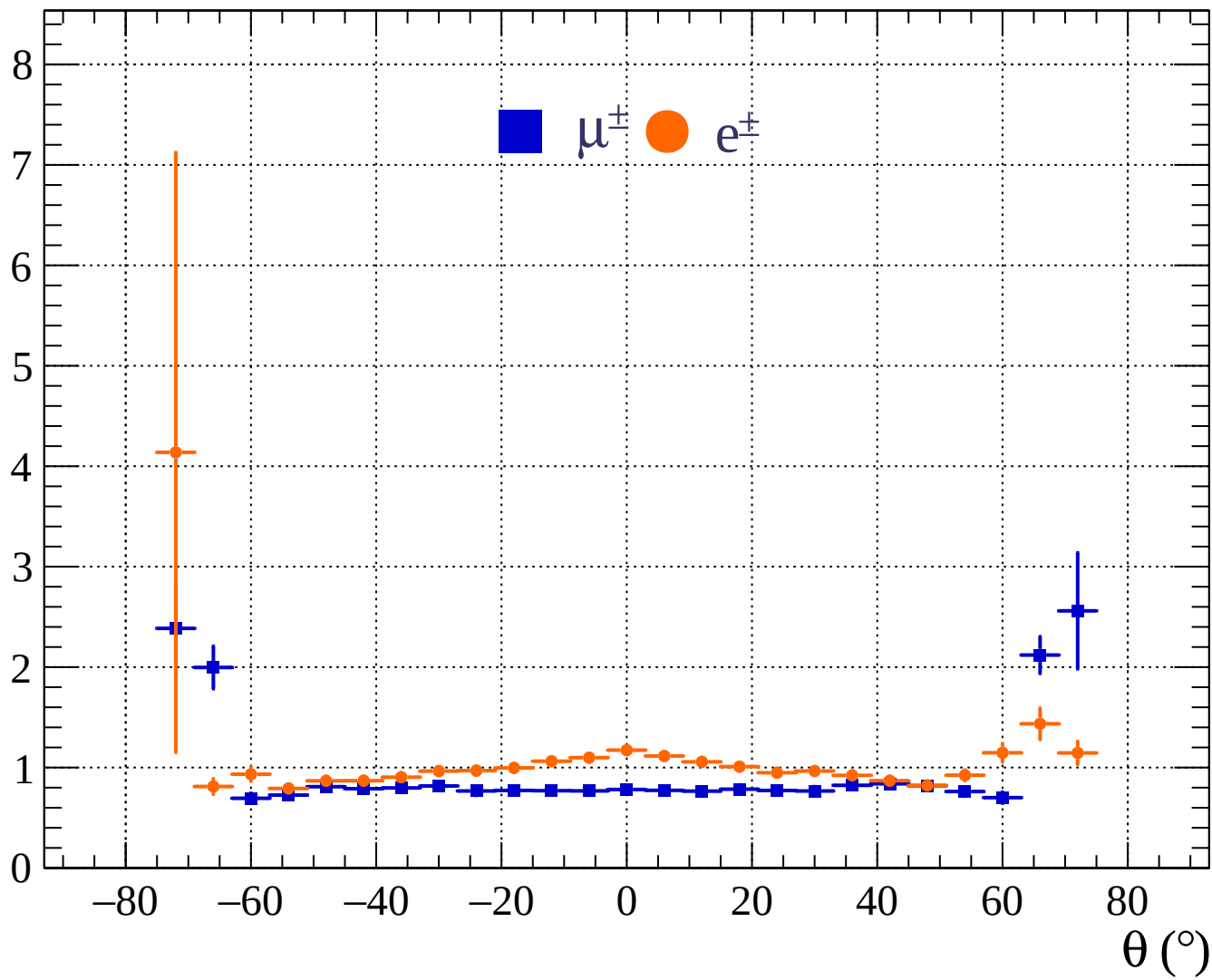


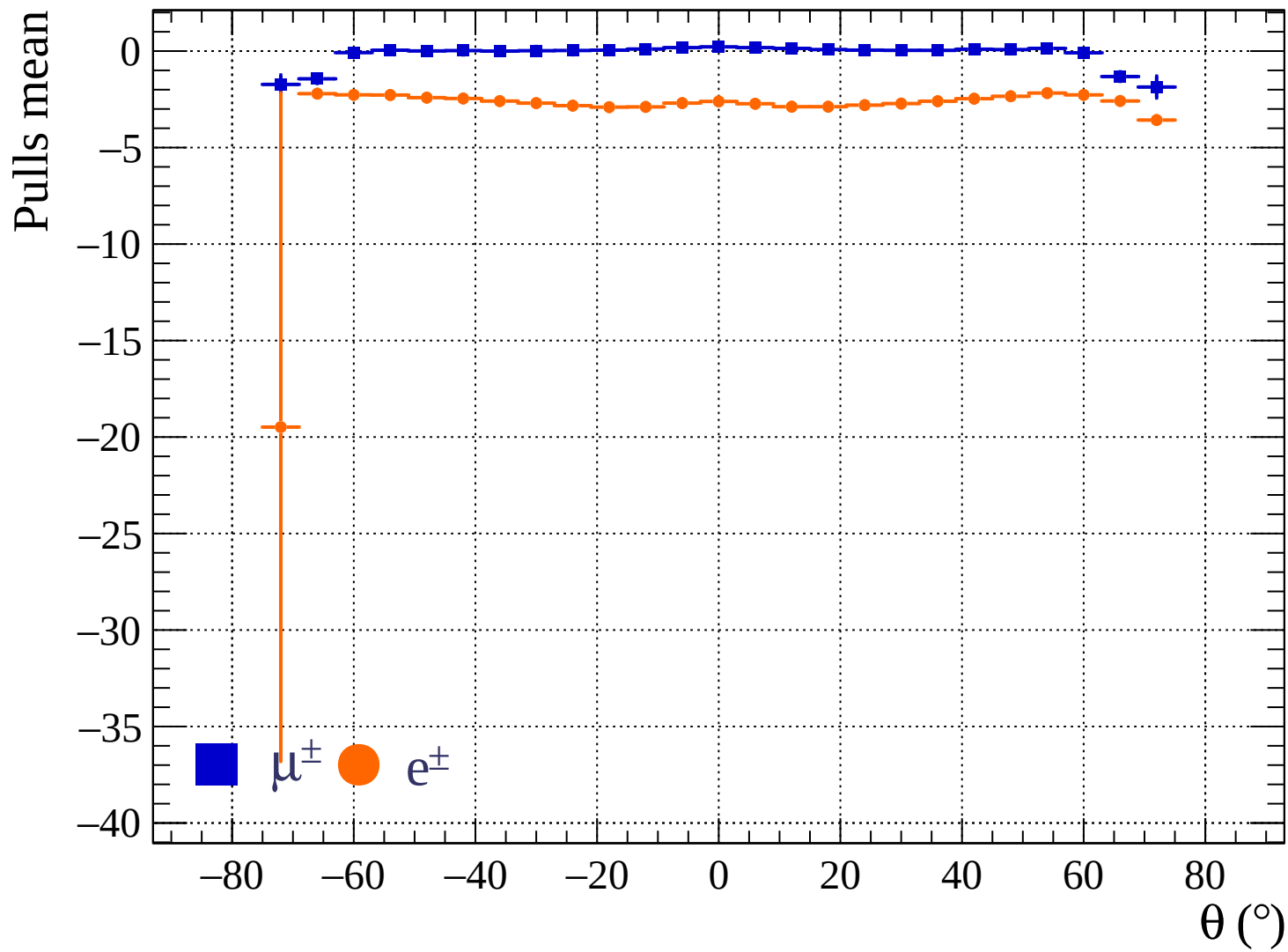




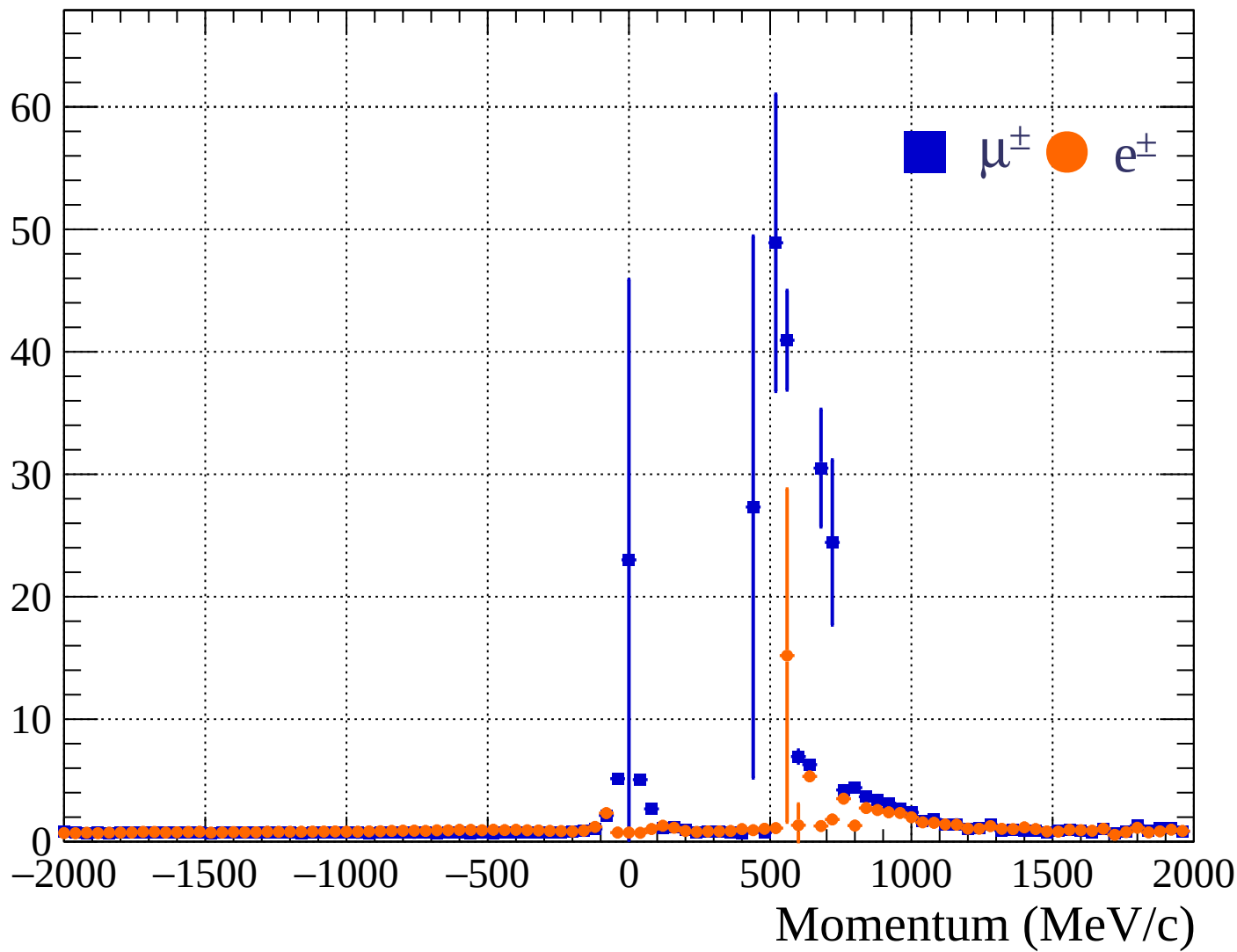


Pulls standard deviation

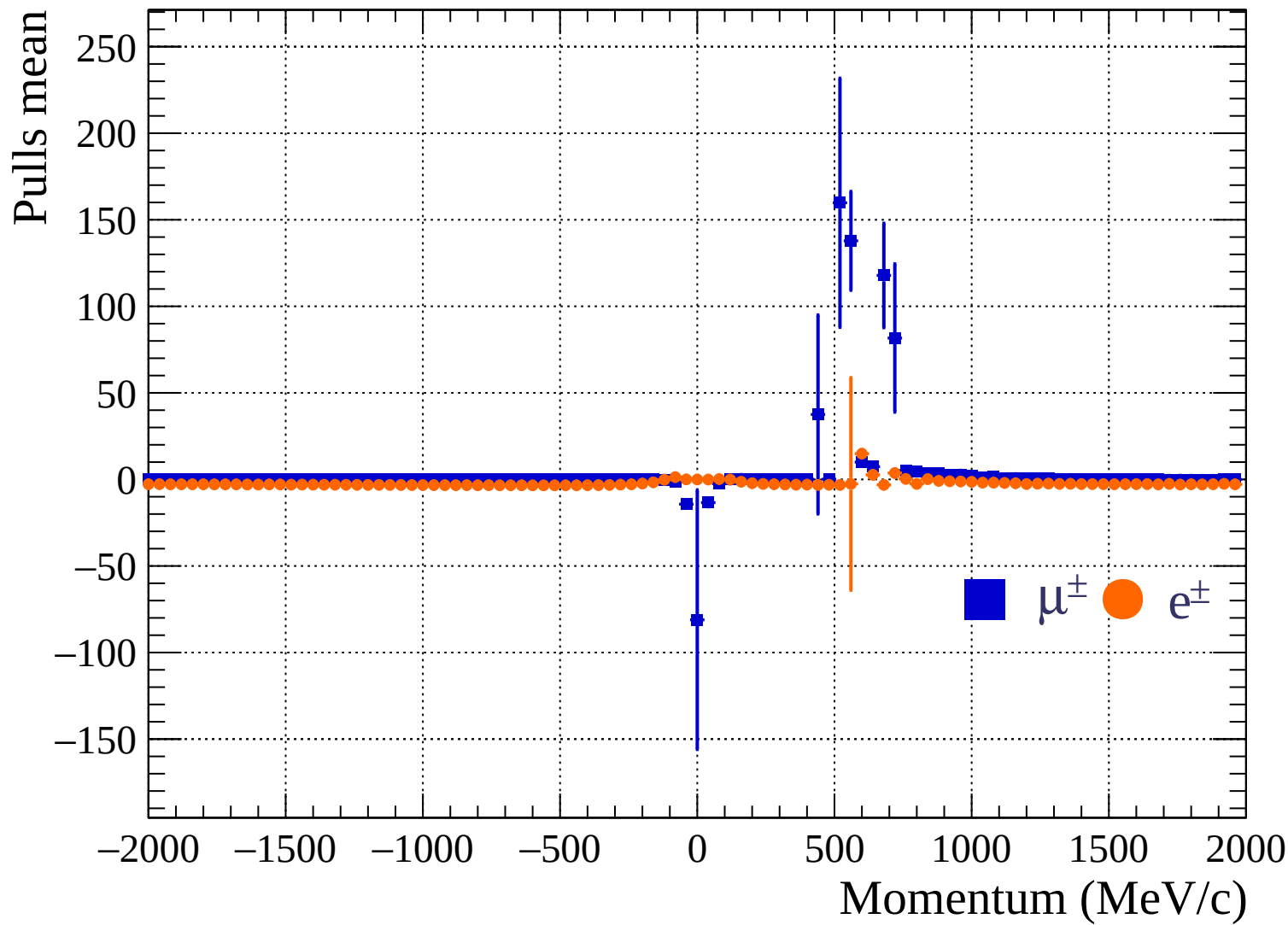


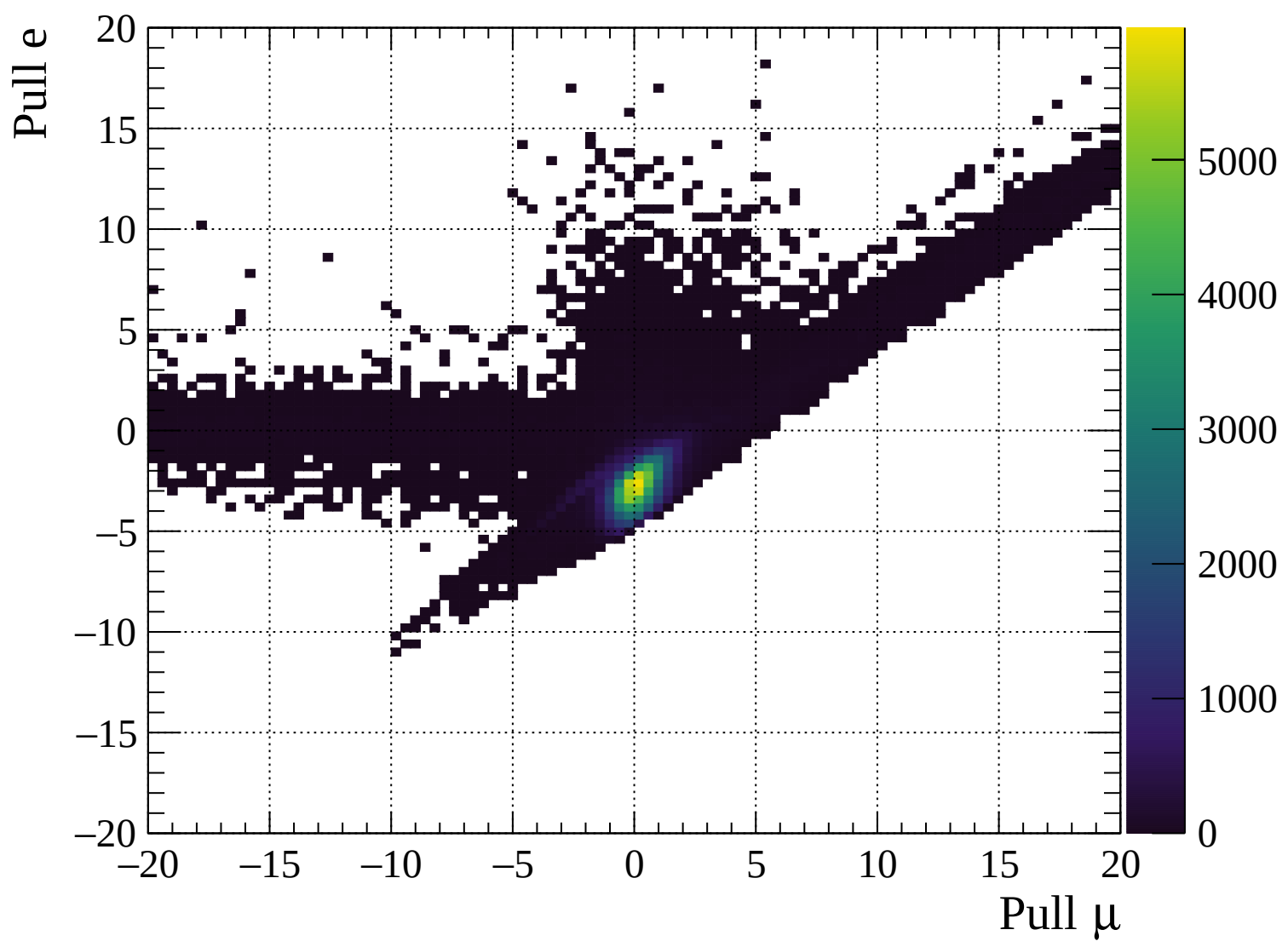


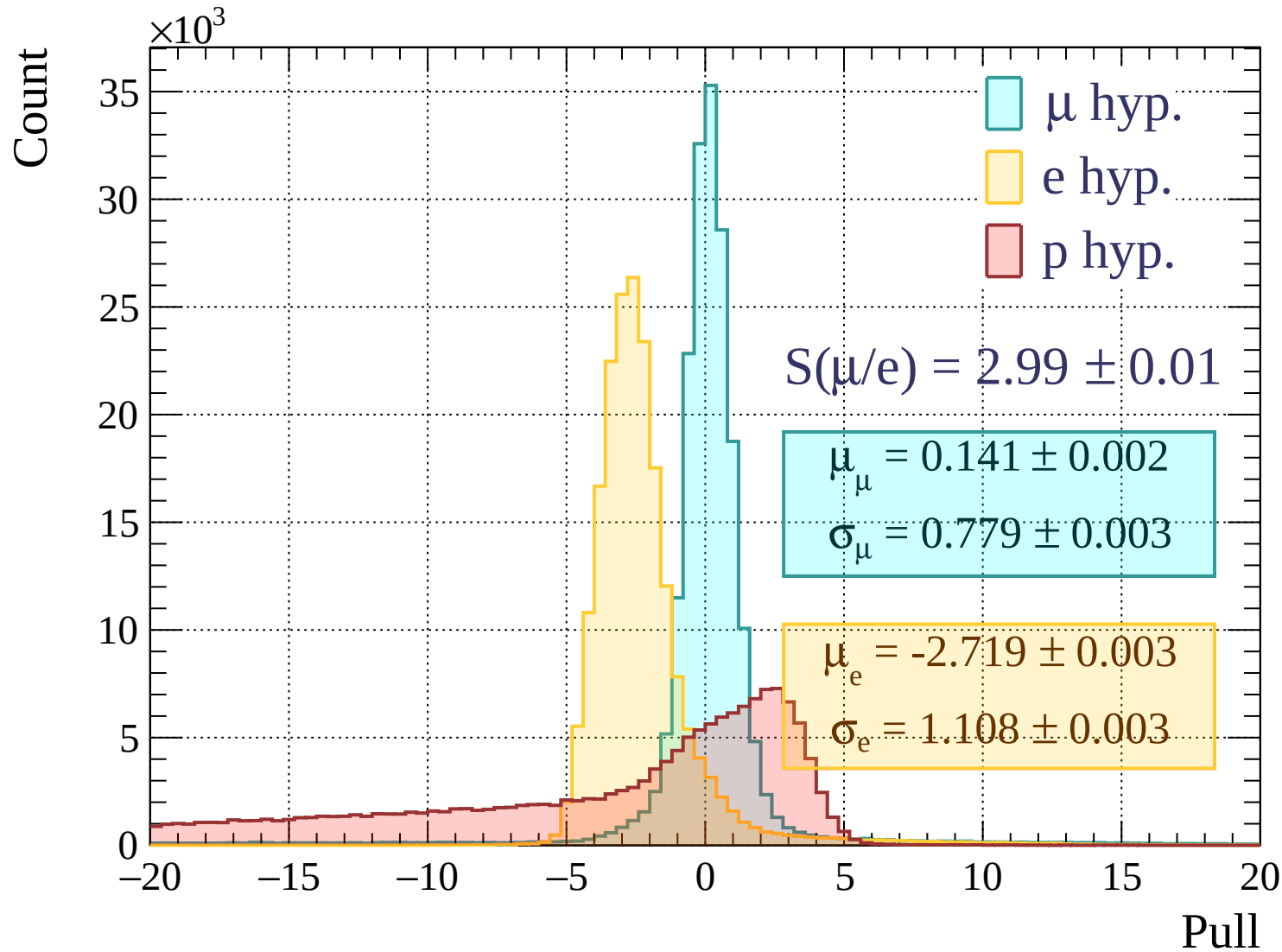
Pulls standard deviation

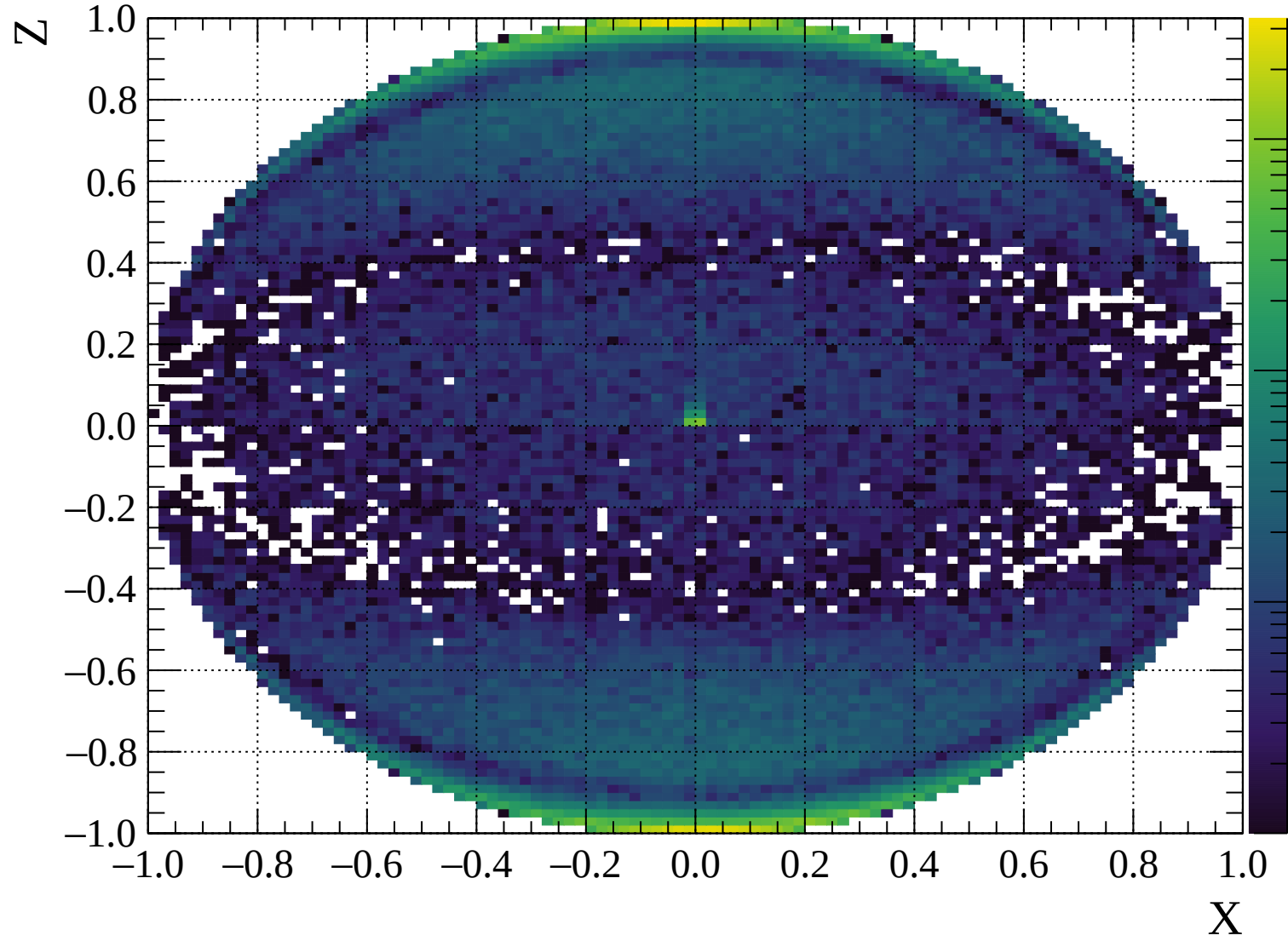


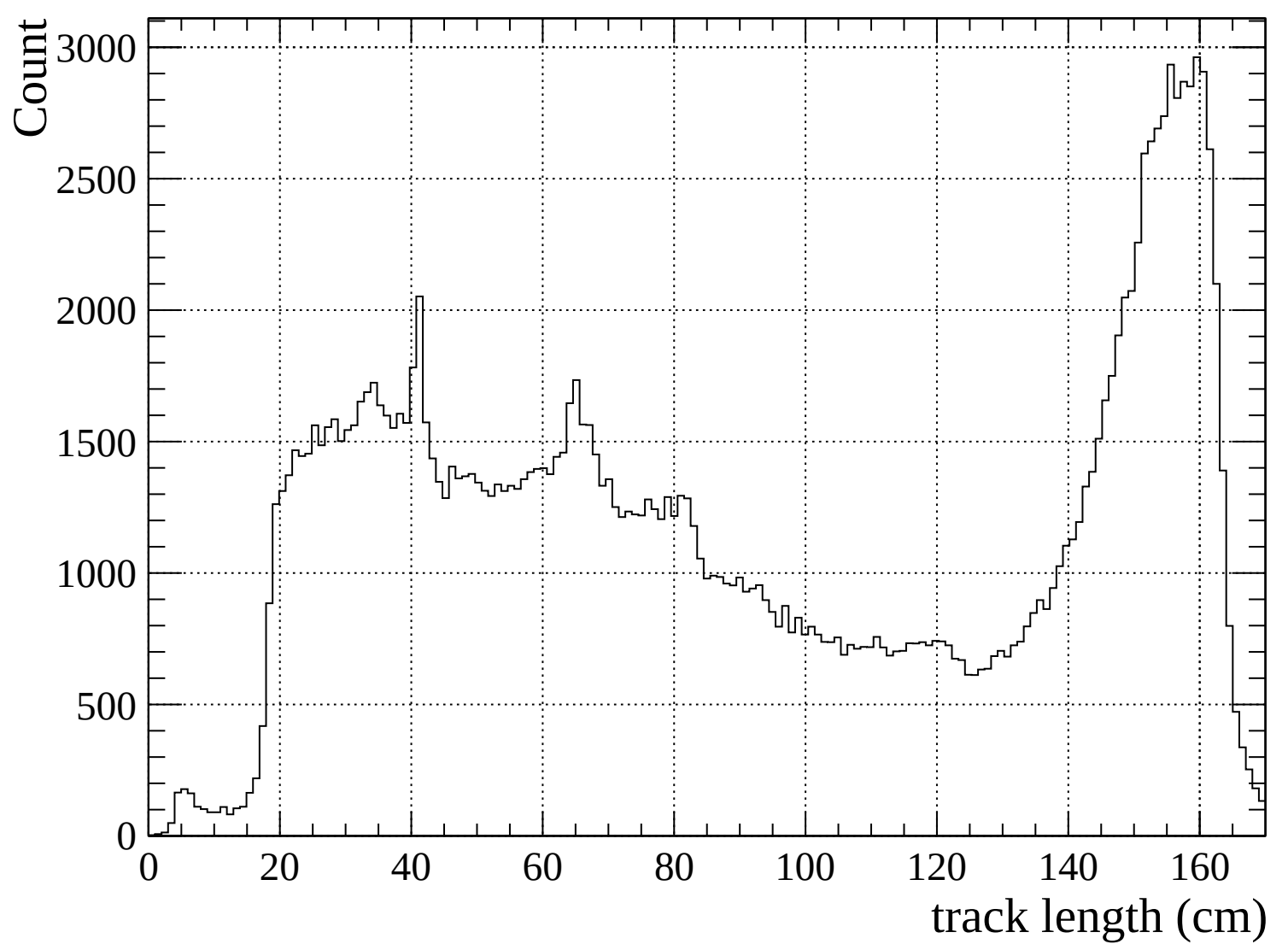


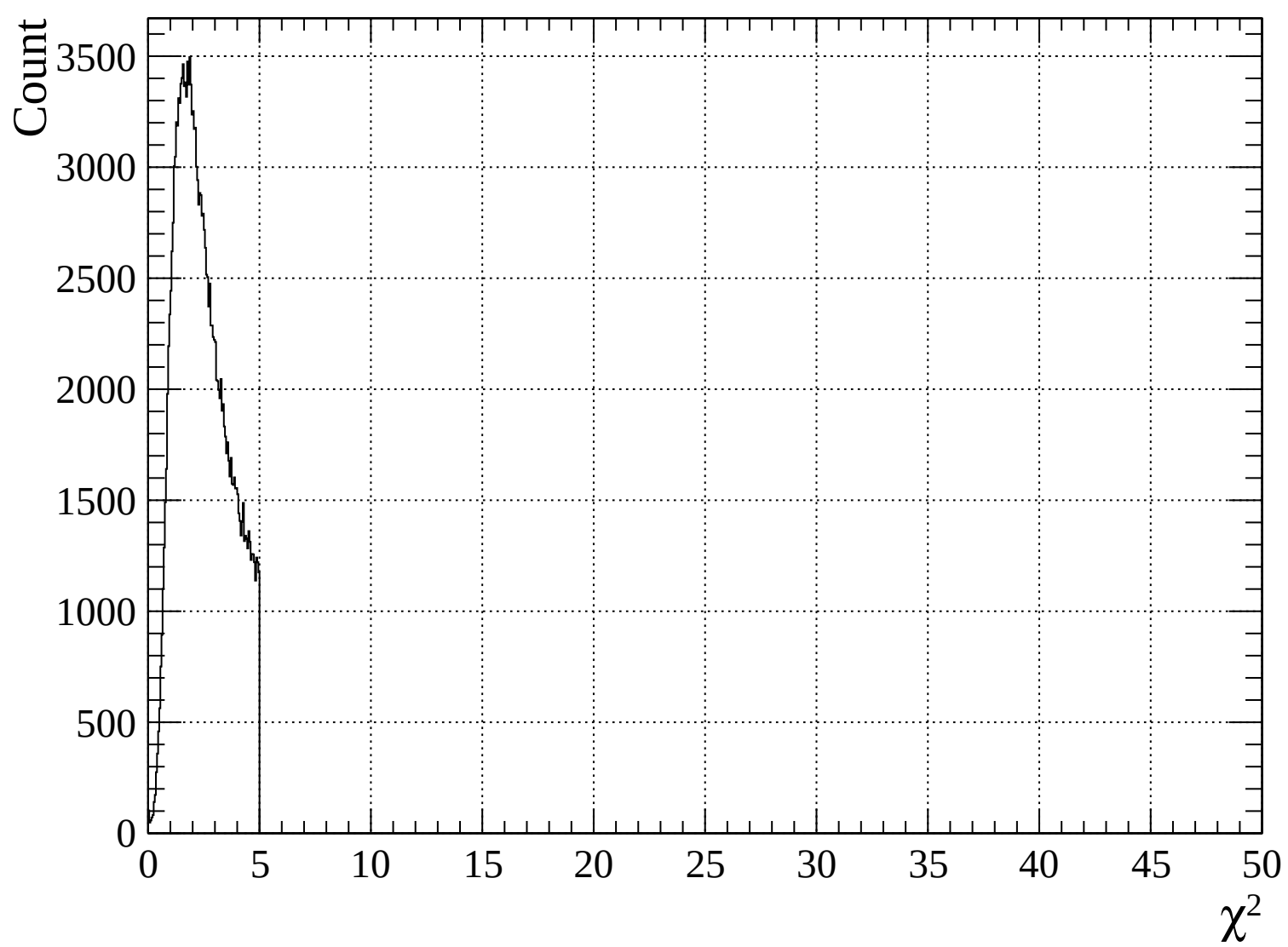


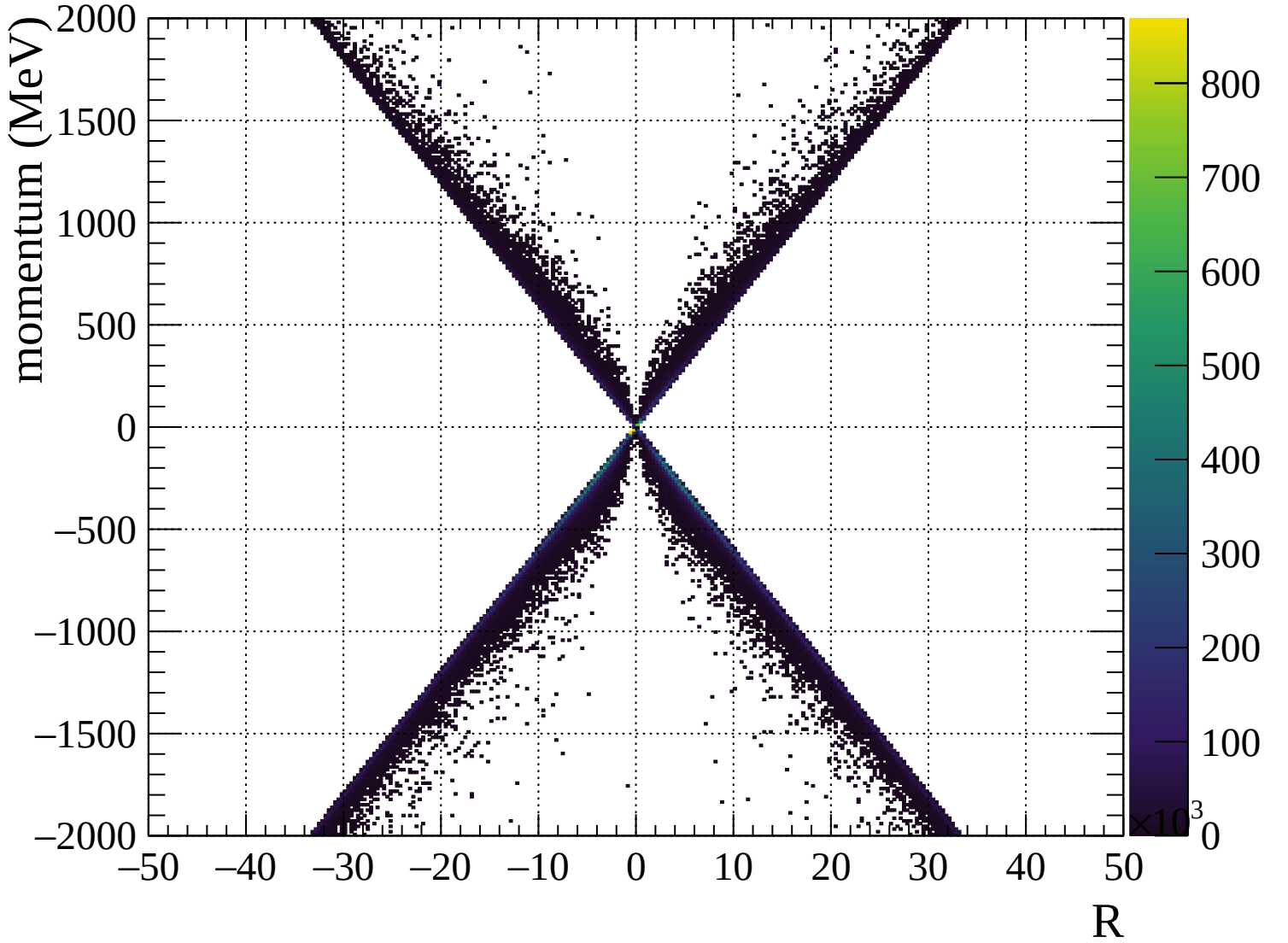


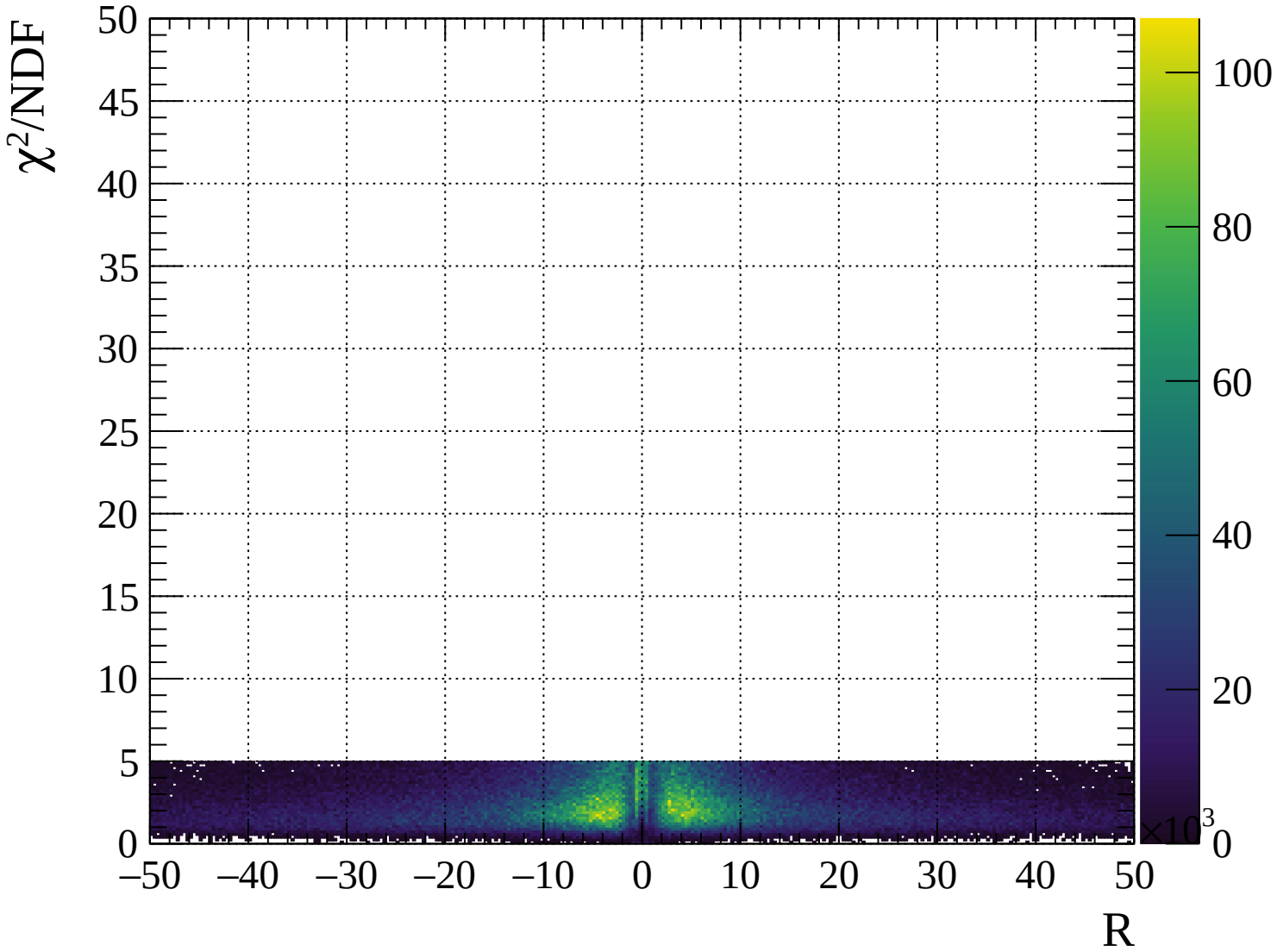








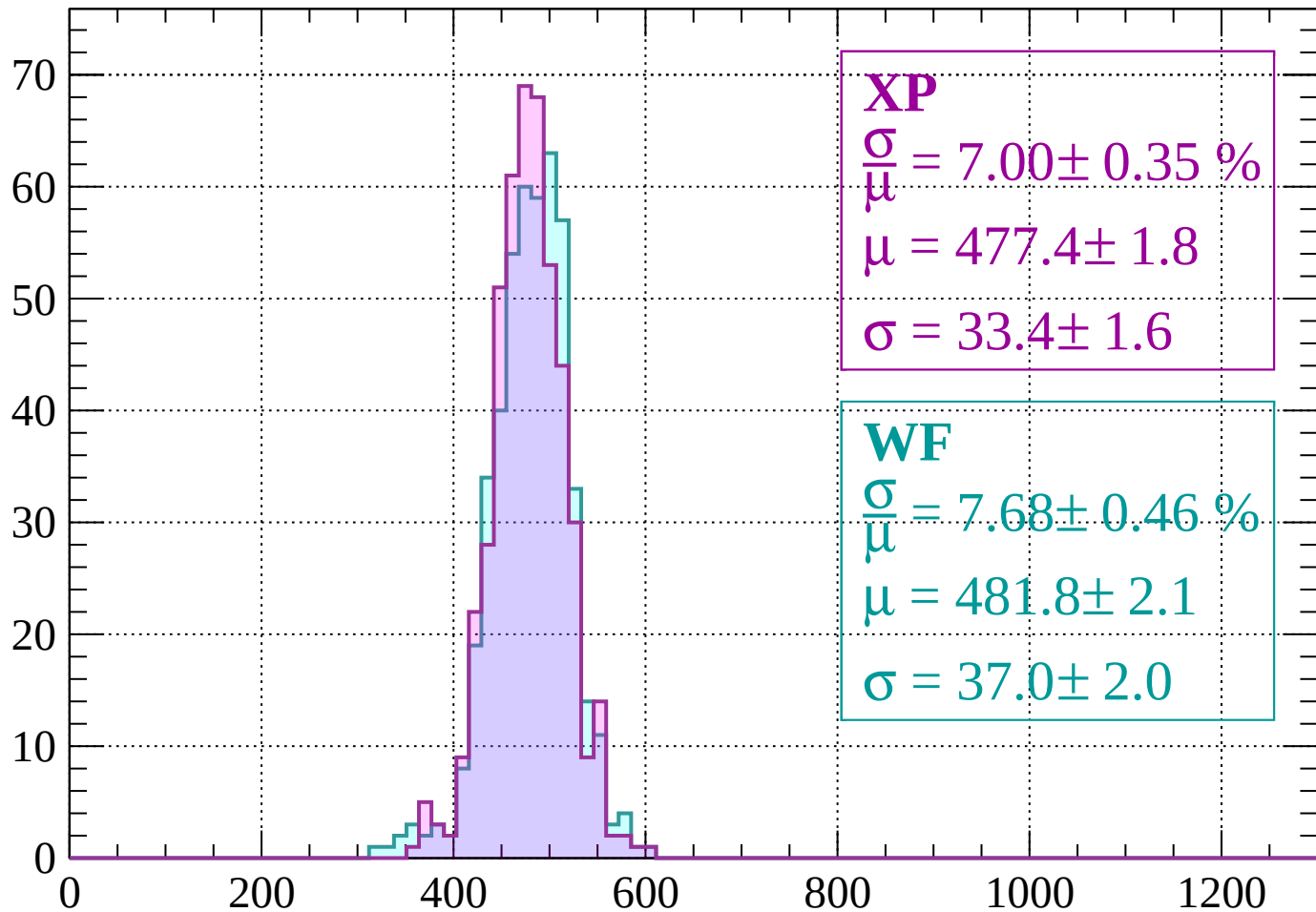






# Energy loss | $-2000 < p < -1960$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.00 \pm 0.35 \%$$

$$\mu = 477.4 \pm 1.8$$

$$\sigma = 33.4 \pm 1.6$$

**WF**

$$\frac{\sigma}{\mu} = 7.68 \pm 0.46 \%$$

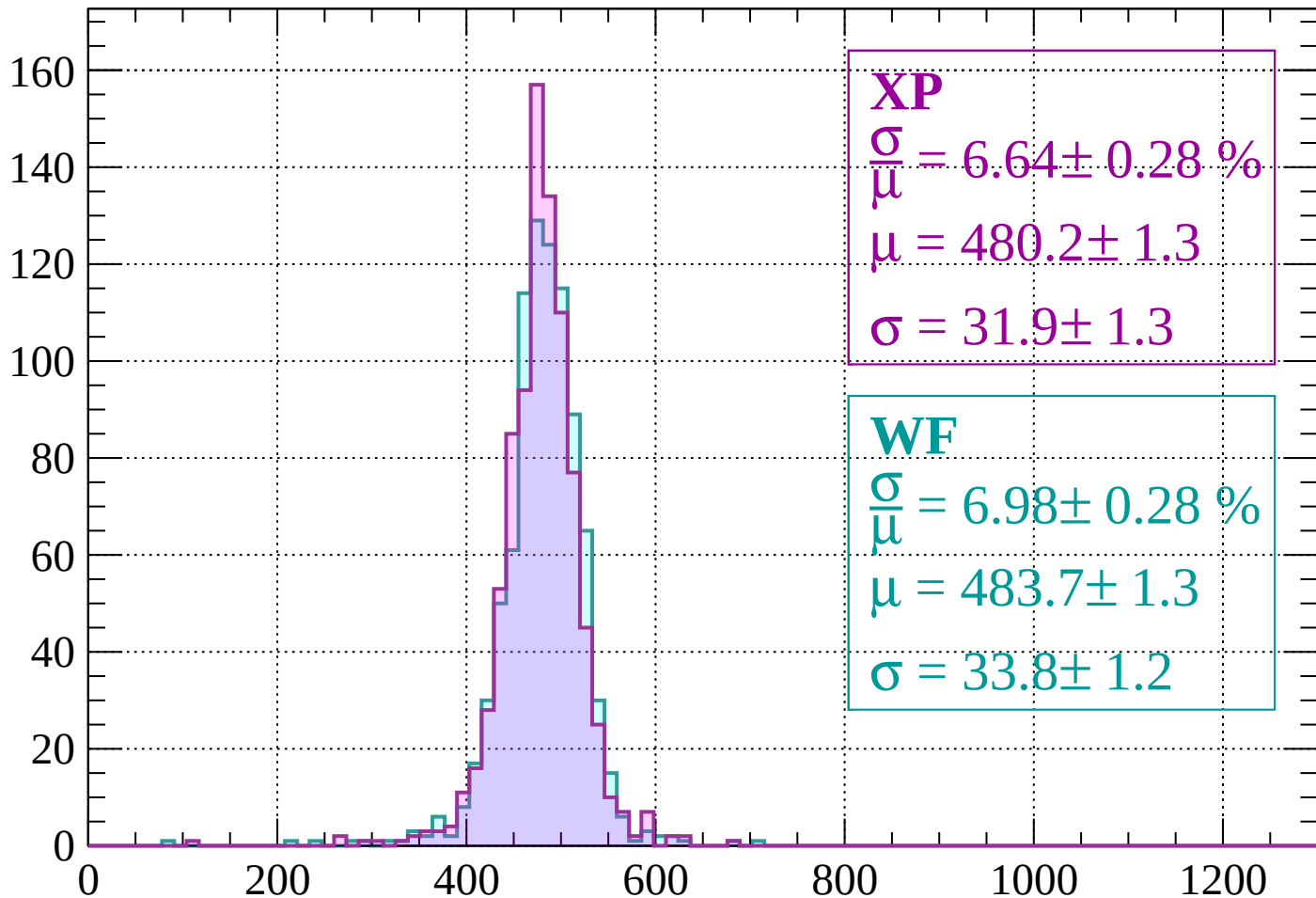
$$\mu = 481.8 \pm 2.1$$

$$\sigma = 37.0 \pm 2.0$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1960 < p < -1920$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.64 \pm 0.28 \%$$

$$\mu = 480.2 \pm 1.3$$

$$\sigma = 31.9 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.98 \pm 0.28 \%$$

$$\mu = 483.7 \pm 1.3$$

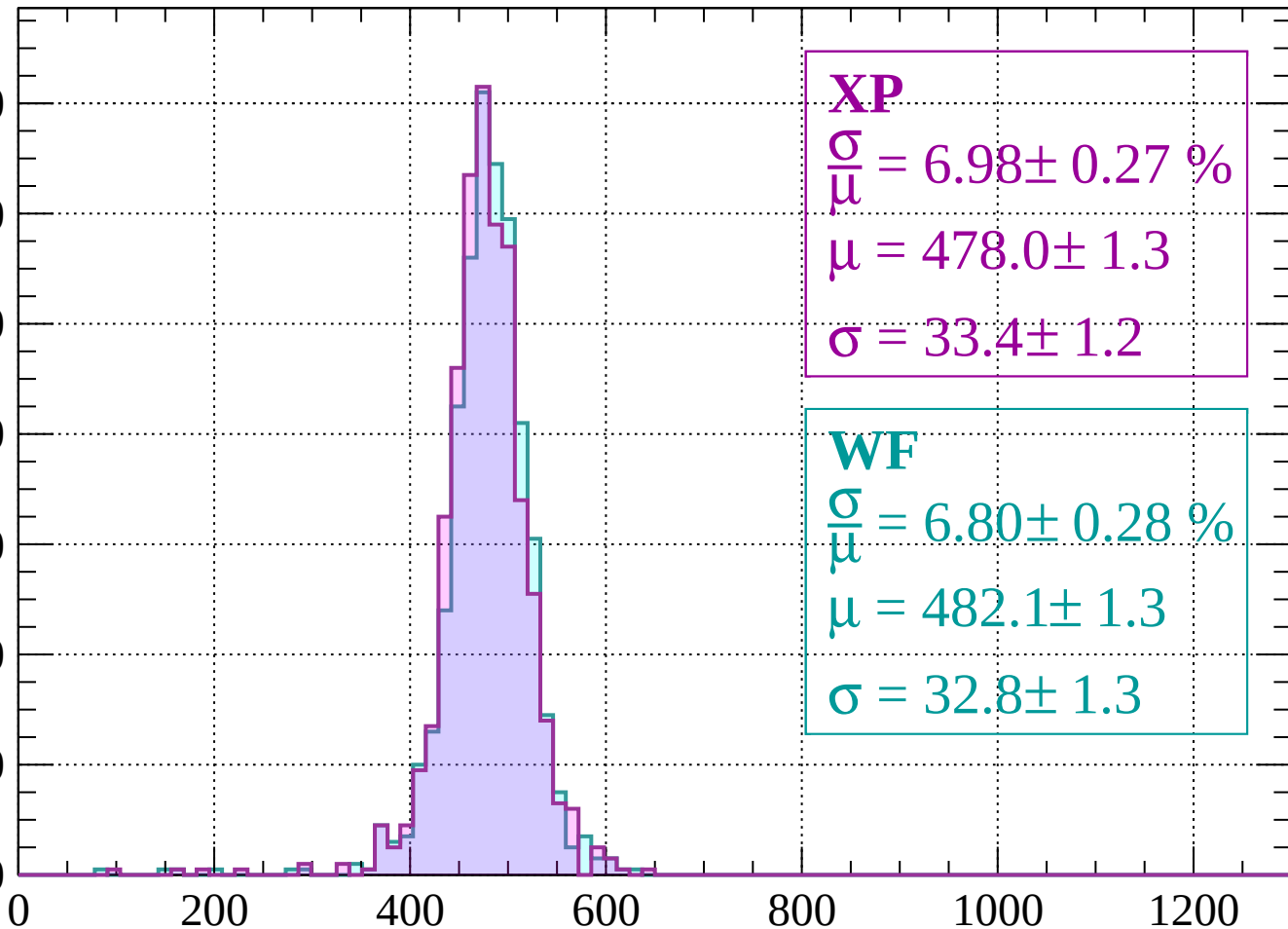
$$\sigma = 33.8 \pm 1.2$$

dE/dx (ADC counts/cm)

# Energy loss | -1920 < p < -1880

Count

140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 6.98 \pm 0.27 \%$$

$$\mu = 478.0 \pm 1.3$$

$$\sigma = 33.4 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.80 \pm 0.28 \%$$

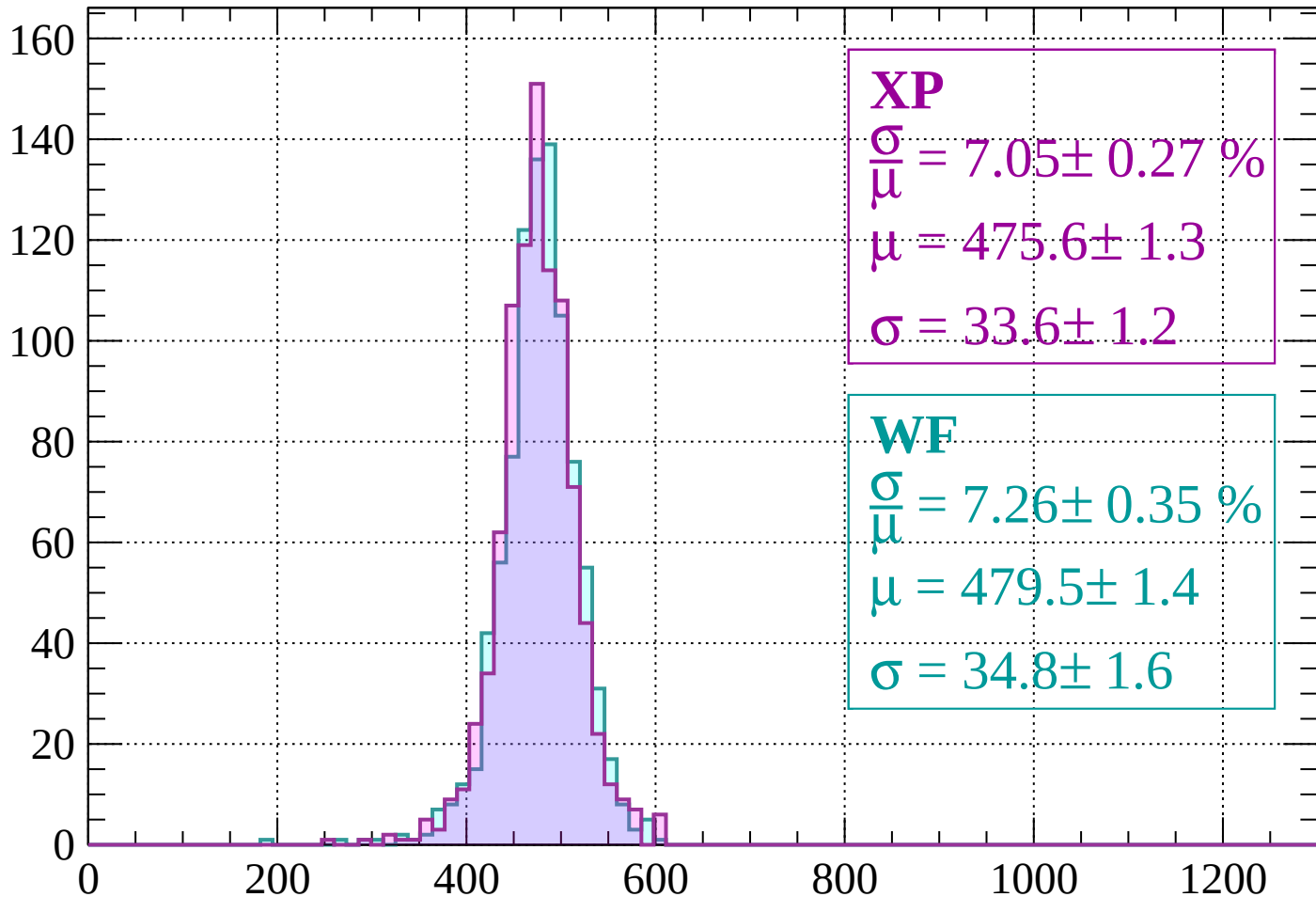
$$\mu = 482.1 \pm 1.3$$

$$\sigma = 32.8 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss |  $-1880 < p < -1840$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.05 \pm 0.27 \%$$

$$\mu = 475.6 \pm 1.3$$

$$\sigma = 33.6 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 7.26 \pm 0.35 \%$$

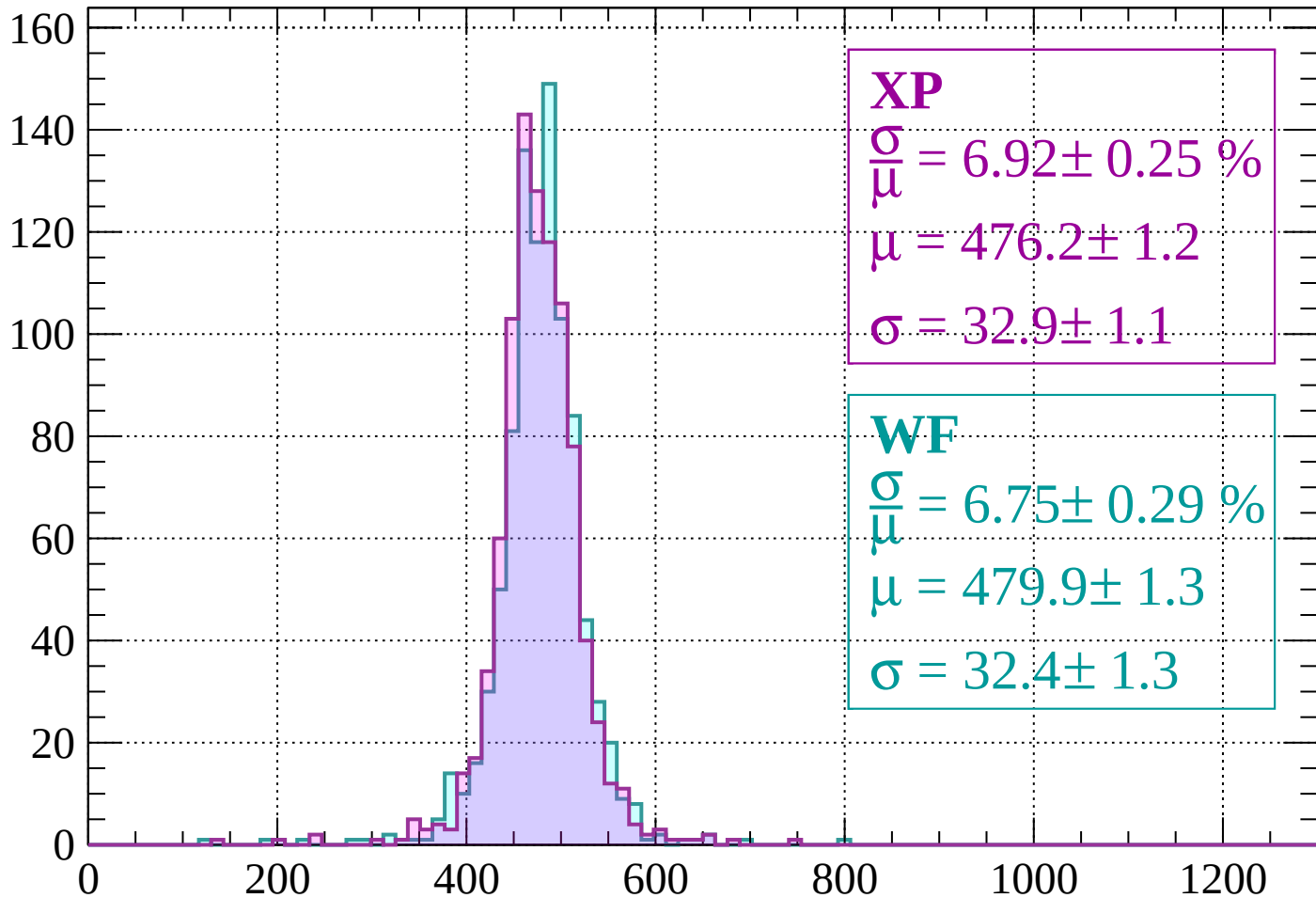
$$\mu = 479.5 \pm 1.4$$

$$\sigma = 34.8 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-1840 < p < -1800$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.92 \pm 0.25 \%$$

$$\mu = 476.2 \pm 1.2$$

$$\sigma = 32.9 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 6.75 \pm 0.29 \%$$

$$\mu = 479.9 \pm 1.3$$

$$\sigma = 32.4 \pm 1.3$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1800 < p < -1760$

Count

140

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP**

$$\frac{\sigma}{\mu} = 7.17 \pm 0.28 \%$$

$$\mu = 474.1 \pm 1.3$$

$$\sigma = 34.0 \pm 1.2$$

**WF**

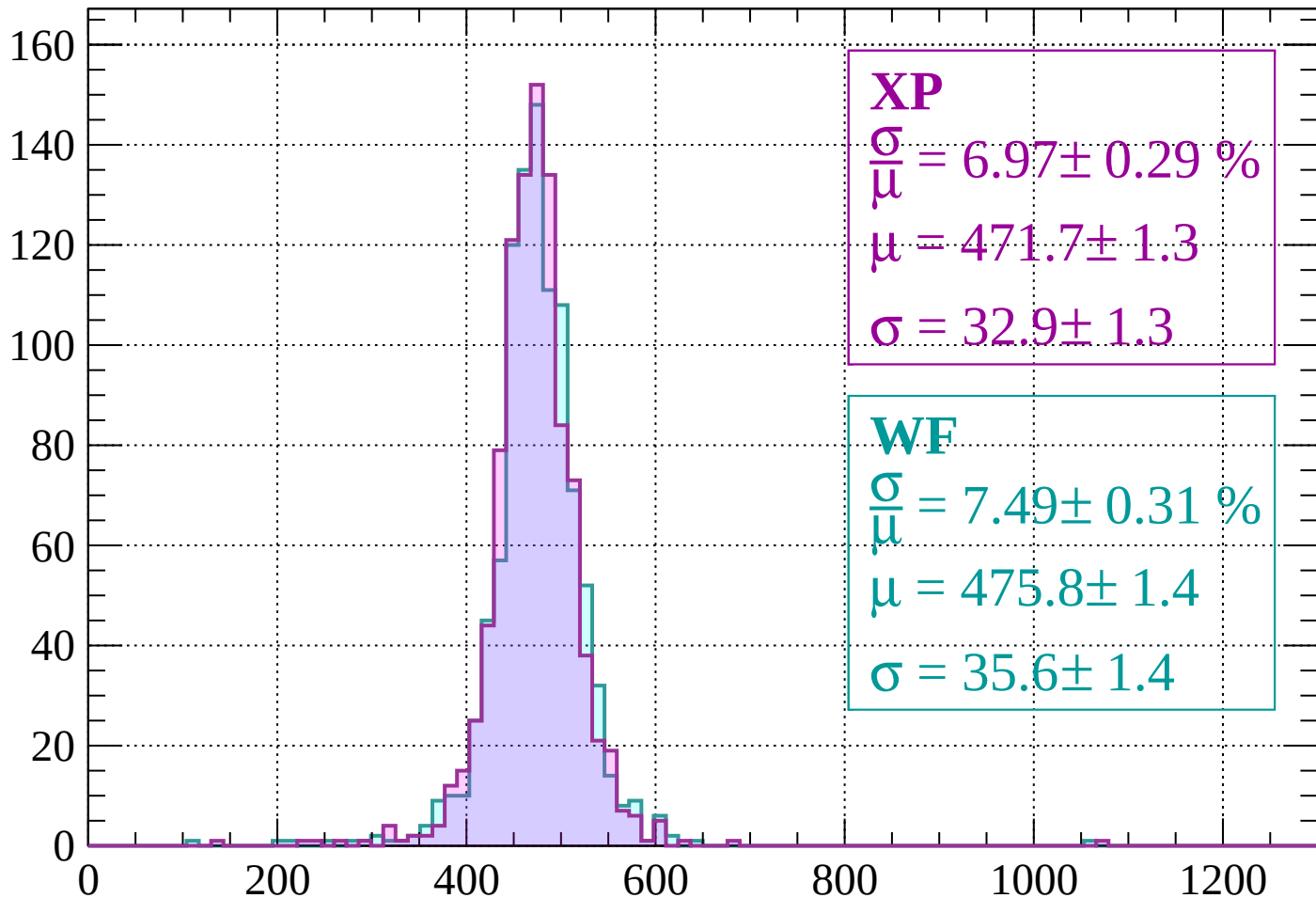
$$\frac{\sigma}{\mu} = 6.79 \pm 0.28 \%$$

$$\mu = 479.3 \pm 1.3$$

$$\sigma = 32.5 \pm 1.2$$

# Energy loss | -1760 < p < -1720

Count



**XP**

$$\frac{\sigma}{\mu} = 6.97 \pm 0.29 \%$$

$$\mu = 471.7 \pm 1.3$$

$$\sigma = 32.9 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 7.49 \pm 0.31 \%$$

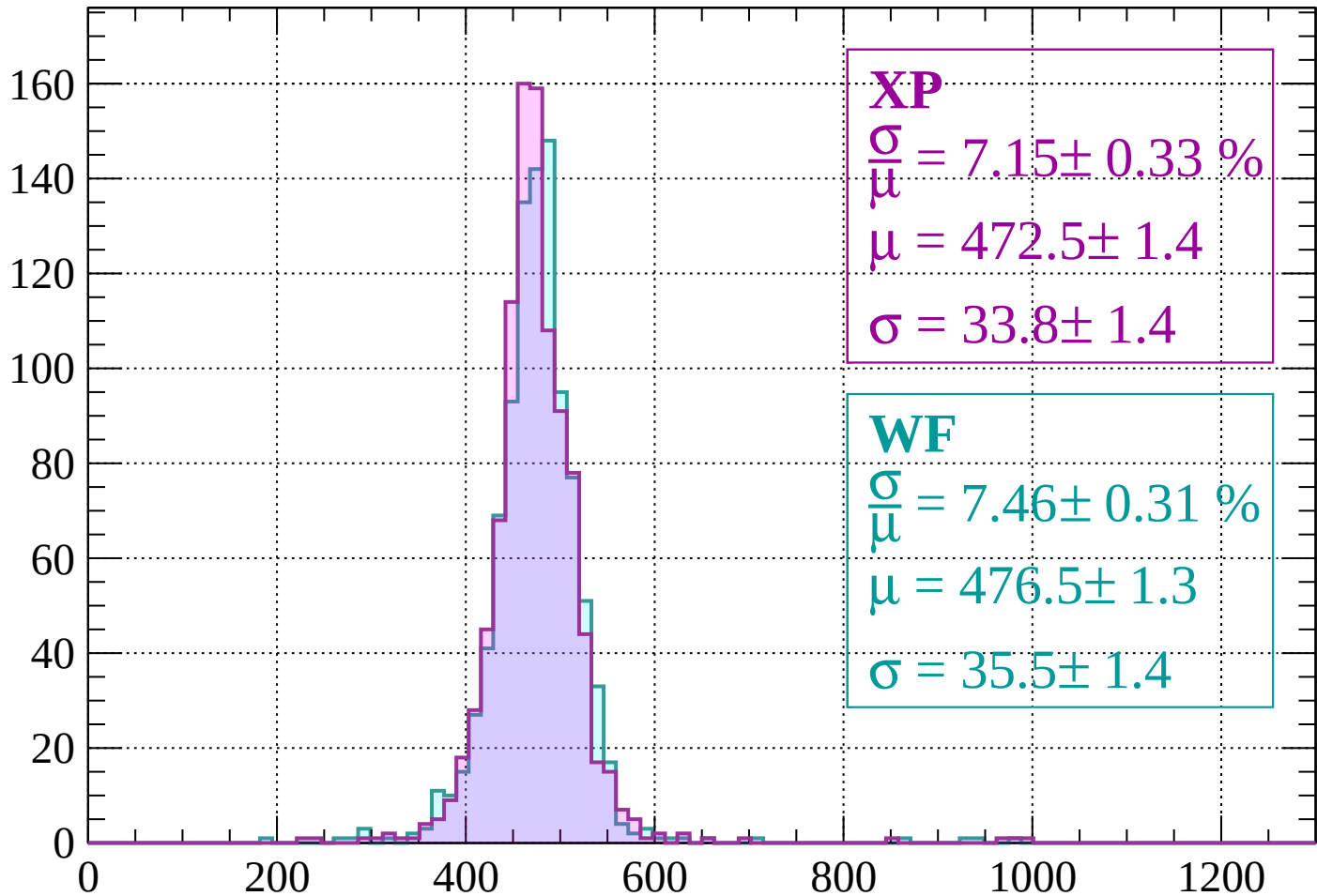
$$\mu = 475.8 \pm 1.4$$

$$\sigma = 35.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss |  $-1720 < p < -1680$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.15 \pm 0.33 \%$$

$$\mu = 472.5 \pm 1.4$$

$$\sigma = 33.8 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 7.46 \pm 0.31 \%$$

$$\mu = 476.5 \pm 1.3$$

$$\sigma = 35.5 \pm 1.4$$

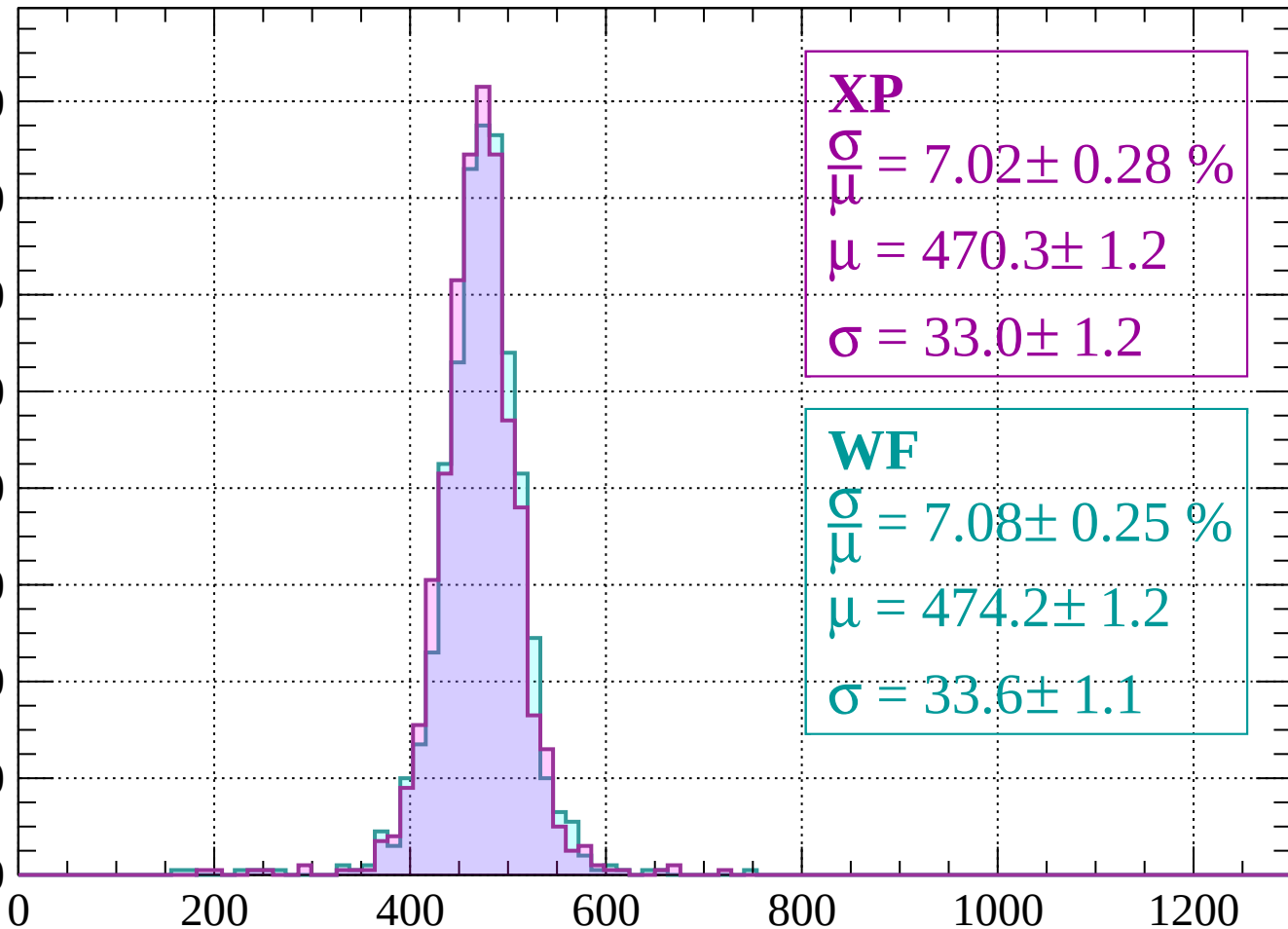
dE/dx (ADC counts/cm)



Energy loss |  $-1680 < p < -1640$

Count

160  
140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 7.02 \pm 0.28 \%$$

$$\mu = 470.3 \pm 1.2$$

$$\sigma = 33.0 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 7.08 \pm 0.25 \%$$

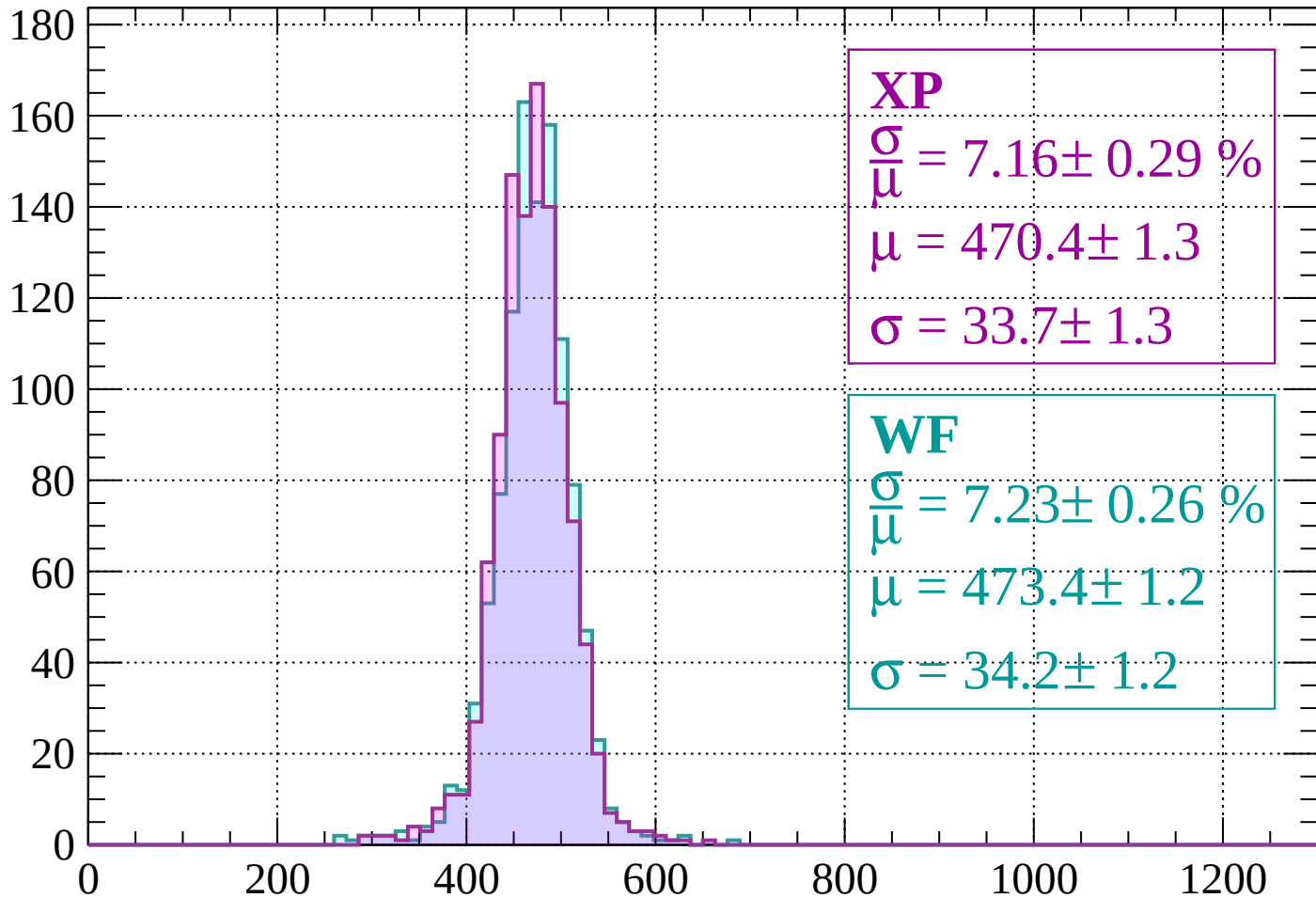
$$\mu = 474.2 \pm 1.2$$

$$\sigma = 33.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-1640 < p < -1600$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.16 \pm 0.29 \%$$

$$\mu = 470.4 \pm 1.3$$

$$\sigma = 33.7 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 7.23 \pm 0.26 \%$$

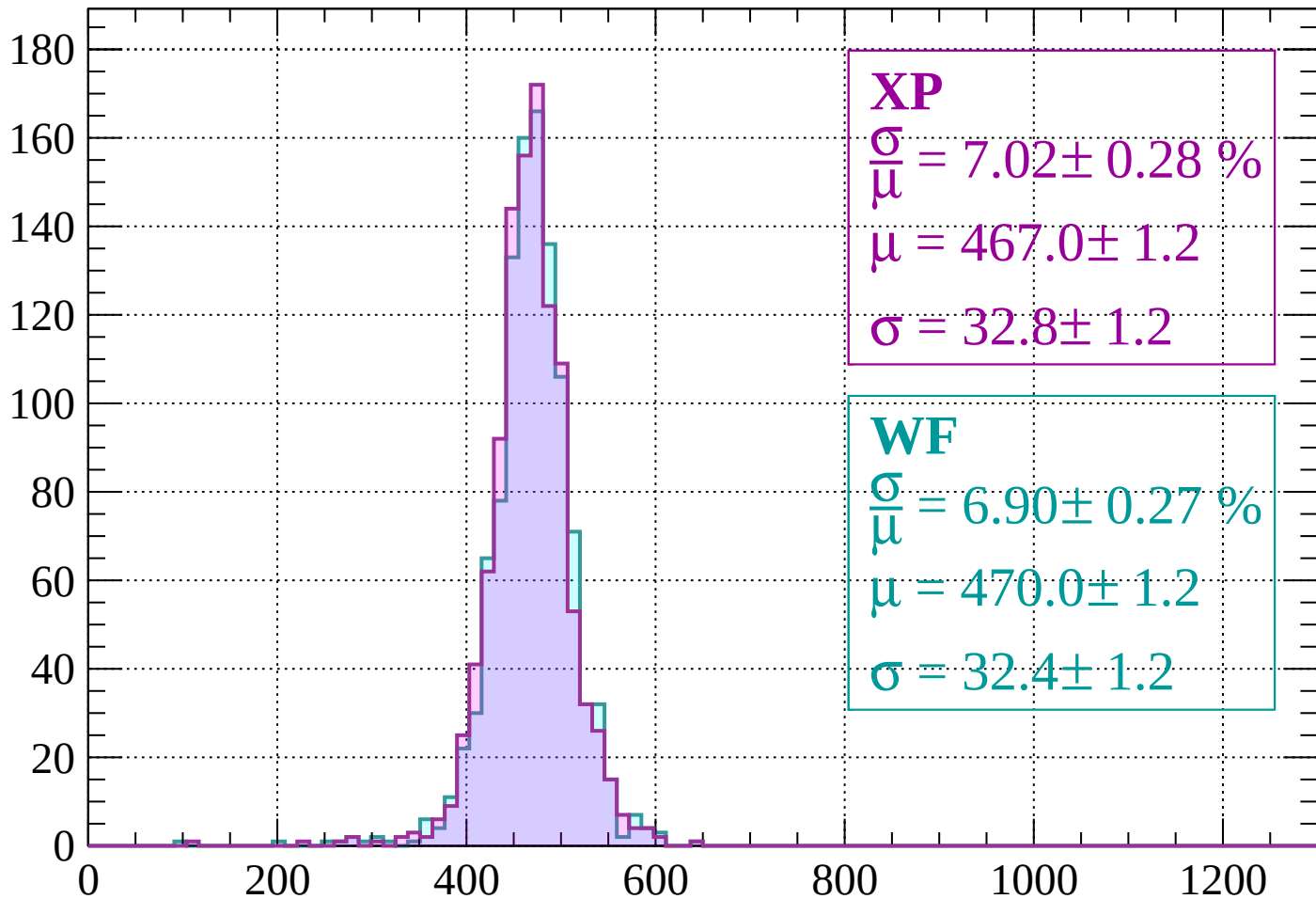
$$\mu = 473.4 \pm 1.2$$

$$\sigma = 34.2 \pm 1.2$$

$dE/dx$  (ADC counts/cm)

# Energy loss | -1600 < p < -1560

Count



**XP**

$$\frac{\sigma}{\mu} = 7.02 \pm 0.28 \%$$

$$\mu = 467.0 \pm 1.2$$

$$\sigma = 32.8 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.90 \pm 0.27 \%$$

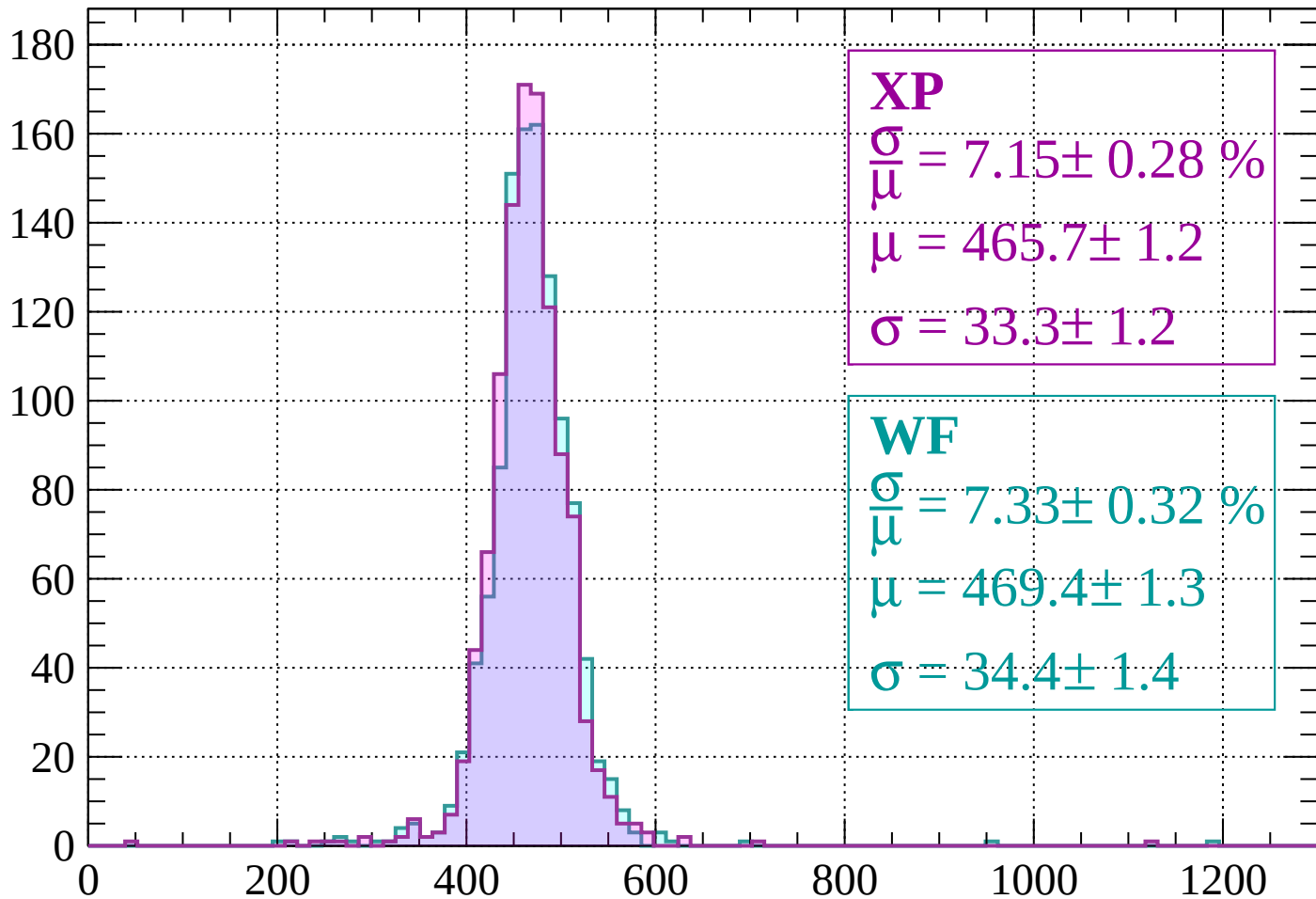
$$\mu = 470.0 \pm 1.2$$

$$\sigma = 32.4 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss |  $-1560 < p < -1520$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.15 \pm 0.28 \%$$

$$\mu = 465.7 \pm 1.2$$

$$\sigma = 33.3 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 7.33 \pm 0.32 \%$$

$$\mu = 469.4 \pm 1.3$$

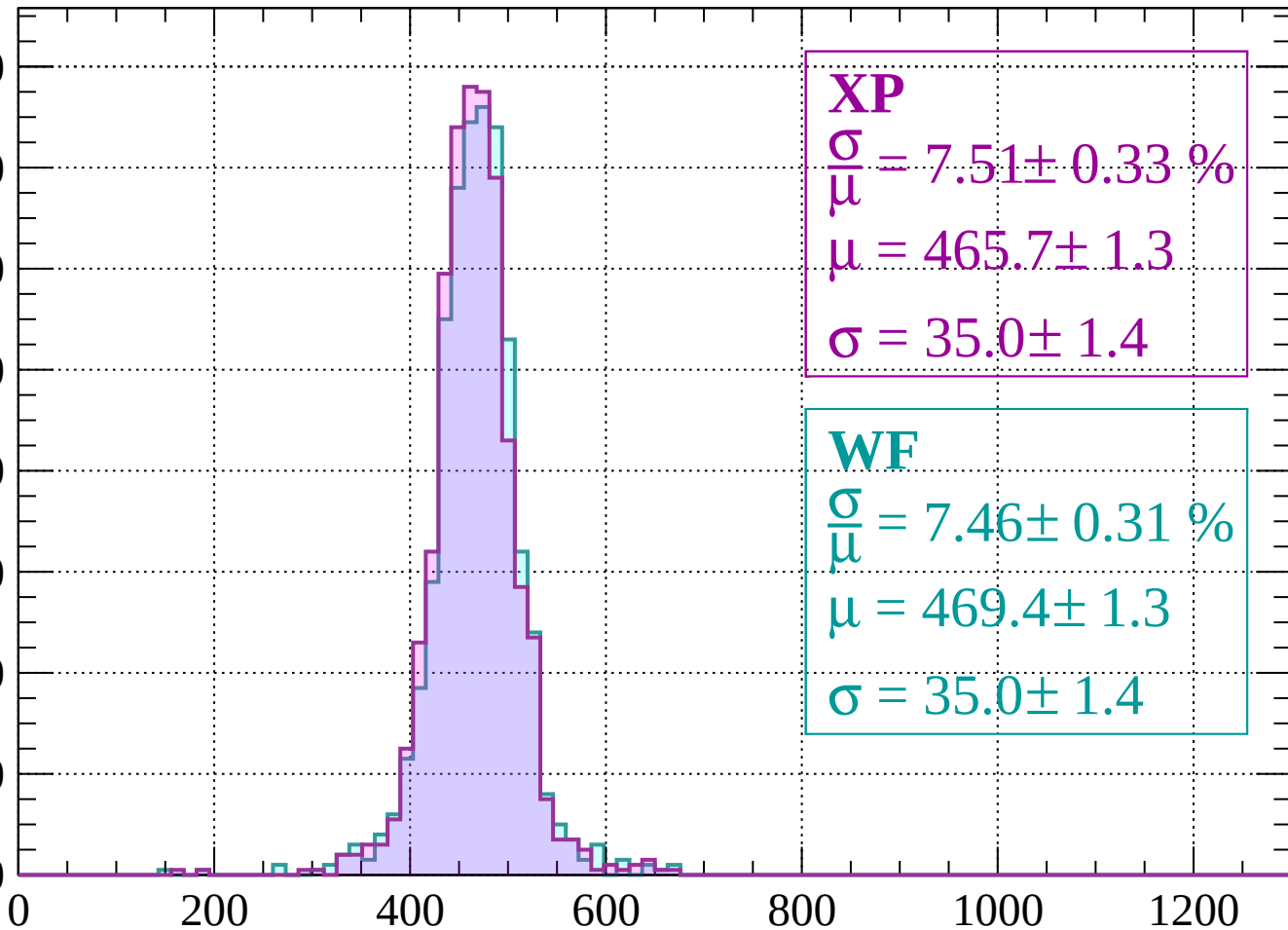
$$\sigma = 34.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss |  $-1520 < p < -1480$

Count

160  
140  
120  
100  
80  
60  
40  
20  
0



**XP**

$$\frac{\sigma}{\mu} = 7.51 \pm 0.33 \%$$

$$\mu = 465.7 \pm 1.3$$

$$\sigma = 35.0 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 7.46 \pm 0.31 \%$$

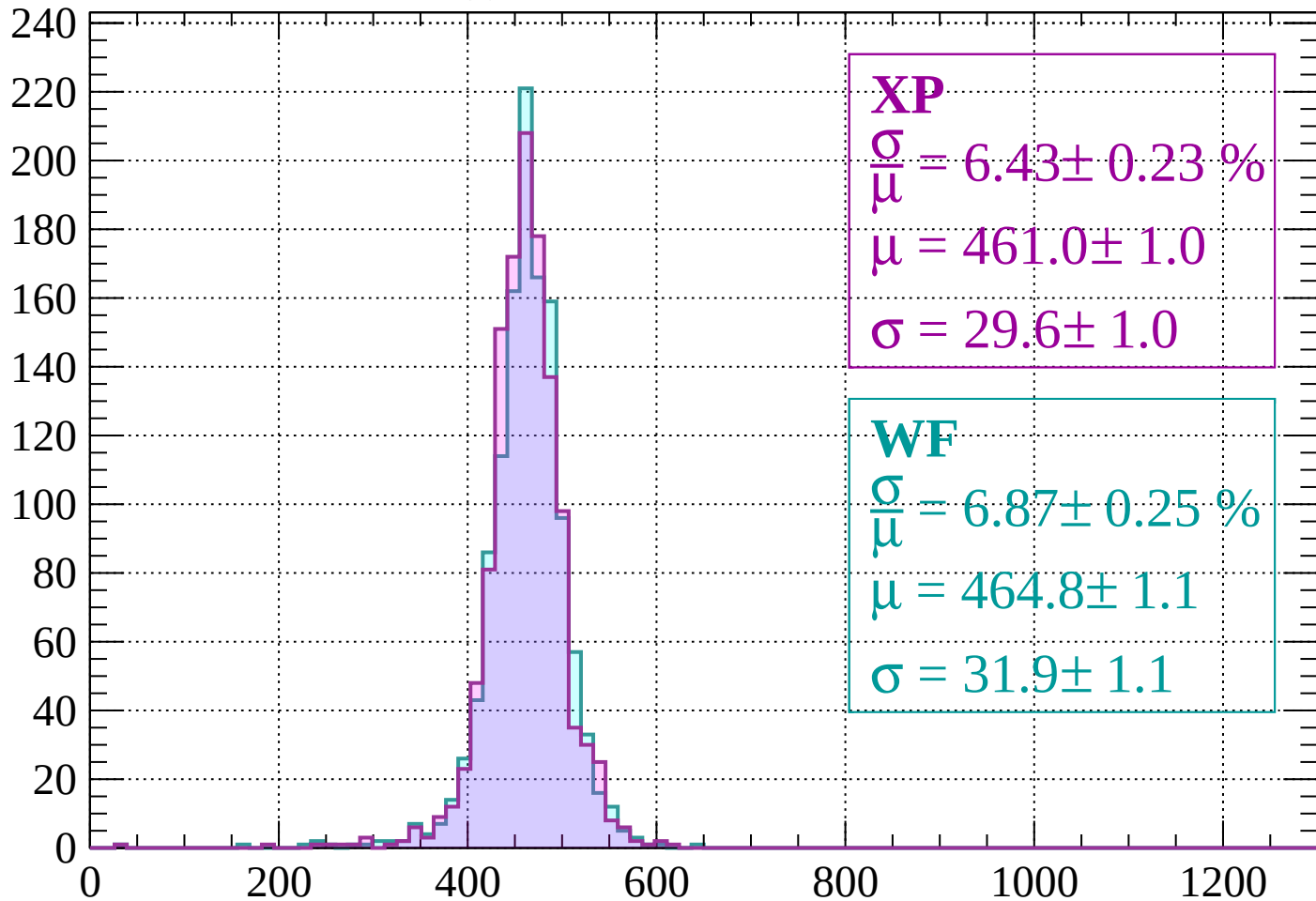
$$\mu = 469.4 \pm 1.3$$

$$\sigma = 35.0 \pm 1.4$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1480 < p < -1440$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.43 \pm 0.23 \%$$

$$\mu = 461.0 \pm 1.0$$

$$\sigma = 29.6 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 6.87 \pm 0.25 \%$$

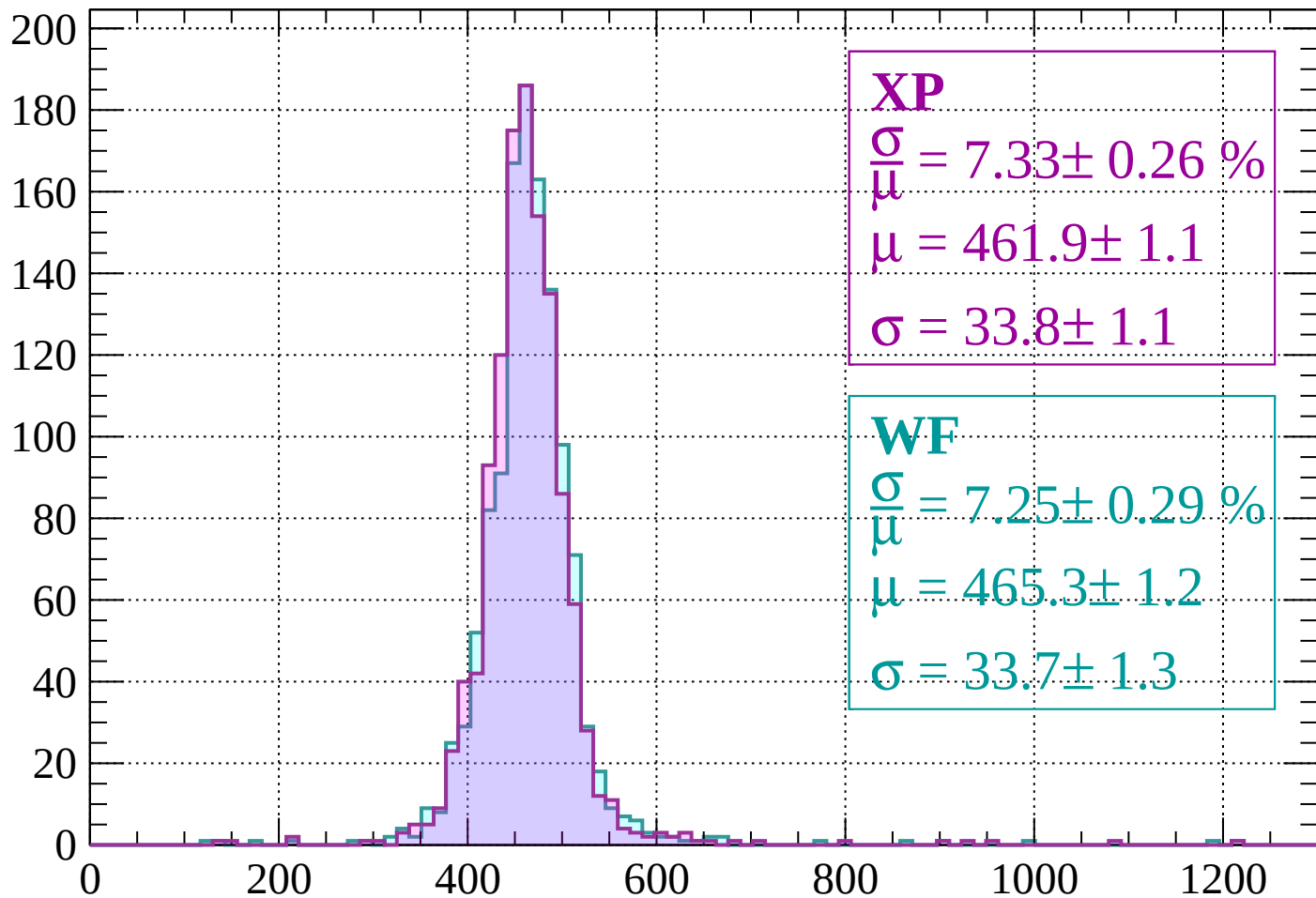
$$\mu = 464.8 \pm 1.1$$

$$\sigma = 31.9 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-1440 < p < -1400$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.33 \pm 0.26 \%$$

$$\mu = 461.9 \pm 1.1$$

$$\sigma = 33.8 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 7.25 \pm 0.29 \%$$

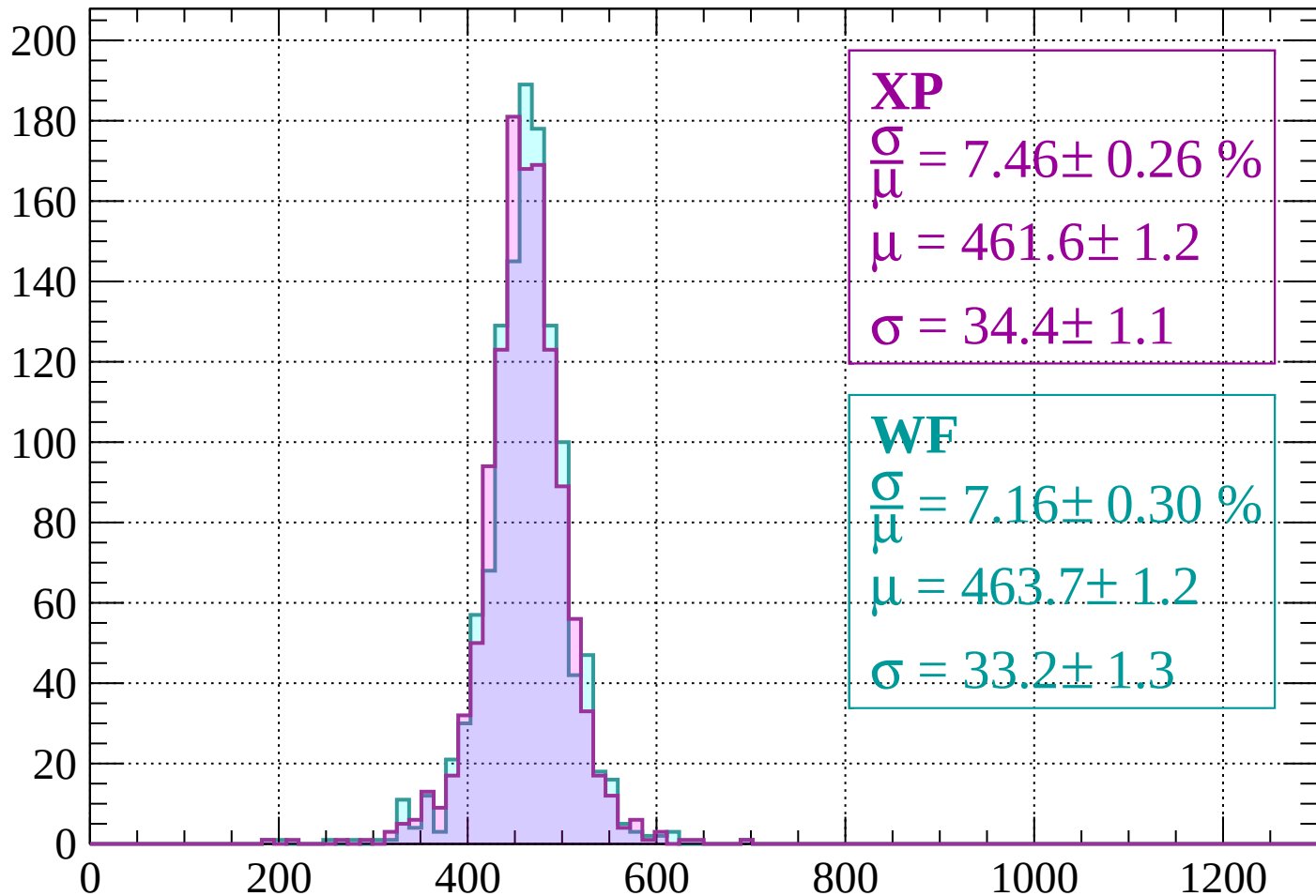
$$\mu = 465.3 \pm 1.2$$

$$\sigma = 33.7 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss |  $-1400 < p < -1360$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.46 \pm 0.26 \%$$

$$\mu = 461.6 \pm 1.2$$

$$\sigma = 34.4 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 7.16 \pm 0.30 \%$$

$$\mu = 463.7 \pm 1.2$$

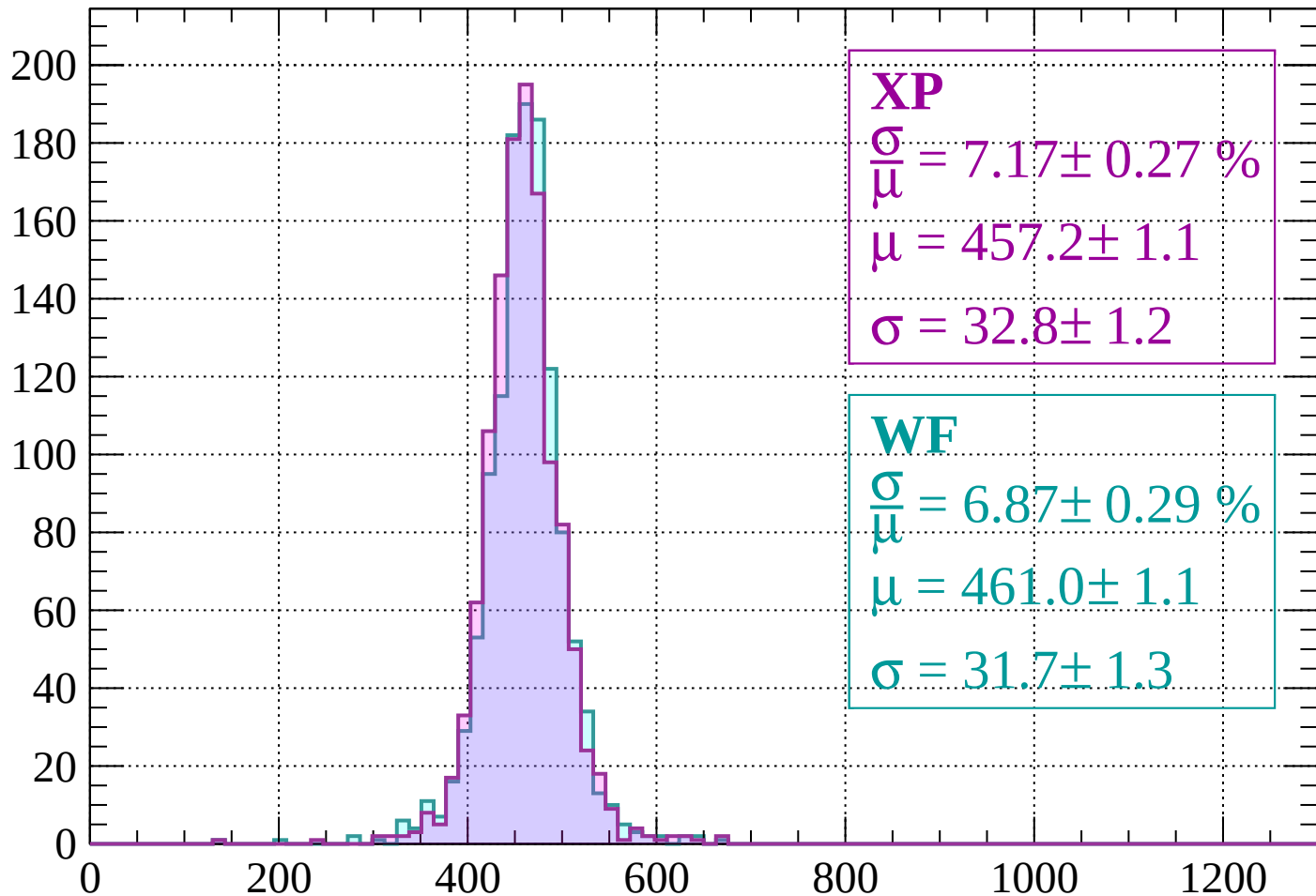
$$\sigma = 33.2 \pm 1.3$$

$dE/dx$  (ADC counts/cm)



Energy loss |  $-1360 < p < -1320$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.17 \pm 0.27 \%$$

$$\mu = 457.2 \pm 1.1$$

$$\sigma = 32.8 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.87 \pm 0.29 \%$$

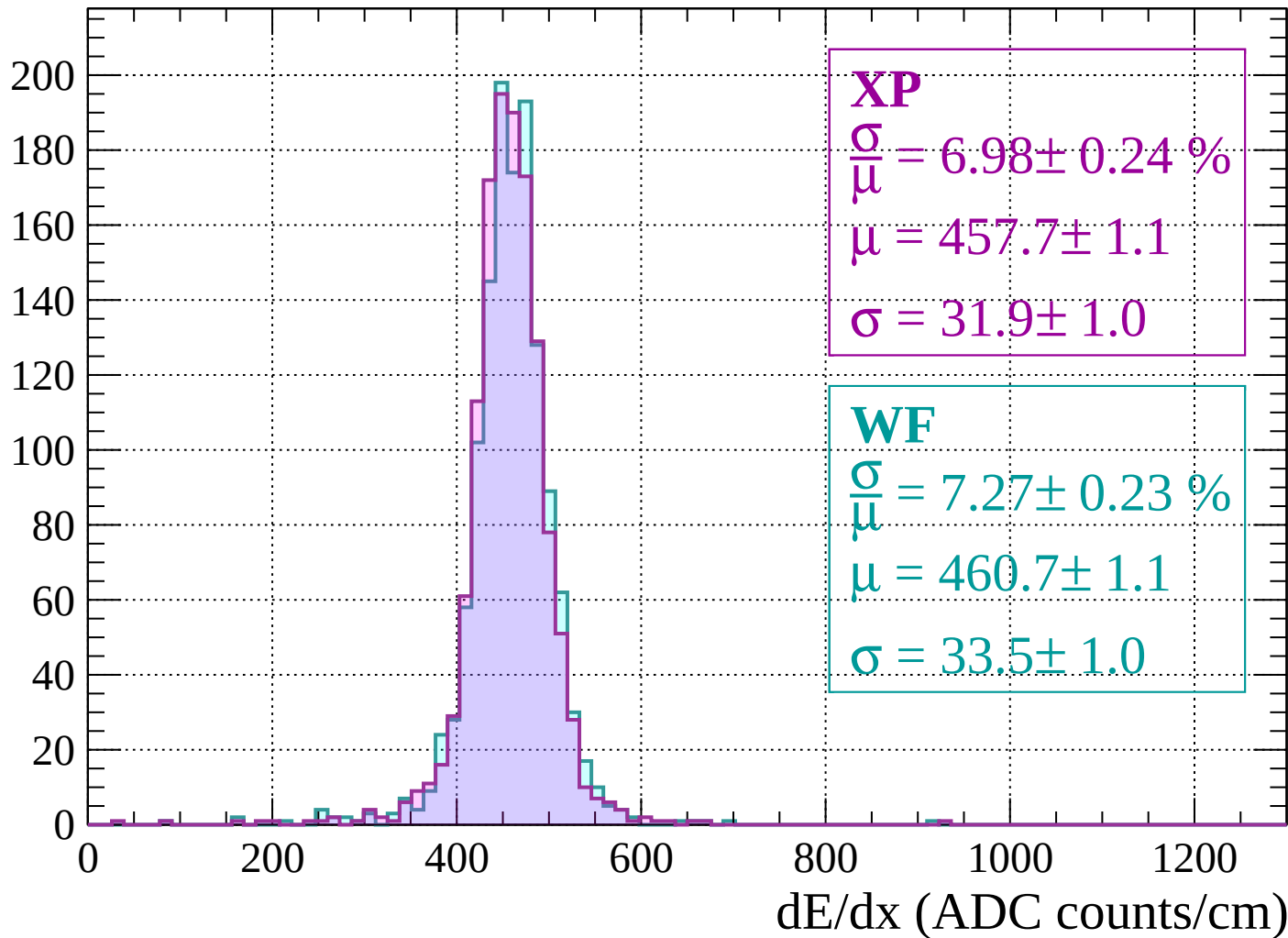
$$\mu = 461.0 \pm 1.1$$

$$\sigma = 31.7 \pm 1.3$$

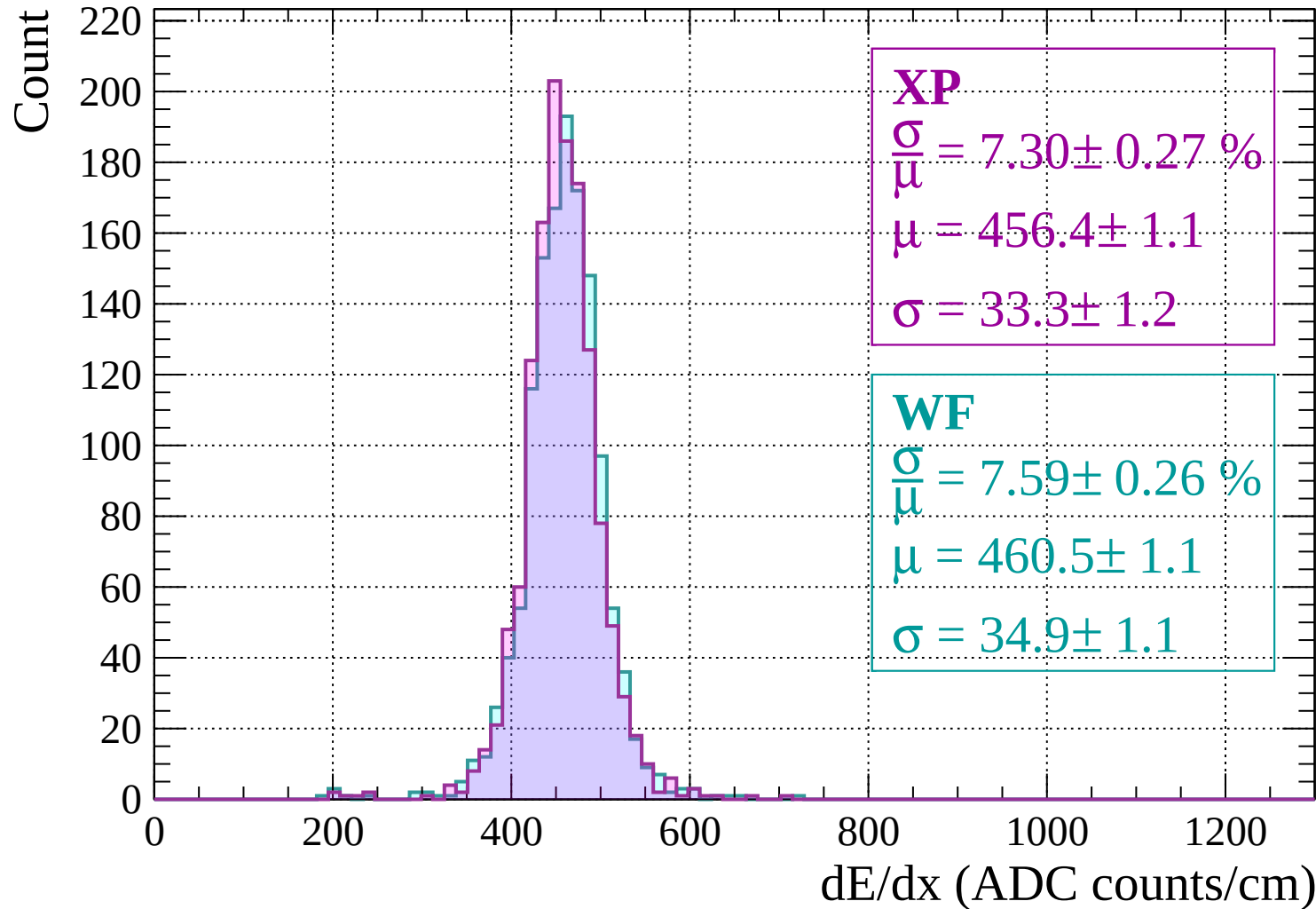
dE/dx (ADC counts/cm)

Energy loss |  $-1320 < p < -1280$

Count

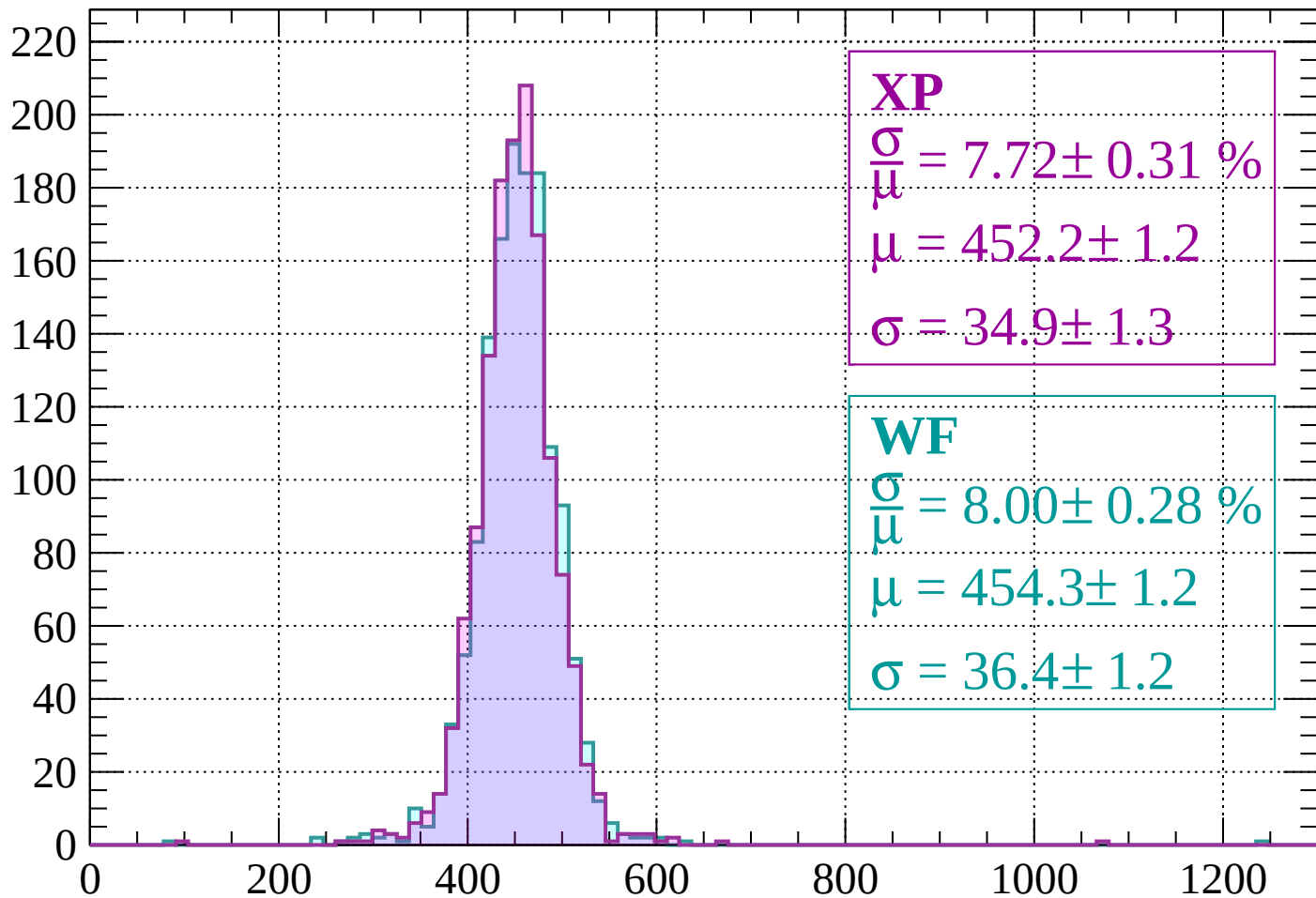


# Energy loss | -1280 < p < -1240



Energy loss |  $-1240 < p < -1200$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.72 \pm 0.31 \%$$

$$\mu = 452.2 \pm 1.2$$

$$\sigma = 34.9 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 8.00 \pm 0.28 \%$$

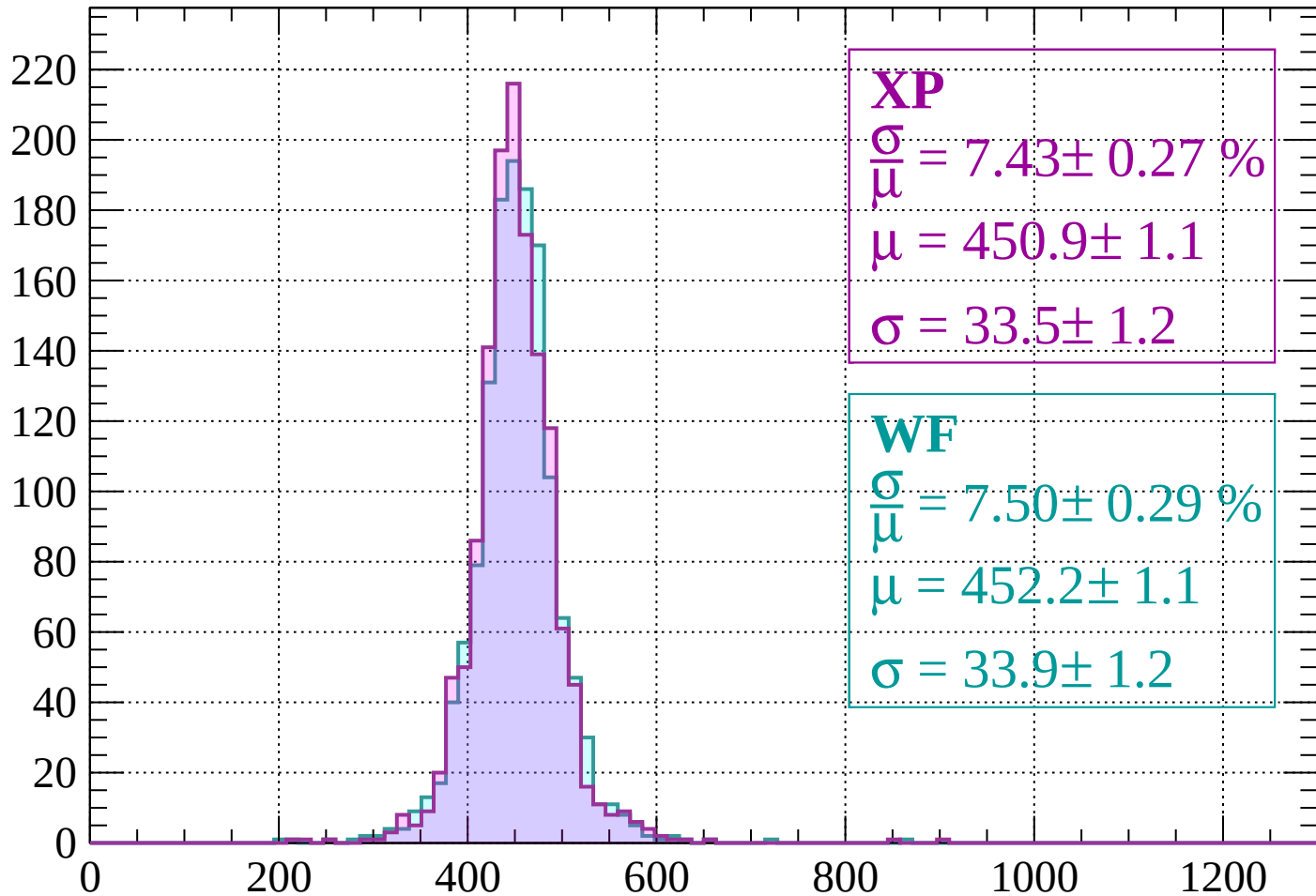
$$\mu = 454.3 \pm 1.2$$

$$\sigma = 36.4 \pm 1.2$$

$dE/dx$  (ADC counts/cm)

# Energy loss | -1200 < p < -1160

Count



**XP**

$$\frac{\sigma}{\mu} = 7.43 \pm 0.27 \%$$

$$\mu = 450.9 \pm 1.1$$

$$\sigma = 33.5 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 7.50 \pm 0.29 \%$$

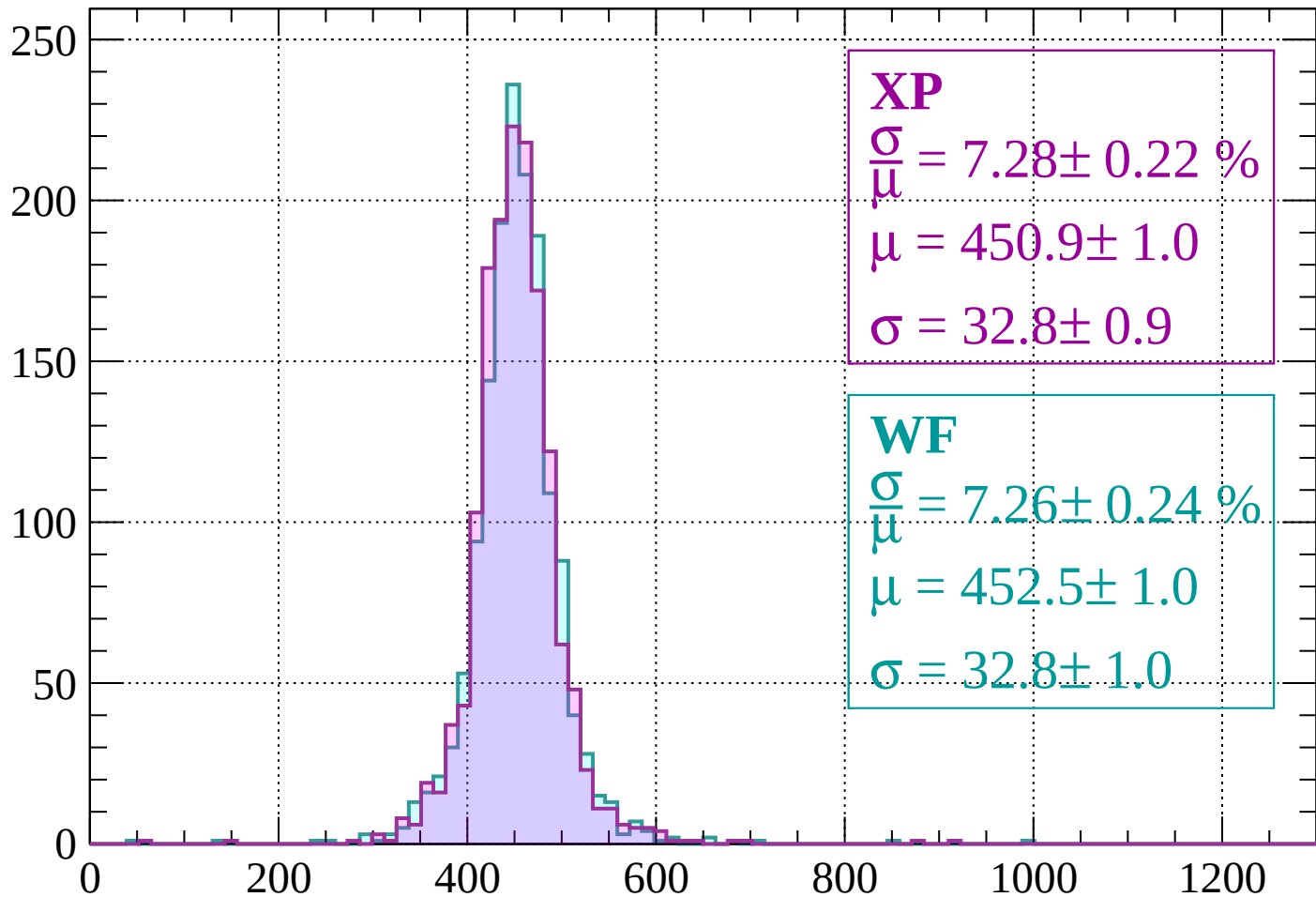
$$\mu = 452.2 \pm 1.1$$

$$\sigma = 33.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss |  $-1160 < p < -1120$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.28 \pm 0.22 \%$$

$$\mu = 450.9 \pm 1.0$$

$$\sigma = 32.8 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.26 \pm 0.24 \%$$

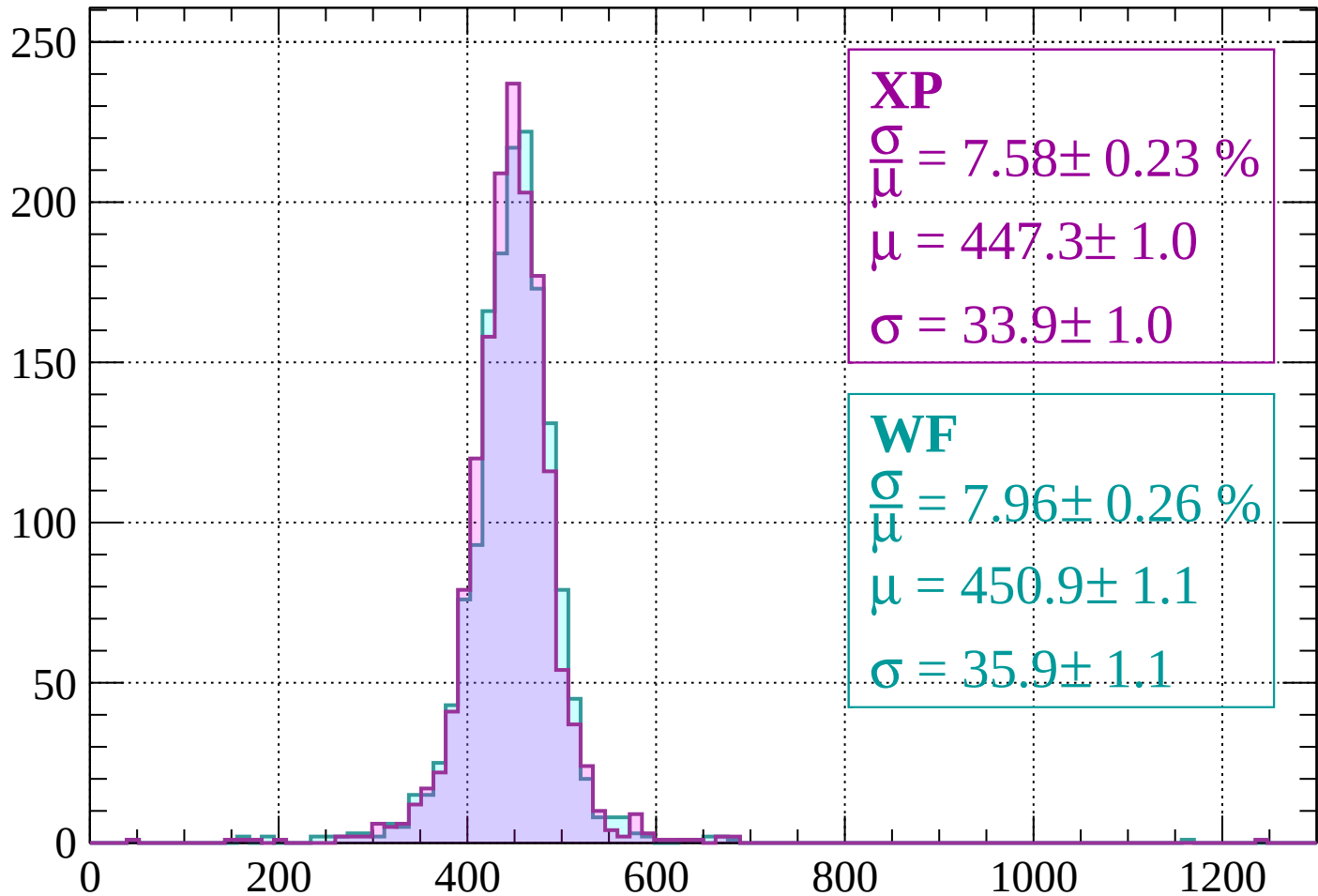
$$\mu = 452.5 \pm 1.0$$

$$\sigma = 32.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $-1120 < p < -1080$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.58 \pm 0.23 \%$$

$$\mu = 447.3 \pm 1.0$$

$$\sigma = 33.9 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 7.96 \pm 0.26 \%$$

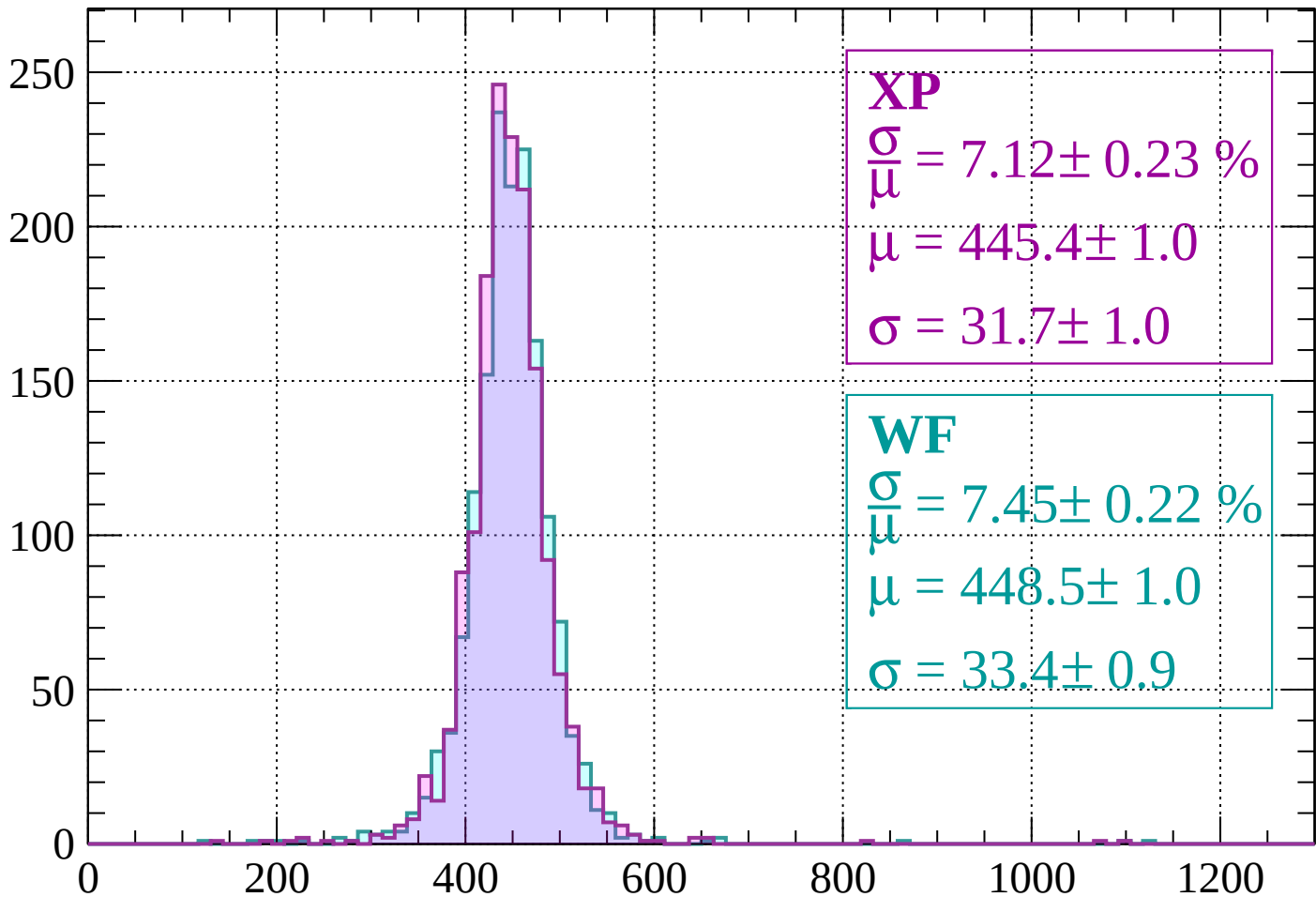
$$\mu = 450.9 \pm 1.1$$

$$\sigma = 35.9 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-1080 < p < -1040$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.12 \pm 0.23 \%$$

$$\mu = 445.4 \pm 1.0$$

$$\sigma = 31.7 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 7.45 \pm 0.22 \%$$

$$\mu = 448.5 \pm 1.0$$

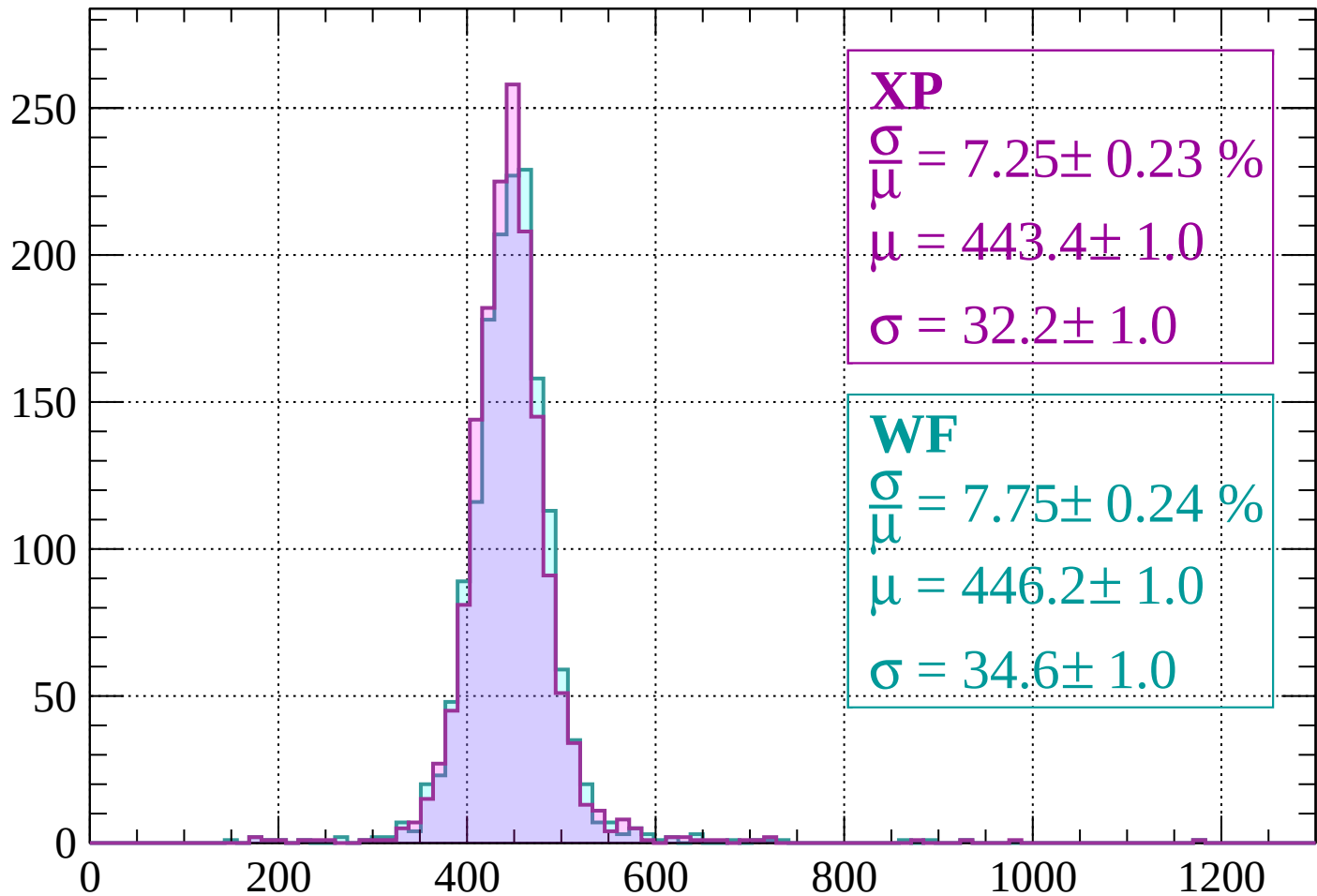
$$\sigma = 33.4 \pm 0.9$$

dE/dx (ADC counts/cm)



Energy loss |  $-1040 < p < -1000$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.25 \pm 0.23 \%$$

$$\mu = 443.4 \pm 1.0$$

$$\sigma = 32.2 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 7.75 \pm 0.24 \%$$

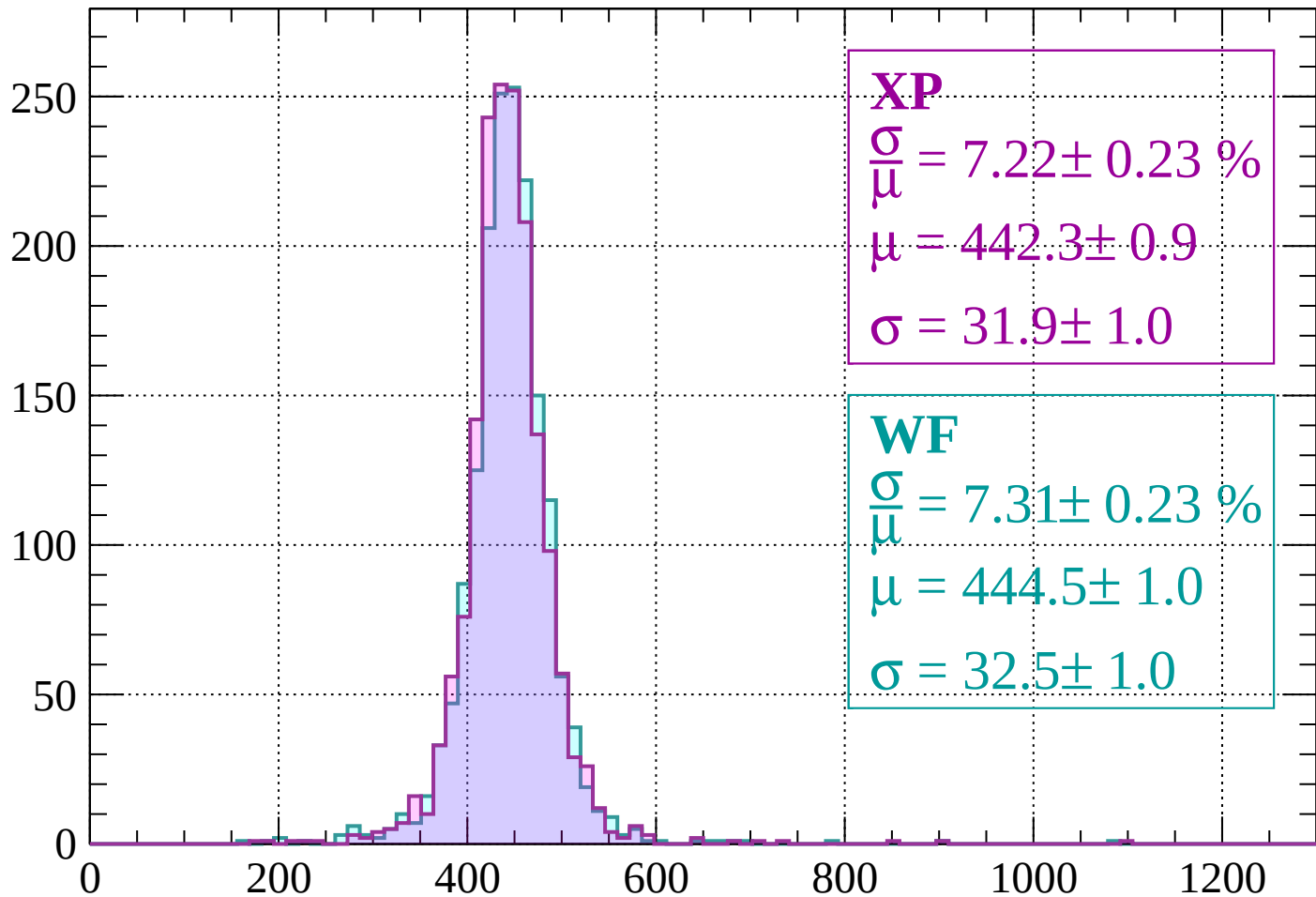
$$\mu = 446.2 \pm 1.0$$

$$\sigma = 34.6 \pm 1.0$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-1000 < p < -960$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.22 \pm 0.23 \%$$

$$\mu = 442.3 \pm 0.9$$

$$\sigma = 31.9 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 7.31 \pm 0.23 \%$$

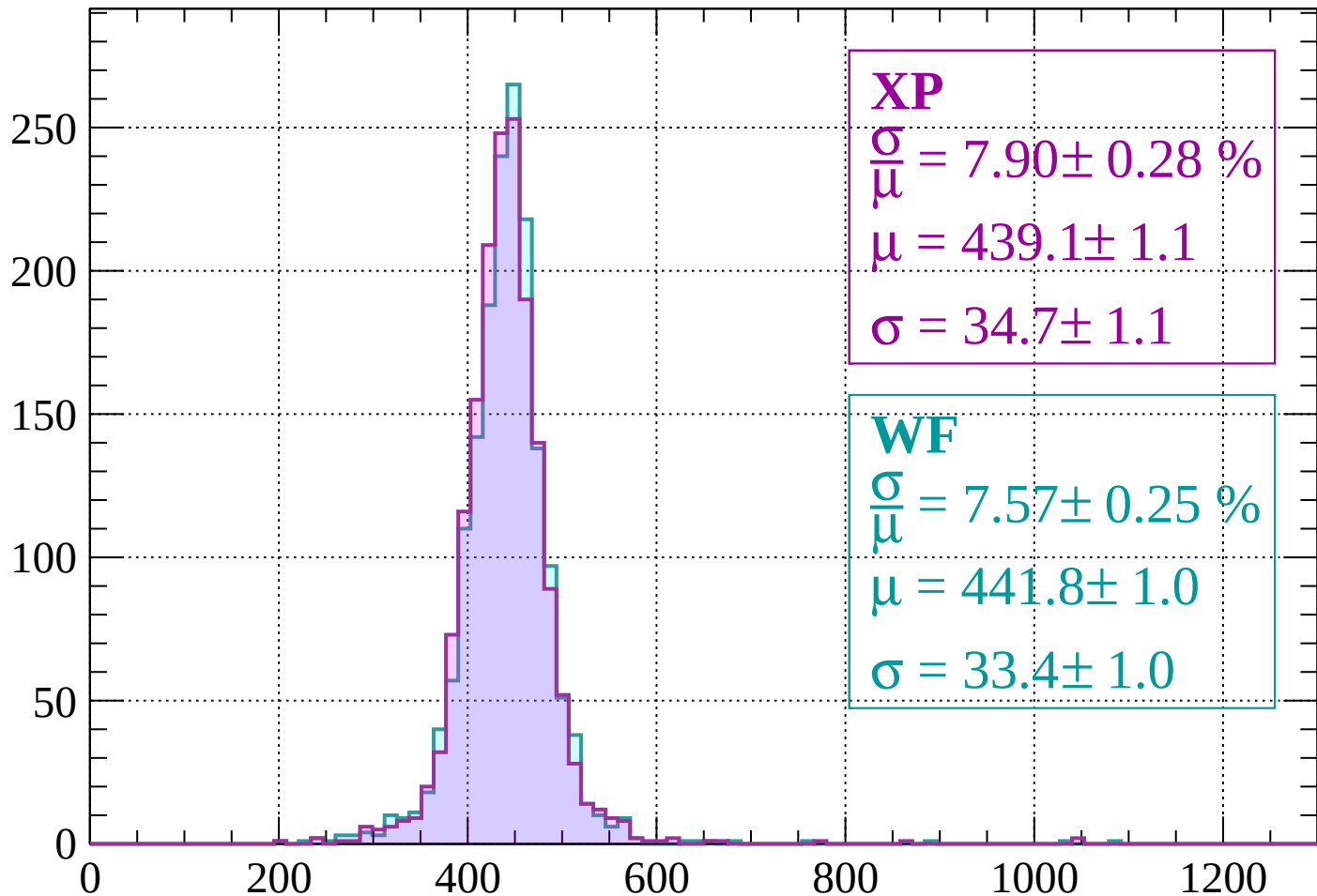
$$\mu = 444.5 \pm 1.0$$

$$\sigma = 32.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $-960 < p < -920$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.90 \pm 0.28 \%$$

$$\mu = 439.1 \pm 1.1$$

$$\sigma = 34.7 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 7.57 \pm 0.25 \%$$

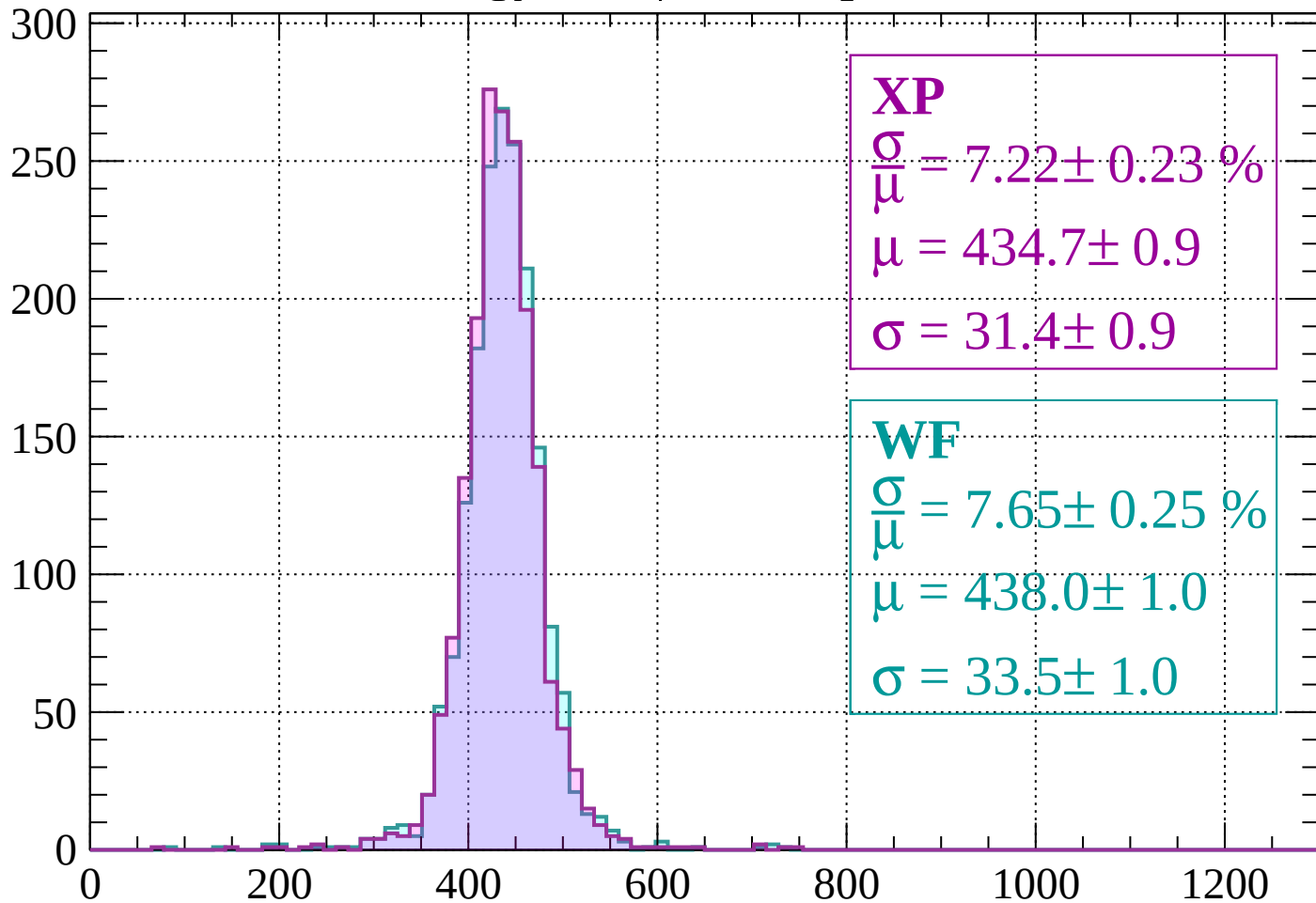
$$\mu = 441.8 \pm 1.0$$

$$\sigma = 33.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $-920 < p < -880$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.22 \pm 0.23 \%$$

$$\mu = 434.7 \pm 0.9$$

$$\sigma = 31.4 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.65 \pm 0.25 \%$$

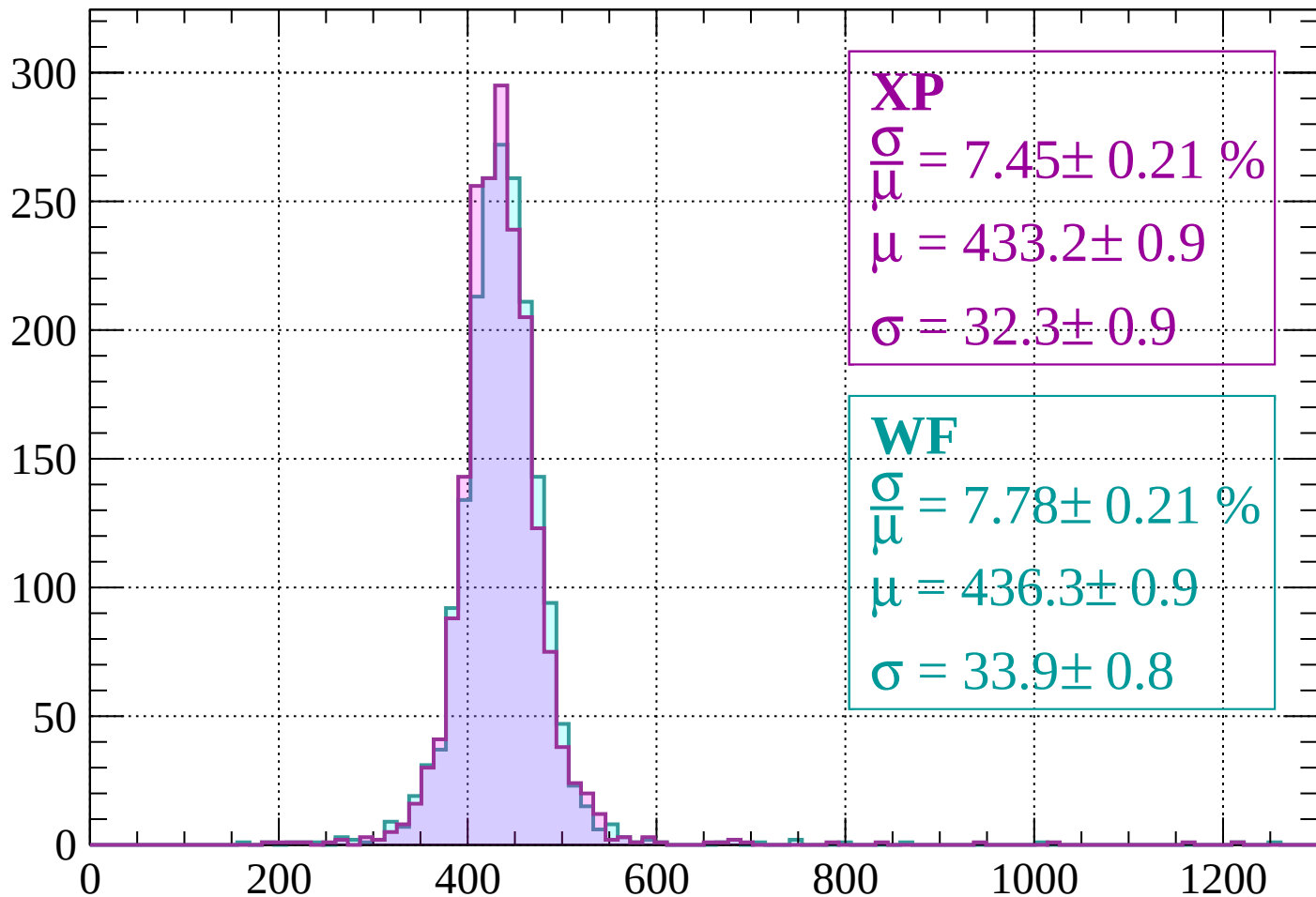
$$\mu = 438.0 \pm 1.0$$

$$\sigma = 33.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $-880 < p < -840$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.45 \pm 0.21 \%$$

$$\mu = 433.2 \pm 0.9$$

$$\sigma = 32.3 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.78 \pm 0.21 \%$$

$$\mu = 436.3 \pm 0.9$$

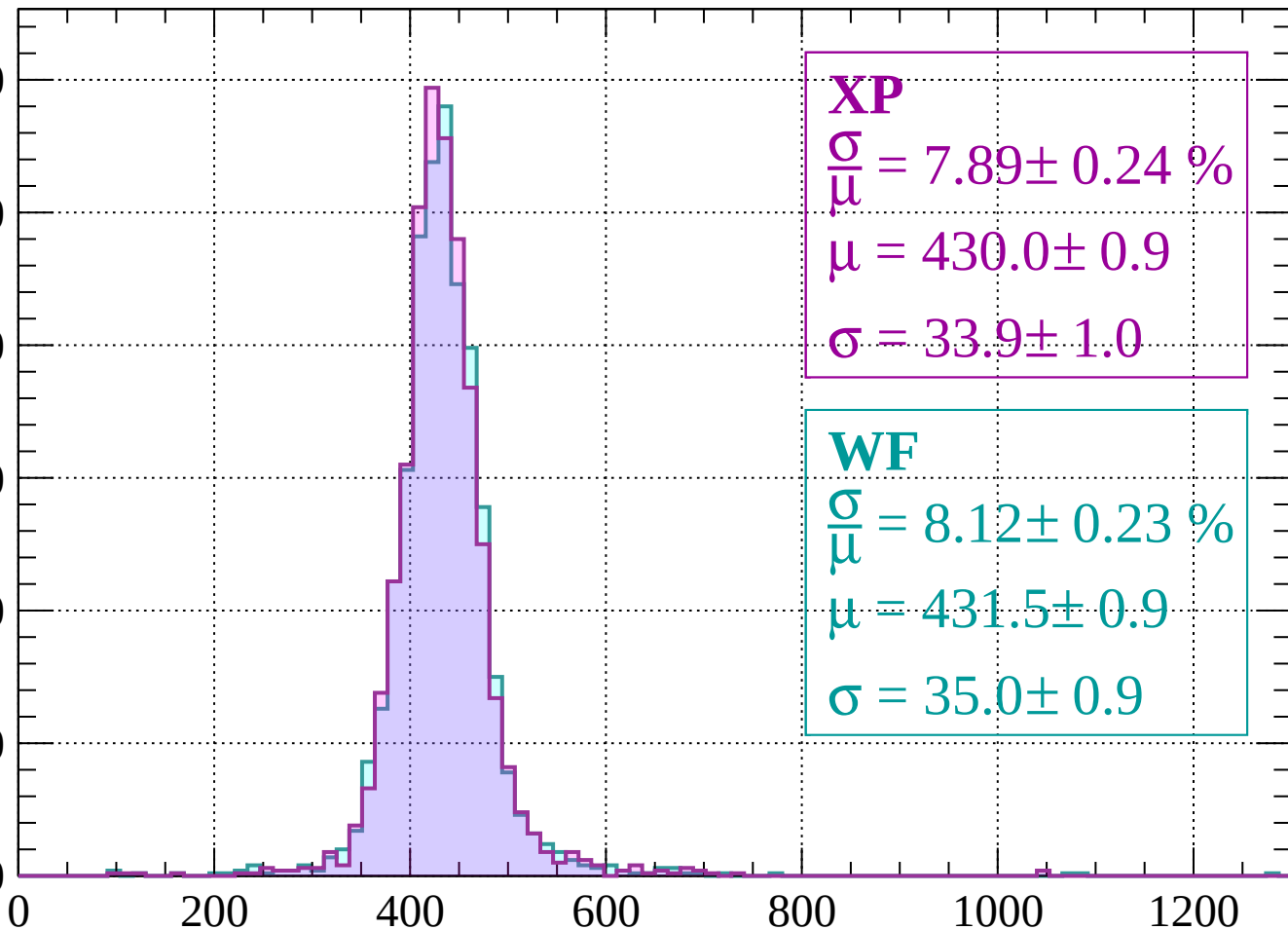
$$\sigma = 33.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-840 < p < -800$

Count

300  
250  
200  
150  
100  
50  
0



**XP**

$$\frac{\sigma}{\mu} = 7.89 \pm 0.24 \%$$

$$\mu = 430.0 \pm 0.9$$

$$\sigma = 33.9 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 8.12 \pm 0.23 \%$$

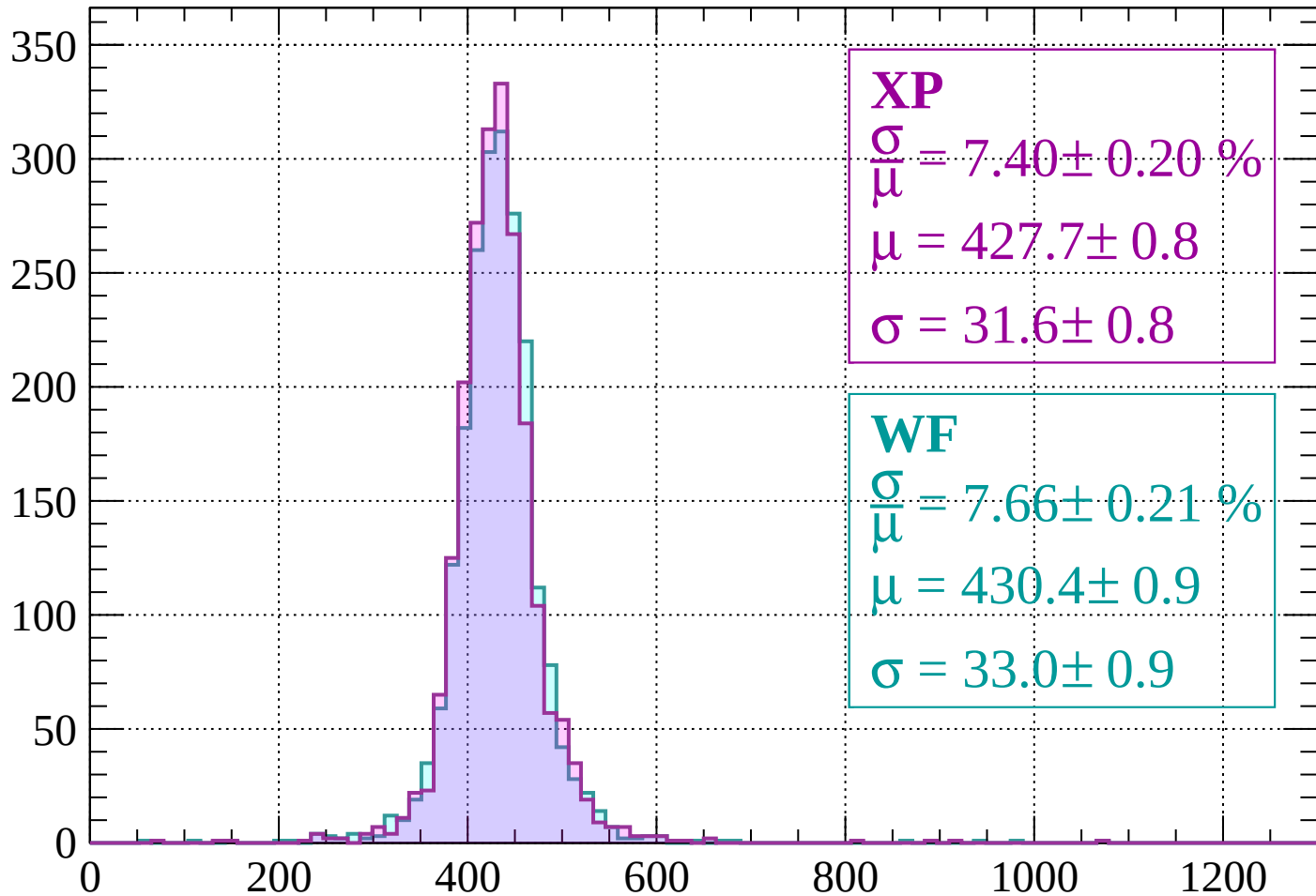
$$\mu = 431.5 \pm 0.9$$

$$\sigma = 35.0 \pm 0.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $-800 < p < -760$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.40 \pm 0.20 \%$$

$$\mu = 427.7 \pm 0.8$$

$$\sigma = 31.6 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.66 \pm 0.21 \%$$

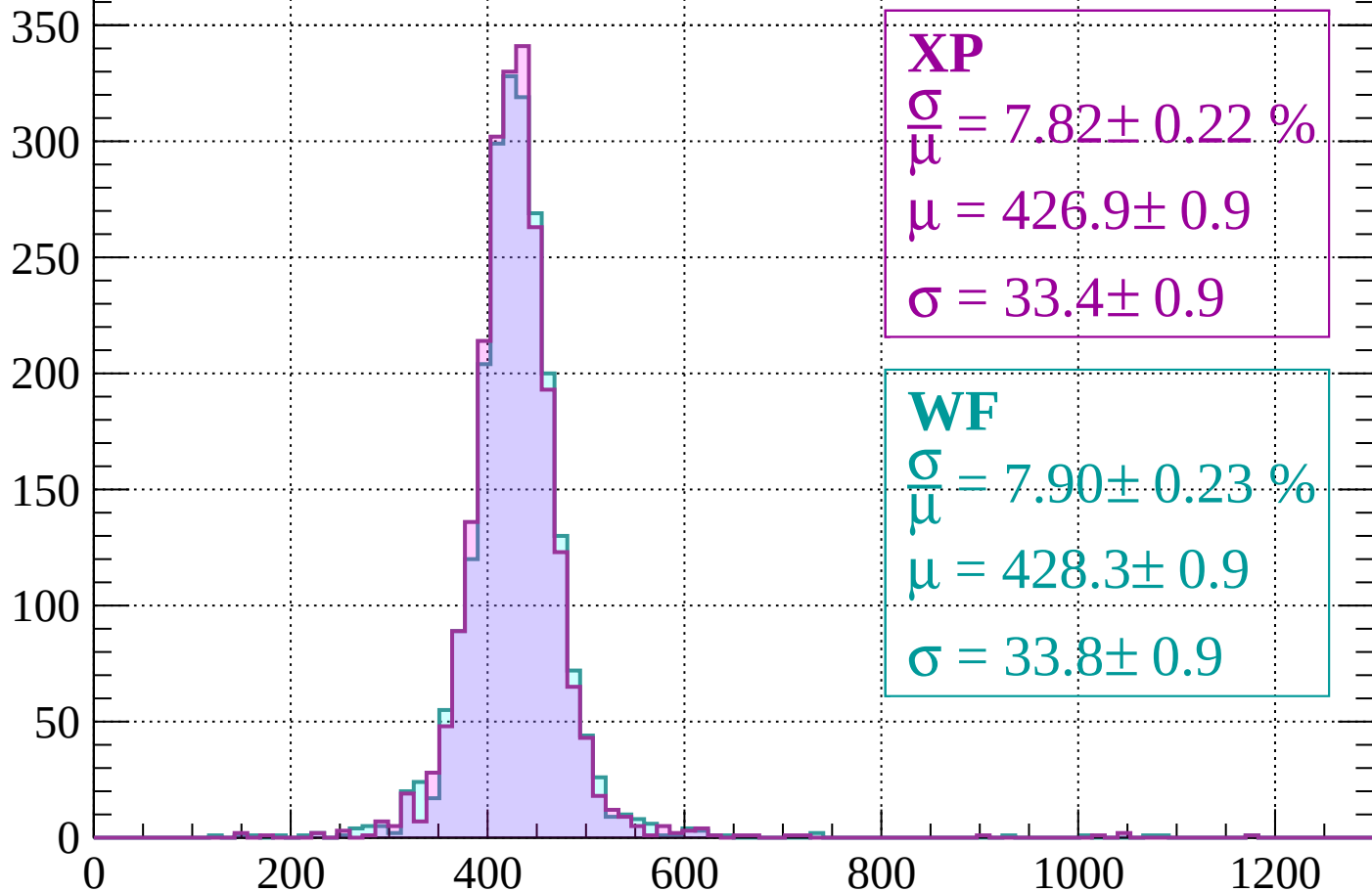
$$\mu = 430.4 \pm 0.9$$

$$\sigma = 33.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-760 < p < -720$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.82 \pm 0.22 \%$$

$$\mu = 426.9 \pm 0.9$$

$$\sigma = 33.4 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.90 \pm 0.23 \%$$

$$\mu = 428.3 \pm 0.9$$

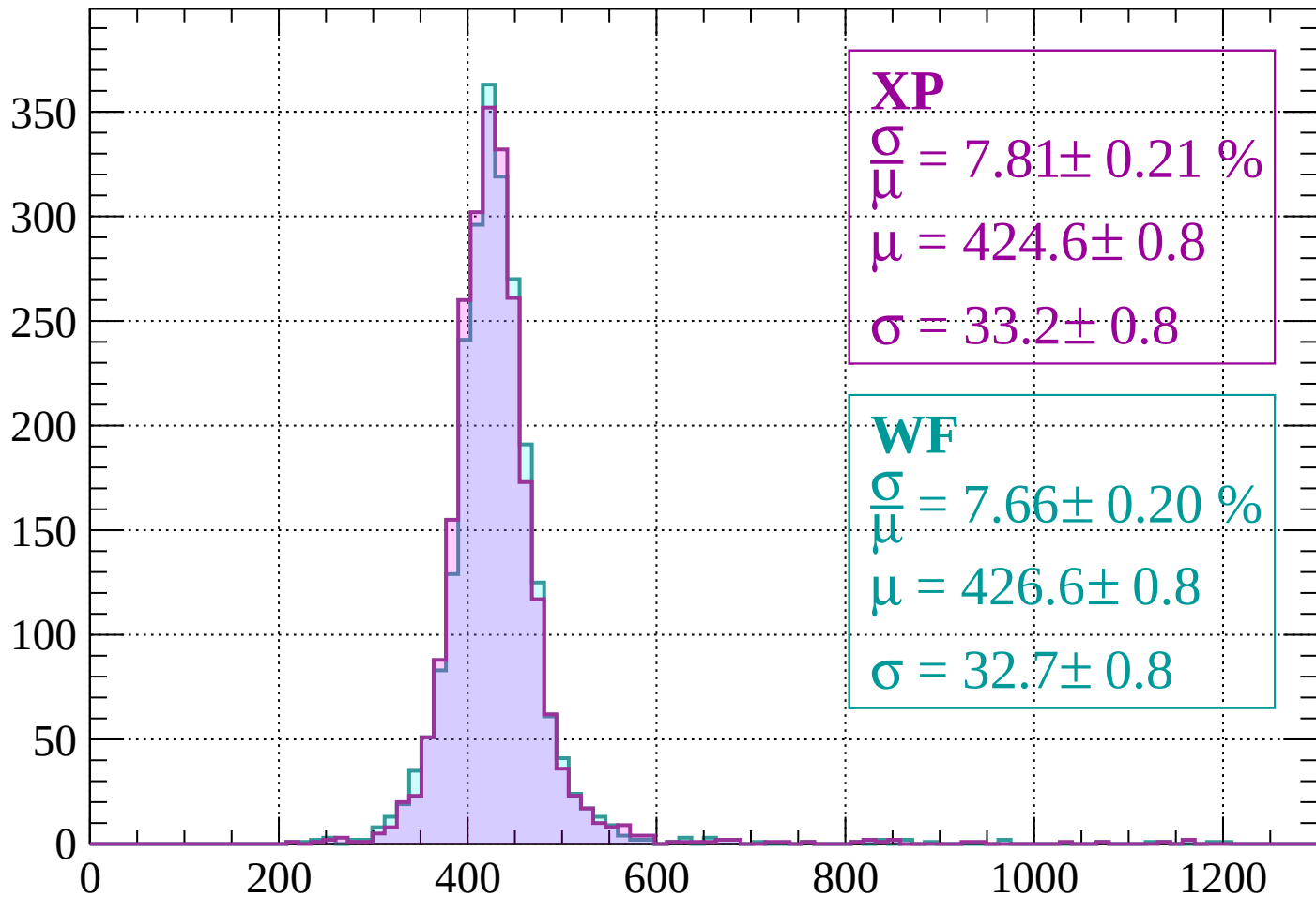
$$\sigma = 33.8 \pm 0.9$$

dE/dx (ADC counts/cm)



Energy loss |  $-720 < p < -680$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.81 \pm 0.21 \%$$

$$\mu = 424.6 \pm 0.8$$

$$\sigma = 33.2 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.66 \pm 0.20 \%$$

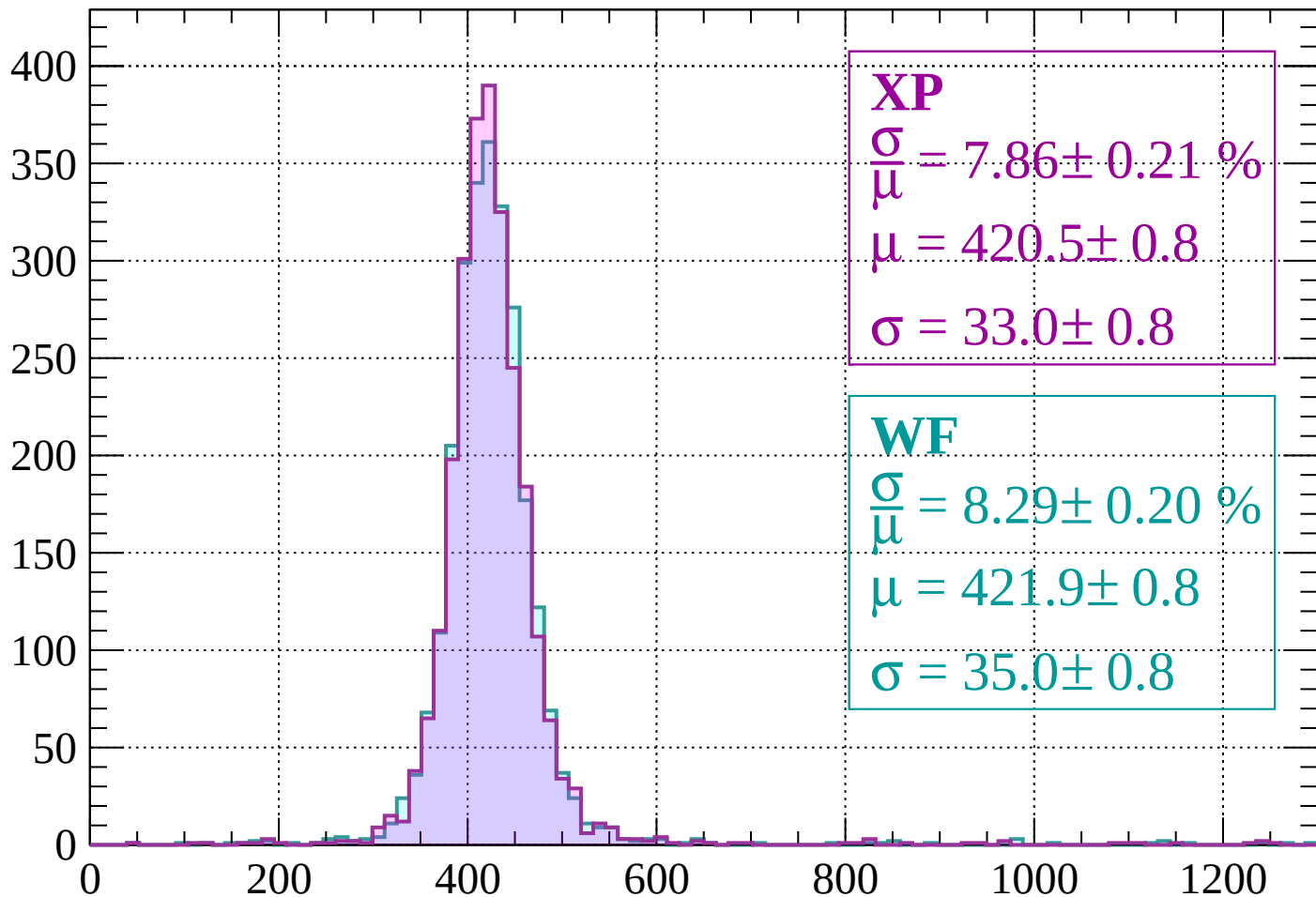
$$\mu = 426.6 \pm 0.8$$

$$\sigma = 32.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-680 < p < -640$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.86 \pm 0.21 \%$$

$$\mu = 420.5 \pm 0.8$$

$$\sigma = 33.0 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 8.29 \pm 0.20 \%$$

$$\mu = 421.9 \pm 0.8$$

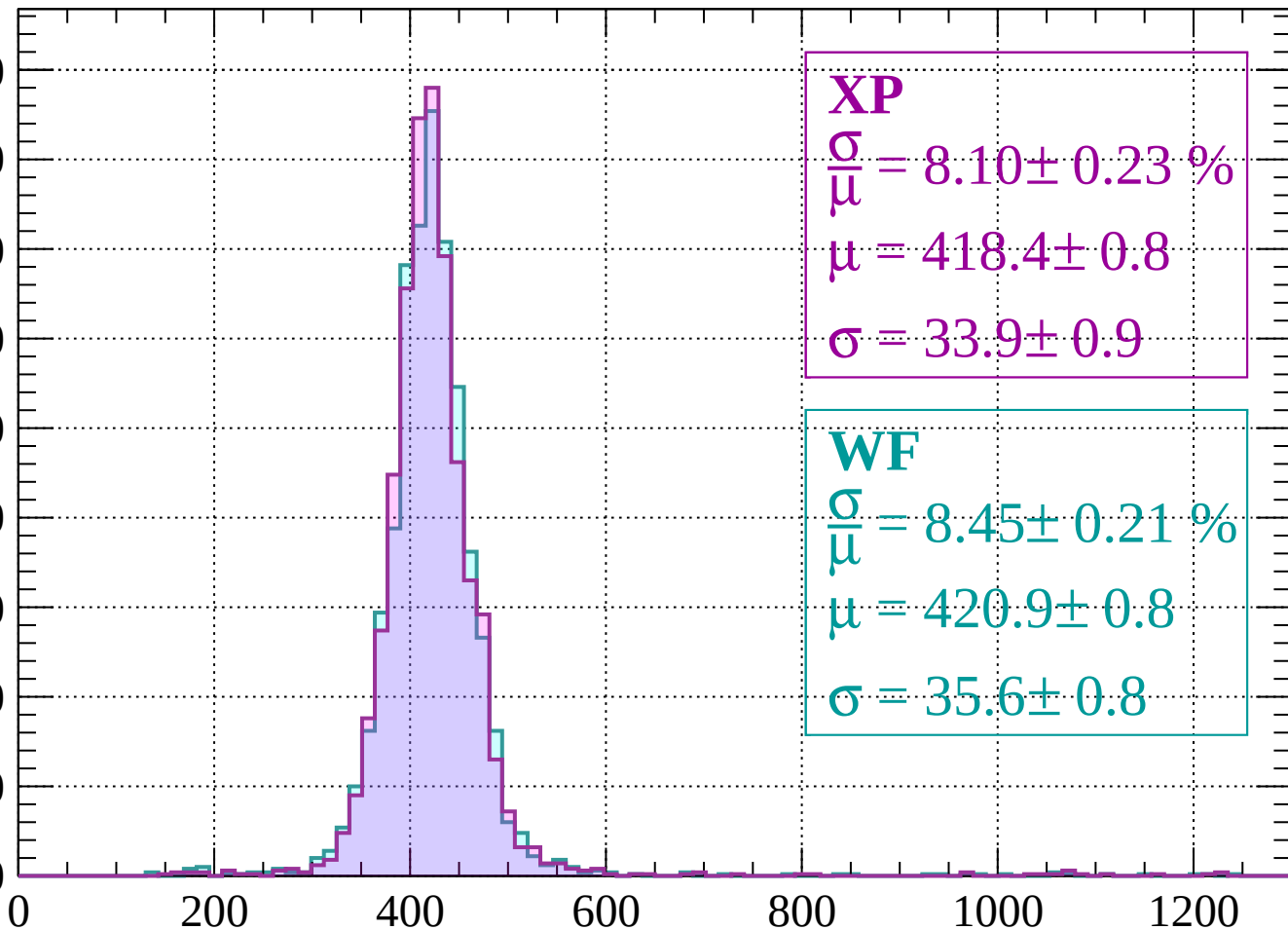
$$\sigma = 35.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-640 < p < -600$

Count

450  
400  
350  
300  
250  
200  
150  
100  
50  
0



**XP**

$$\frac{\sigma}{\bar{\mu}} = 8.10 \pm 0.23 \%$$

$$\mu = 418.4 \pm 0.8$$

$$\sigma = 33.9 \pm 0.9$$

**WF**

$$\frac{\sigma}{\bar{\mu}} = 8.45 \pm 0.21 \%$$

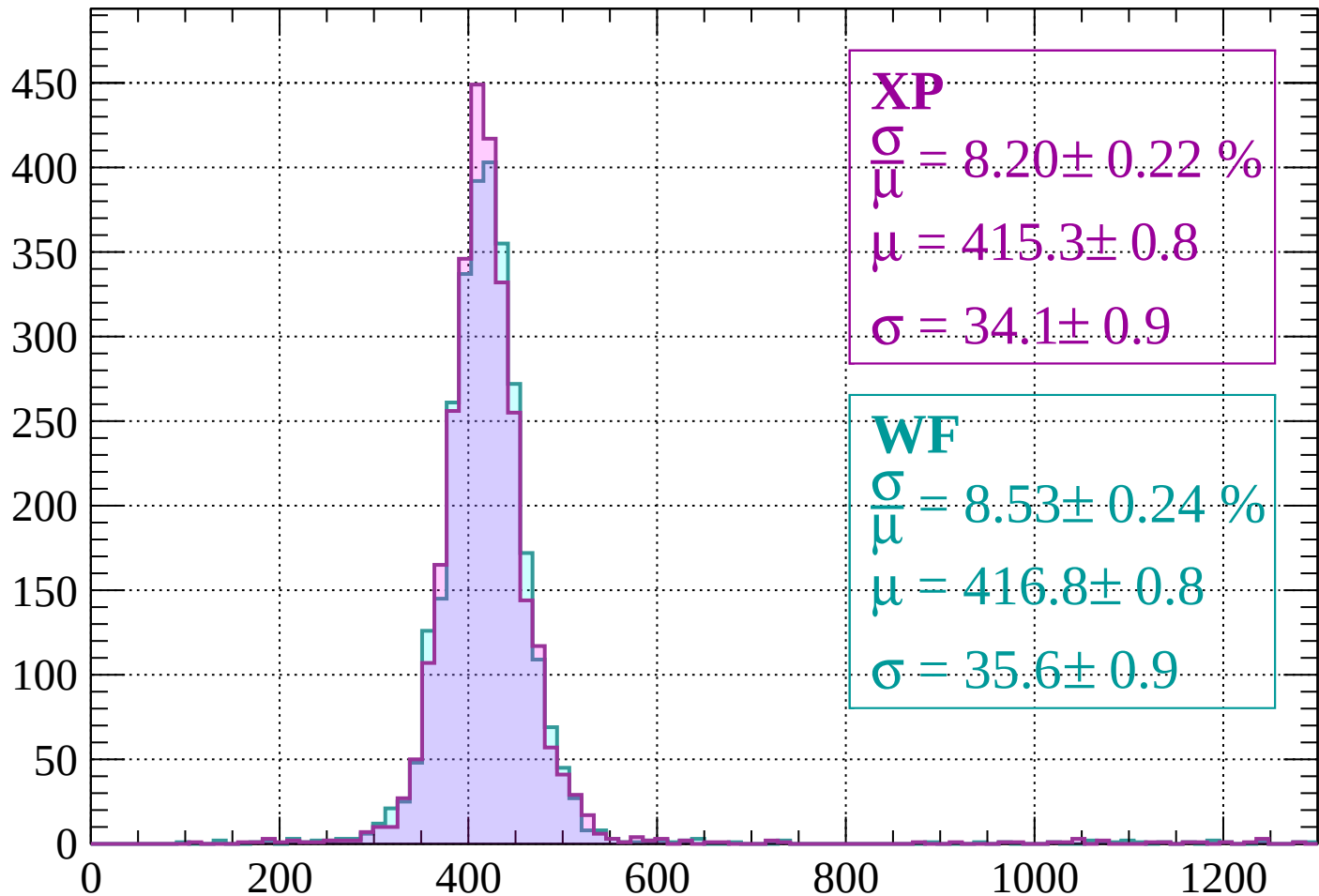
$$\mu = 420.9 \pm 0.8$$

$$\sigma = 35.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-600 < p < -560$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.20 \pm 0.22 \%$$

$$\mu = 415.3 \pm 0.8$$

$$\sigma = 34.1 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 8.53 \pm 0.24 \%$$

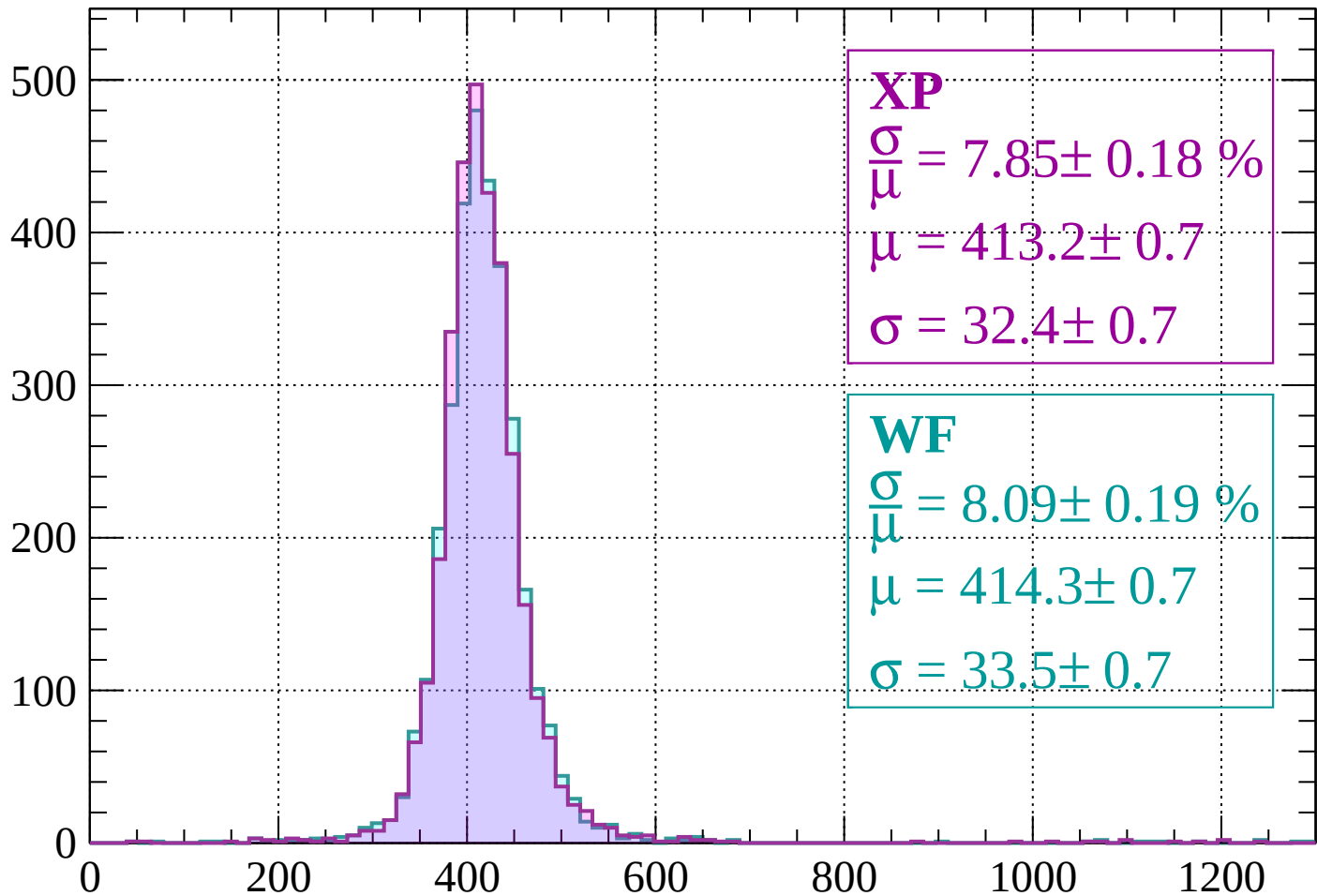
$$\mu = 416.8 \pm 0.8$$

$$\sigma = 35.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $-560 < p < -520$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.85 \pm 0.18 \%$$

$$\mu = 413.2 \pm 0.7$$

$$\sigma = 32.4 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.09 \pm 0.19 \%$$

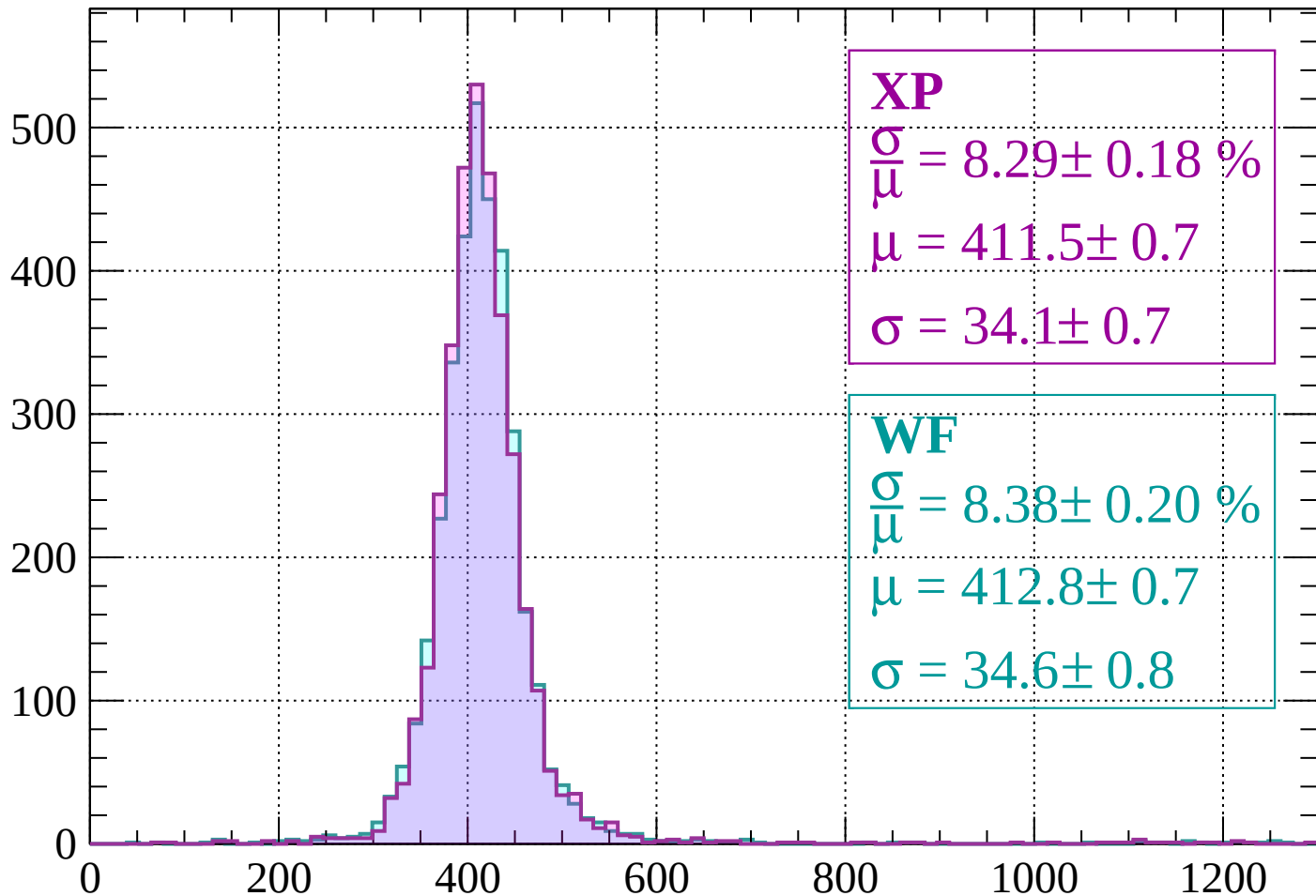
$$\mu = 414.3 \pm 0.7$$

$$\sigma = 33.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-520 < p < -480$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.29 \pm 0.18 \%$$

$$\mu = 411.5 \pm 0.7$$

$$\sigma = 34.1 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 8.38 \pm 0.20 \%$$

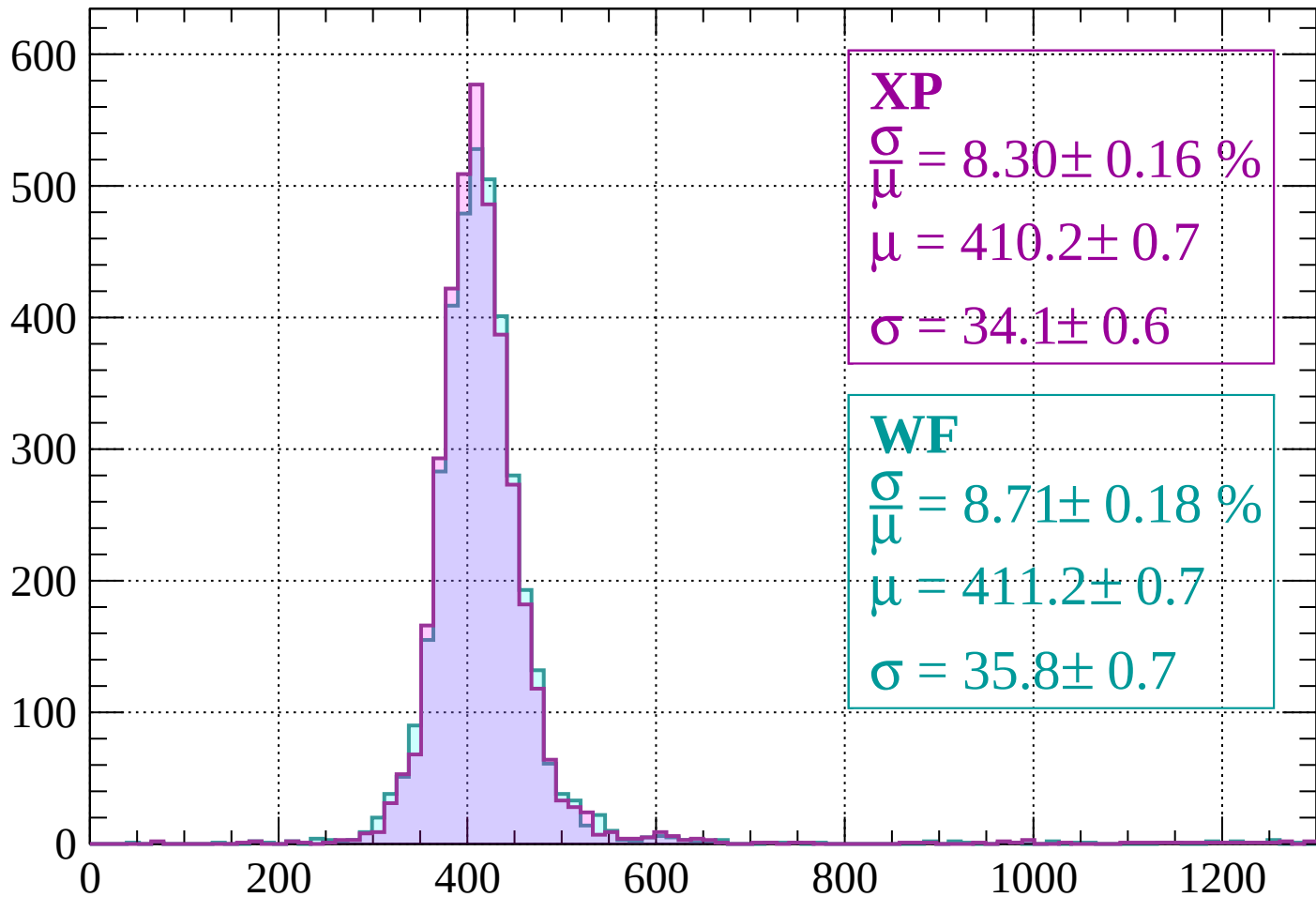
$$\mu = 412.8 \pm 0.7$$

$$\sigma = 34.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-480 < p < -440$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.30 \pm 0.16 \%$$

$$\mu = 410.2 \pm 0.7$$

$$\sigma = 34.1 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.71 \pm 0.18 \%$$

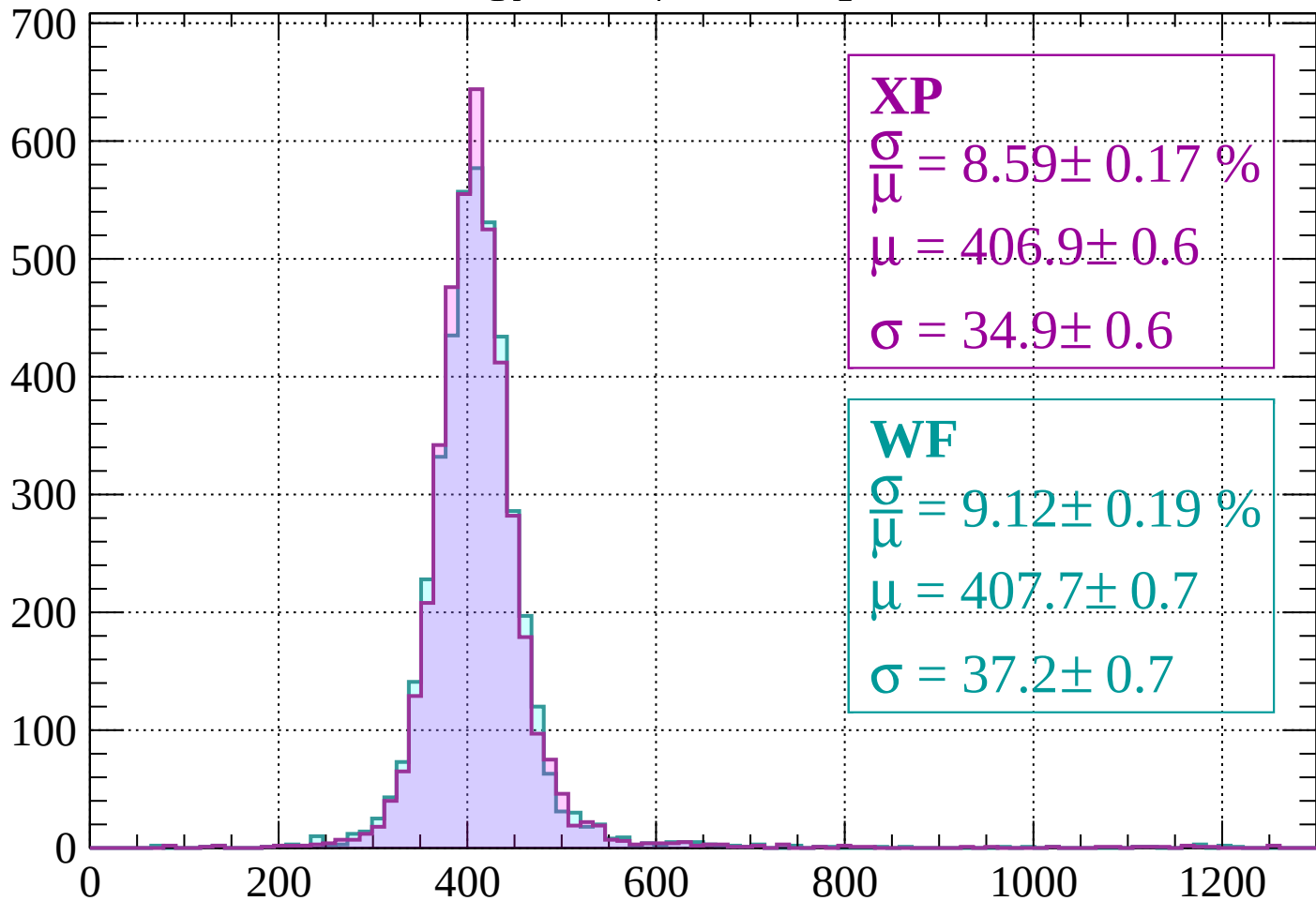
$$\mu = 411.2 \pm 0.7$$

$$\sigma = 35.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-440 < p < -400$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.59 \pm 0.17 \%$$

$$\mu = 406.9 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 9.12 \pm 0.19 \%$$

$$\mu = 407.7 \pm 0.7$$

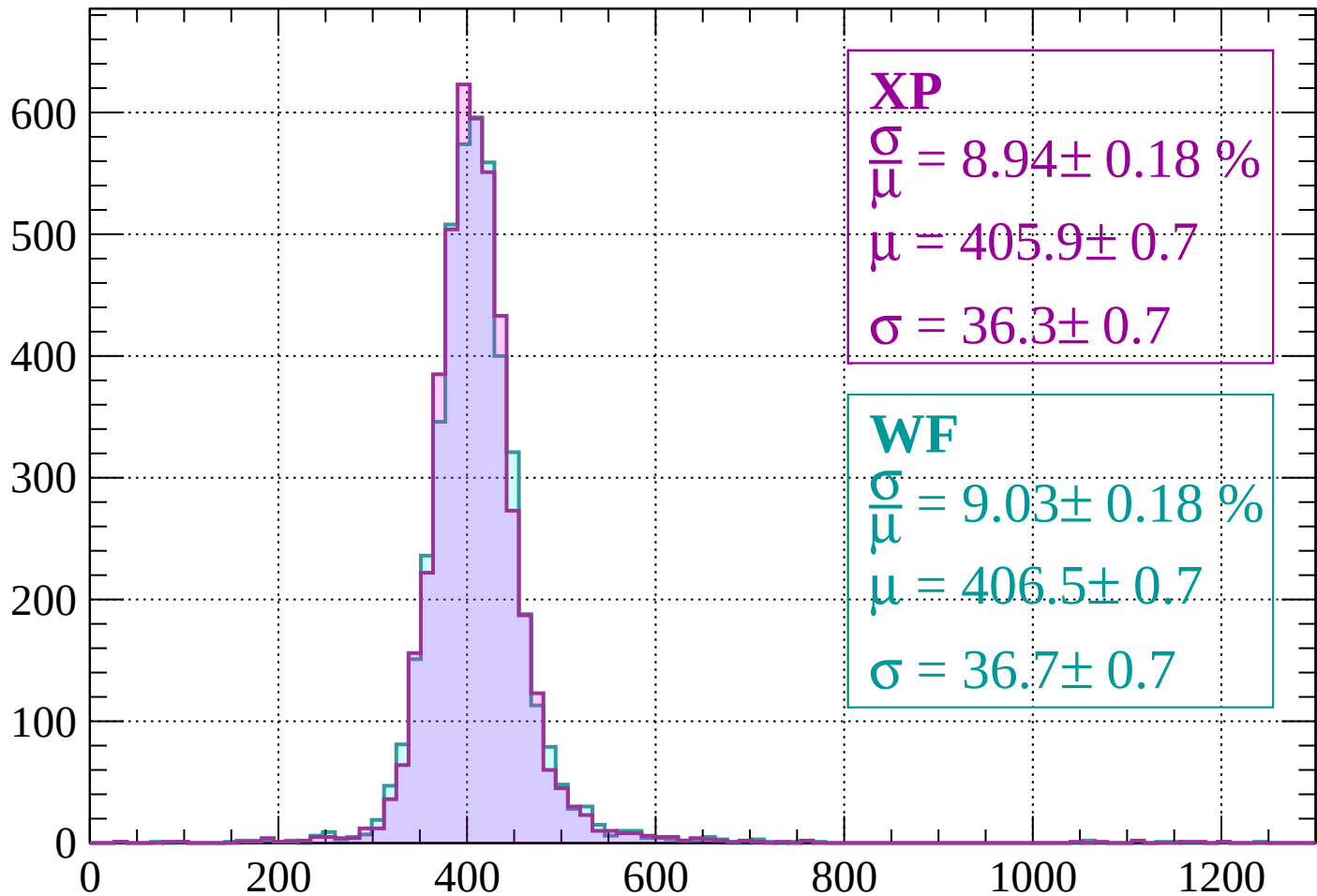
$$\sigma = 37.2 \pm 0.7$$

dE/dx (ADC counts/cm)



Energy loss |  $-400 < p < -360$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.94 \pm 0.18 \%$$

$$\mu = 405.9 \pm 0.7$$

$$\sigma = 36.3 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 9.03 \pm 0.18 \%$$

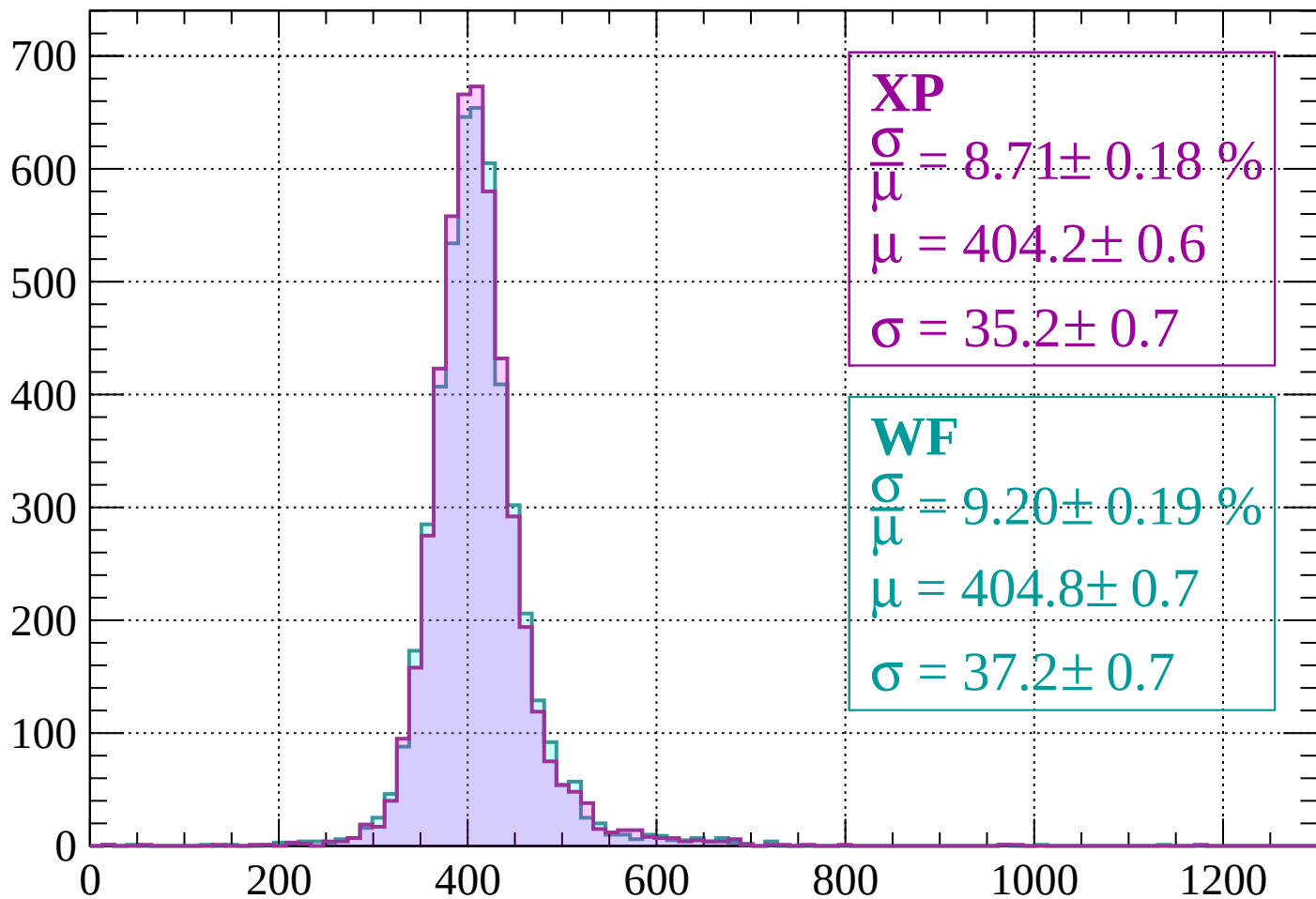
$$\mu = 406.5 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-360 < p < -320$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.71 \pm 0.18 \%$$

$$\mu = 404.2 \pm 0.6$$

$$\sigma = 35.2 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 9.20 \pm 0.19 \%$$

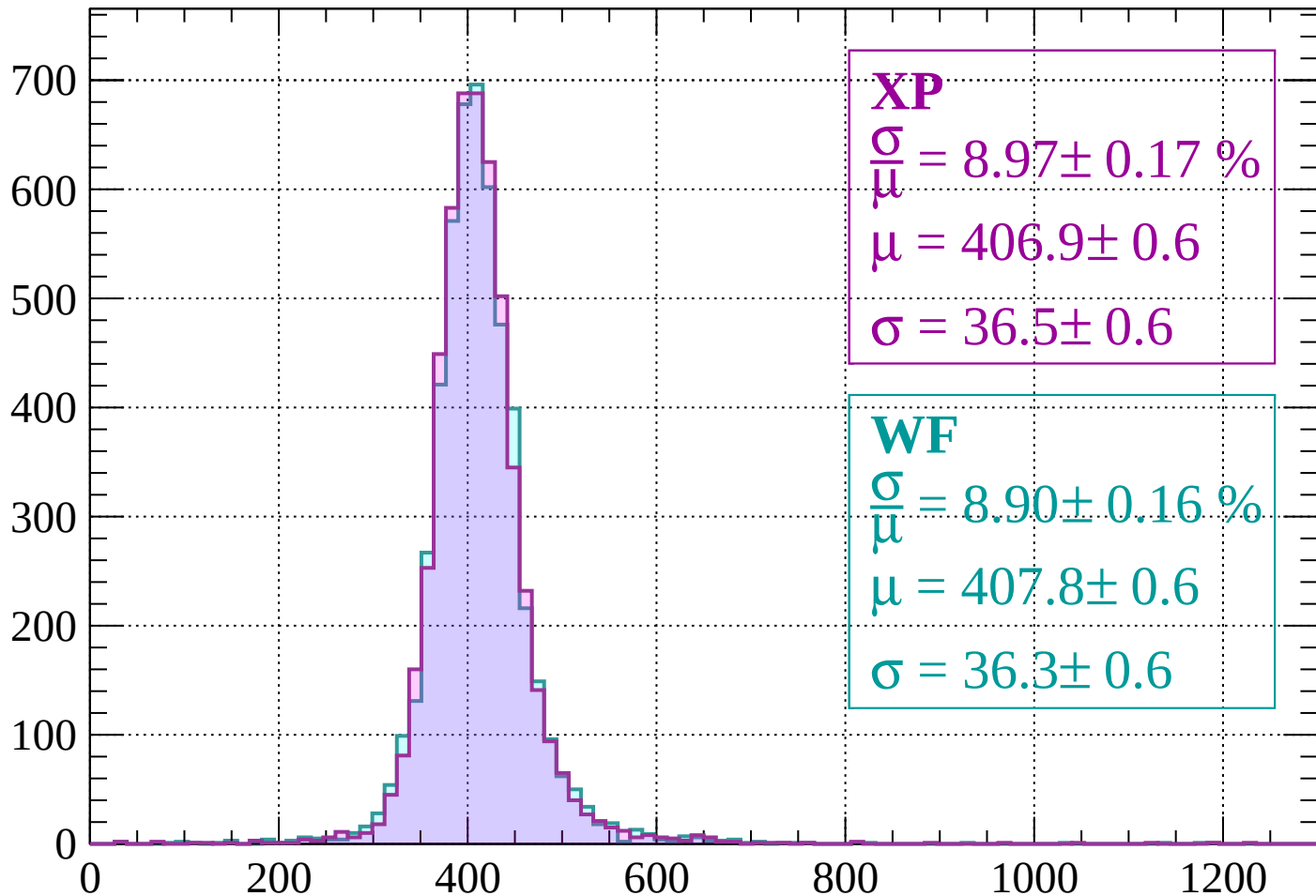
$$\mu = 404.8 \pm 0.7$$

$$\sigma = 37.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-320 < p < -280$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.97 \pm 0.17 \%$$

$$\mu = 406.9 \pm 0.6$$

$$\sigma = 36.5 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 8.90 \pm 0.16 \%$$

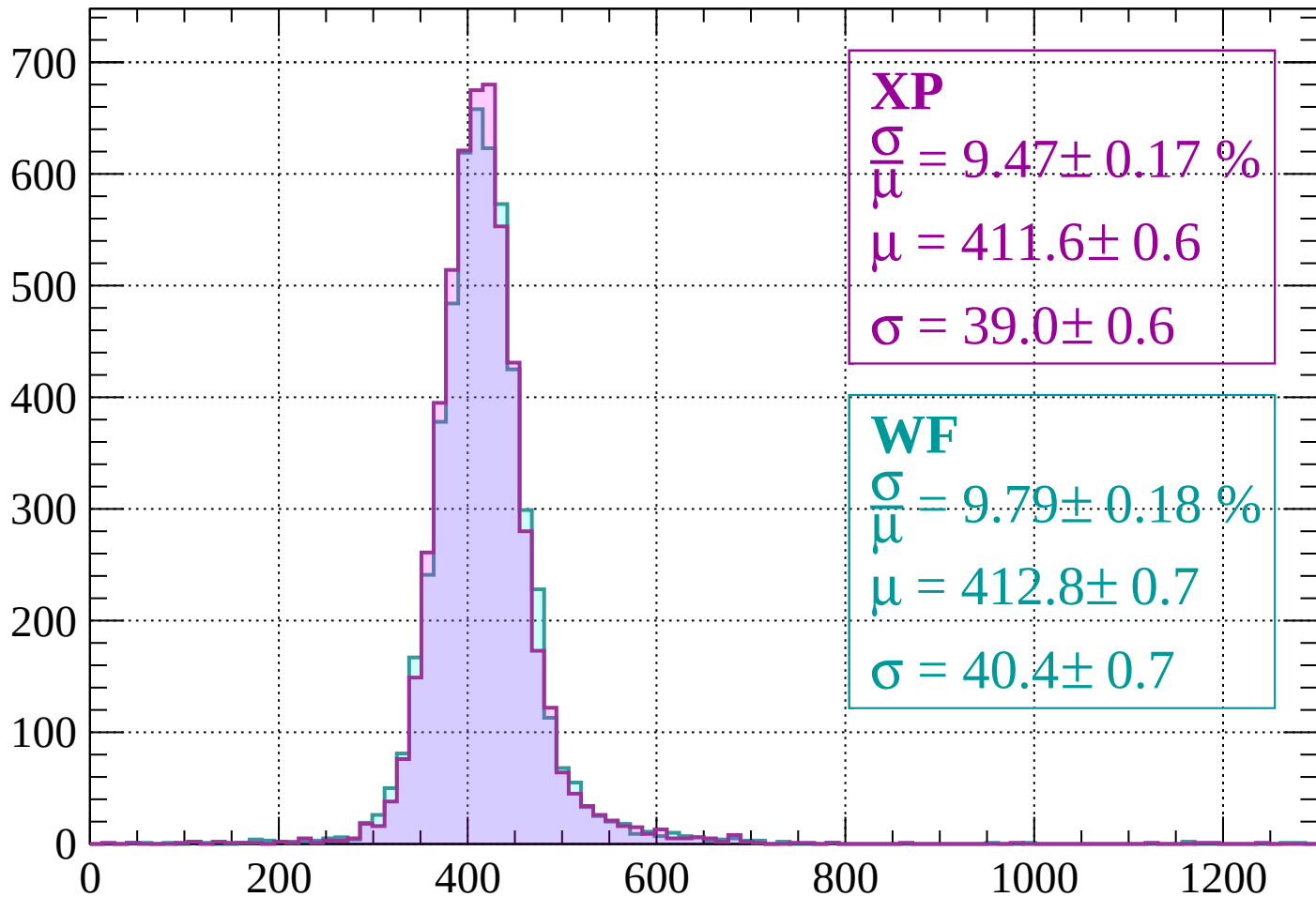
$$\mu = 407.8 \pm 0.6$$

$$\sigma = 36.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $-280 < p < -240$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.47 \pm 0.17 \%$$

$$\mu = 411.6 \pm 0.6$$

$$\sigma = 39.0 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 9.79 \pm 0.18 \%$$

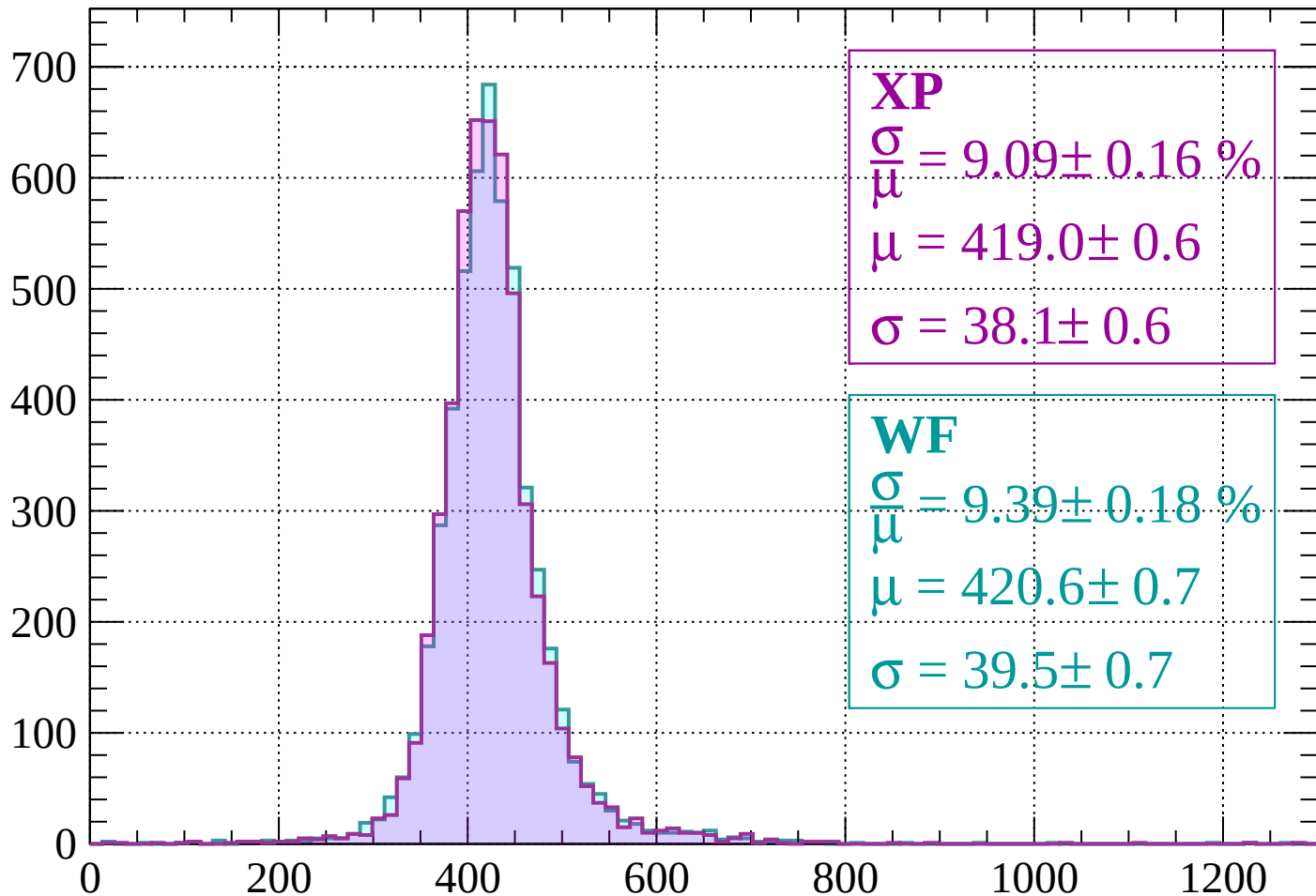
$$\mu = 412.8 \pm 0.7$$

$$\sigma = 40.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-240 < p < -200$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.09 \pm 0.16 \%$$

$$\mu = 419.0 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 9.39 \pm 0.18 \%$$

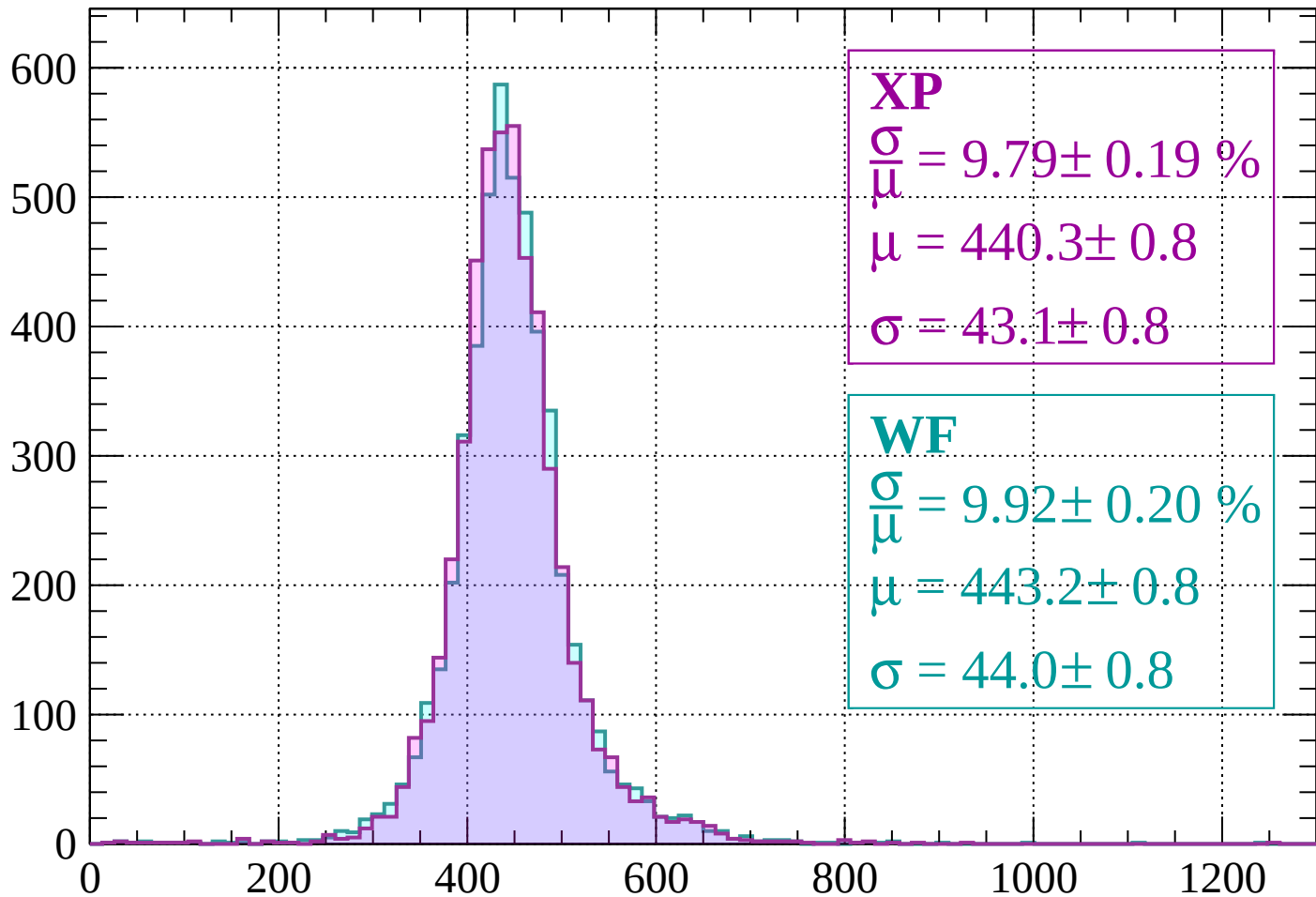
$$\mu = 420.6 \pm 0.7$$

$$\sigma = 39.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-200 < p < -160$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.79 \pm 0.19 \%$$

$$\mu = 440.3 \pm 0.8$$

$$\sigma = 43.1 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 9.92 \pm 0.20 \%$$

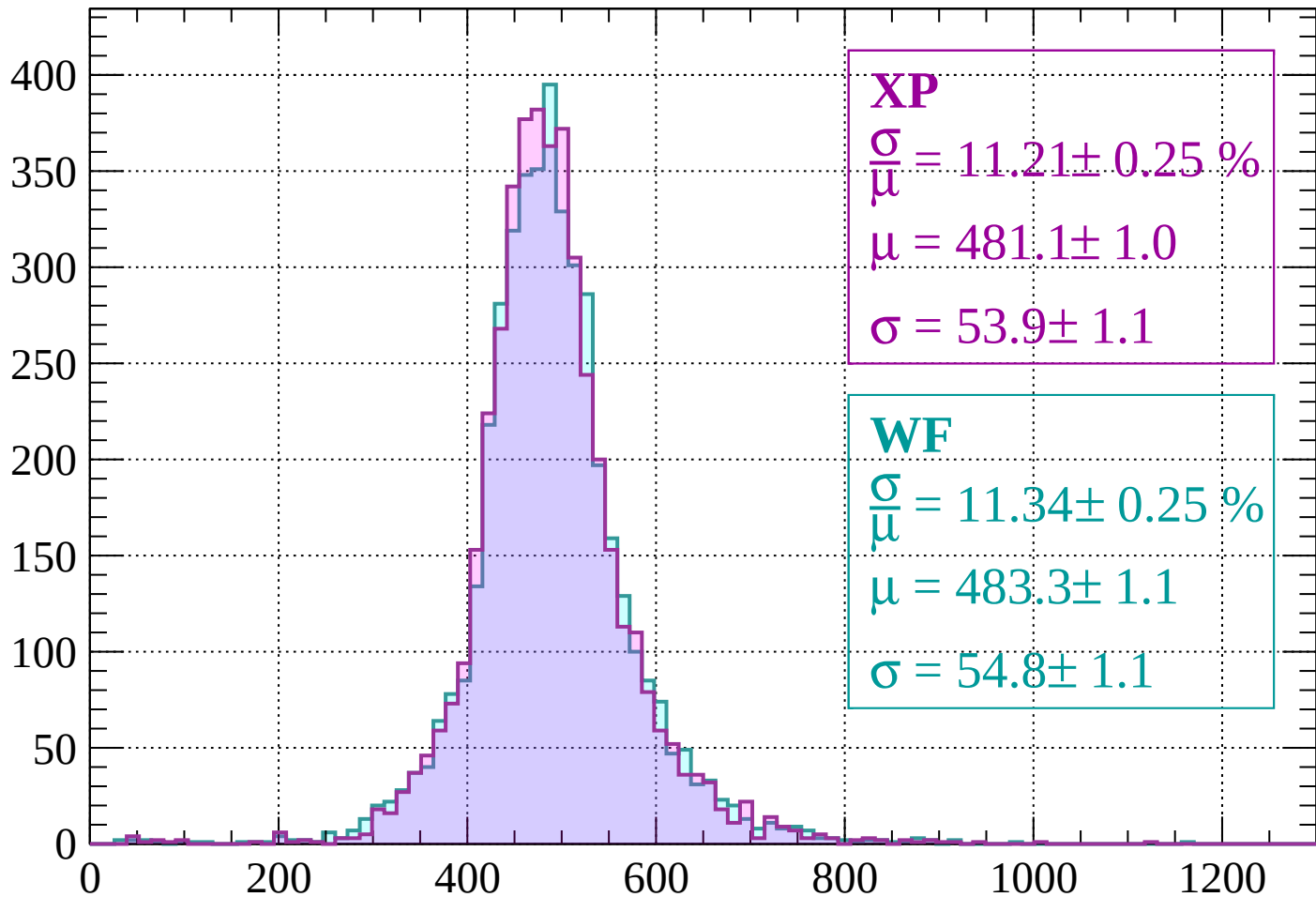
$$\mu = 443.2 \pm 0.8$$

$$\sigma = 44.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-160 < p < -120$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.21 \pm 0.25 \%$$

$$\mu = 481.1 \pm 1.0$$

$$\sigma = 53.9 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 11.34 \pm 0.25 \%$$

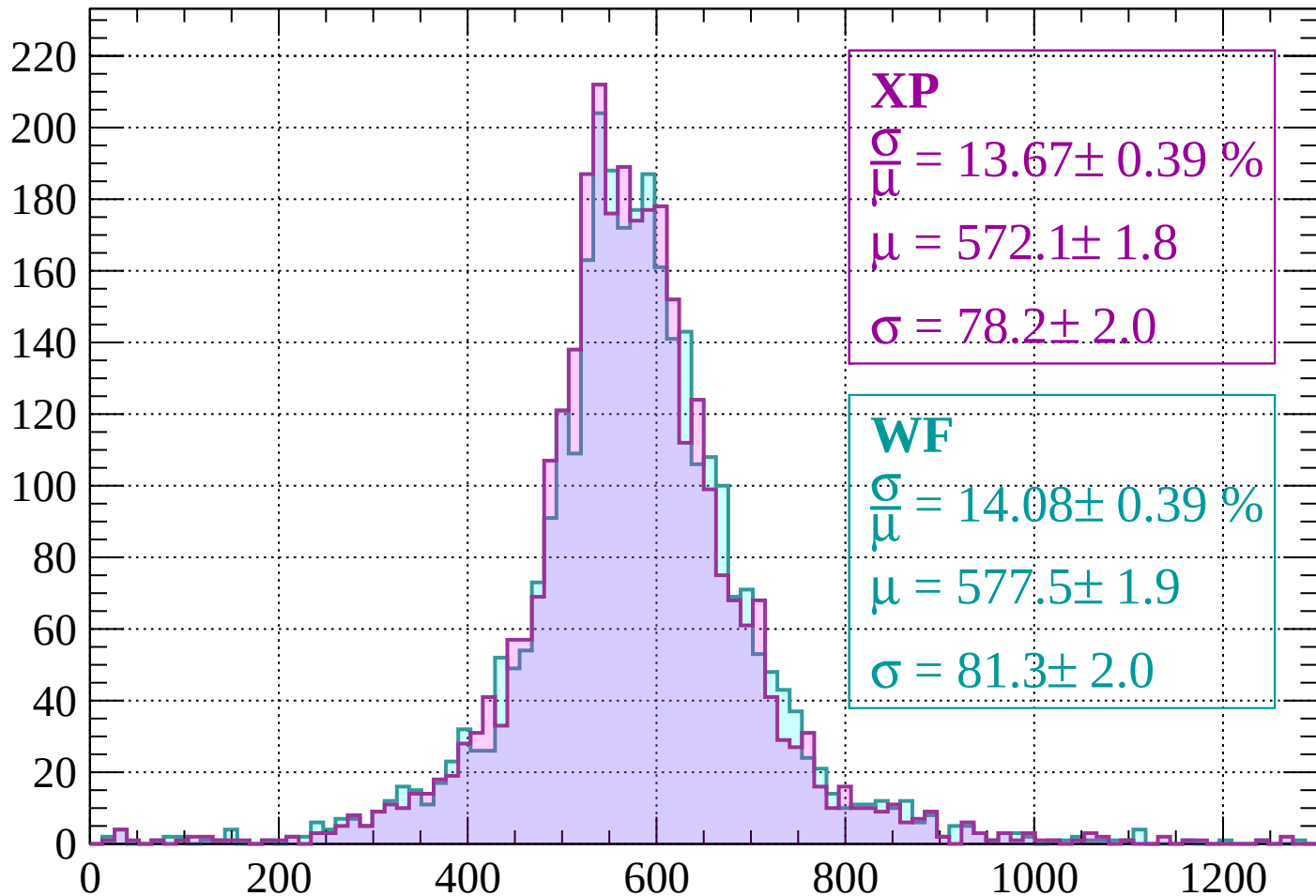
$$\mu = 483.3 \pm 1.1$$

$$\sigma = 54.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss |  $-120 < p < -80$

Count



**XP**

$$\frac{\sigma}{\mu} = 13.67 \pm 0.39 \%$$

$$\mu = 572.1 \pm 1.8$$

$$\sigma = 78.2 \pm 2.0$$

**WF**

$$\frac{\sigma}{\mu} = 14.08 \pm 0.39 \%$$

$$\mu = 577.5 \pm 1.9$$

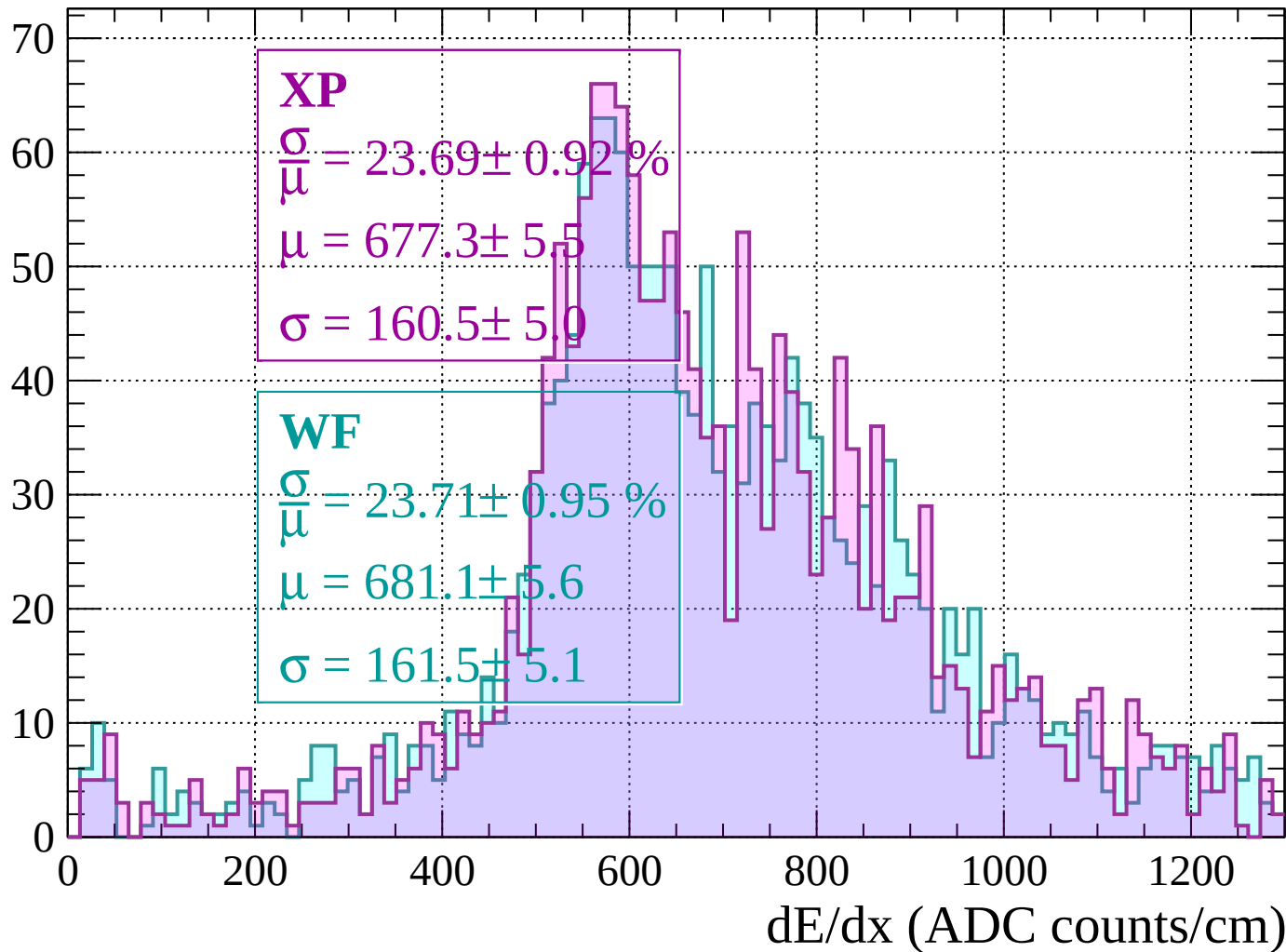
$$\sigma = 81.3 \pm 2.0$$

dE/dx (ADC counts/cm)



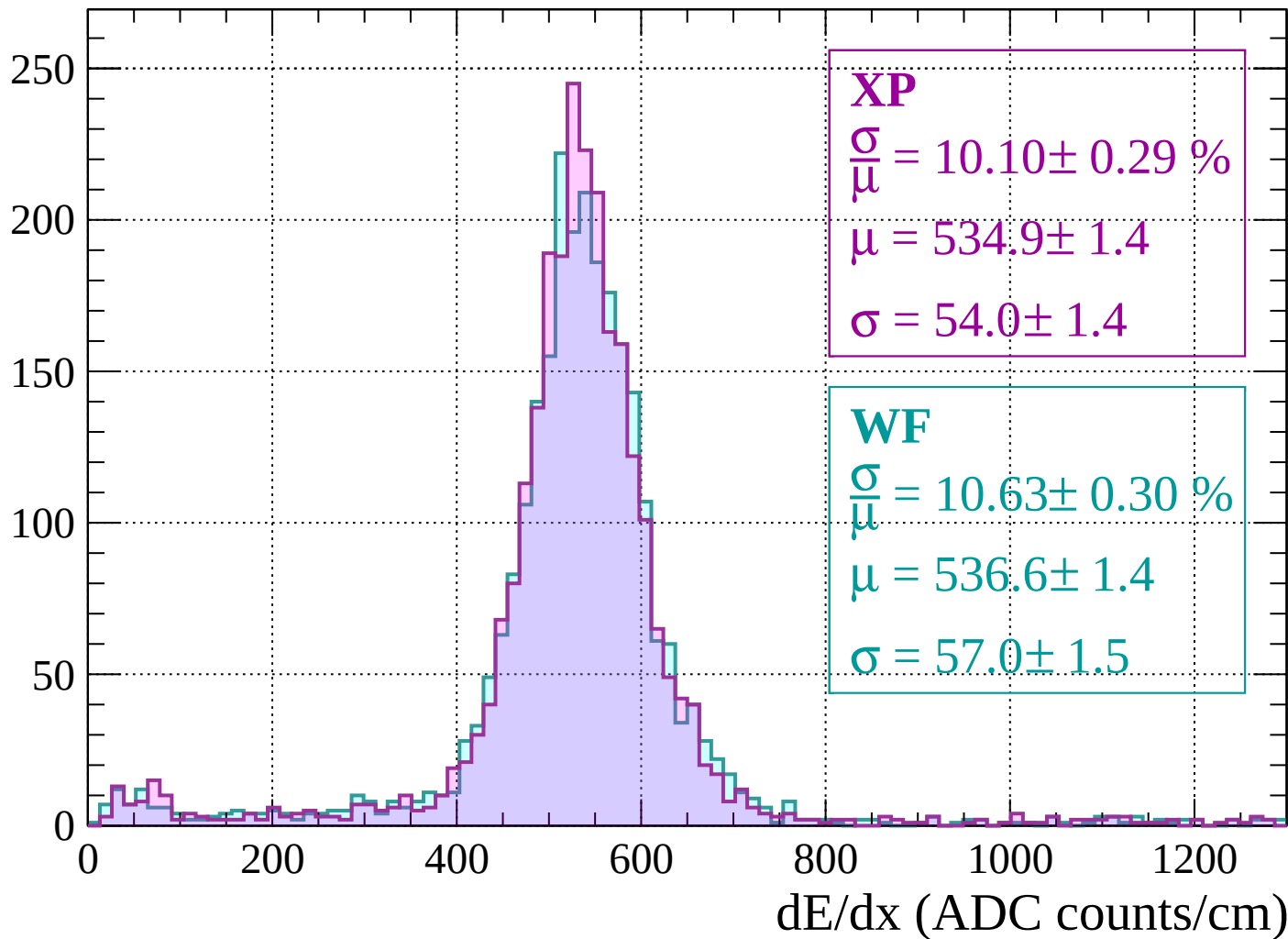
# Energy loss | $-80 < p < -40$

Count



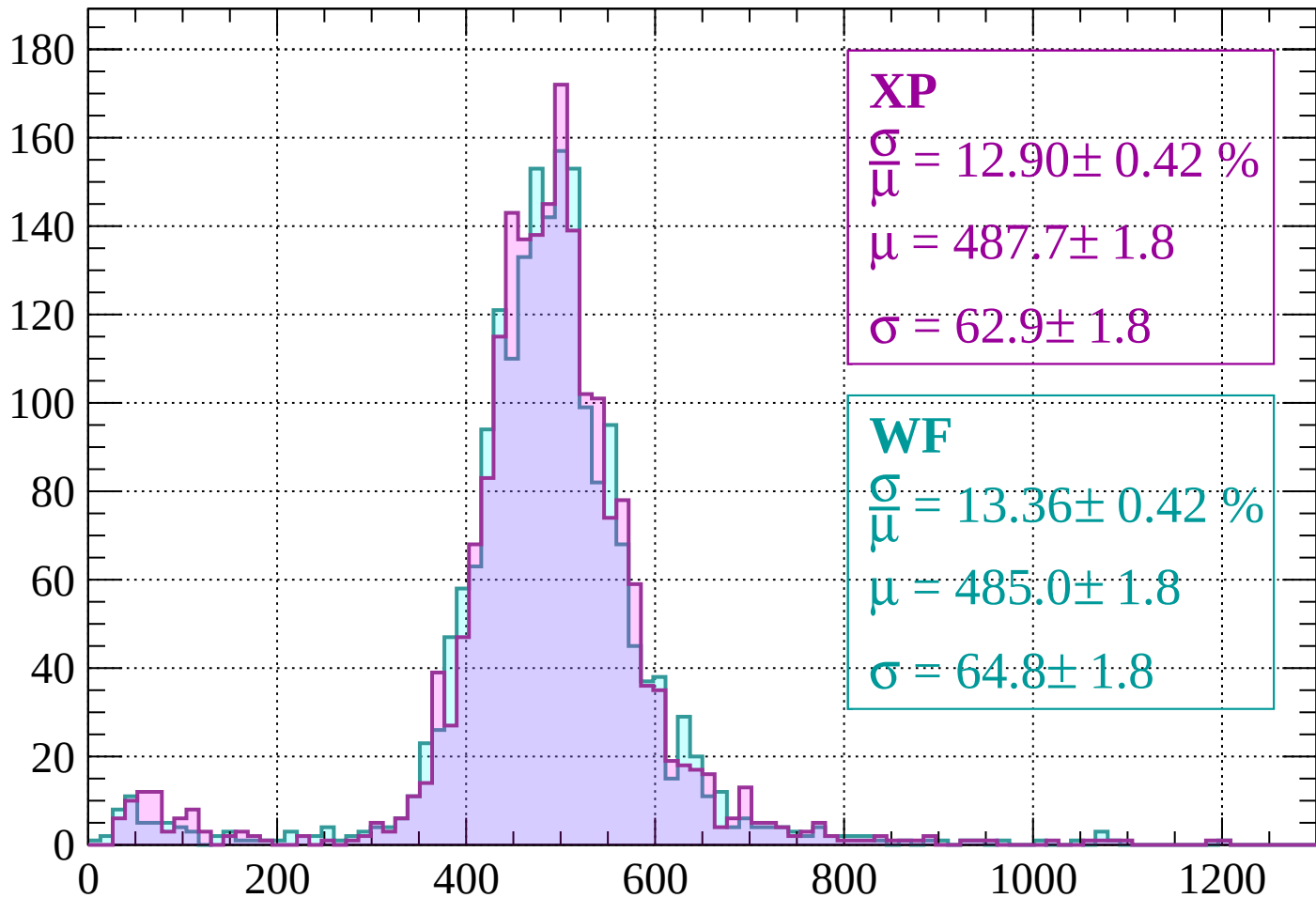
Energy loss |  $-40 < p < 0$

Count



Energy loss |  $0 < p < 40$

Count



**XP**

$$\frac{\sigma}{\mu} = 12.90 \pm 0.42 \%$$

$$\mu = 487.7 \pm 1.8$$

$$\sigma = 62.9 \pm 1.8$$

**WF**

$$\frac{\sigma}{\mu} = 13.36 \pm 0.42 \%$$

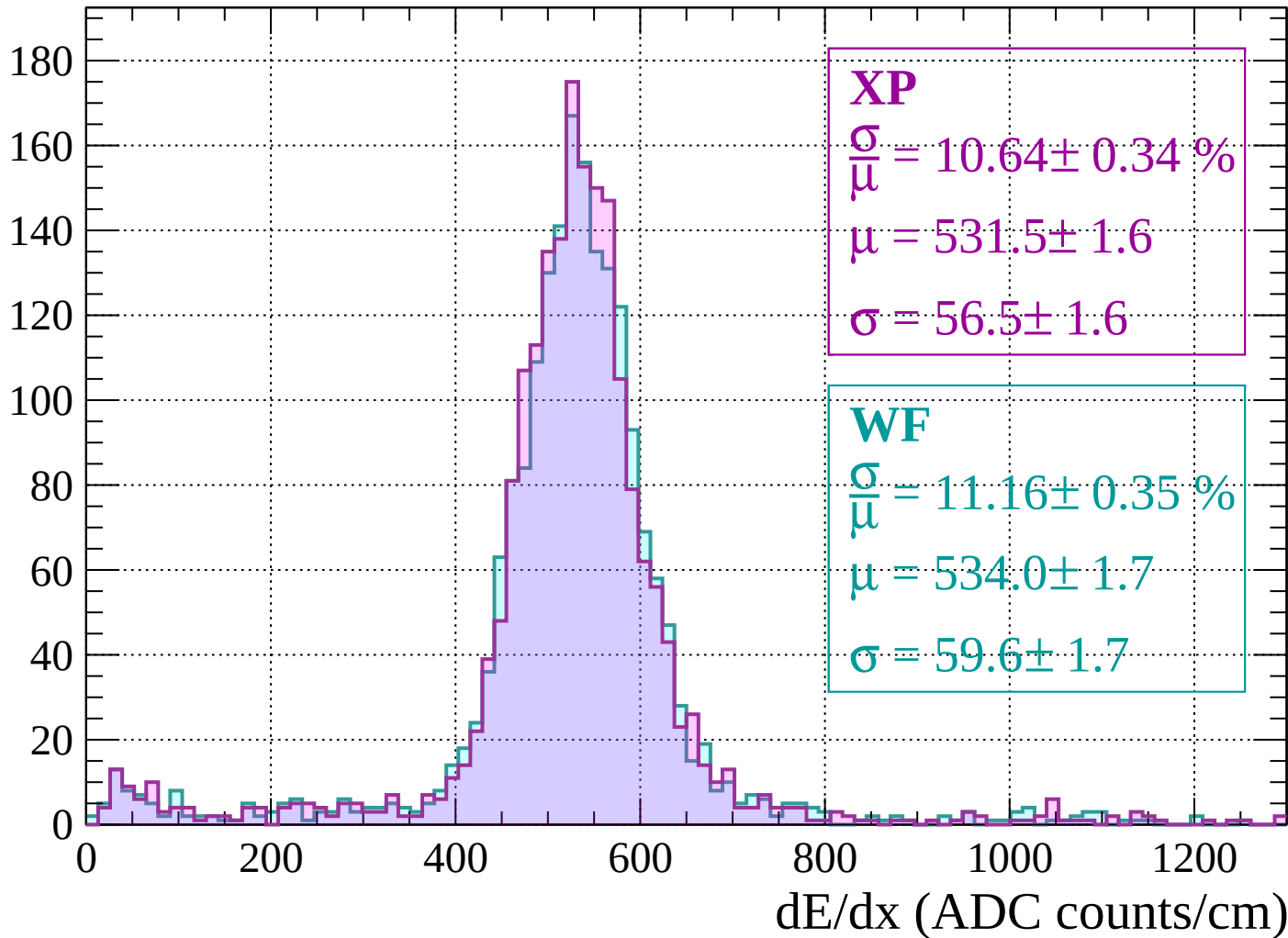
$$\mu = 485.0 \pm 1.8$$

$$\sigma = 64.8 \pm 1.8$$

$dE/dx$  (ADC counts/cm)

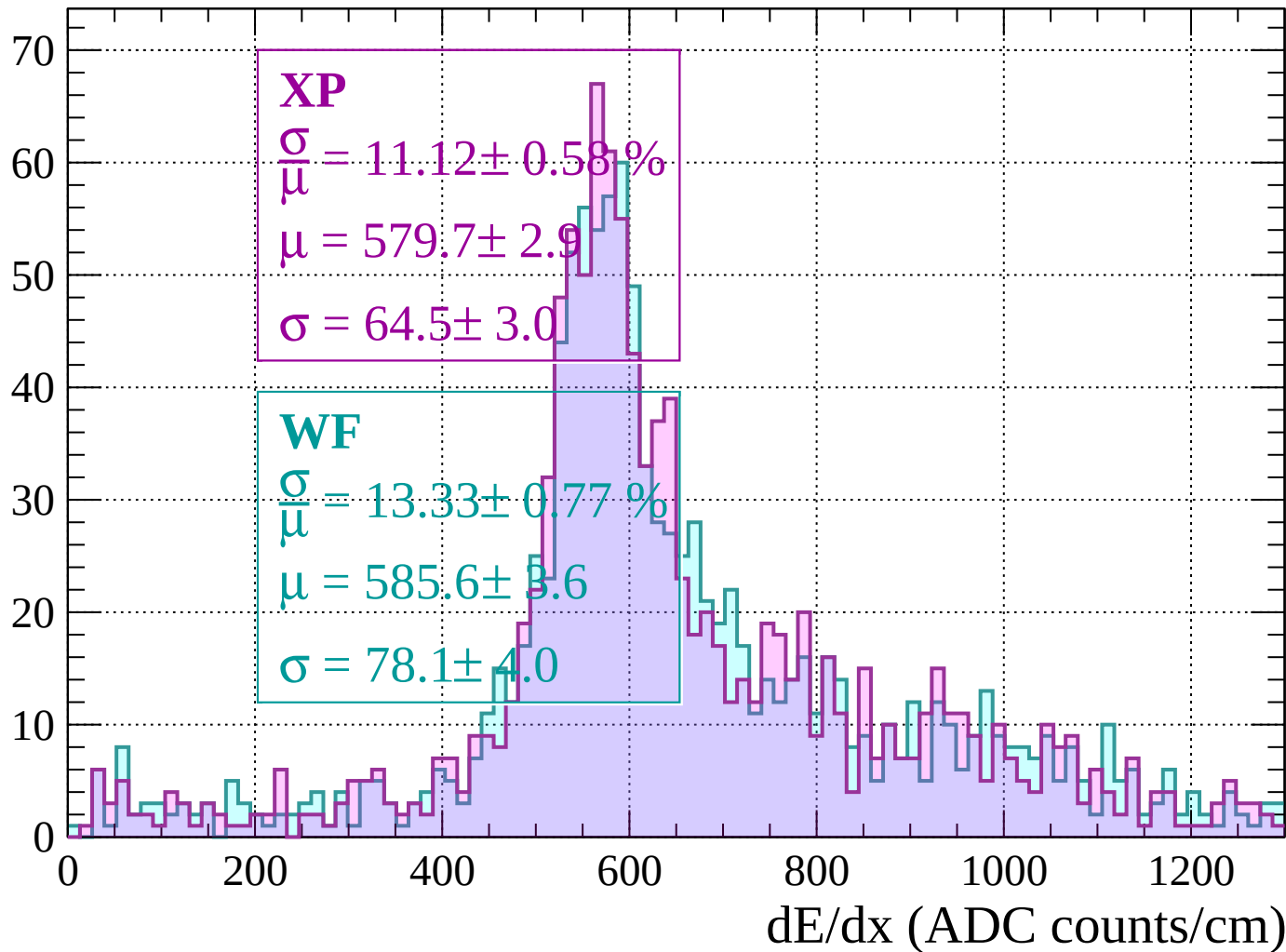
Energy loss |  $40 < p < 80$

Count



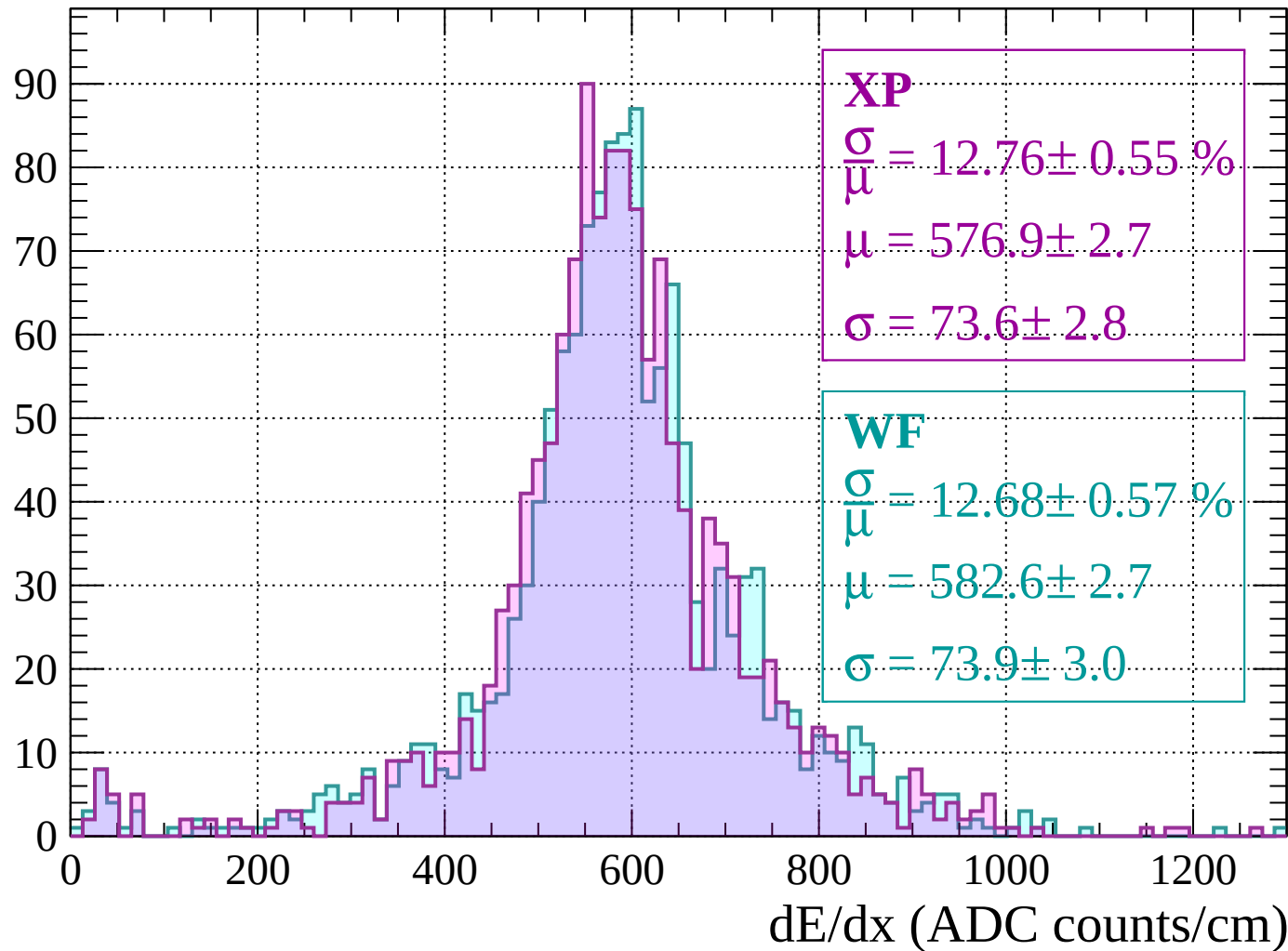
# Energy loss | $80 < p < 120$

Count



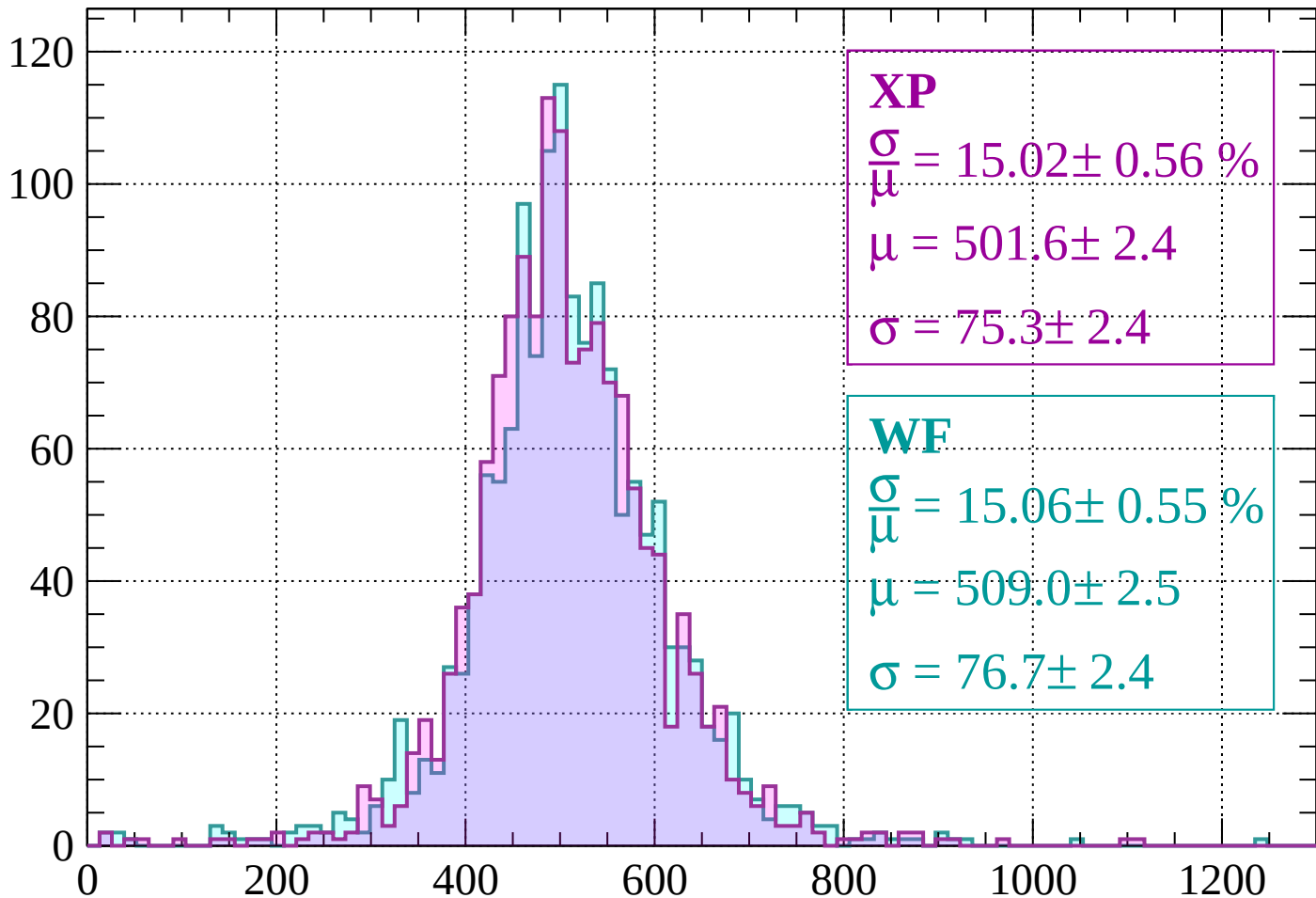
# Energy loss | $120 < p < 160$

Count



Energy loss |  $160 < p < 200$

Count



**XP**

$$\frac{\sigma}{\mu} = 15.02 \pm 0.56 \%$$

$$\mu = 501.6 \pm 2.4$$

$$\sigma = 75.3 \pm 2.4$$

**WF**

$$\frac{\sigma}{\mu} = 15.06 \pm 0.55 \%$$

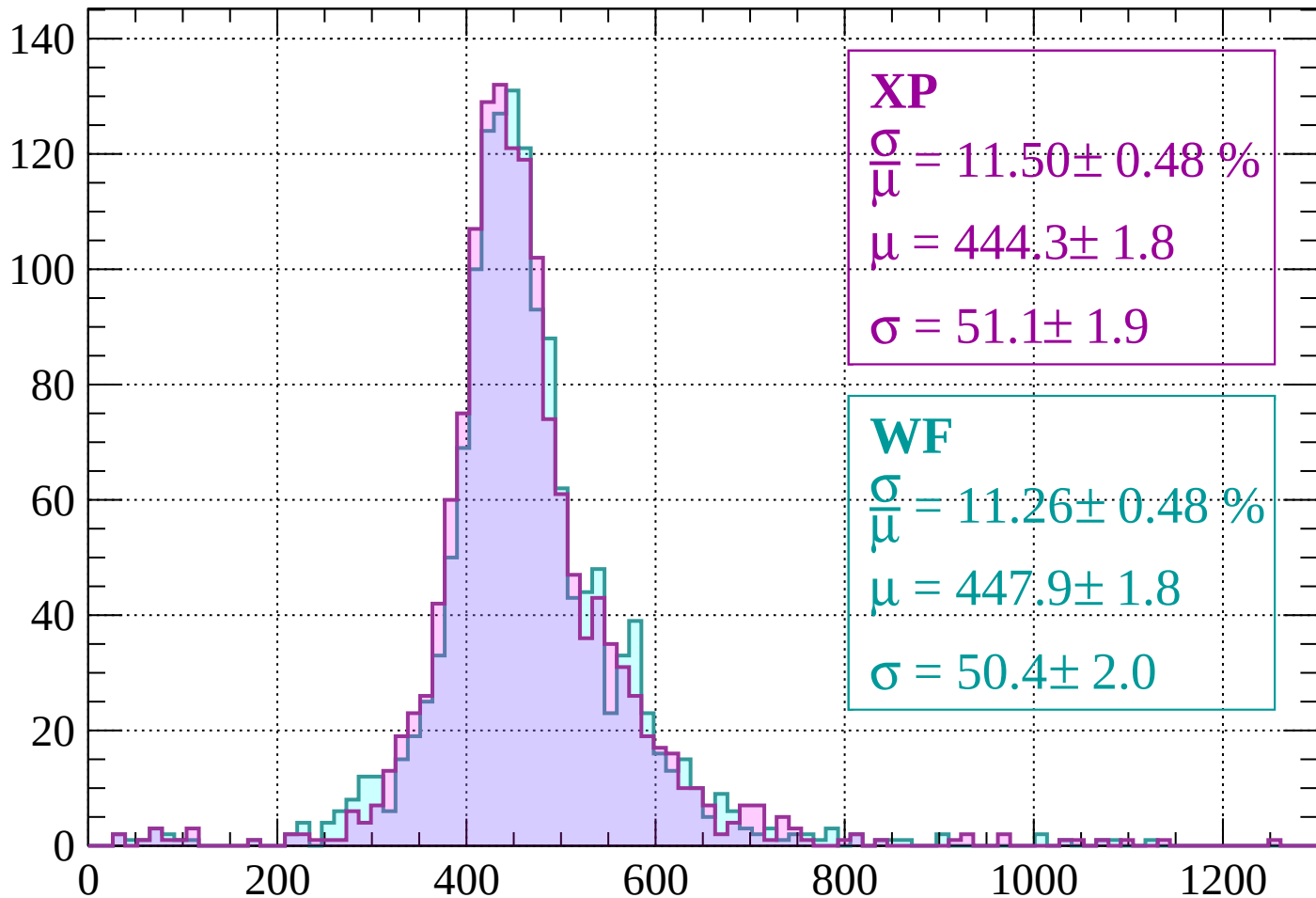
$$\mu = 509.0 \pm 2.5$$

$$\sigma = 76.7 \pm 2.4$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $200 < p < 240$

Count



**XP**

$$\frac{\sigma}{\mu} = 11.50 \pm 0.48 \%$$

$$\mu = 444.3 \pm 1.8$$

$$\sigma = 51.1 \pm 1.9$$

**WF**

$$\frac{\sigma}{\mu} = 11.26 \pm 0.48 \%$$

$$\mu = 447.9 \pm 1.8$$

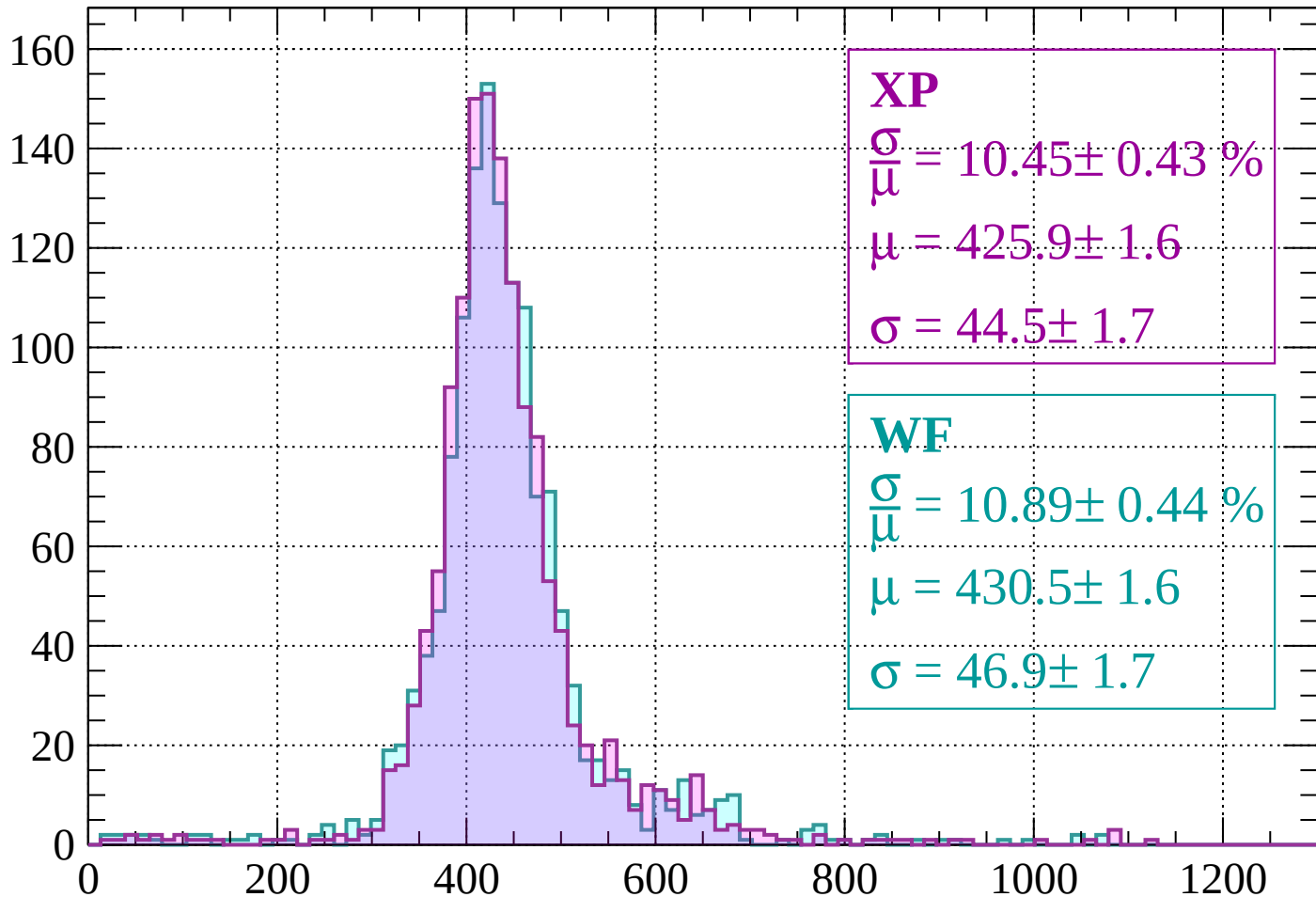
$$\sigma = 50.4 \pm 2.0$$

dE/dx (ADC counts/cm)



Energy loss |  $240 < p < 280$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.45 \pm 0.43 \%$$

$$\mu = 425.9 \pm 1.6$$

$$\sigma = 44.5 \pm 1.7$$

**WF**

$$\frac{\sigma}{\mu} = 10.89 \pm 0.44 \%$$

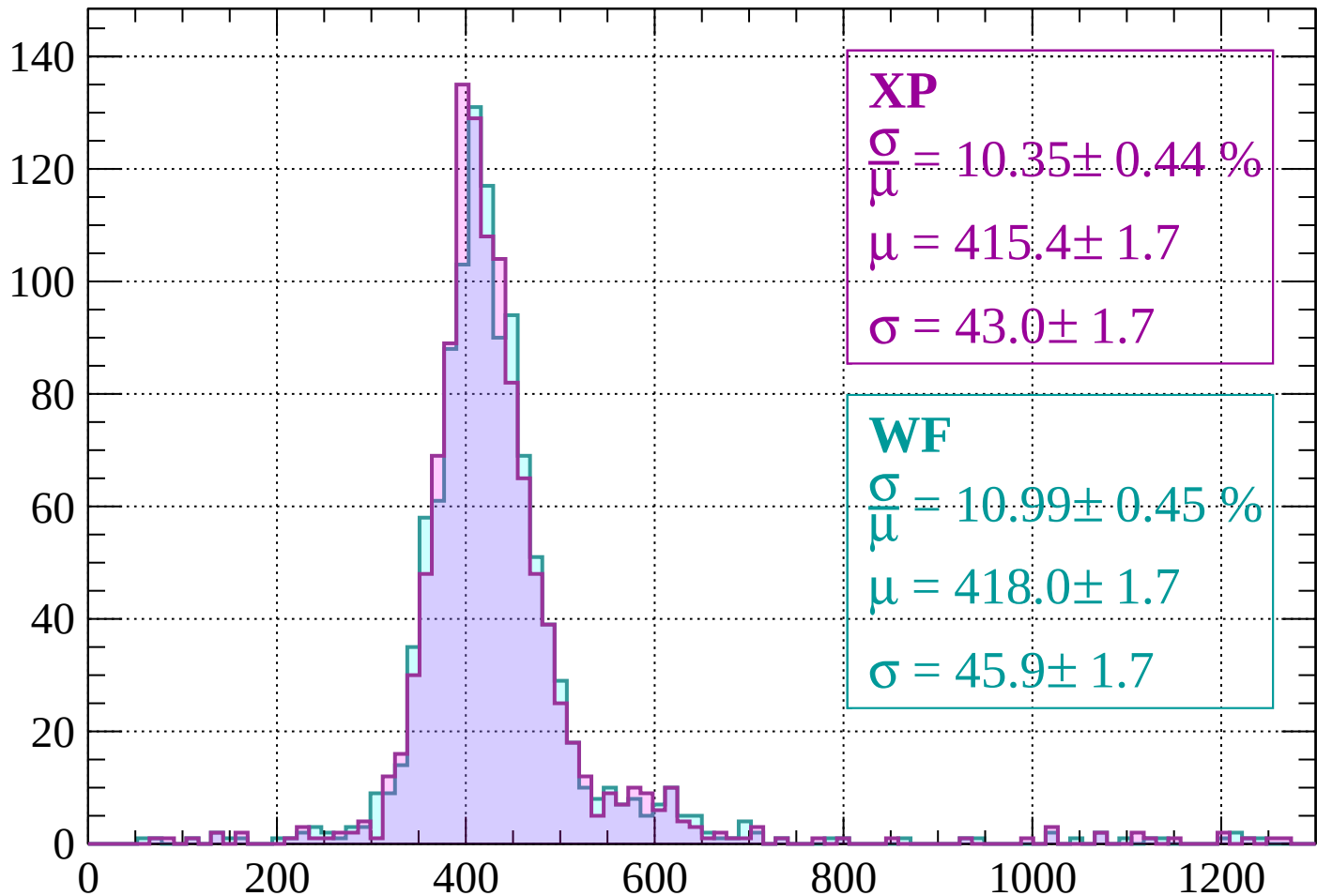
$$\mu = 430.5 \pm 1.6$$

$$\sigma = 46.9 \pm 1.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $280 < p < 320$

Count



**XP**

$$\frac{\sigma}{\mu} = 10.35 \pm 0.44 \%$$

$$\mu = 415.4 \pm 1.7$$

$$\sigma = 43.0 \pm 1.7$$

**WF**

$$\frac{\sigma}{\mu} = 10.99 \pm 0.45 \%$$

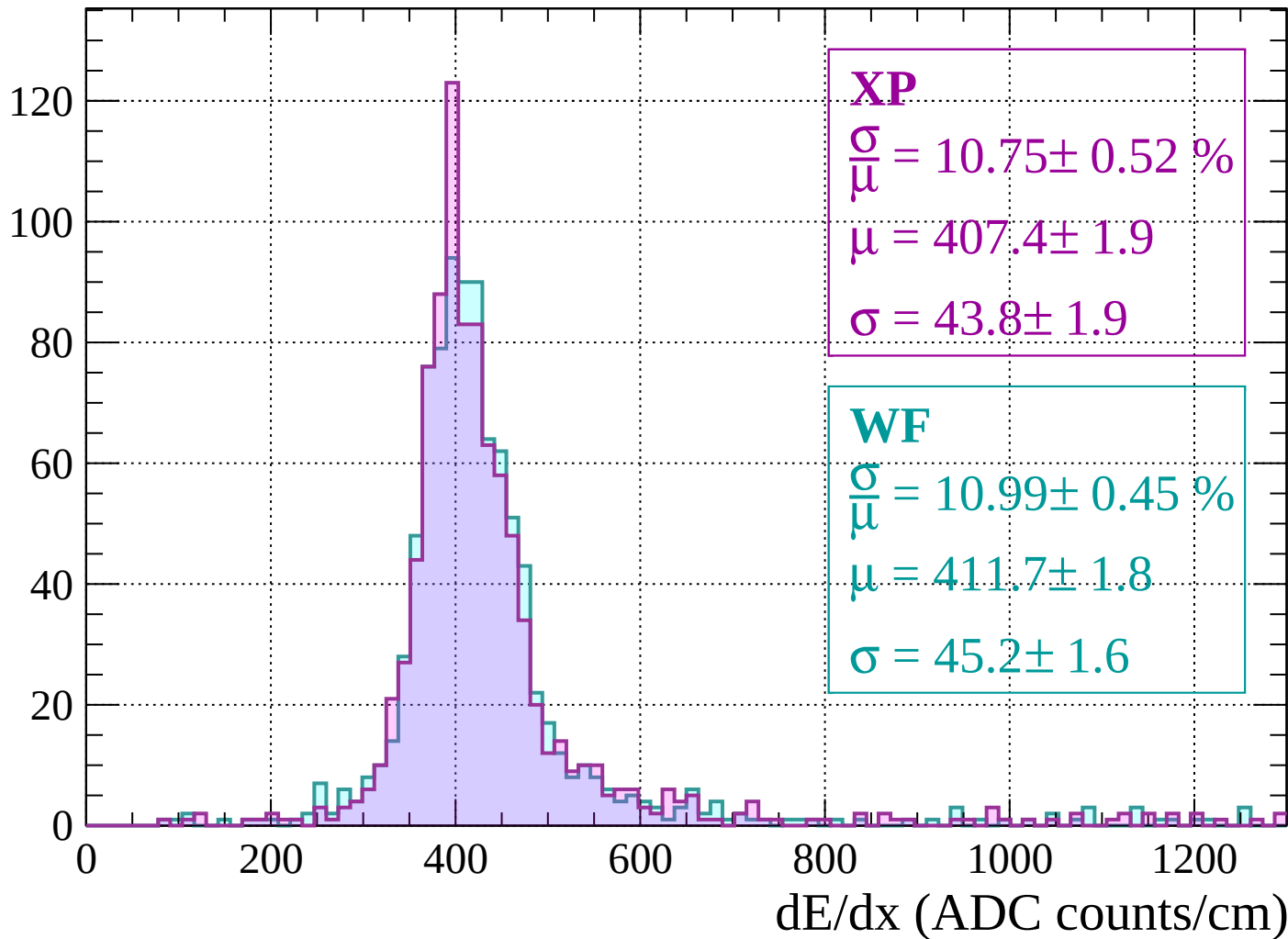
$$\mu = 418.0 \pm 1.7$$

$$\sigma = 45.9 \pm 1.7$$

$dE/dx$  (ADC counts/cm)

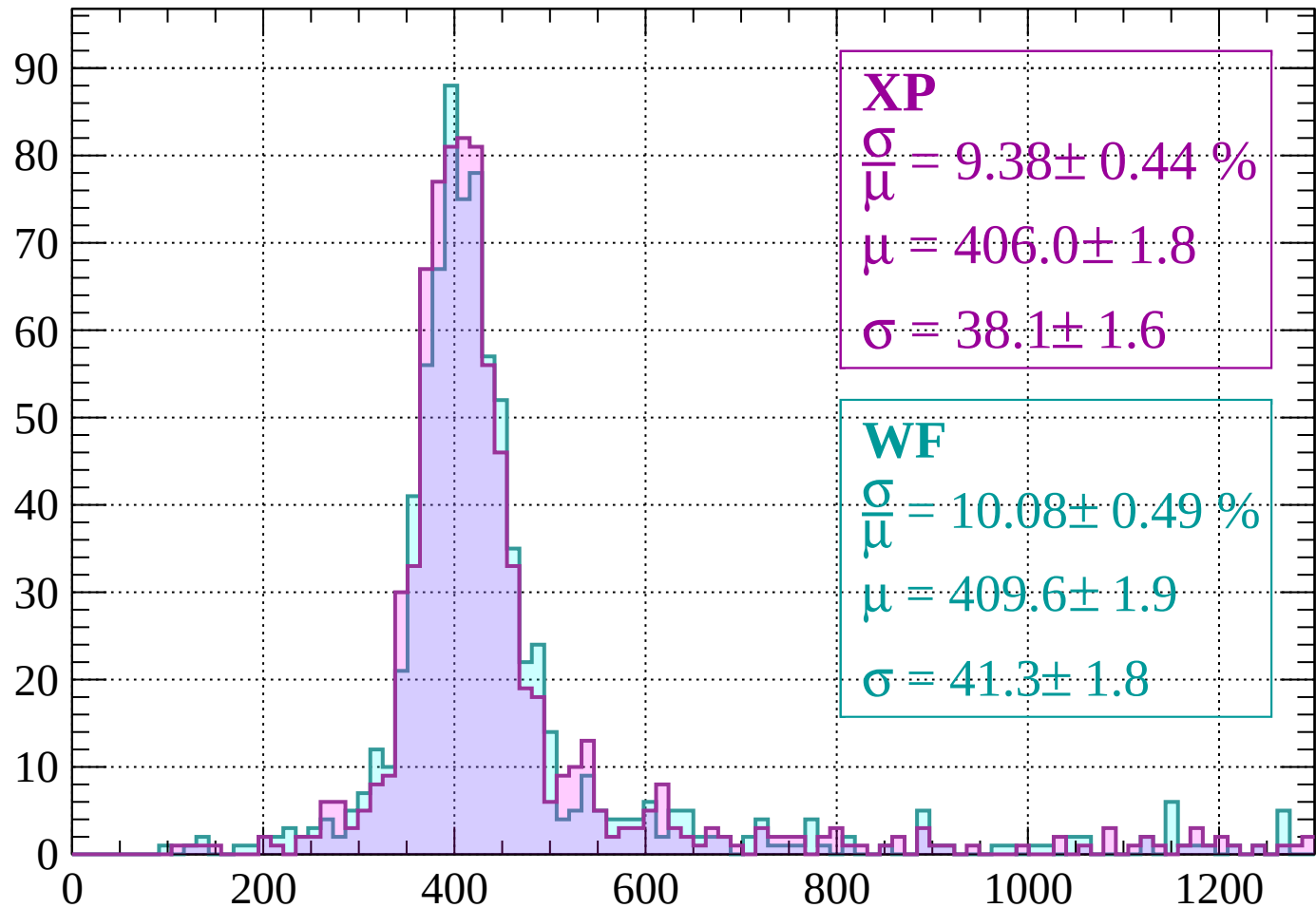
Energy loss |  $320 < p < 360$

Count



Energy loss |  $360 < p < 400$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.38 \pm 0.44 \%$$

$$\mu = 406.0 \pm 1.8$$

$$\sigma = 38.1 \pm 1.6$$

**WF**

$$\frac{\sigma}{\mu} = 10.08 \pm 0.49 \%$$

$$\mu = 409.6 \pm 1.9$$

$$\sigma = 41.3 \pm 1.8$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $400 < p < 440$

Count

100

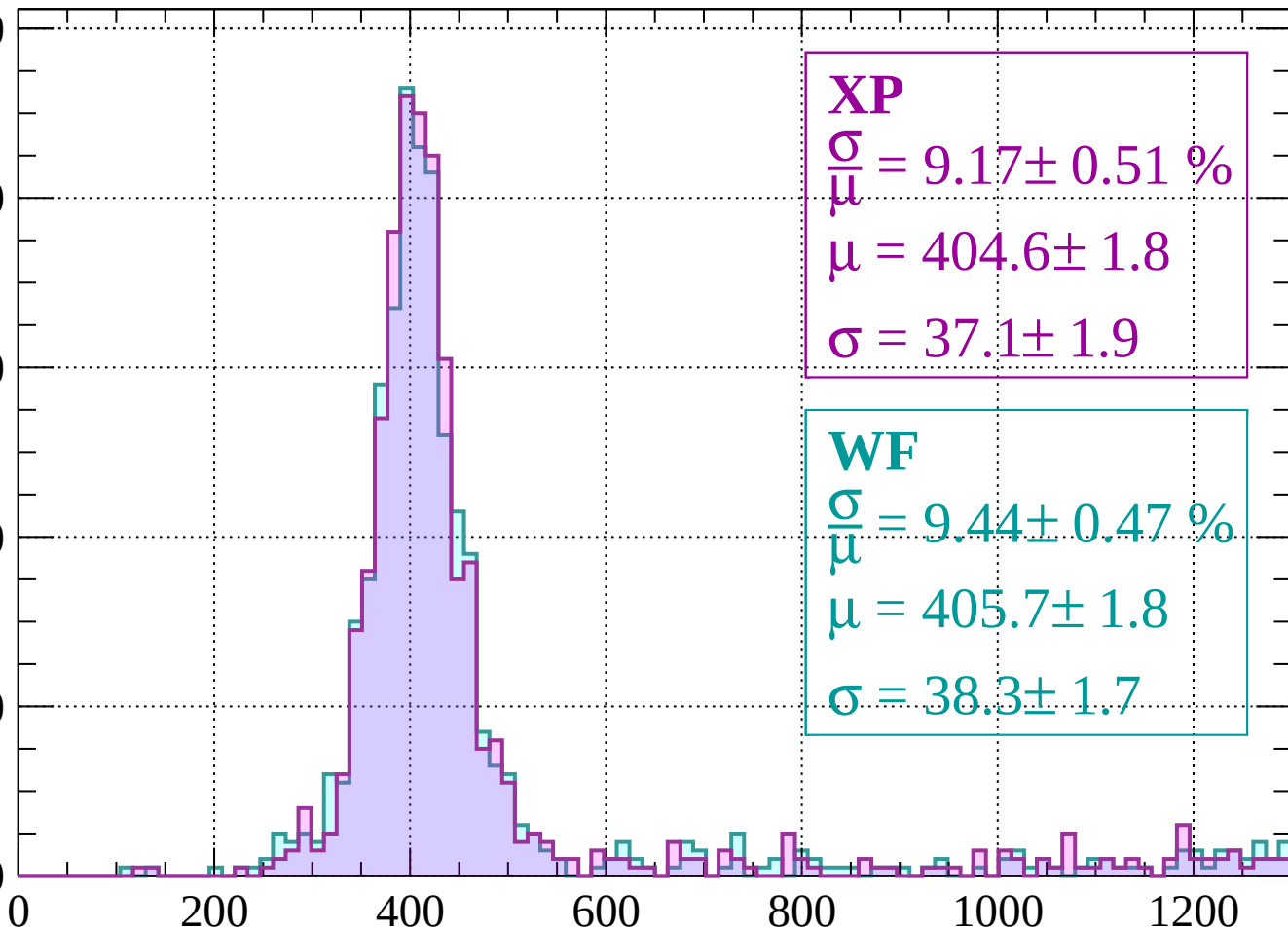
80

60

40

20

0



**XP**

$$\frac{\sigma}{\mu} = 9.17 \pm 0.51 \%$$

$$\mu = 404.6 \pm 1.8$$

$$\sigma = 37.1 \pm 1.9$$

**WF**

$$\frac{\sigma}{\mu} = 9.44 \pm 0.47 \%$$

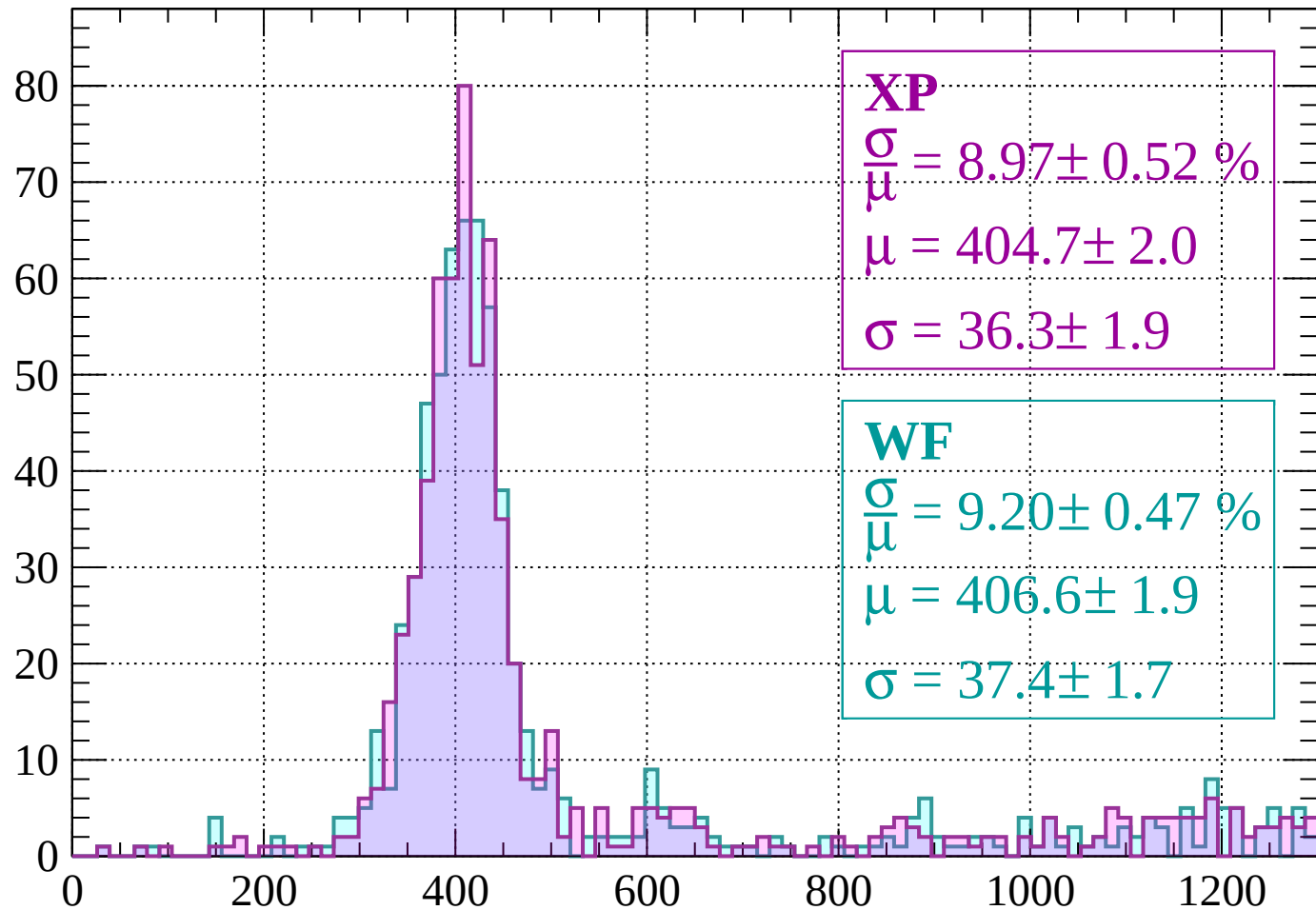
$$\mu = 405.7 \pm 1.8$$

$$\sigma = 38.3 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss |  $440 < p < 480$

Count



**XP**

$$\frac{\sigma}{\mu} = 8.97 \pm 0.52 \%$$

$$\mu = 404.7 \pm 2.0$$

$$\sigma = 36.3 \pm 1.9$$

**WF**

$$\frac{\sigma}{\mu} = 9.20 \pm 0.47 \%$$

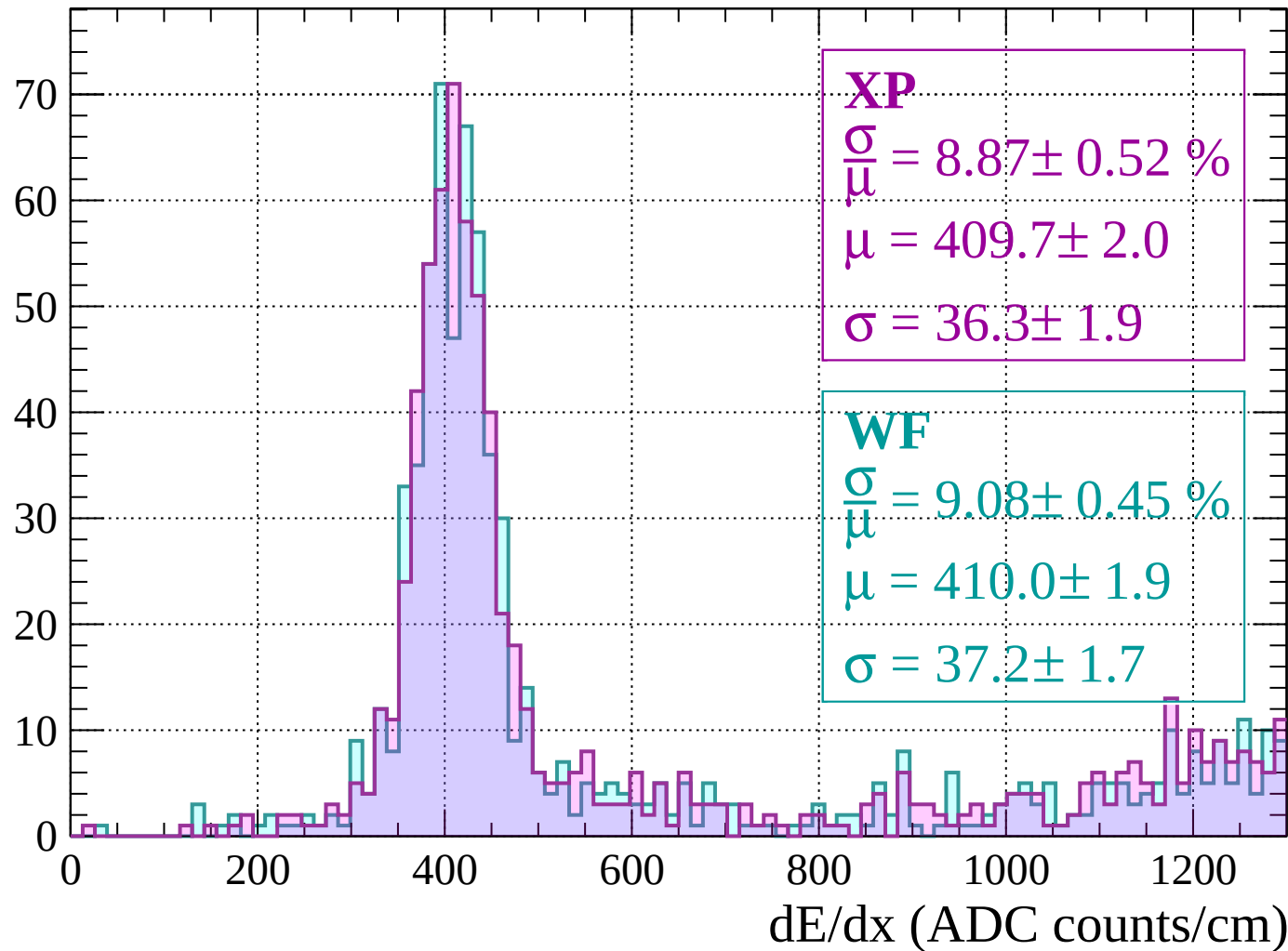
$$\mu = 406.6 \pm 1.9$$

$$\sigma = 37.4 \pm 1.7$$

$dE/dx$  (ADC counts/cm)

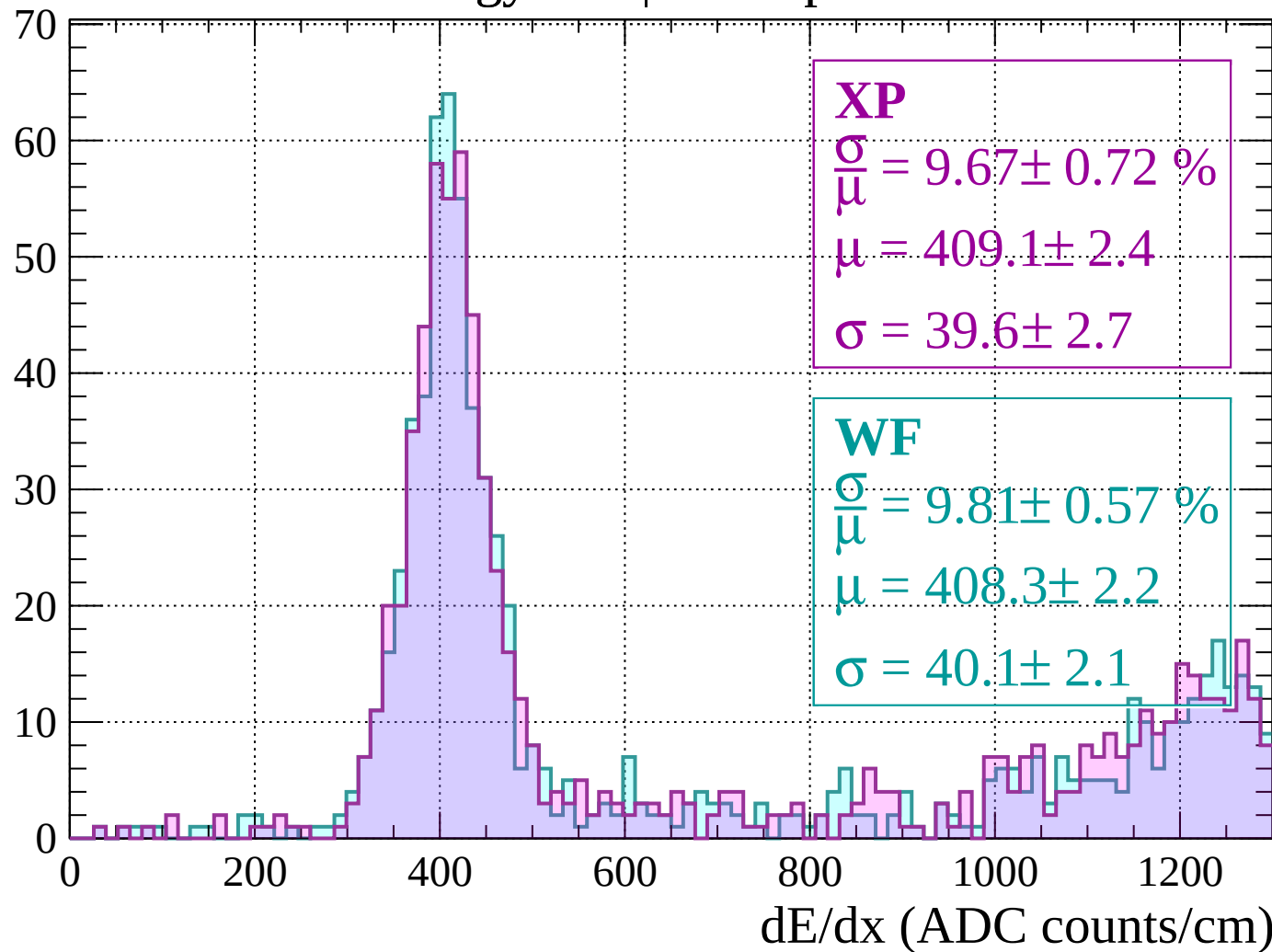
Energy loss |  $480 < p < 520$

Count



Energy loss |  $520 < p < 560$

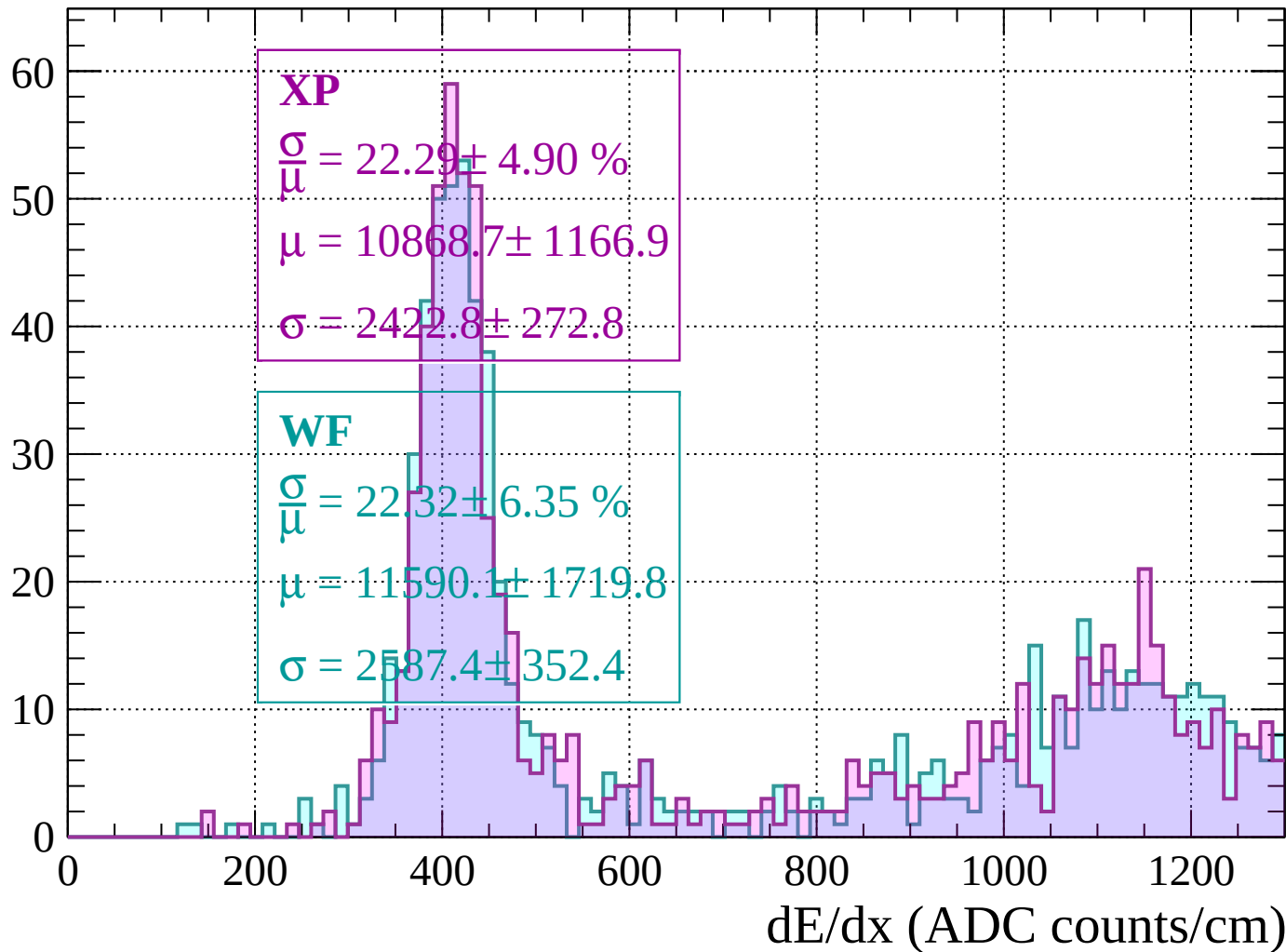
Count





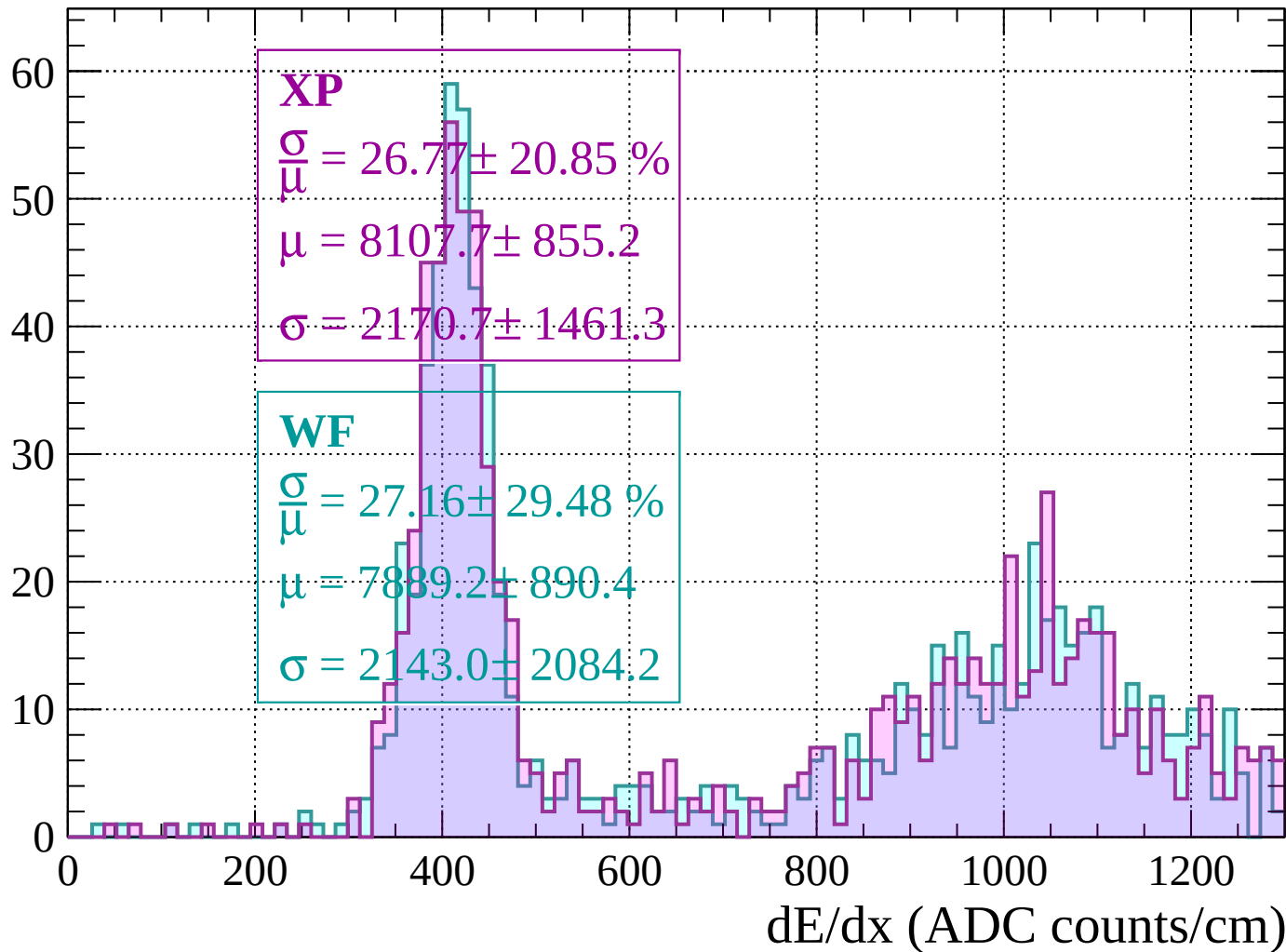
# Energy loss | $560 < p < 600$

Count



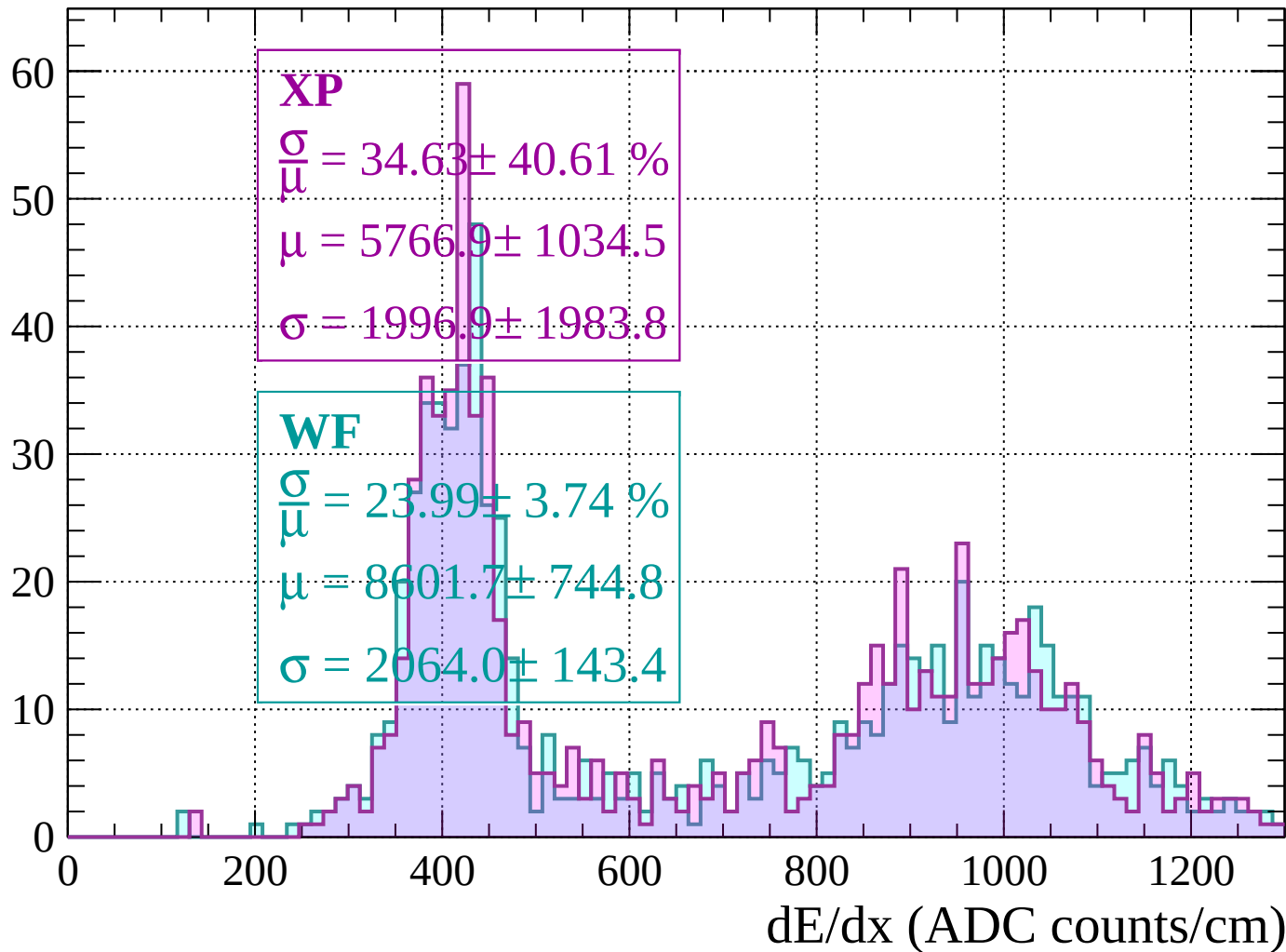
# Energy loss | $600 < p < 640$

Count



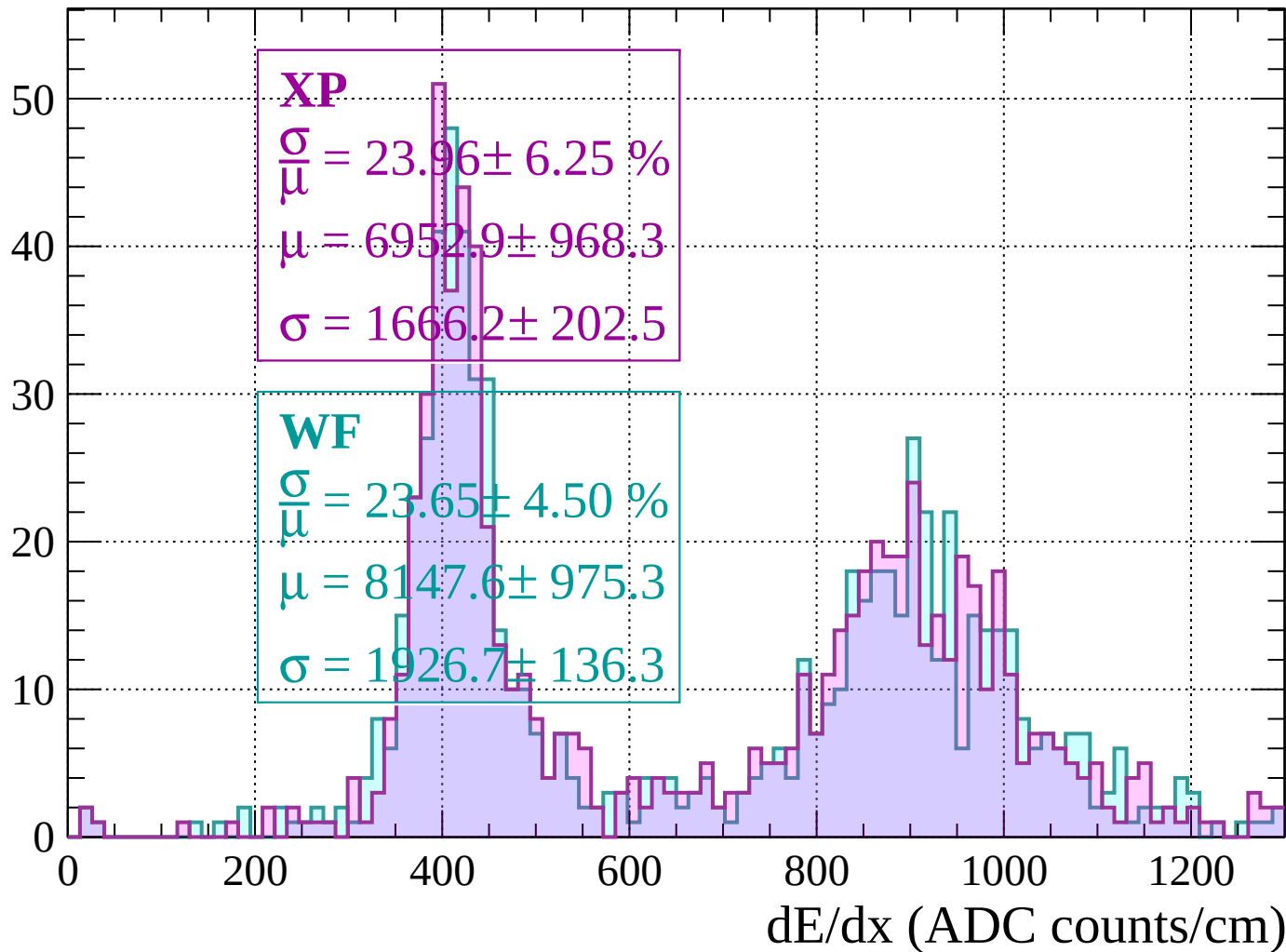
# Energy loss | $640 < p < 680$

Count



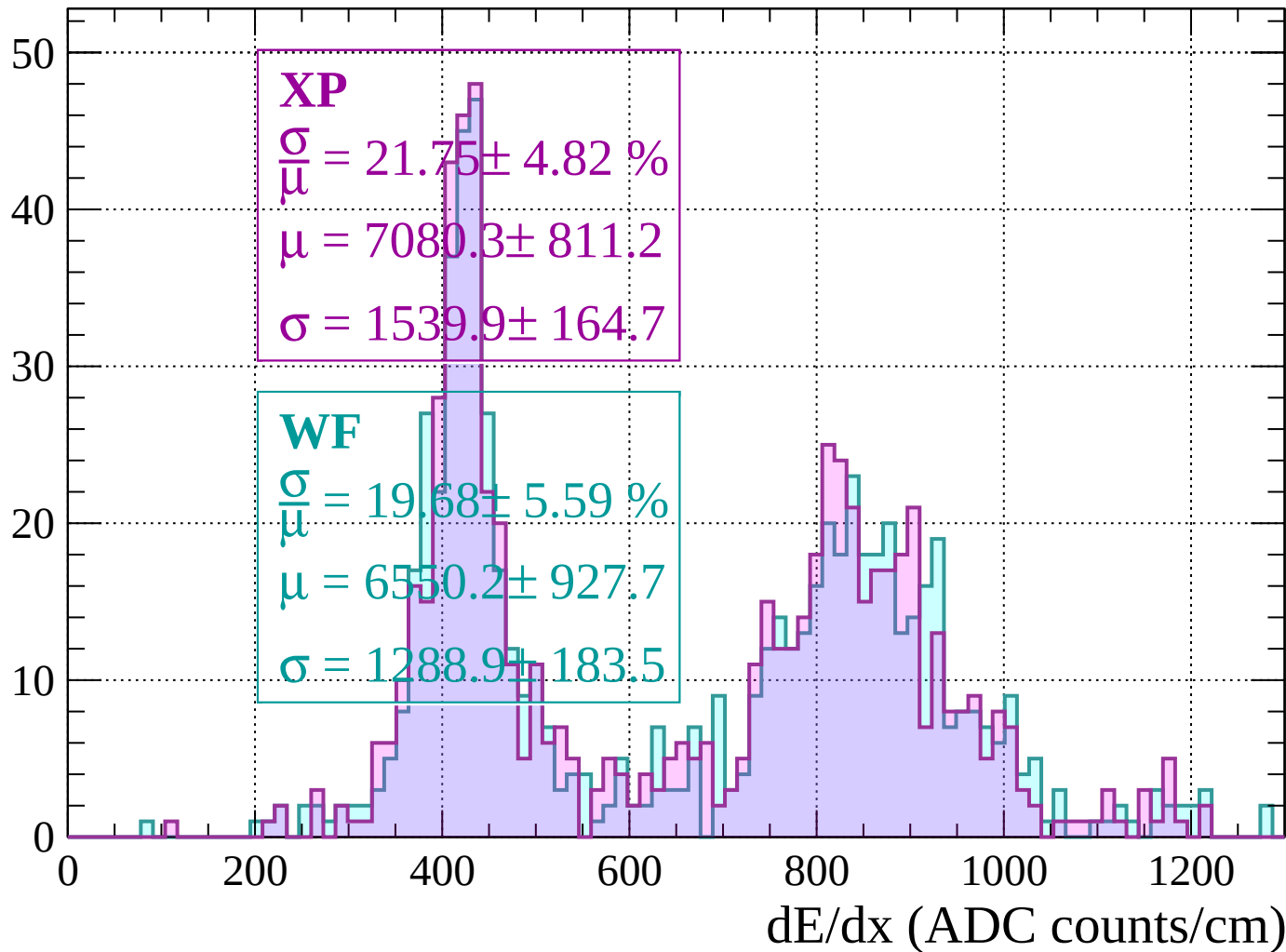
Energy loss |  $680 < p < 720$

Count



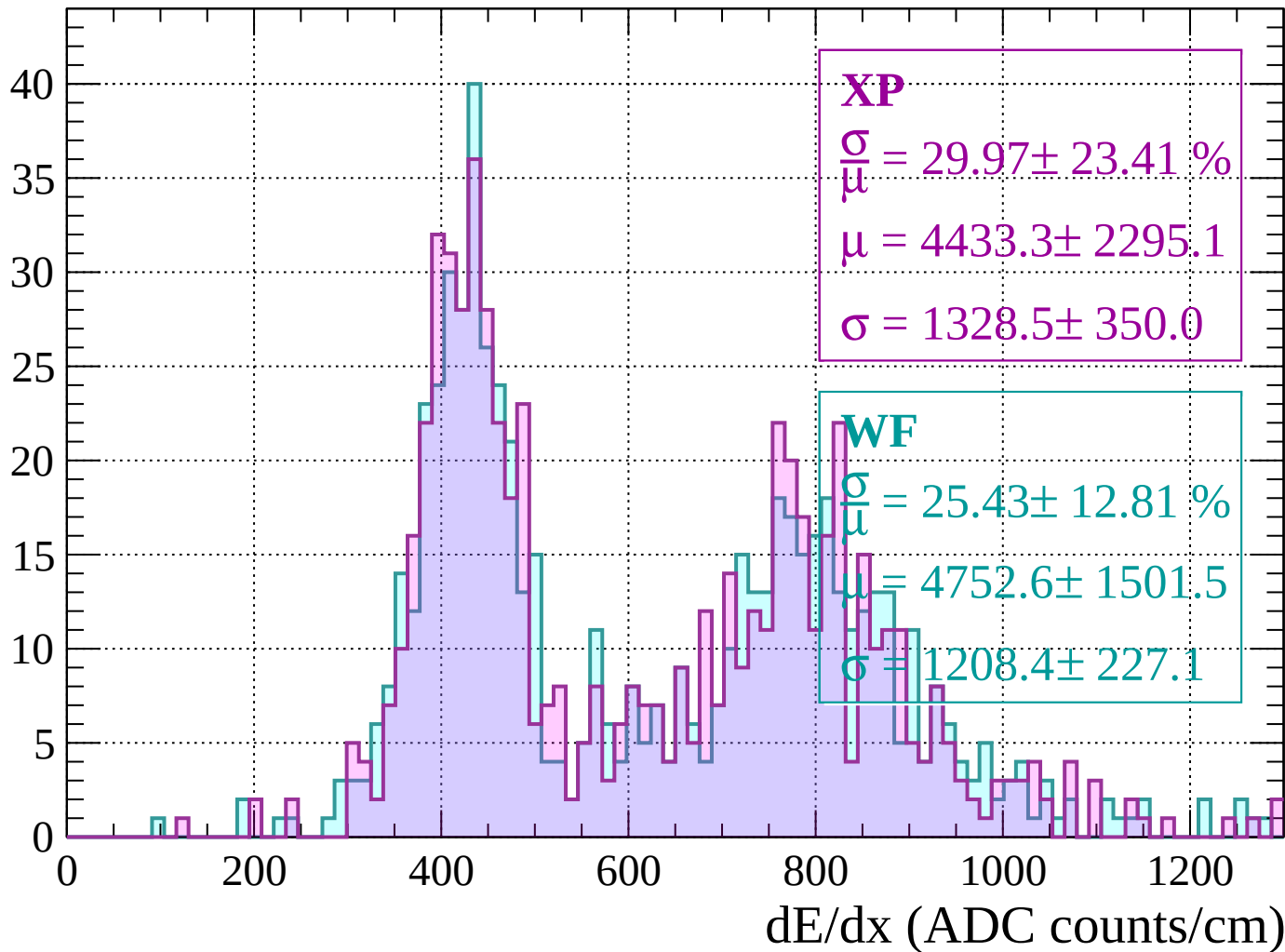
Energy loss |  $720 < p < 760$

Count



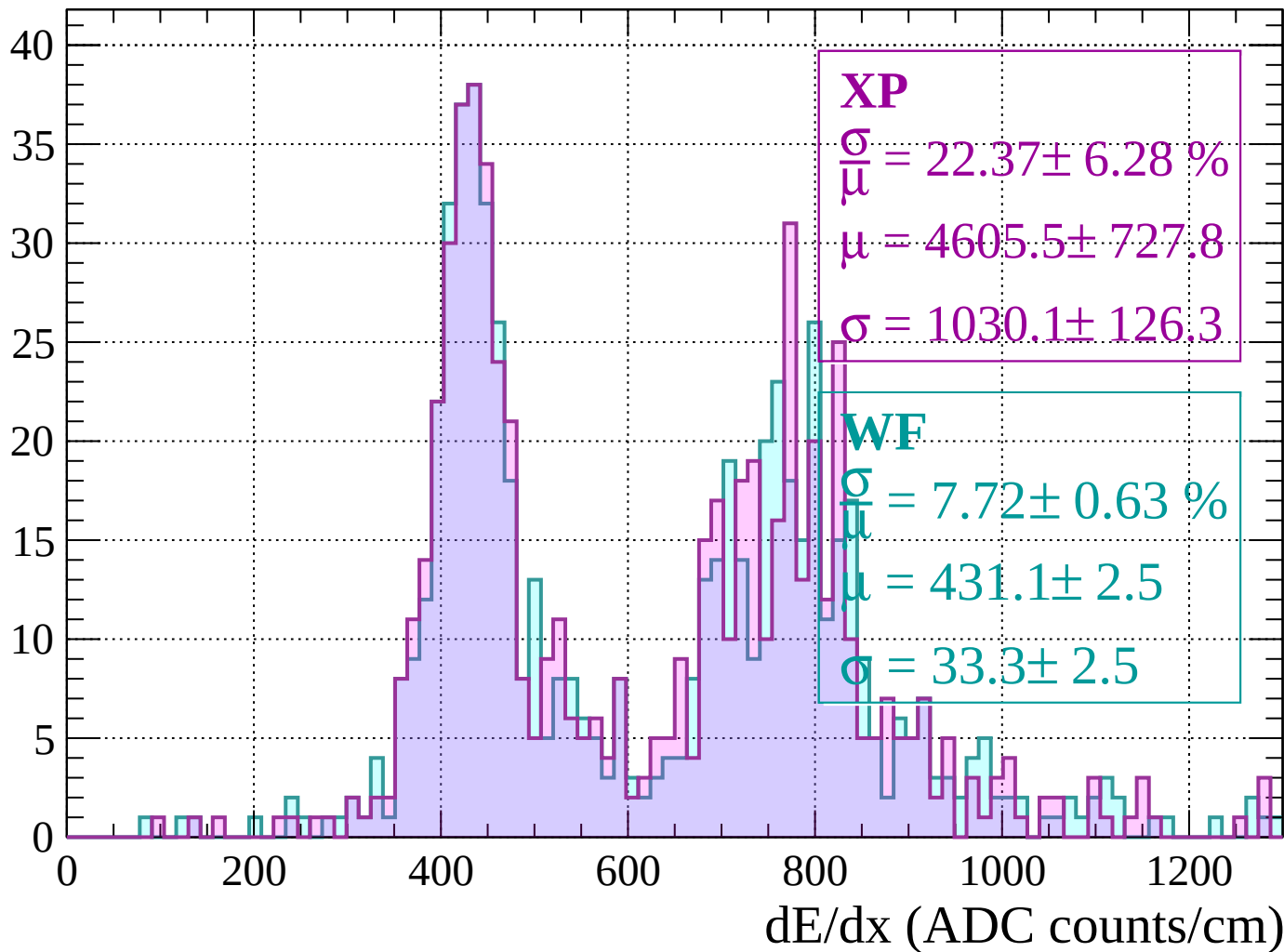
Energy loss |  $760 < p < 800$

Count



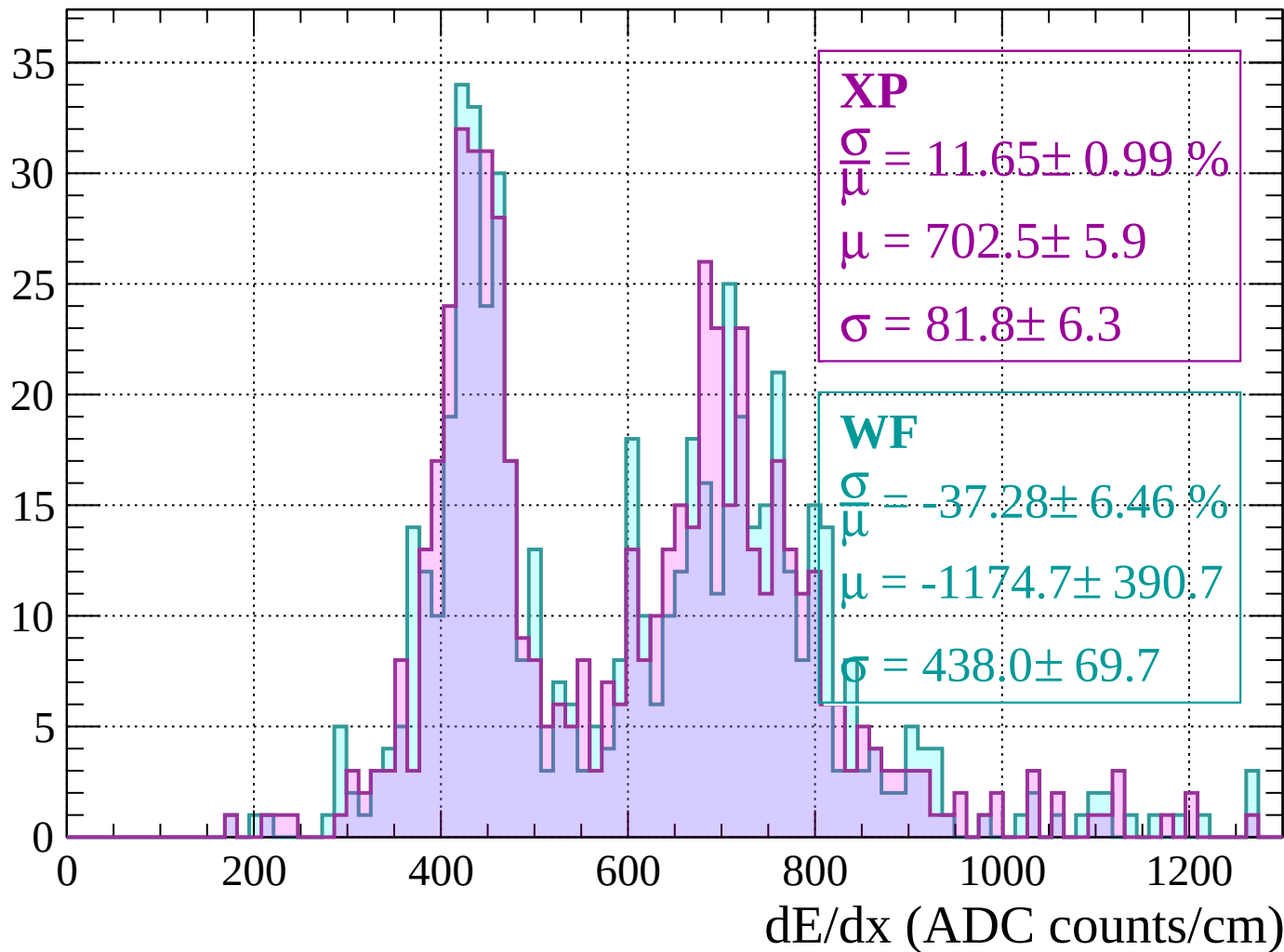
Energy loss |  $800 < p < 840$

Count



Energy loss |  $840 < p < 880$

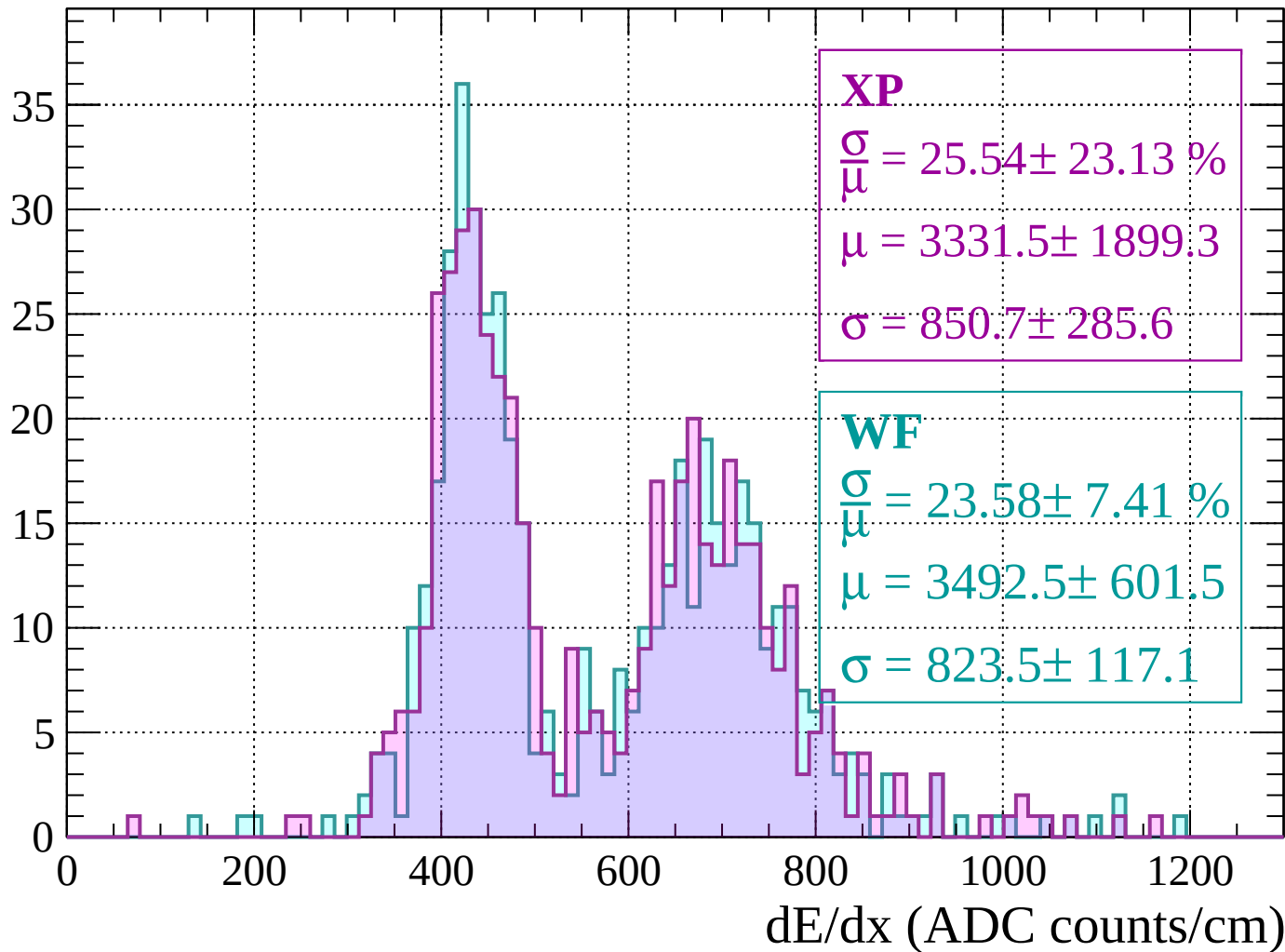
Count





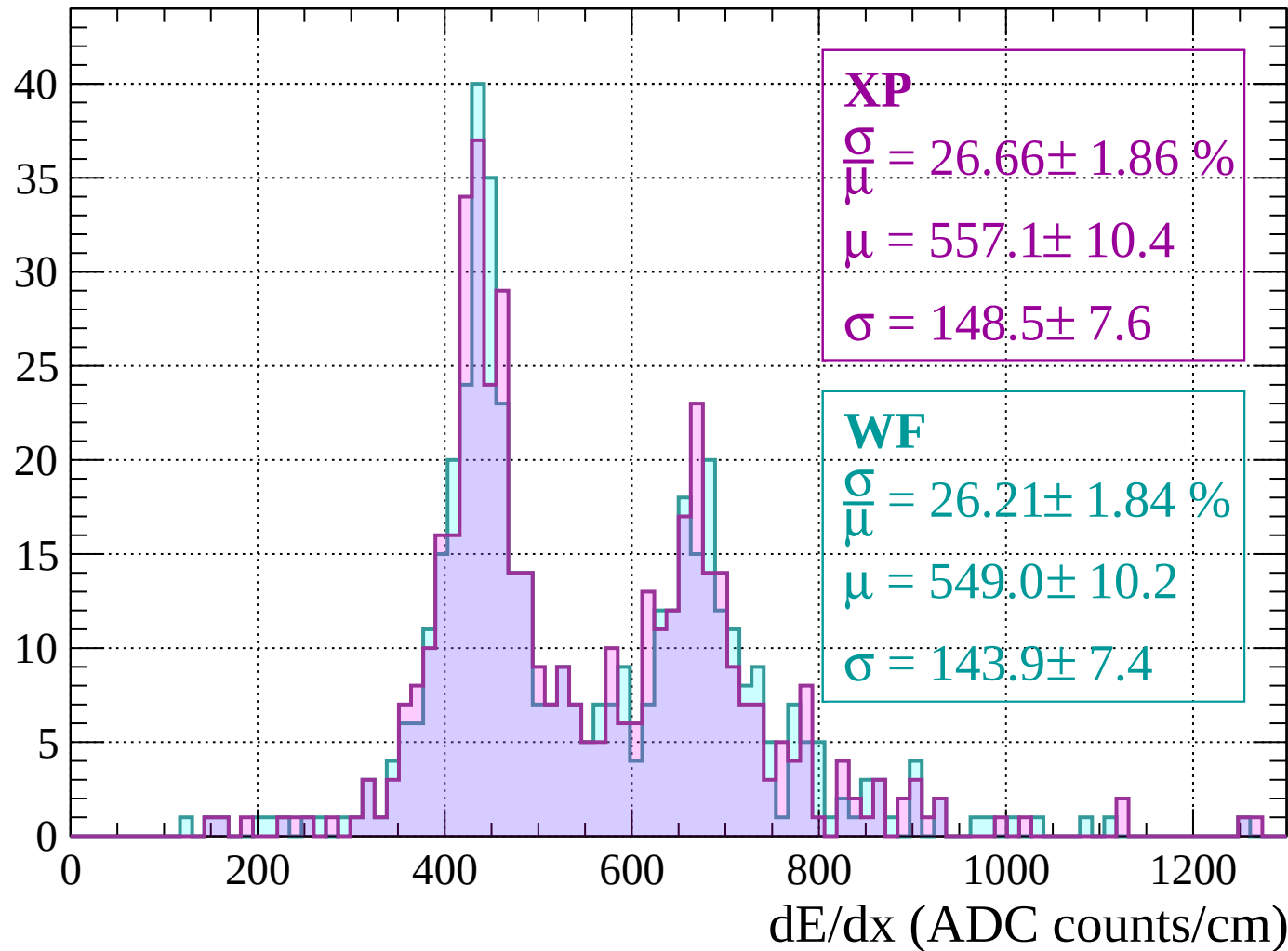
Energy loss |  $880 < p < 920$

Count



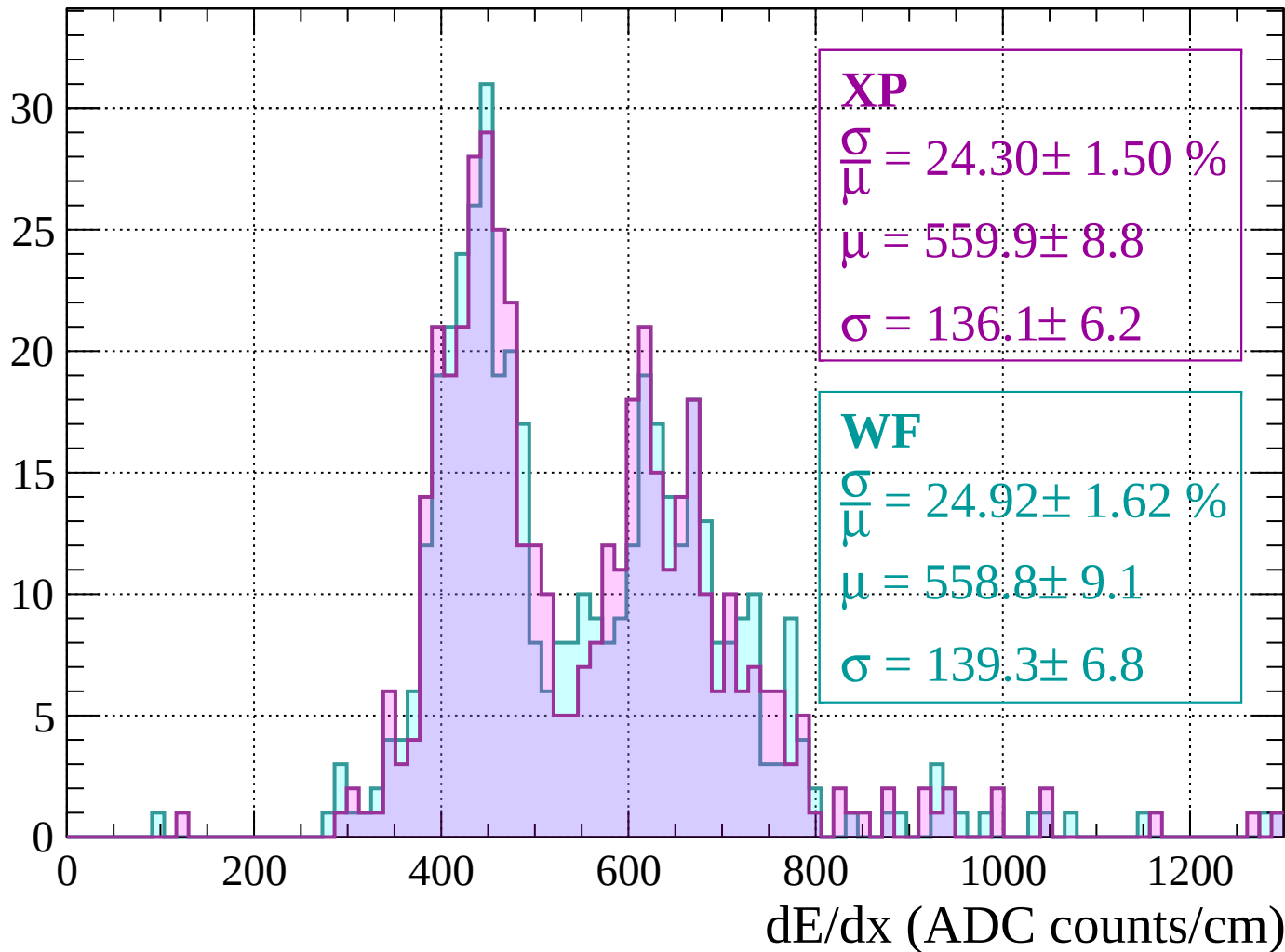
Energy loss |  $920 < p < 960$

Count



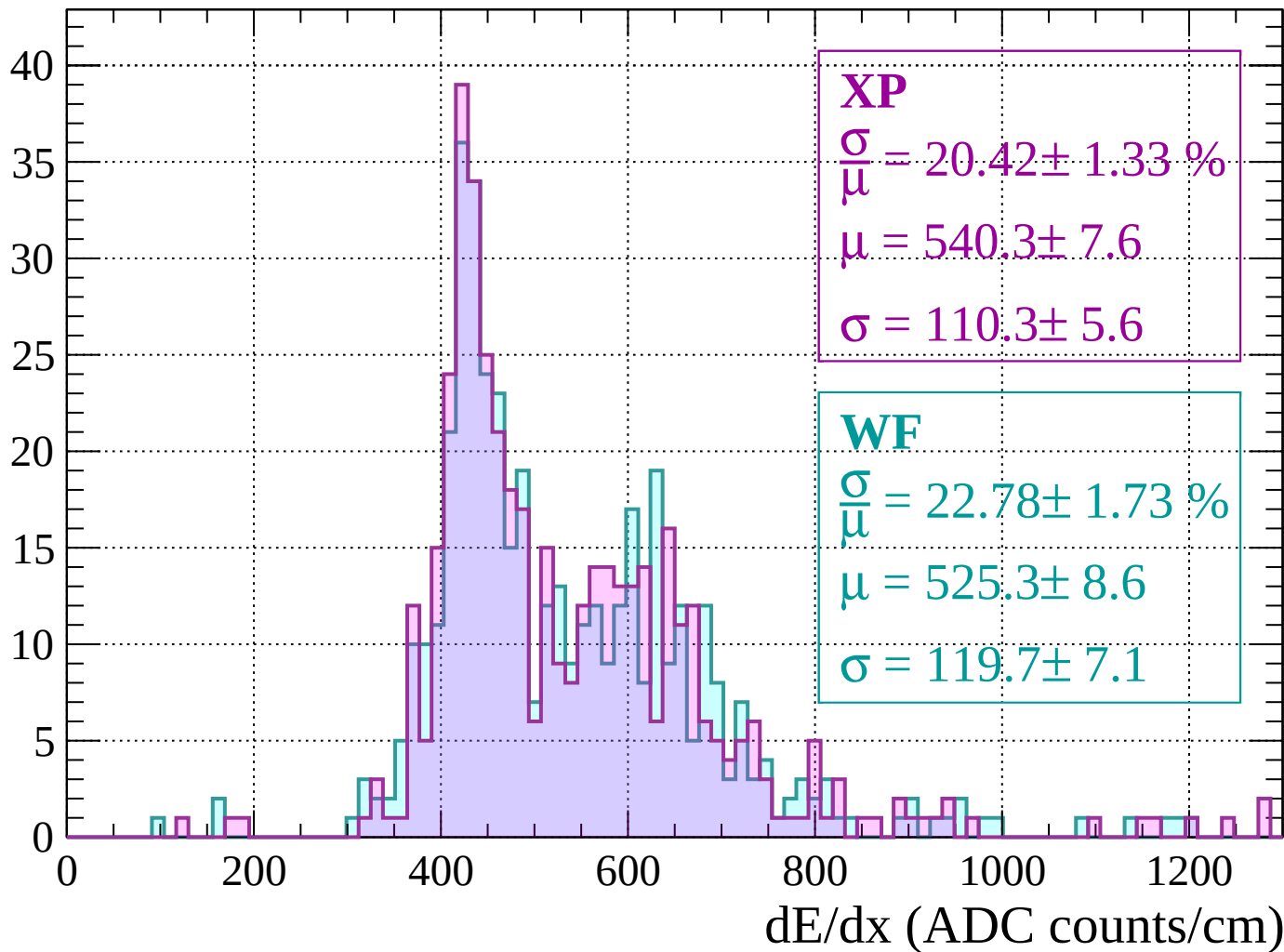
Energy loss |  $960 < p < 1000$

Count



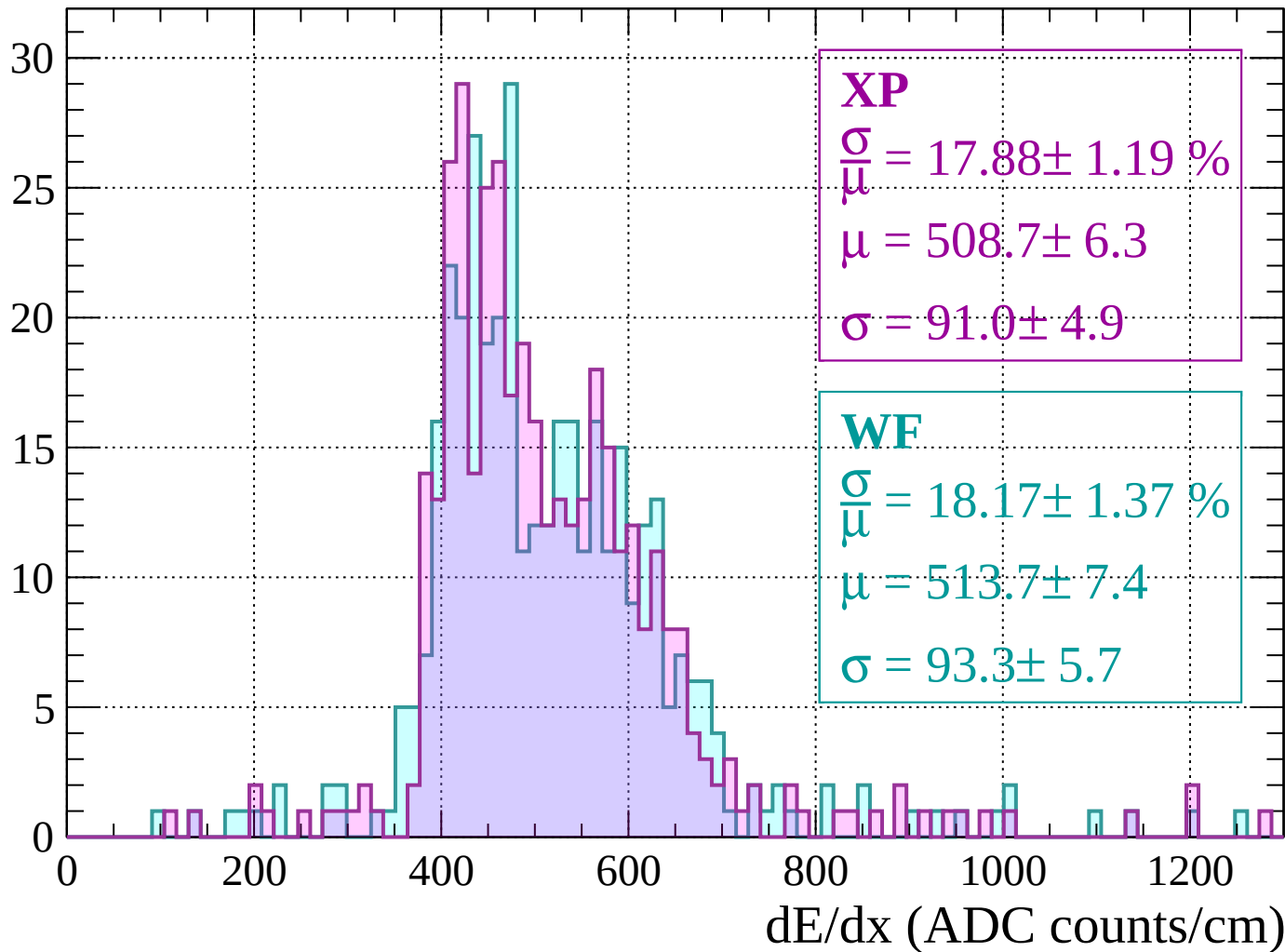
Energy loss |  $1000 < p < 1040$

Count



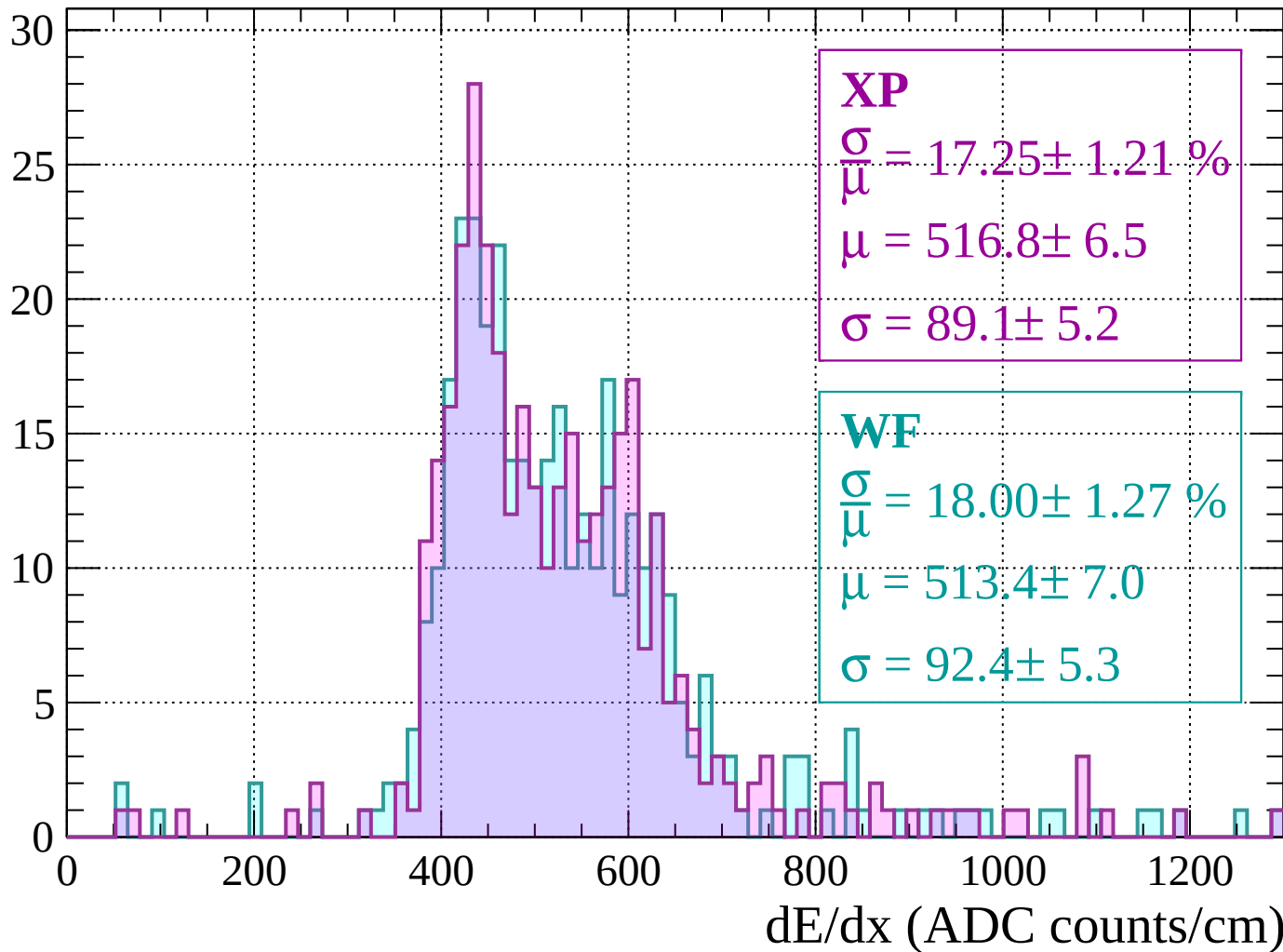
Energy loss |  $1040 < p < 1080$

Count



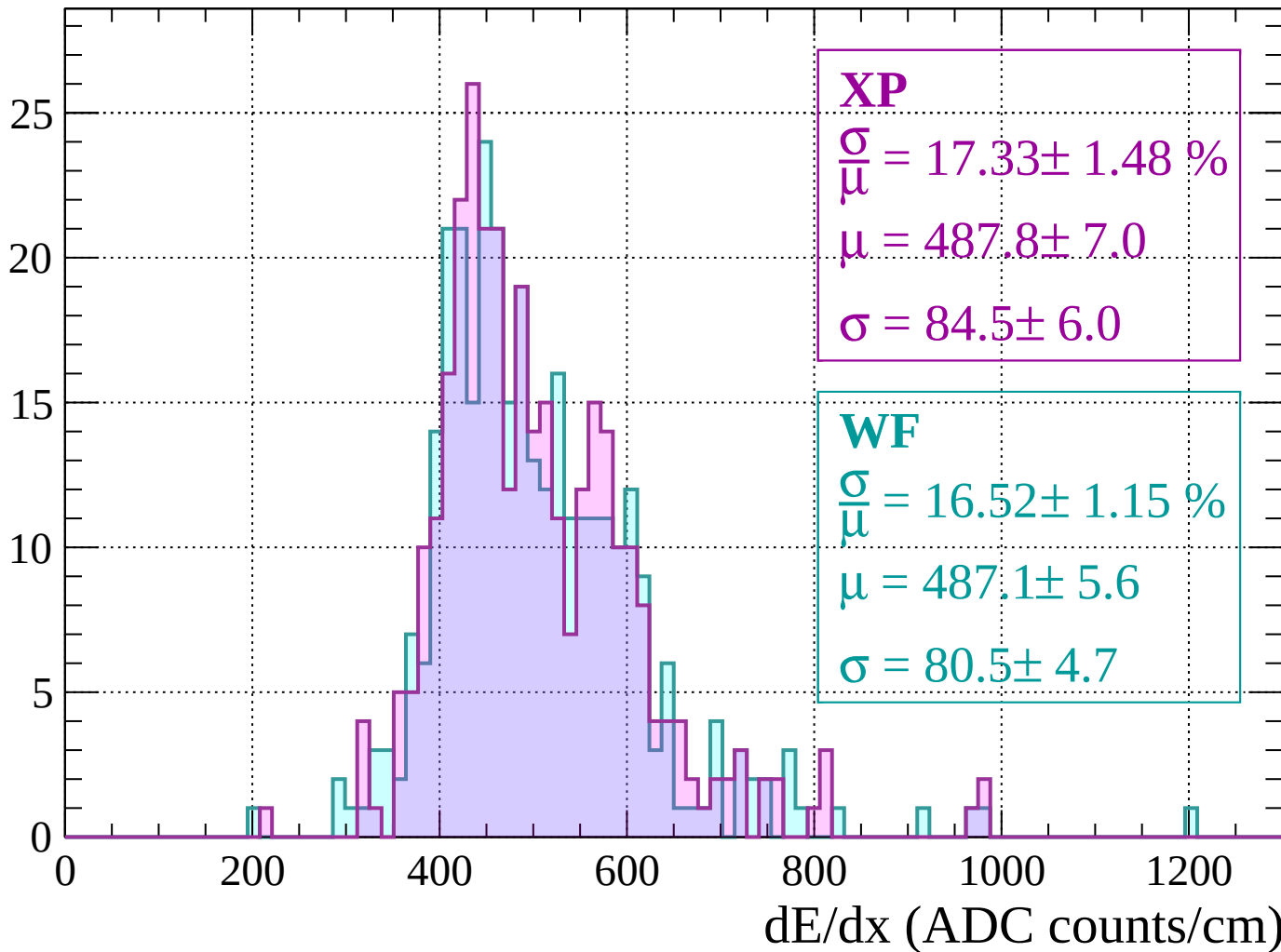
Energy loss |  $1080 < p < 1120$

Count



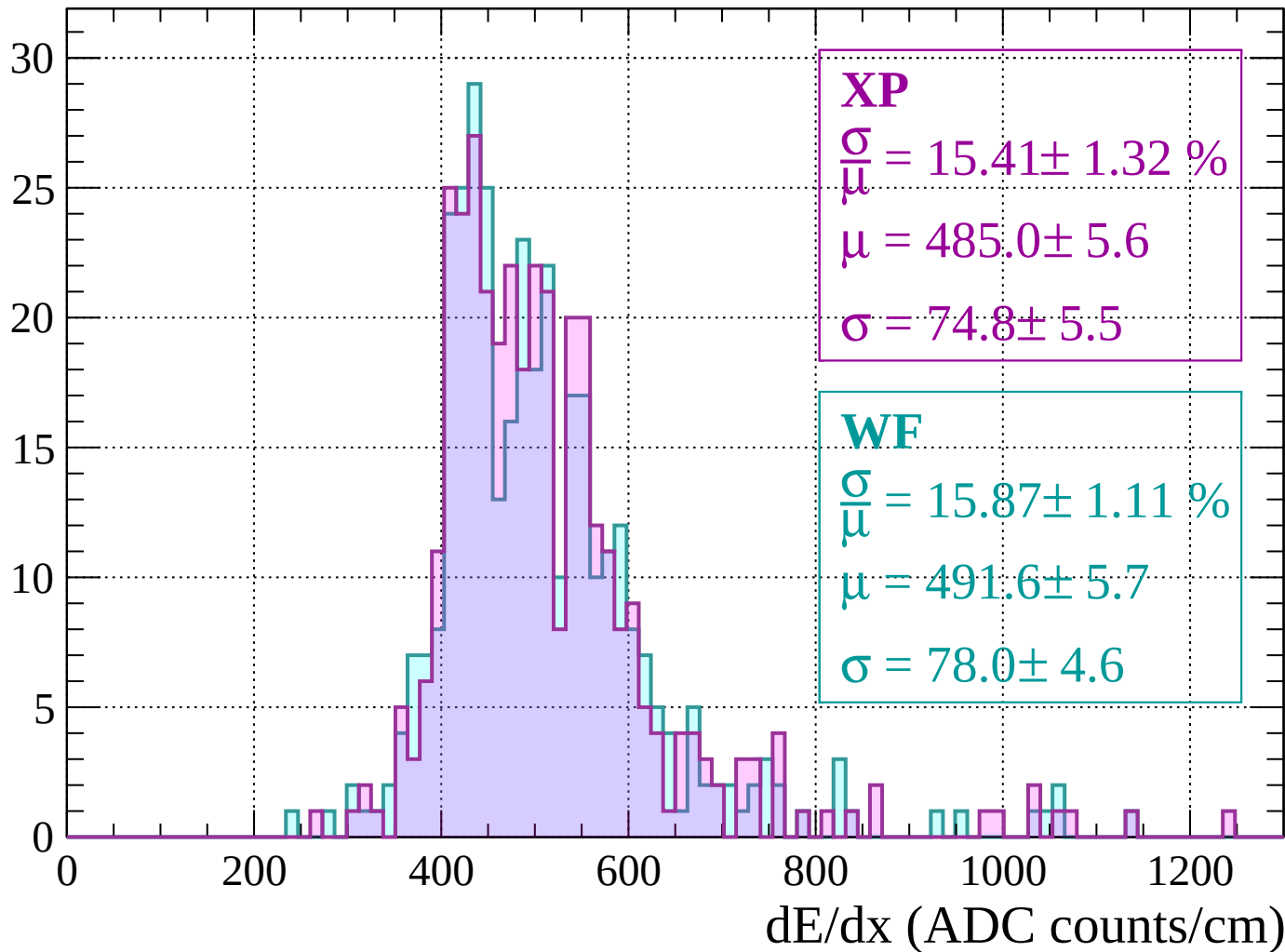
Energy loss |  $1120 < p < 1160$

Count



Energy loss |  $1160 < p < 1200$

Count





Energy loss |  $1200 < p < 1240$

Count

**XP**

$$\frac{\sigma}{\mu} = 10.52 \pm 0.62 \%$$

$$\mu = 465.5 \pm 3.3$$

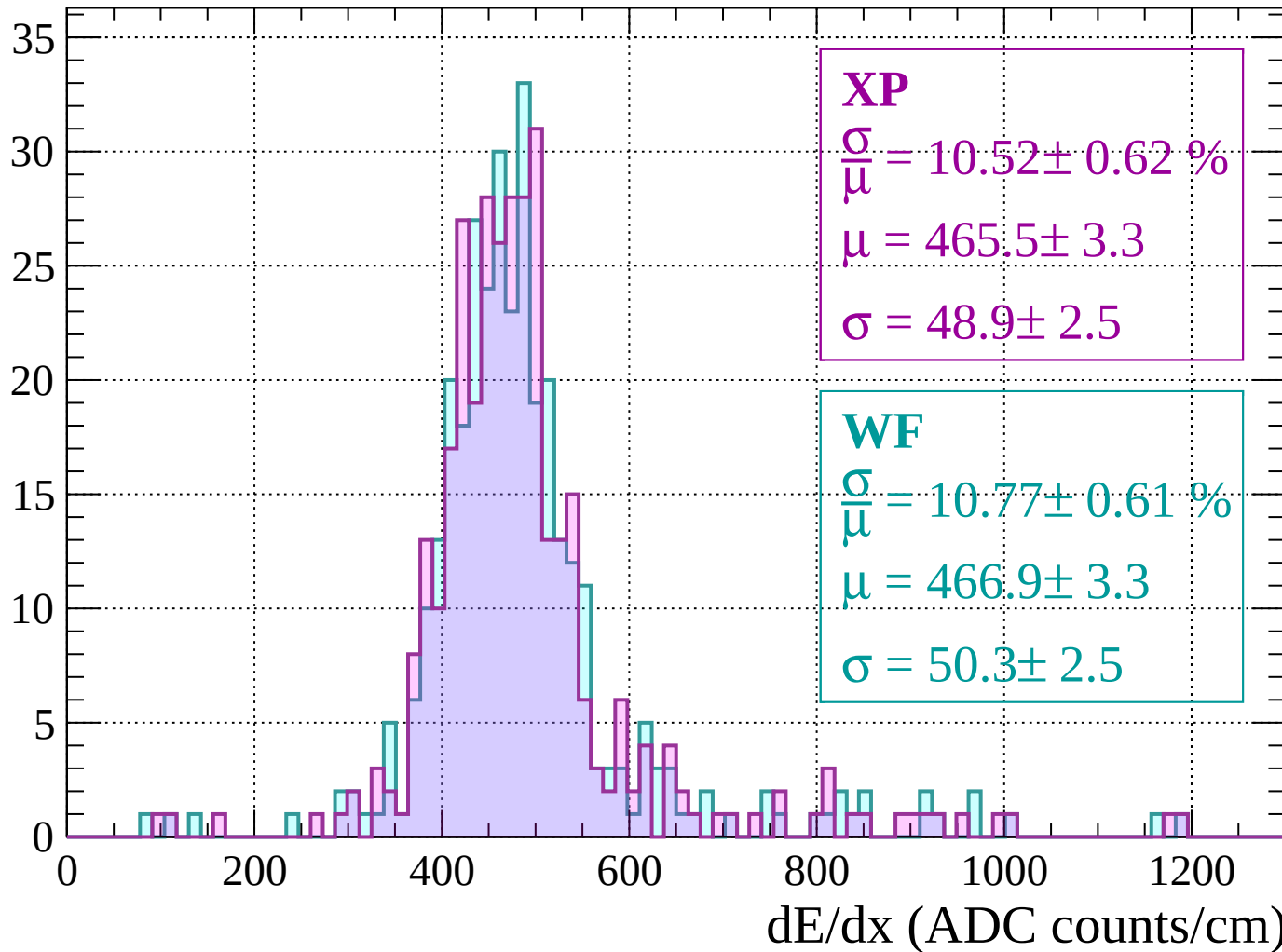
$$\sigma = 48.9 \pm 2.5$$

**WF**

$$\frac{\sigma}{\mu} = 10.77 \pm 0.61 \%$$

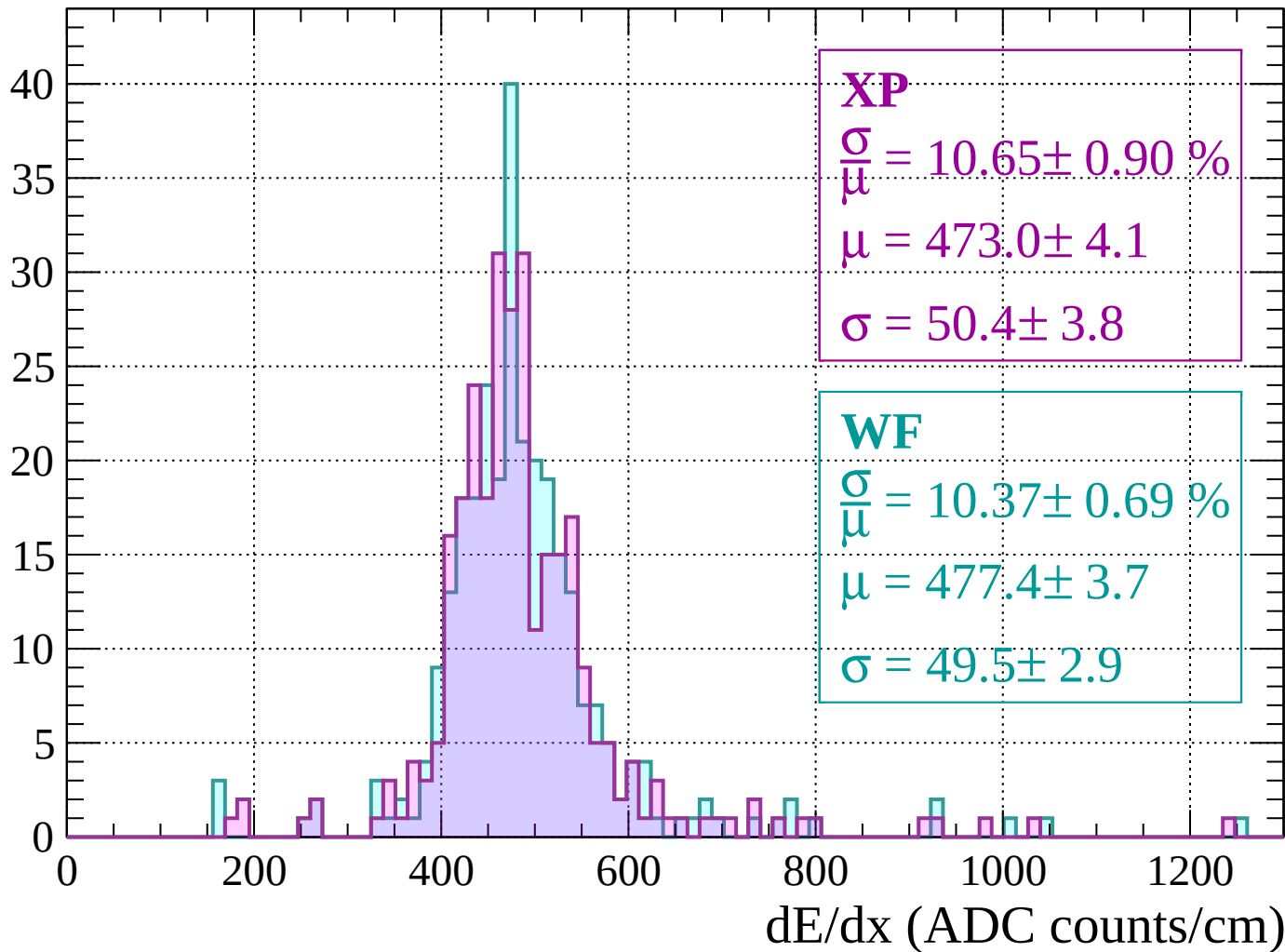
$$\mu = 466.9 \pm 3.3$$

$$\sigma = 50.3 \pm 2.5$$



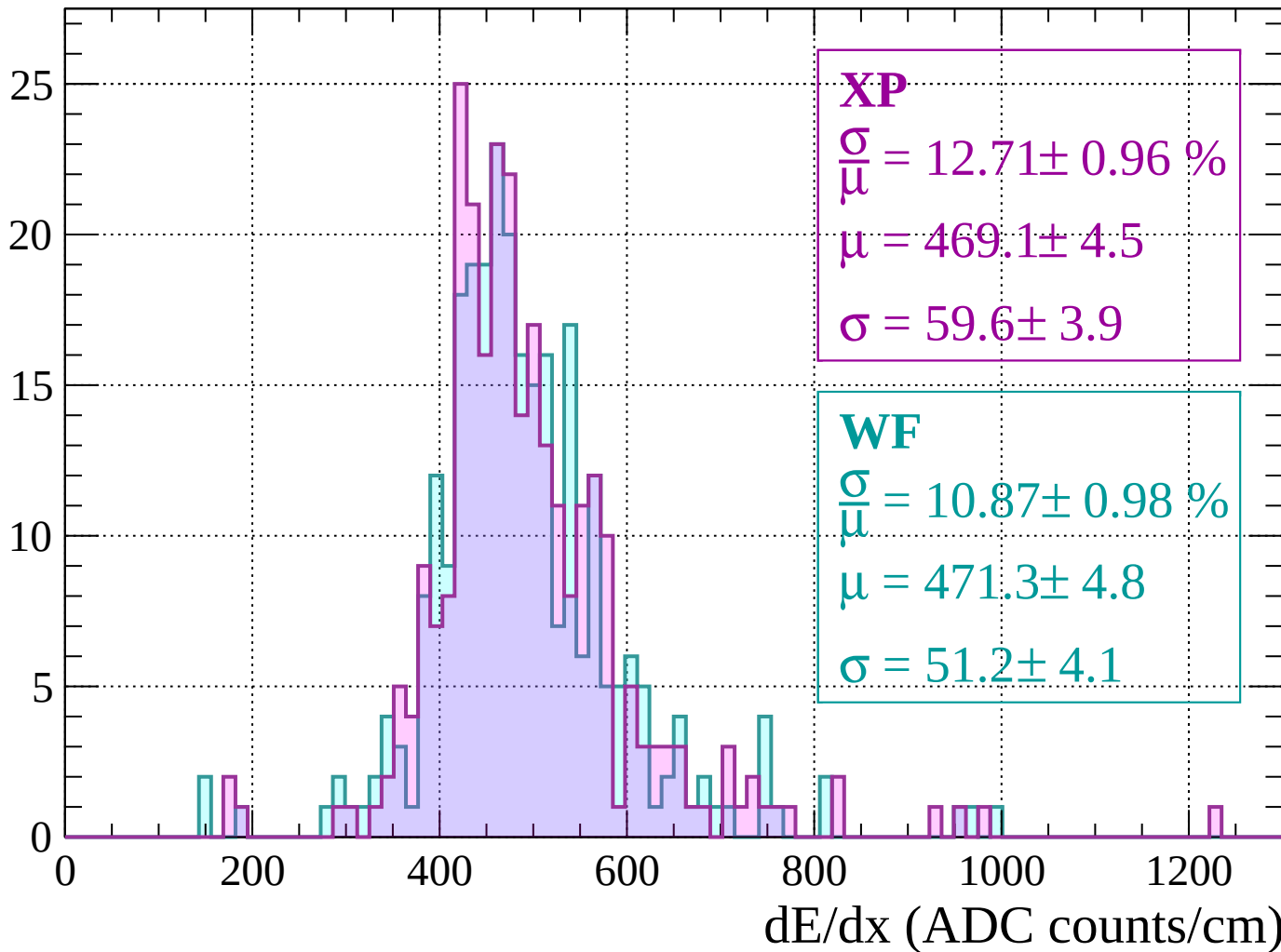
Energy loss |  $1240 < p < 1280$

Count



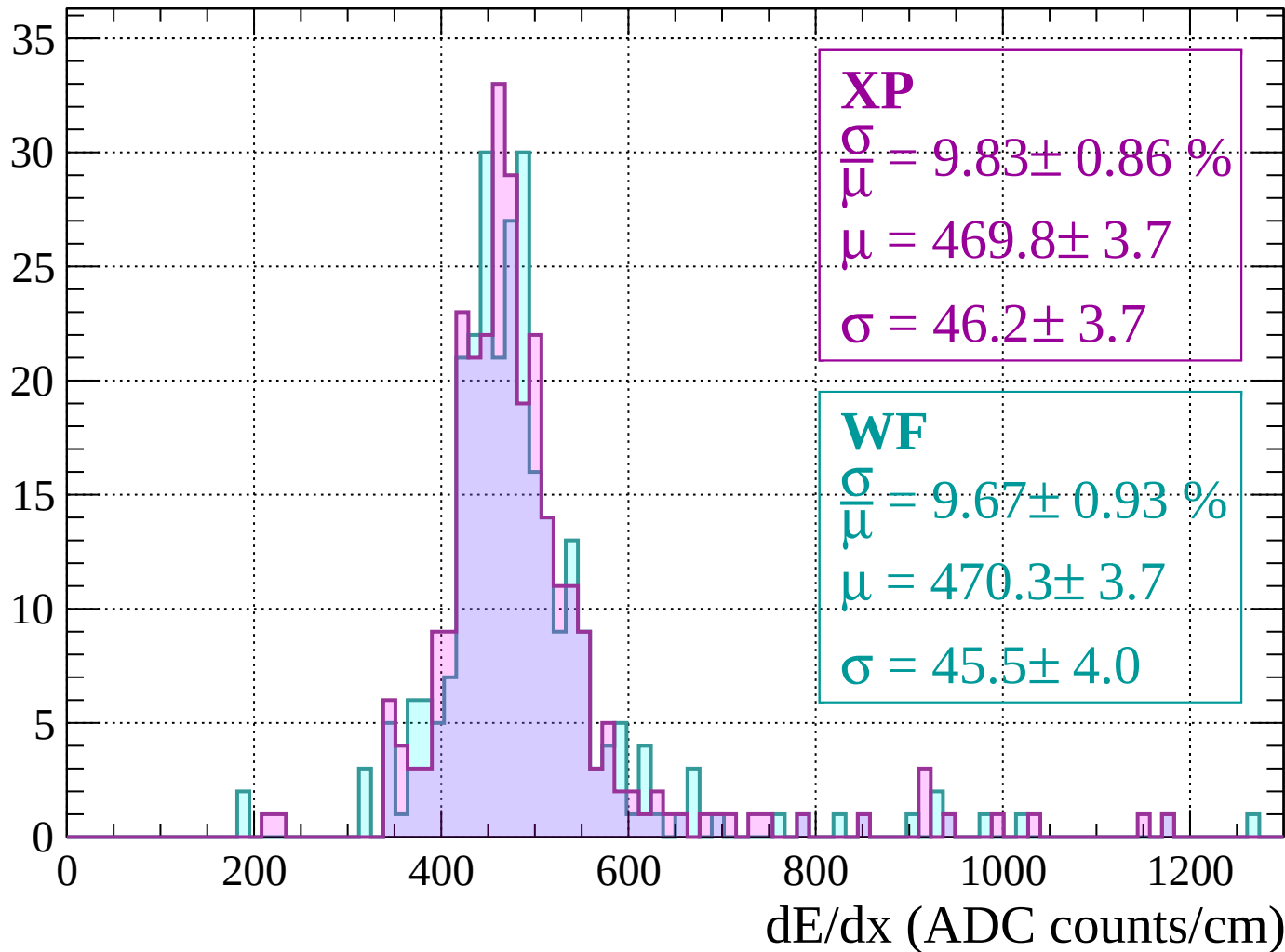
Energy loss |  $1280 < p < 1320$

Count



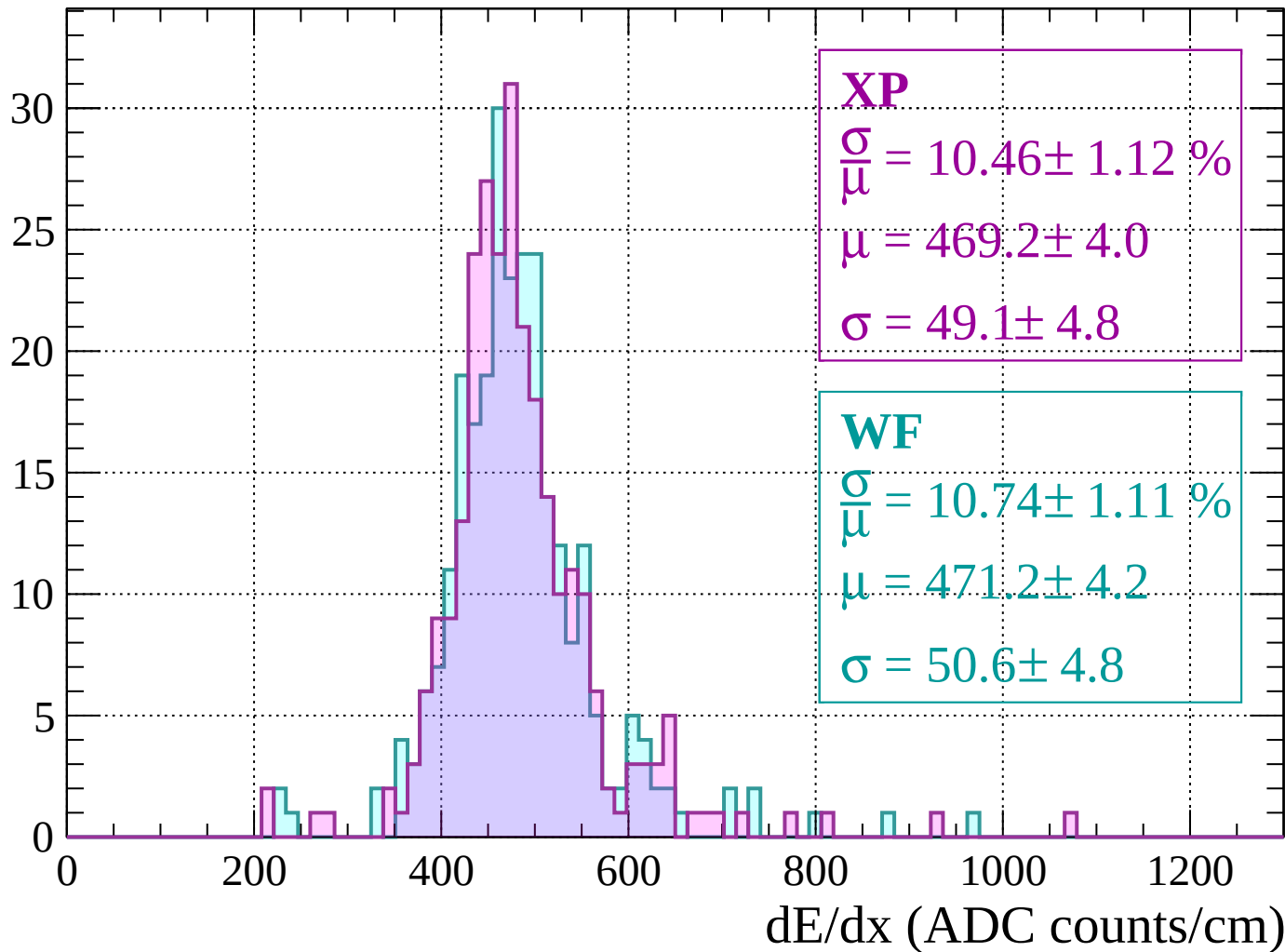
Energy loss |  $1320 < p < 1360$

Count



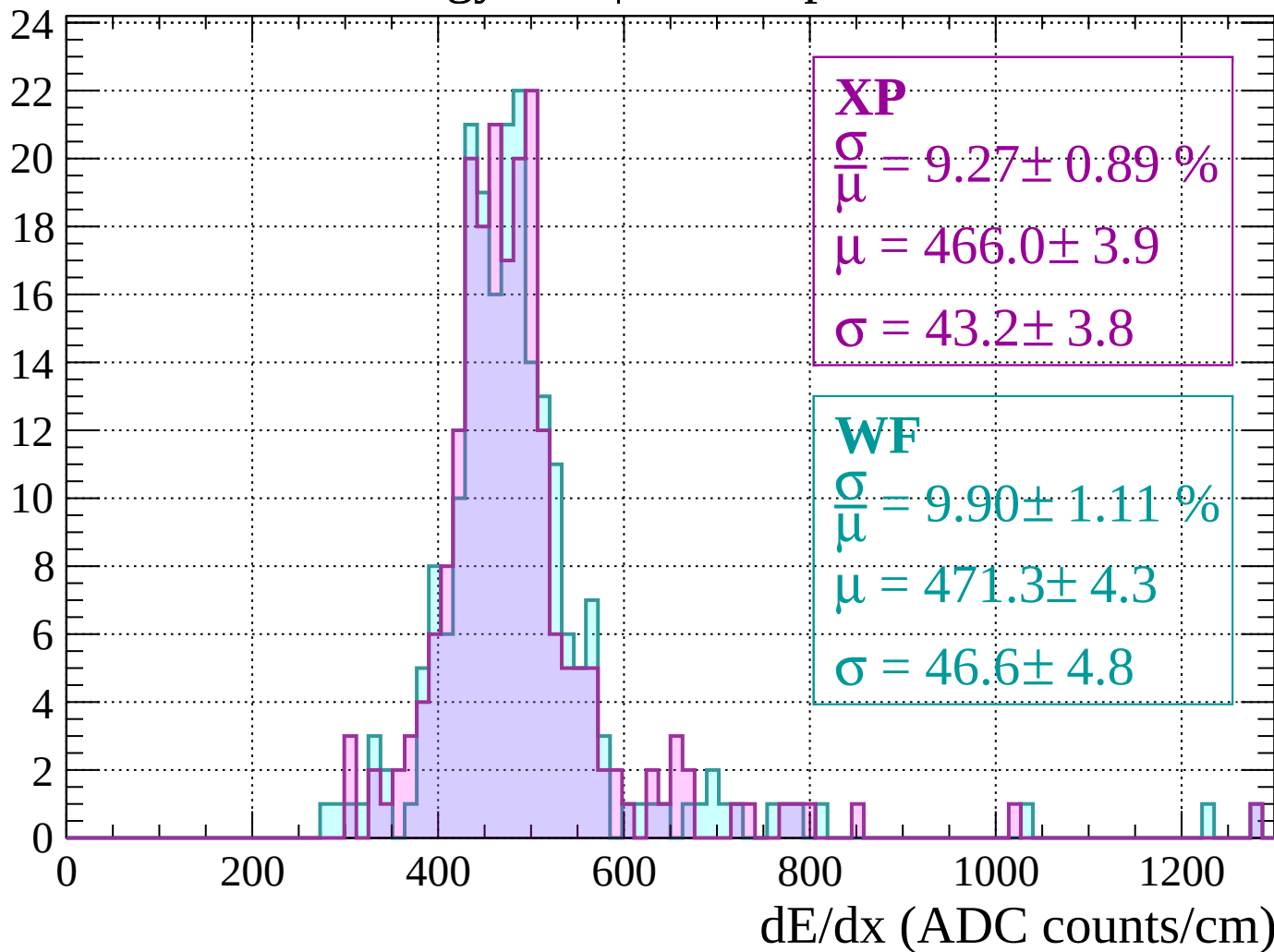
Energy loss |  $1360 < p < 1400$

Count



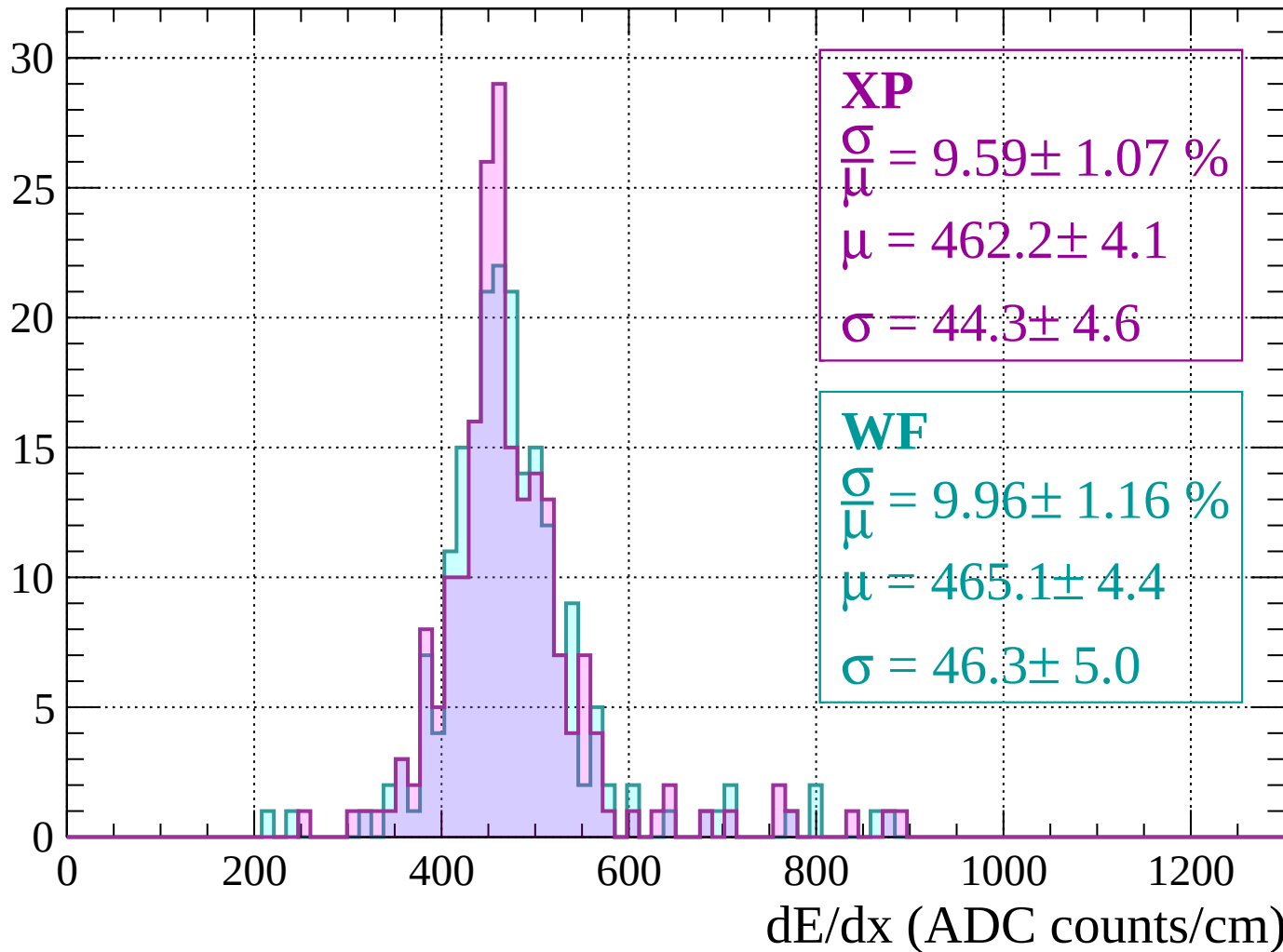
Energy loss |  $1400 < p < 1440$

Count



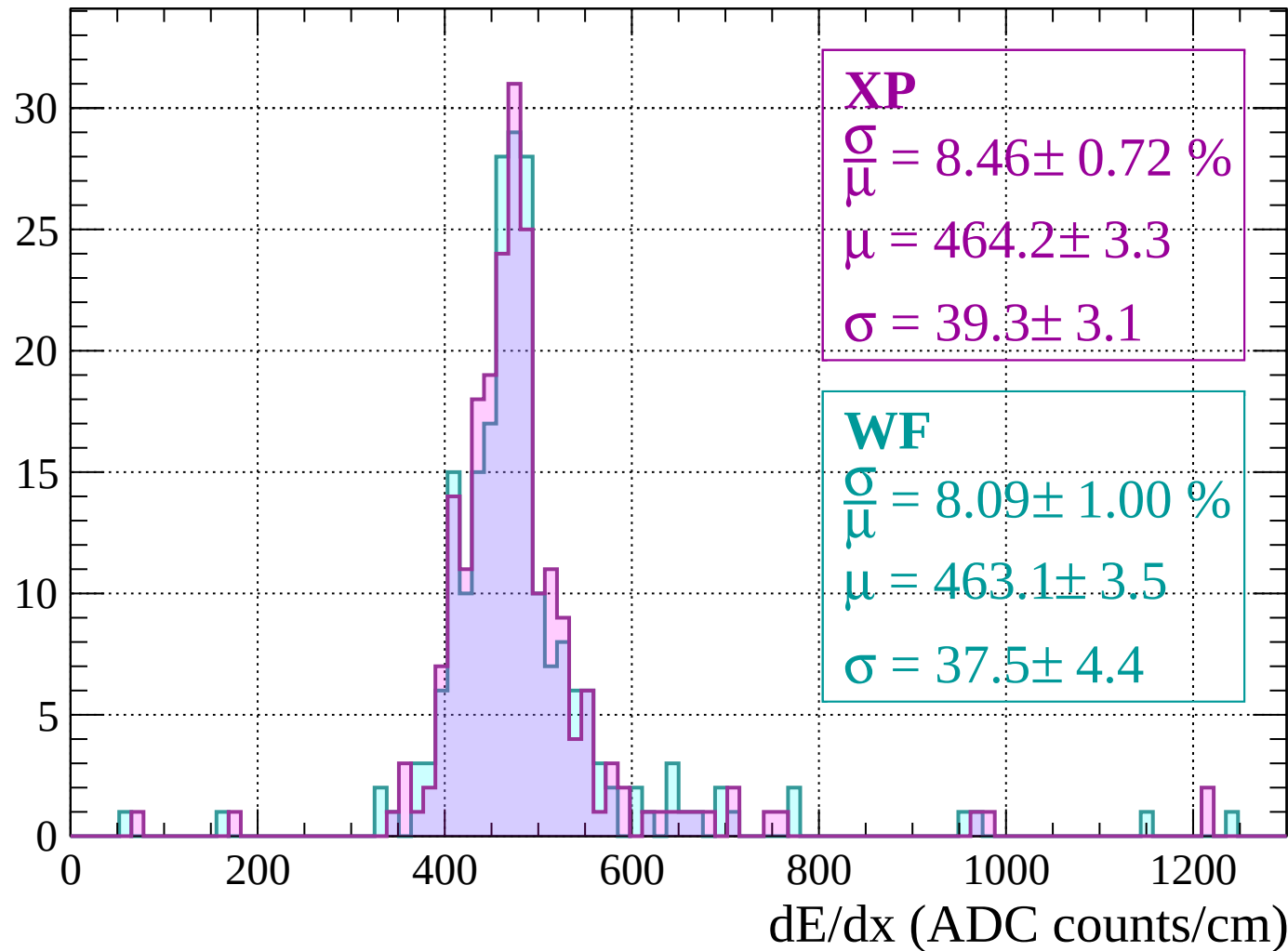
Energy loss |  $1440 < p < 1480$

Count



Energy loss |  $1480 < p < 1520$

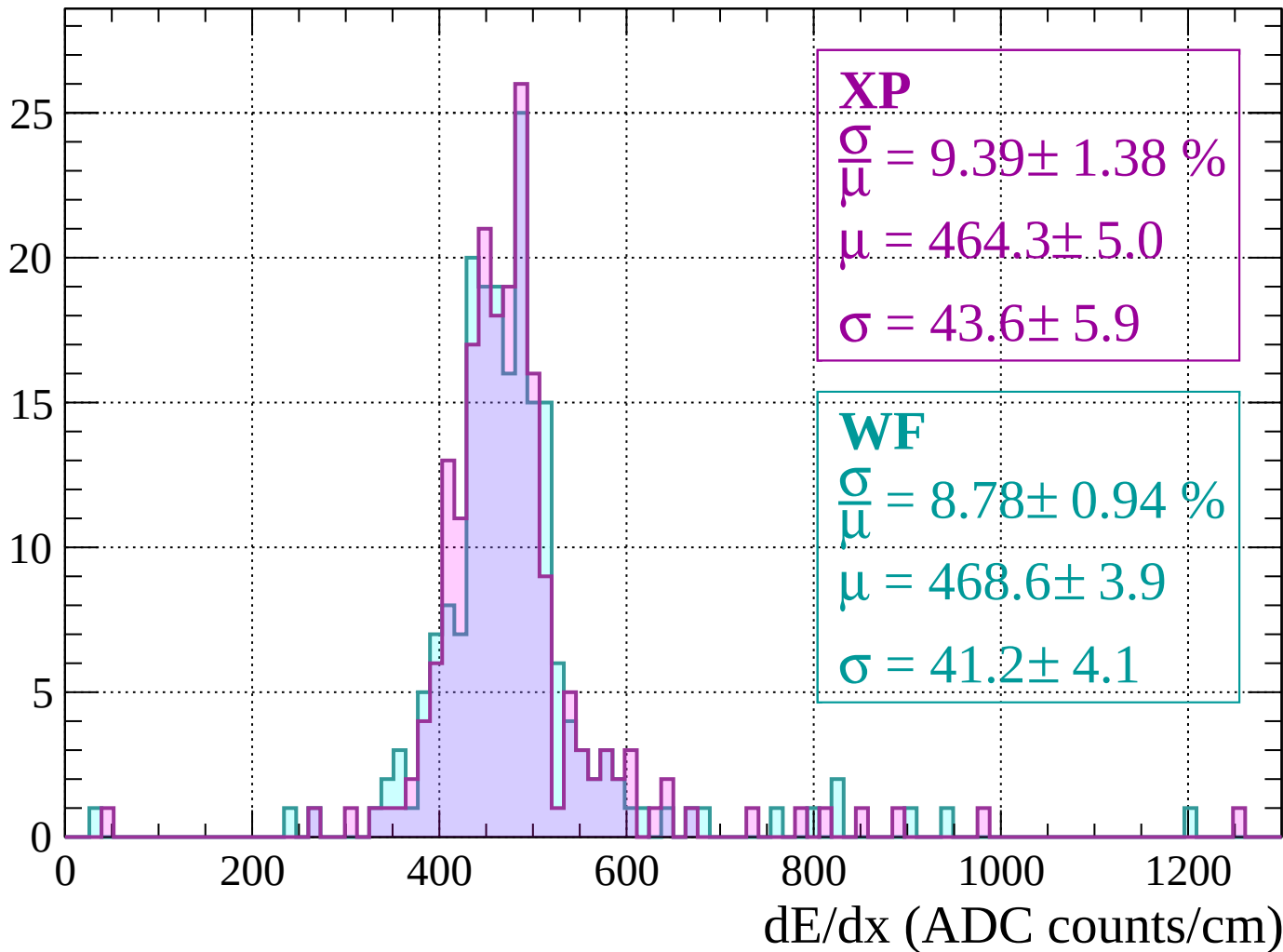
Count





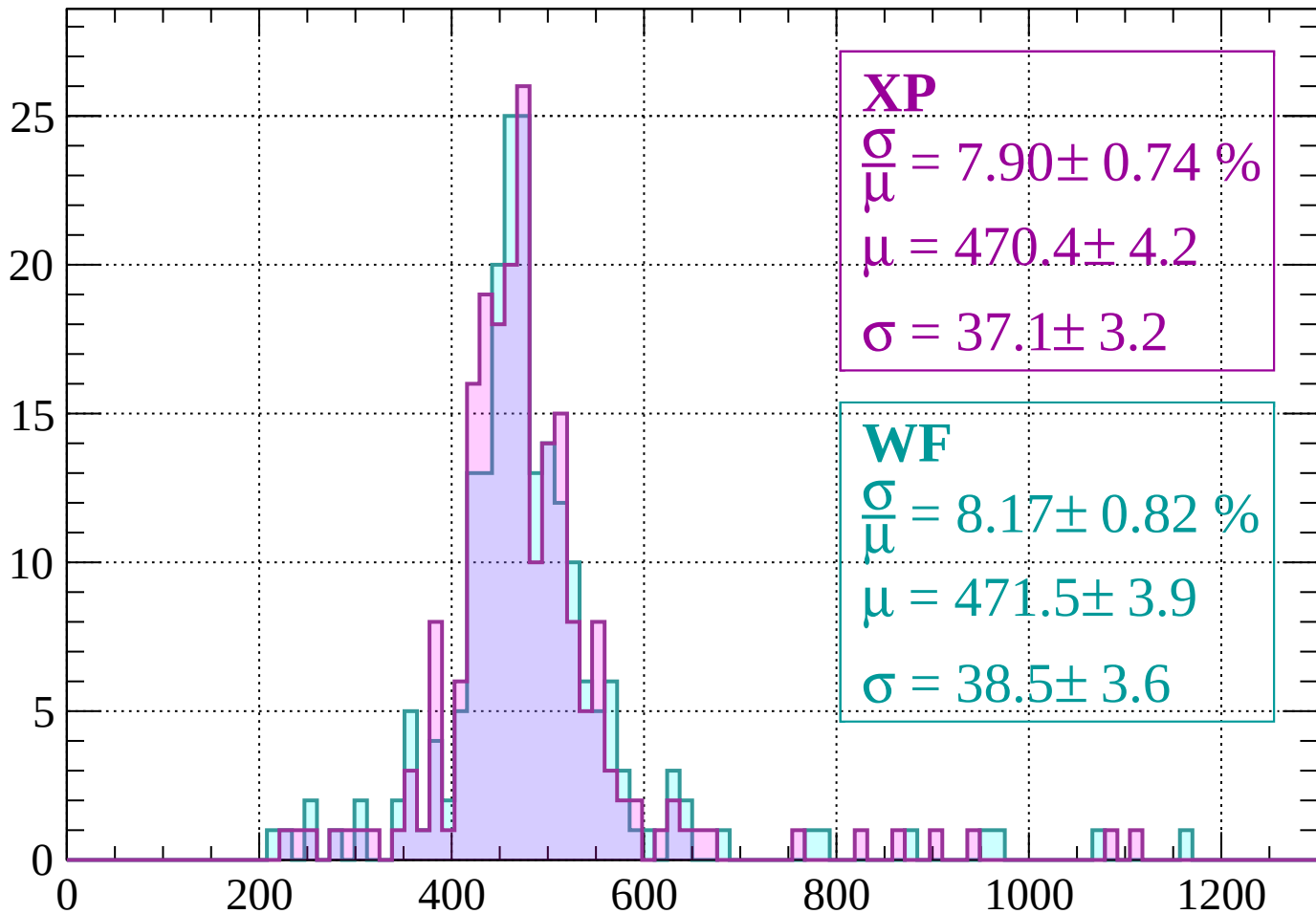
Energy loss |  $1520 < p < 1560$

Count



Energy loss |  $1560 < p < 1600$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.90 \pm 0.74 \%$$

$$\mu = 470.4 \pm 4.2$$

$$\sigma = 37.1 \pm 3.2$$

**WF**

$$\frac{\sigma}{\mu} = 8.17 \pm 0.82 \%$$

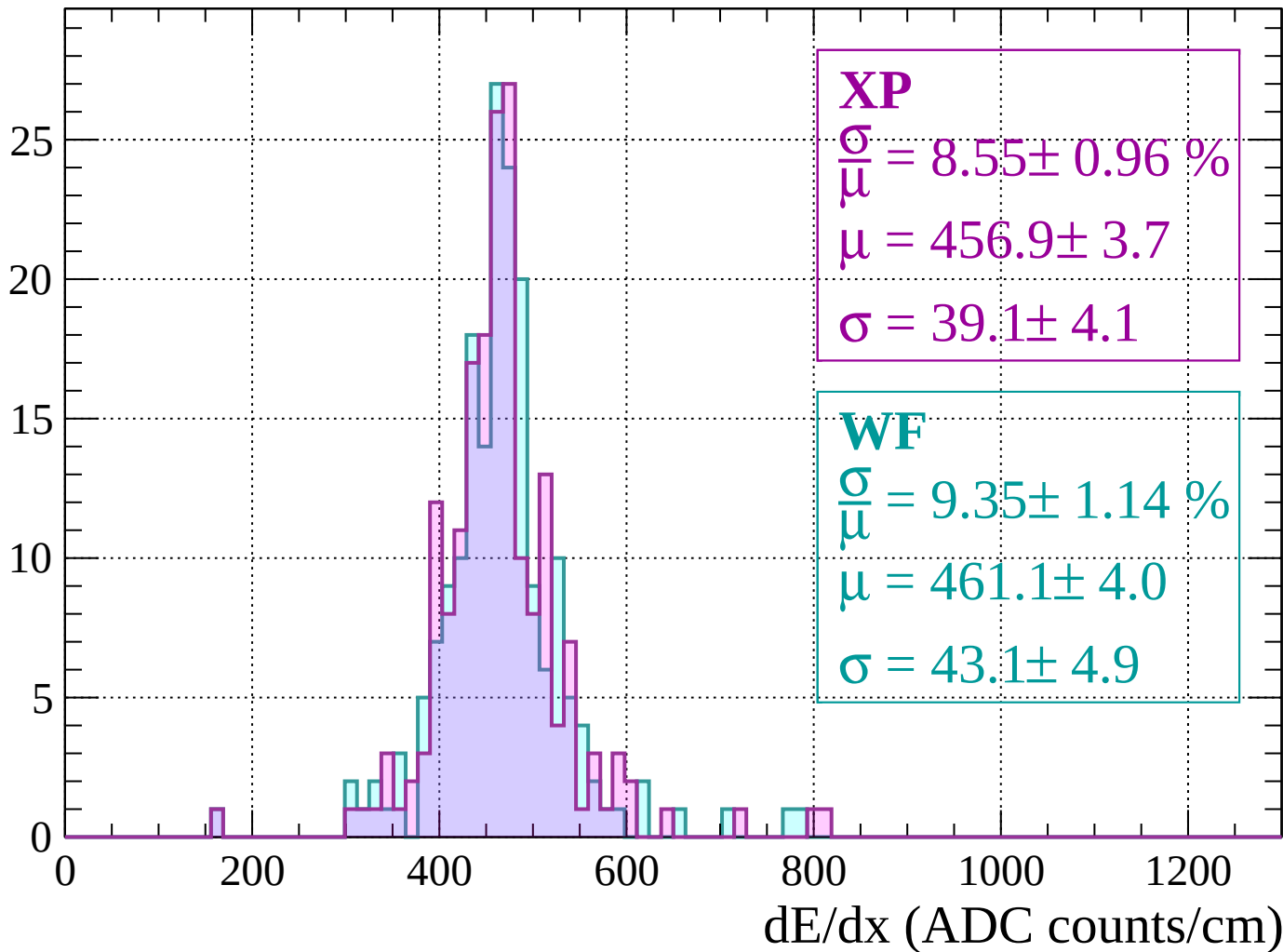
$$\mu = 471.5 \pm 3.9$$

$$\sigma = 38.5 \pm 3.6$$

$dE/dx$  (ADC counts/cm)

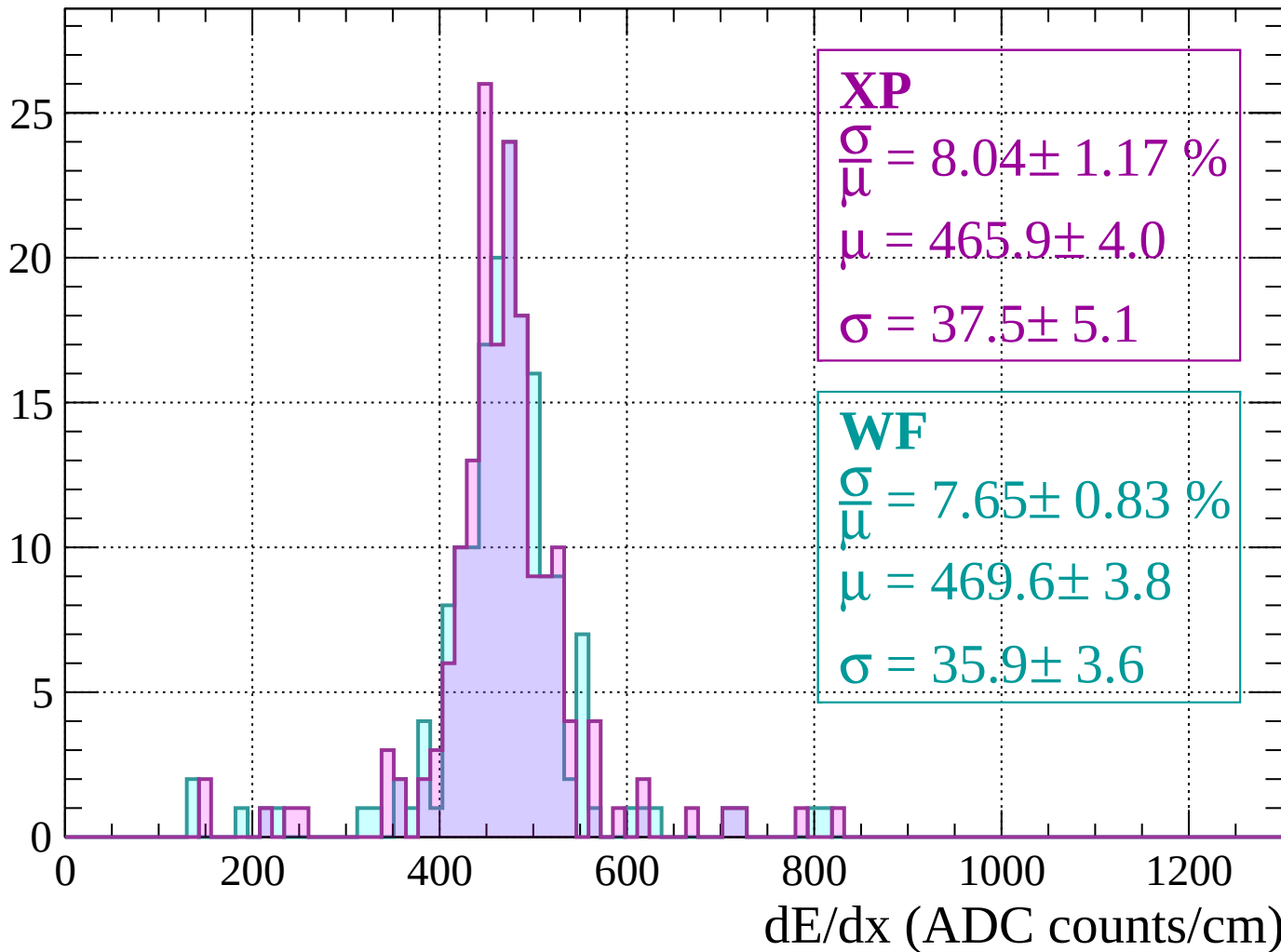
Energy loss |  $1600 < p < 1640$

Count



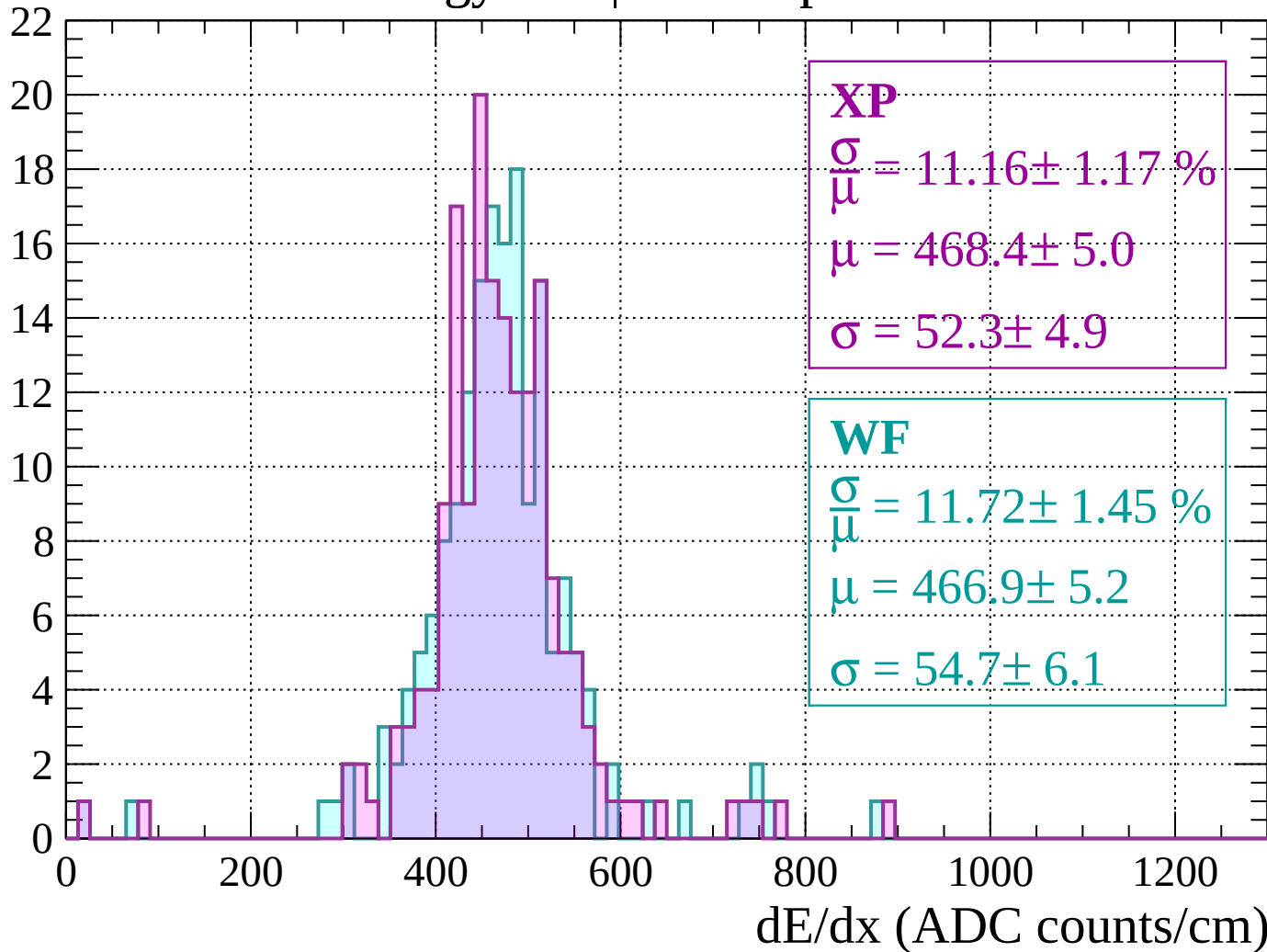
Energy loss |  $1640 < p < 1680$

Count



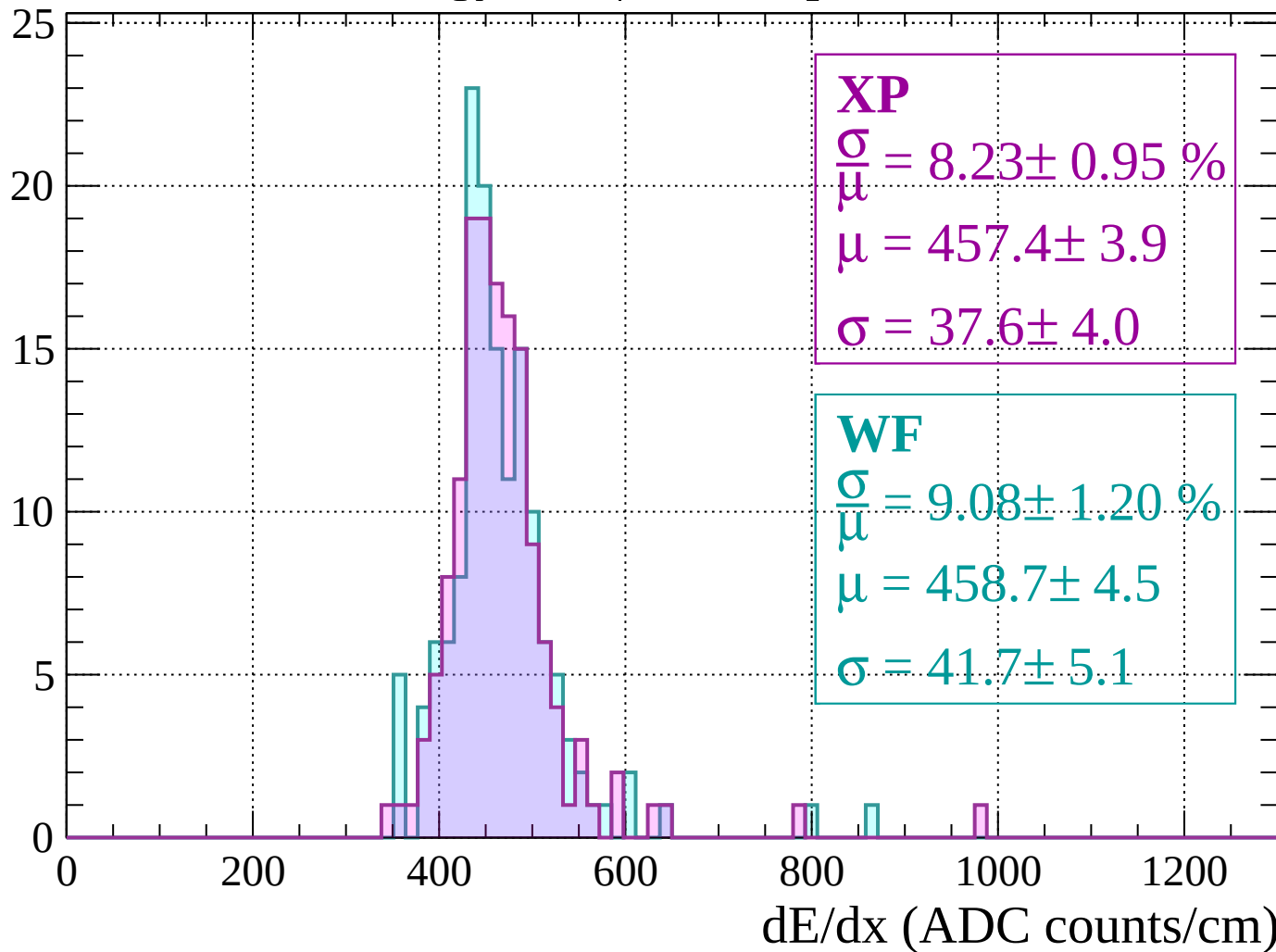
Energy loss |  $1680 < p < 1720$

Count



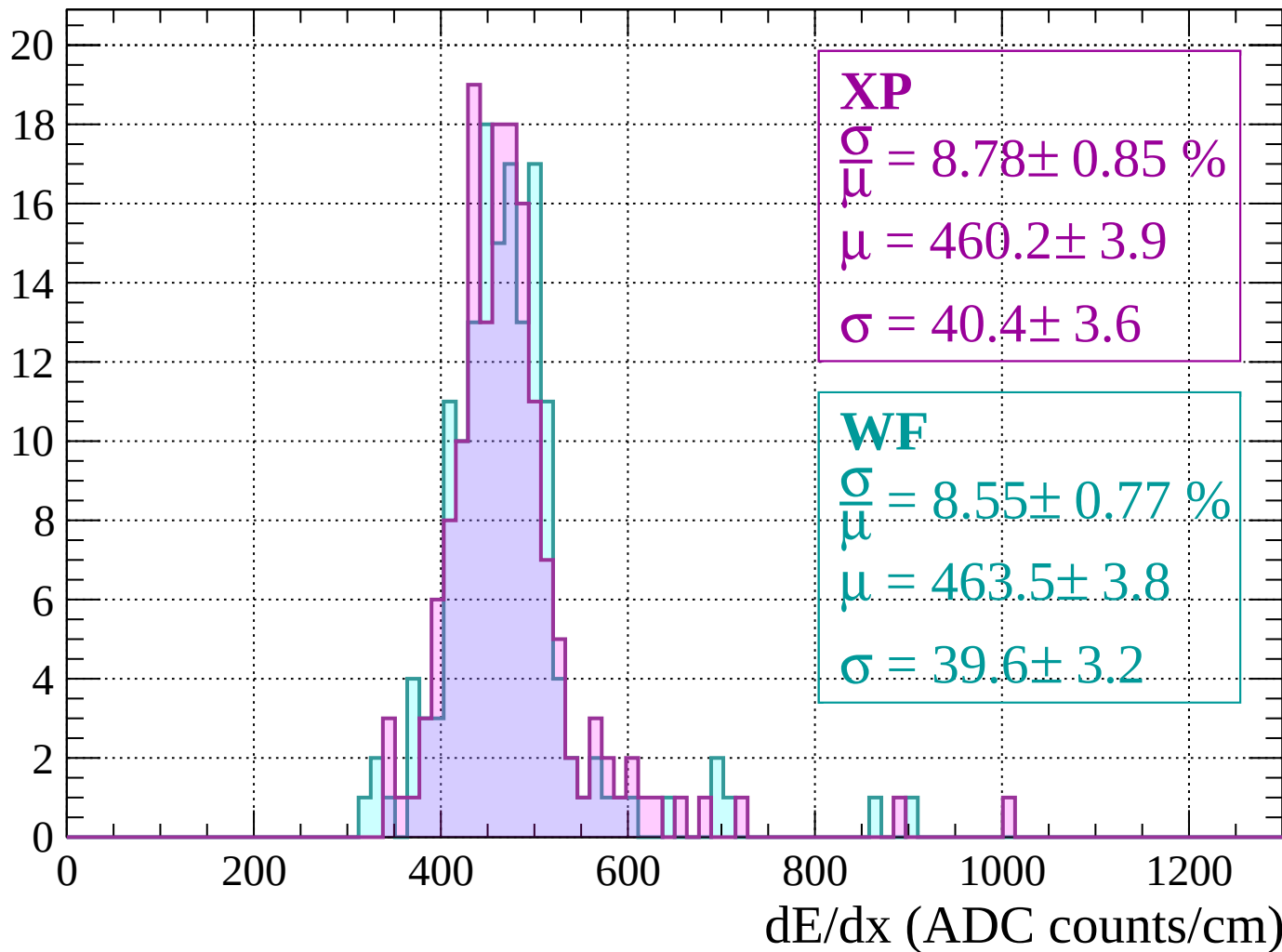
Energy loss |  $1720 < p < 1760$

Count



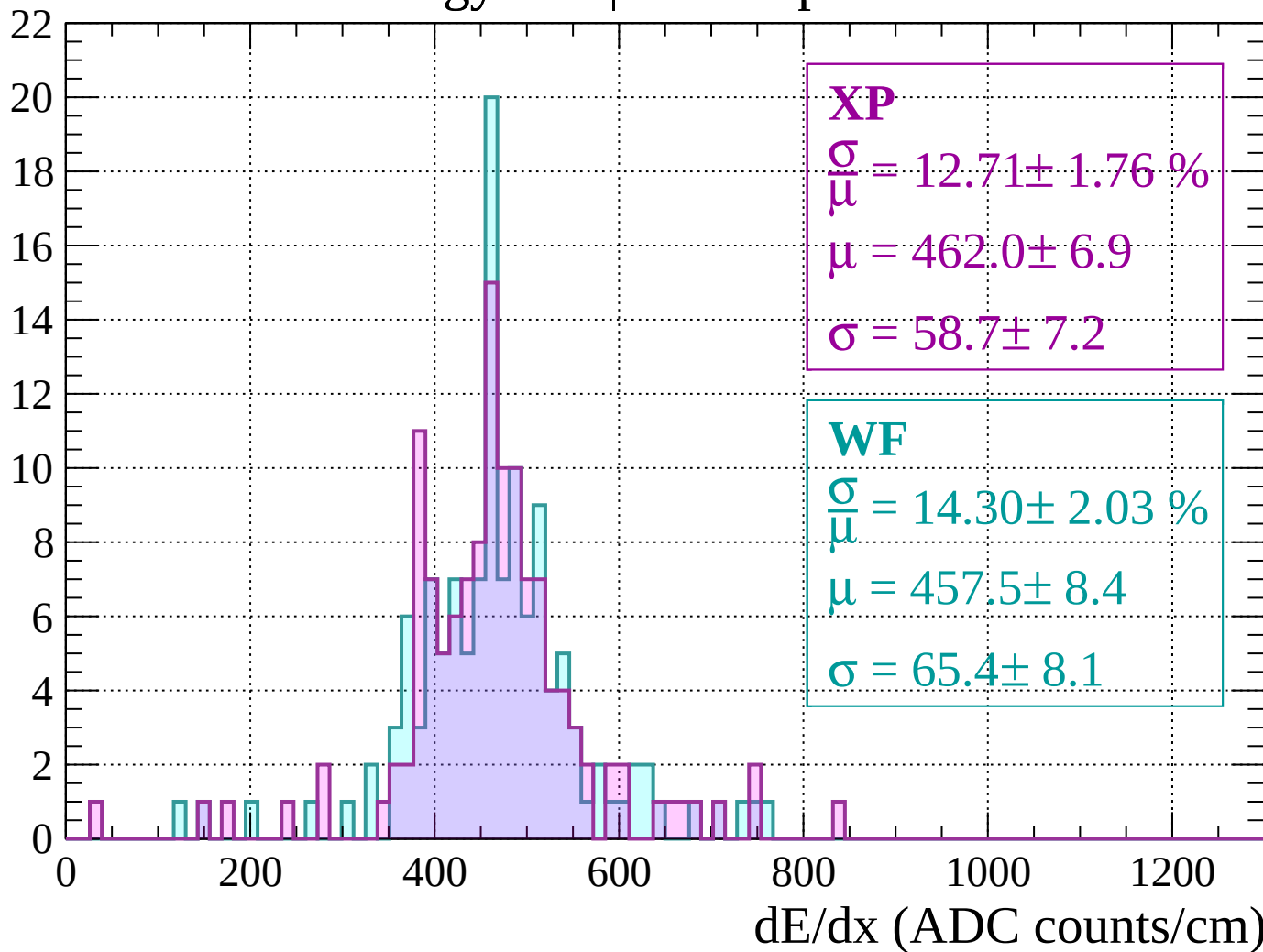
Energy loss |  $1760 < p < 1800$

Count



Energy loss |  $1800 < p < 1840$

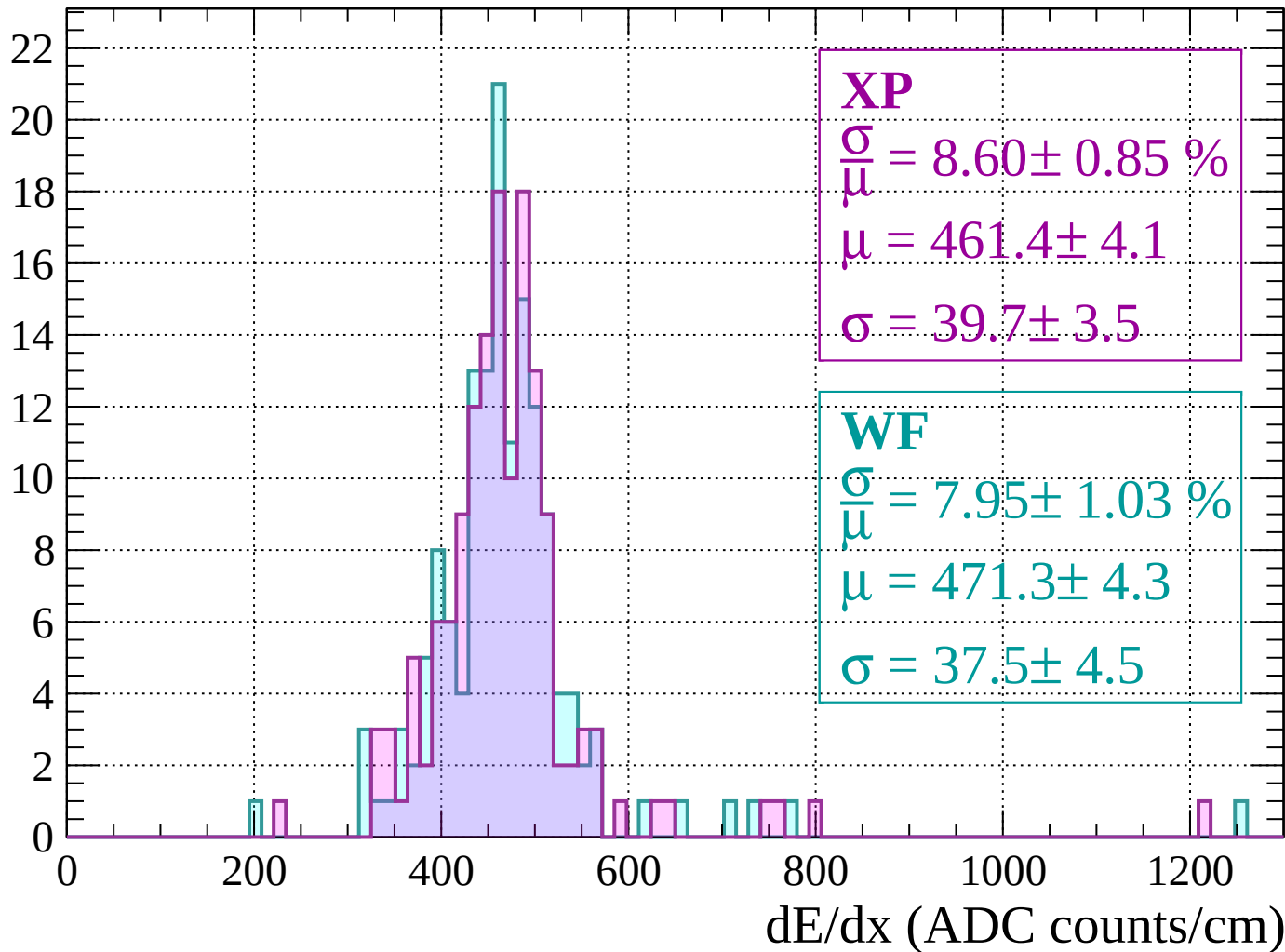
Count





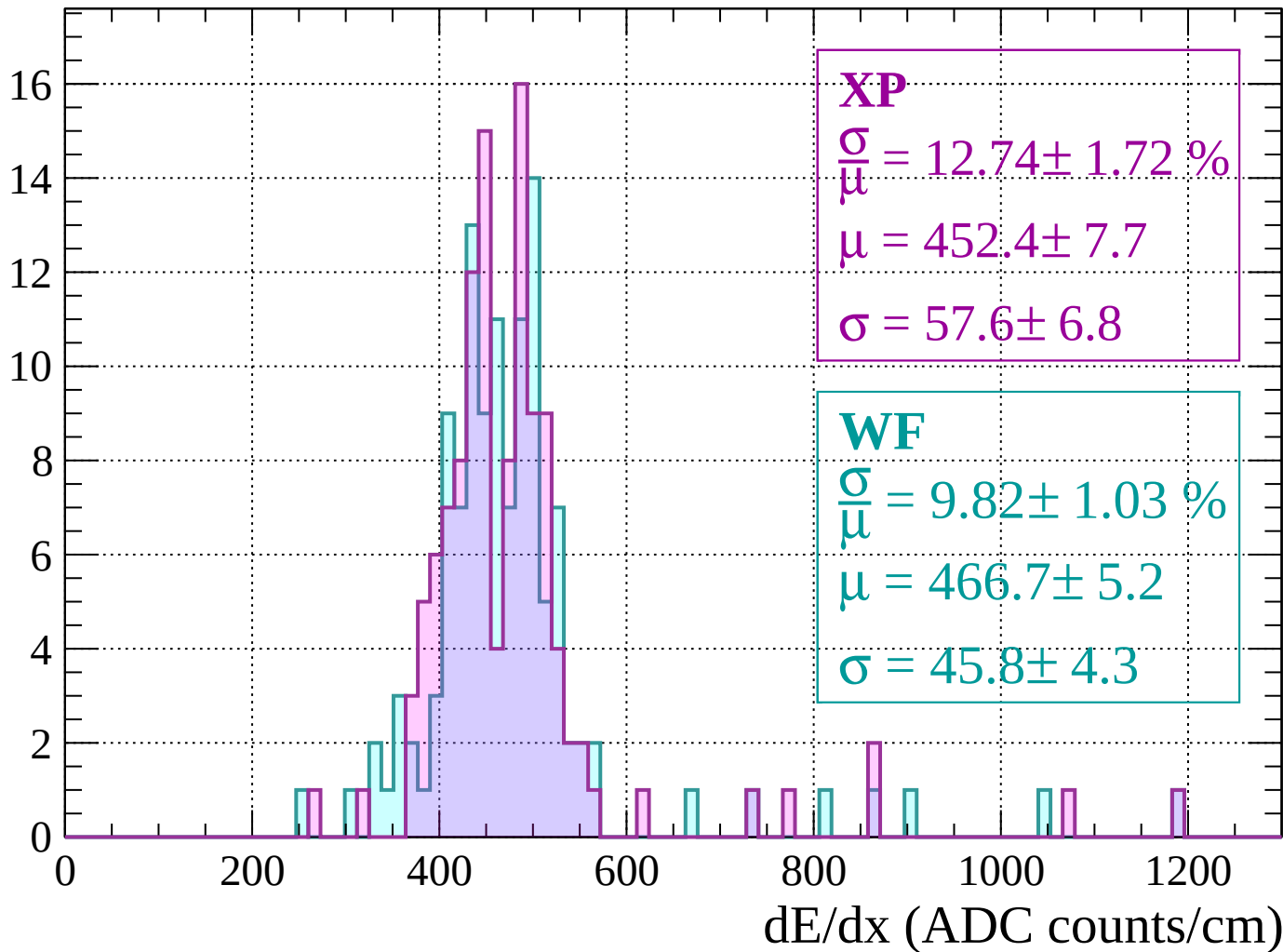
Energy loss |  $1840 < p < 1880$

Count



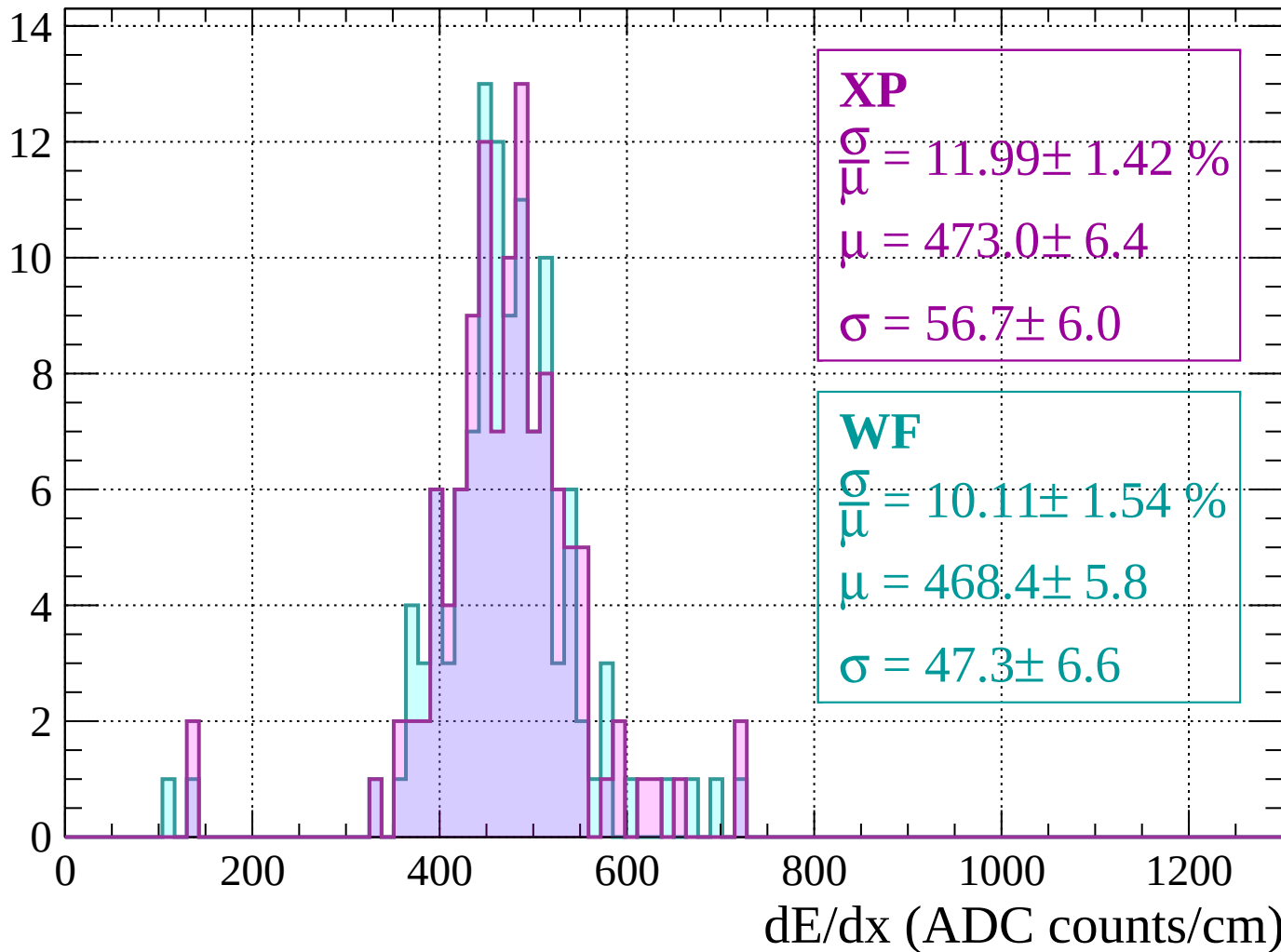
Energy loss |  $1880 < p < 1920$

Count



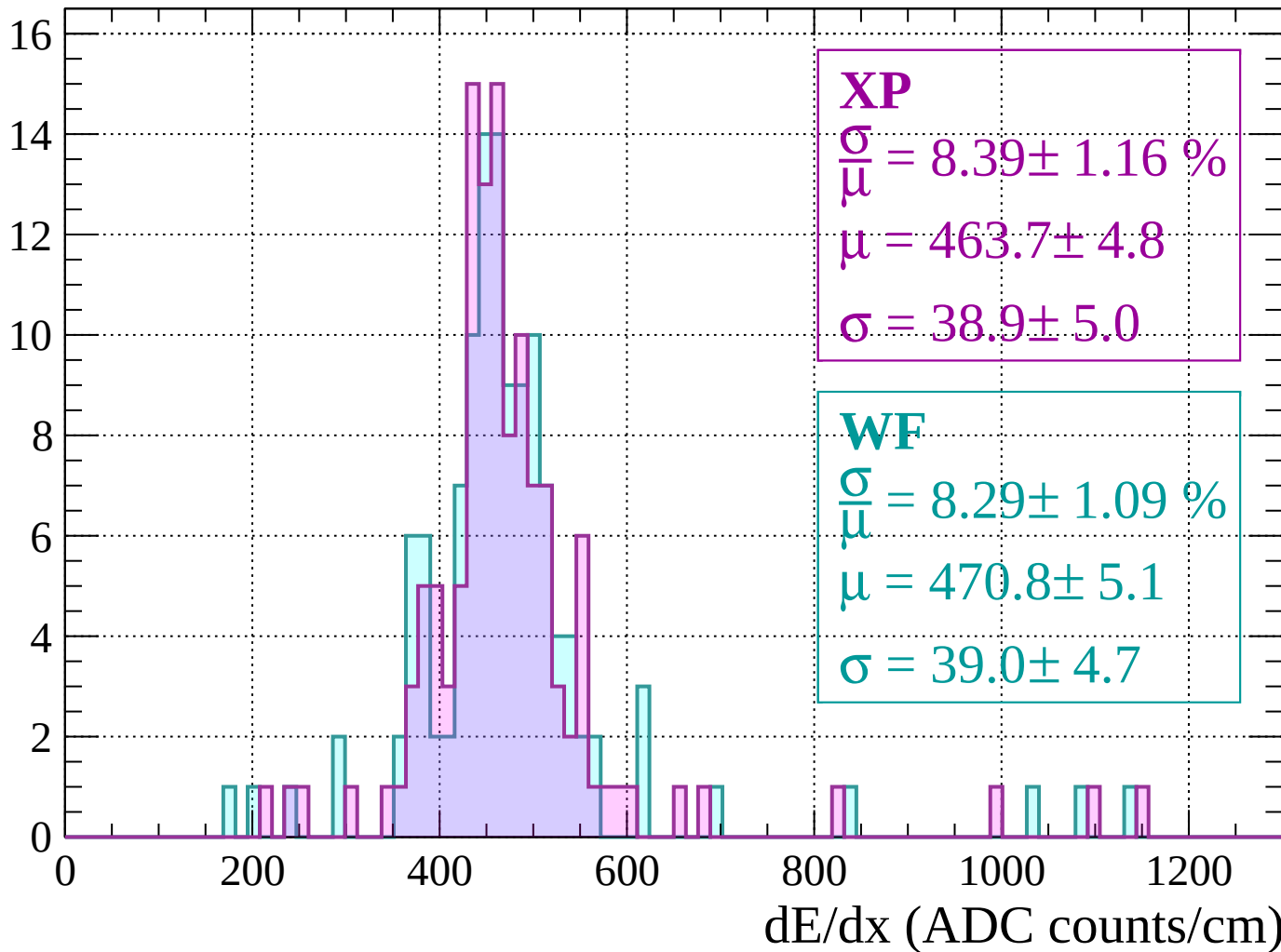
# Energy loss | $1920 < p < 1960$

Count



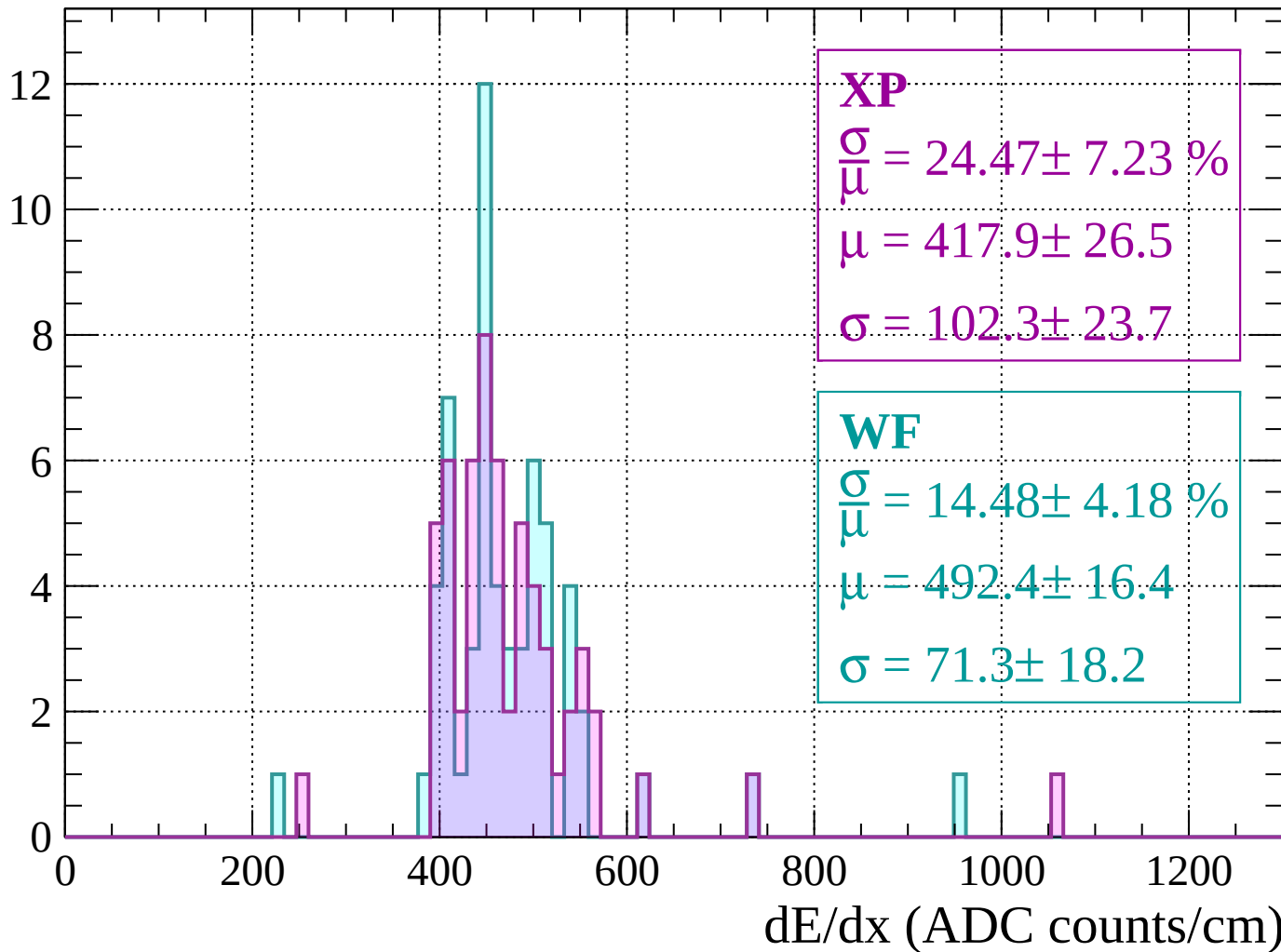
Energy loss |  $1960 < p < 2000$

Count



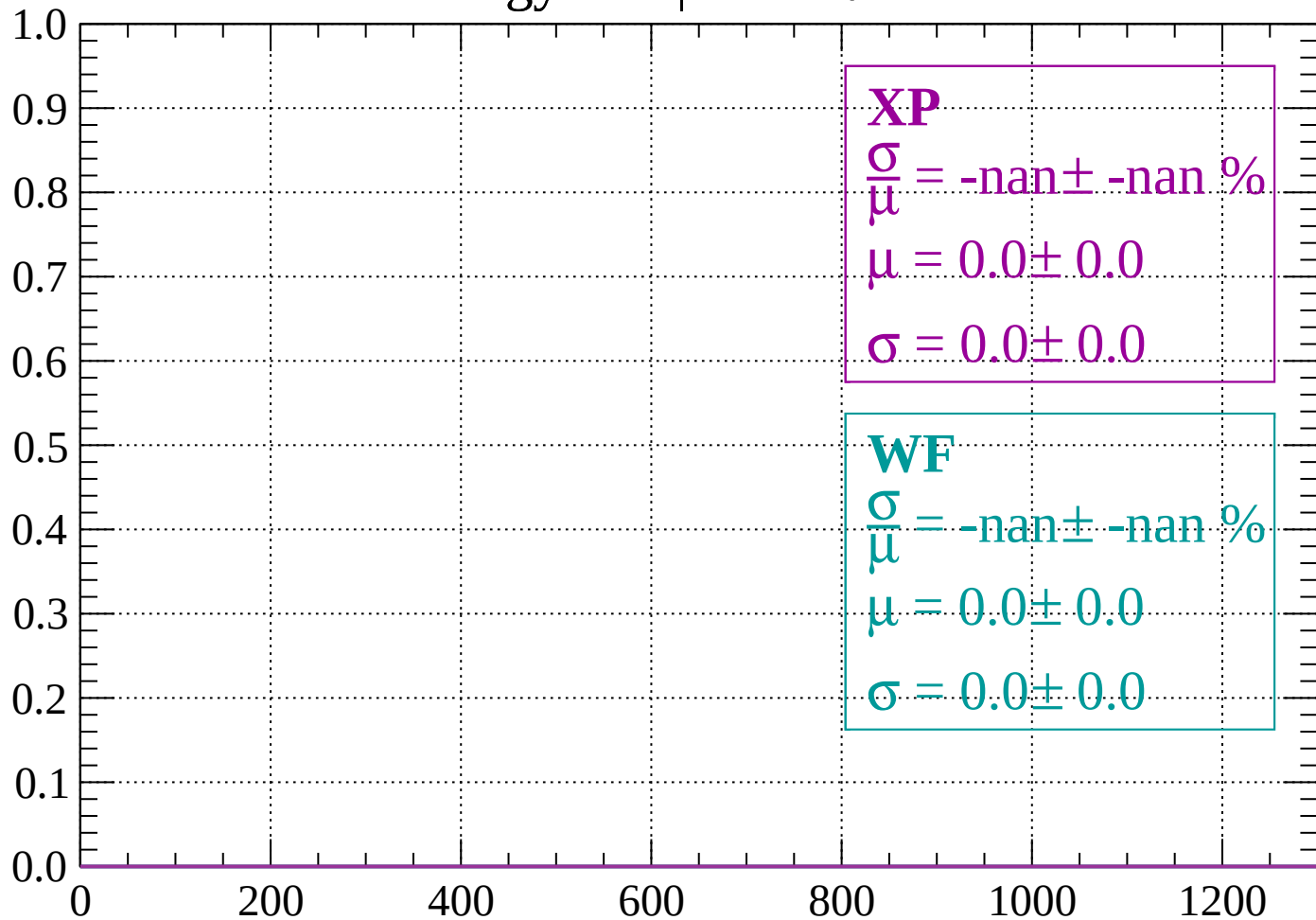
Energy loss |  $2000 < p < 2040$

Count



Energy loss |  $-90 < \theta < -84$

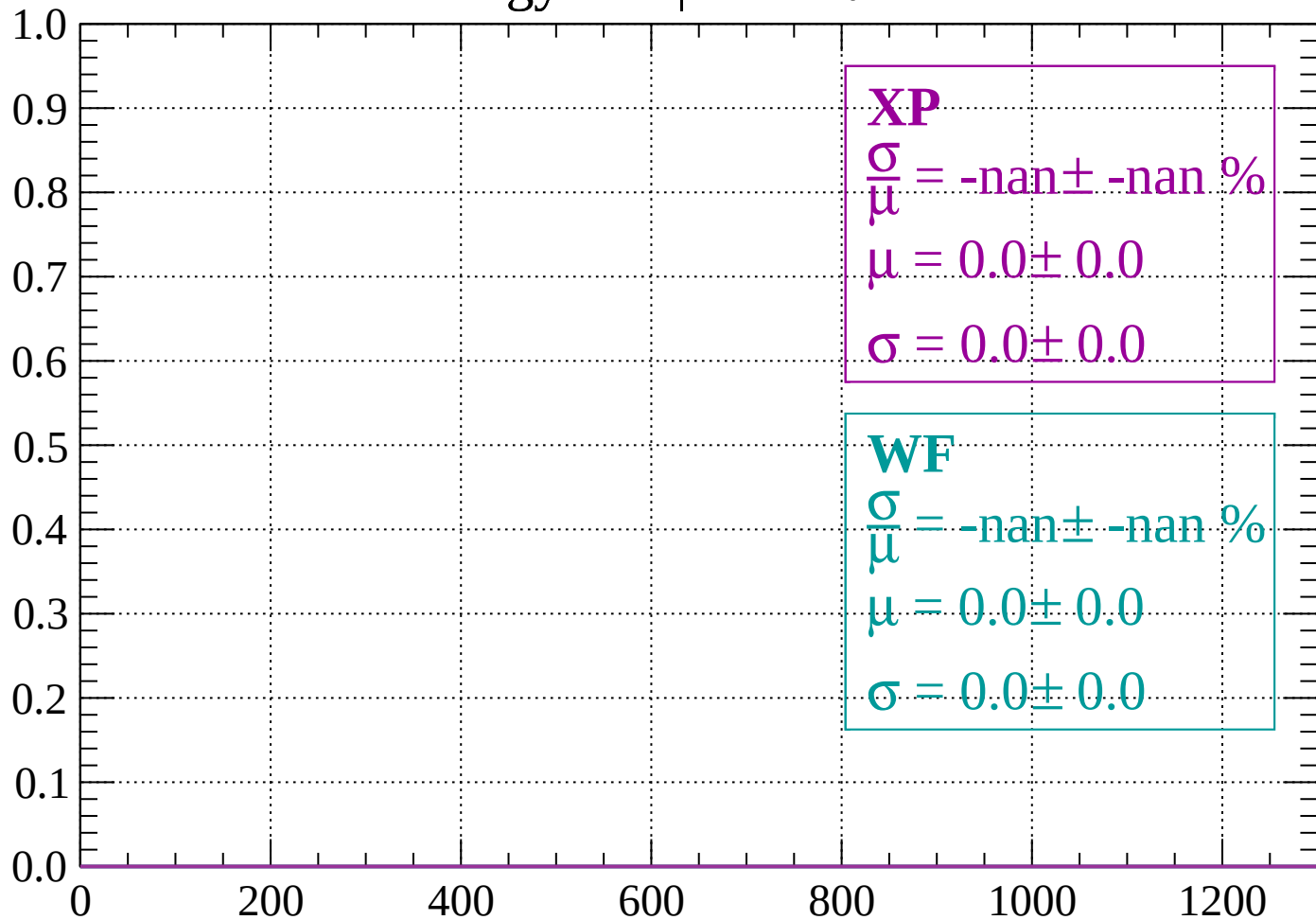
Count



dE/dx (ADC counts/cm)

Energy loss |  $-84 < \theta < -78$

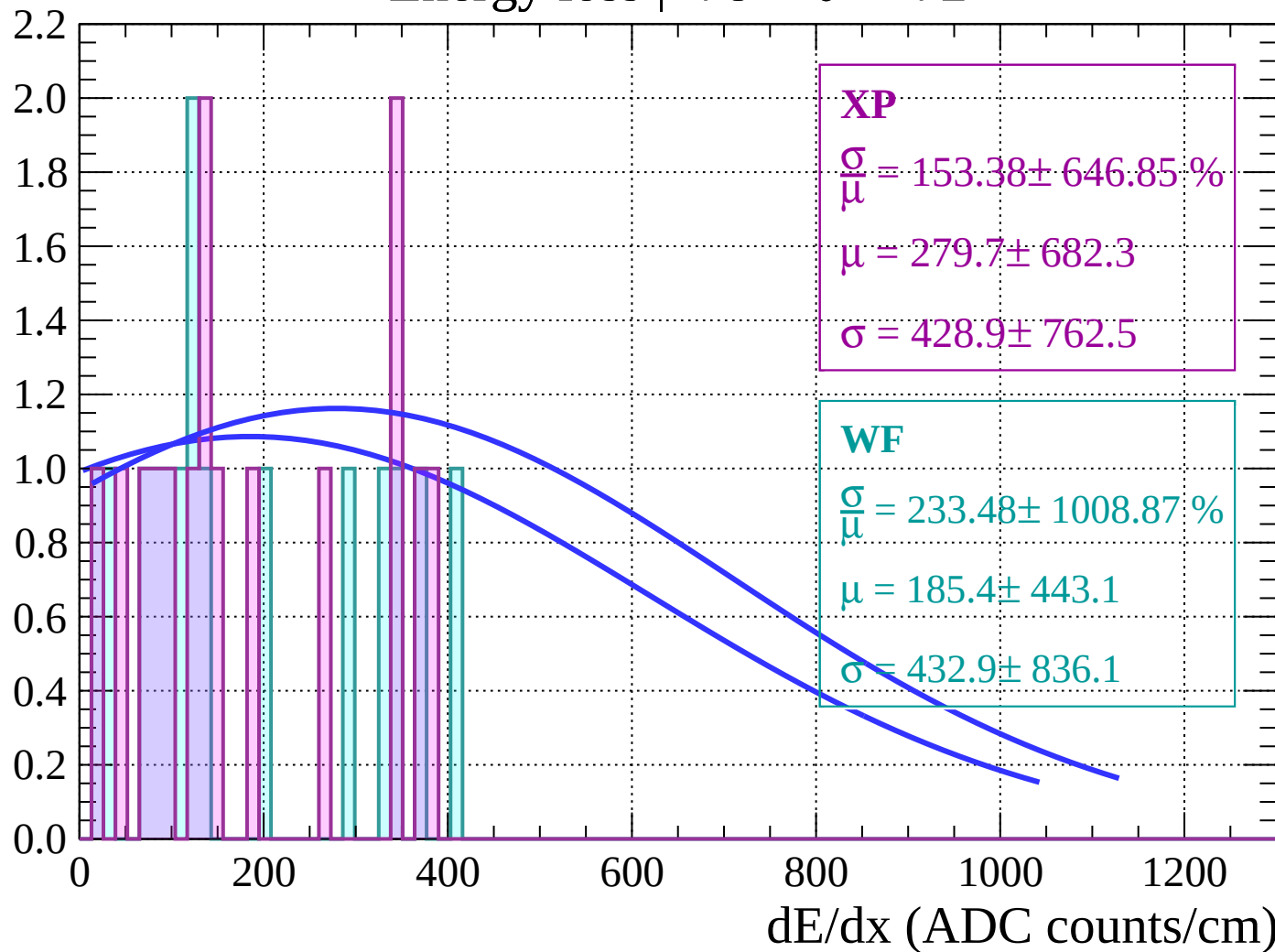
Count



dE/dx (ADC counts/cm)

Energy loss |  $-78 < \theta < -72$

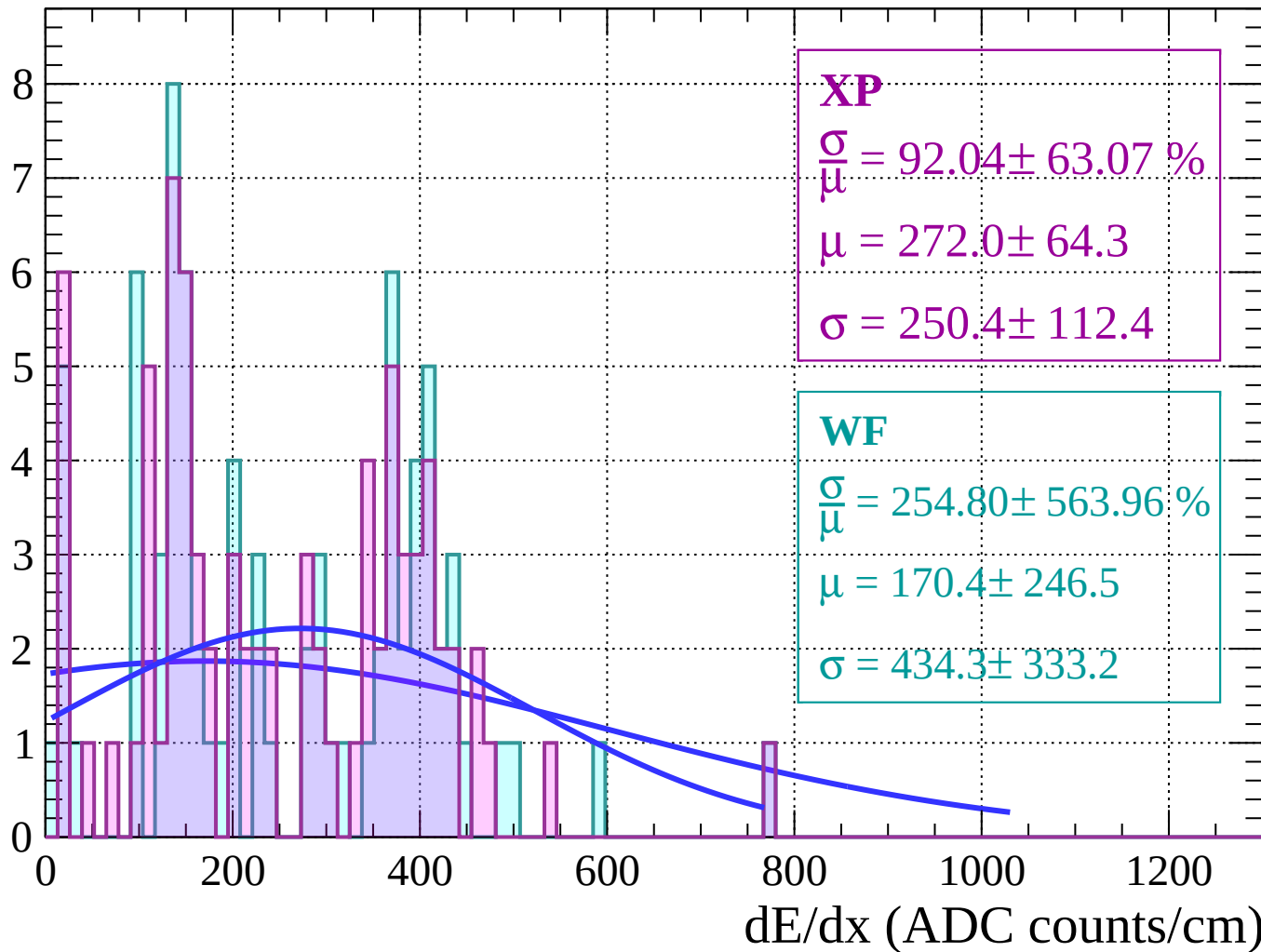
Count





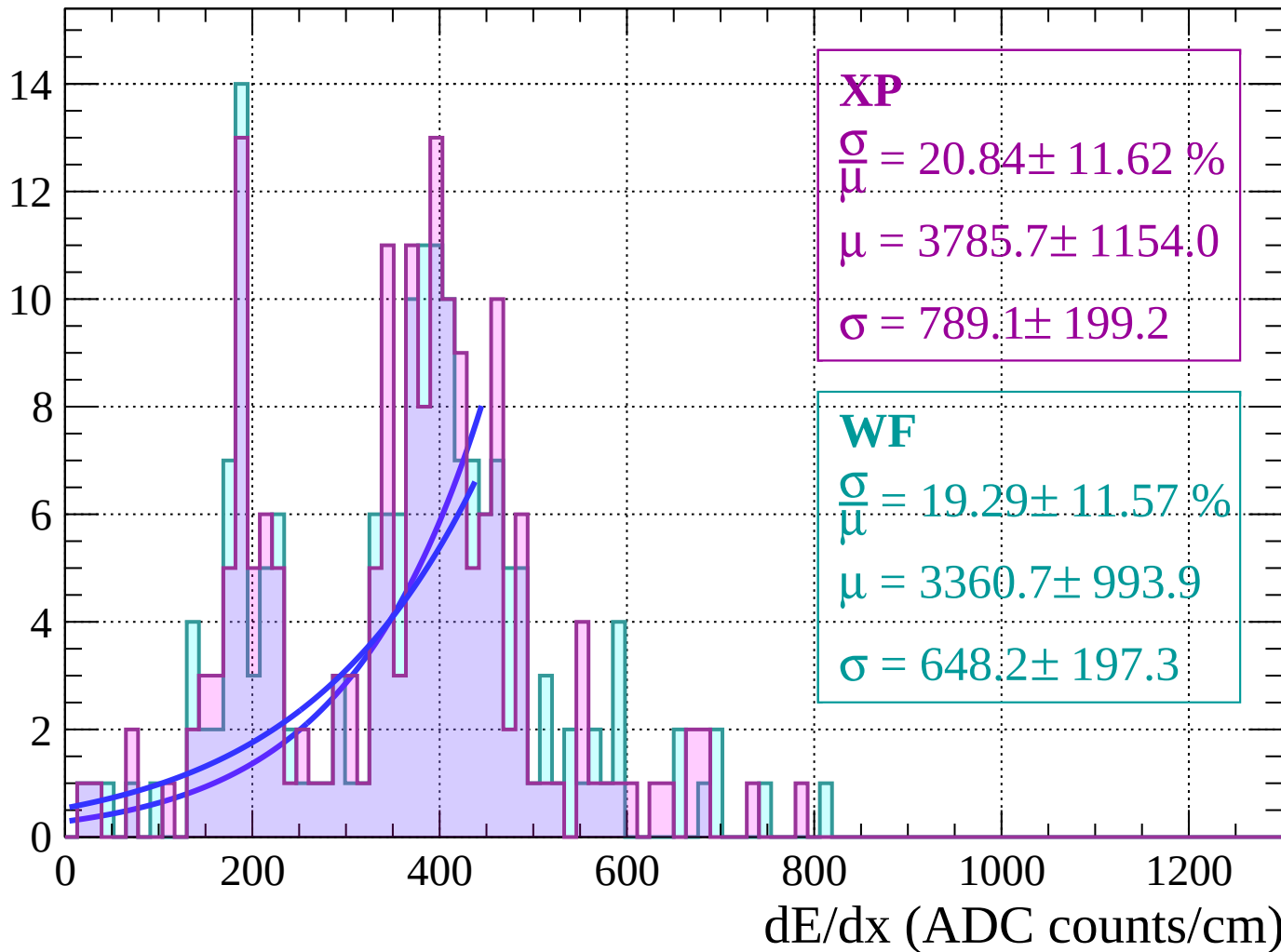
Energy loss |  $-72 < \theta < -66$

Count



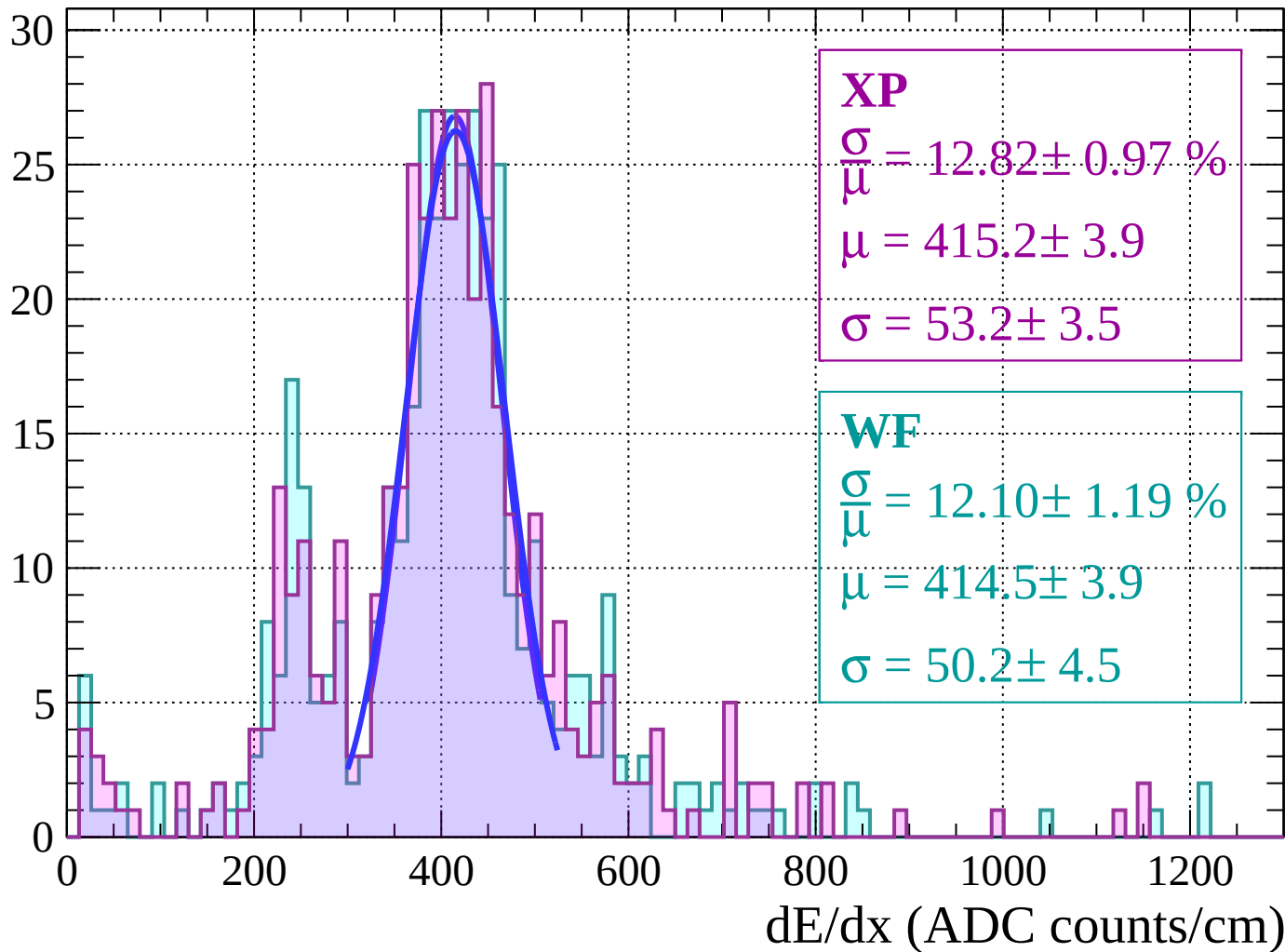
Energy loss |  $-66 < \theta < -60$

Count



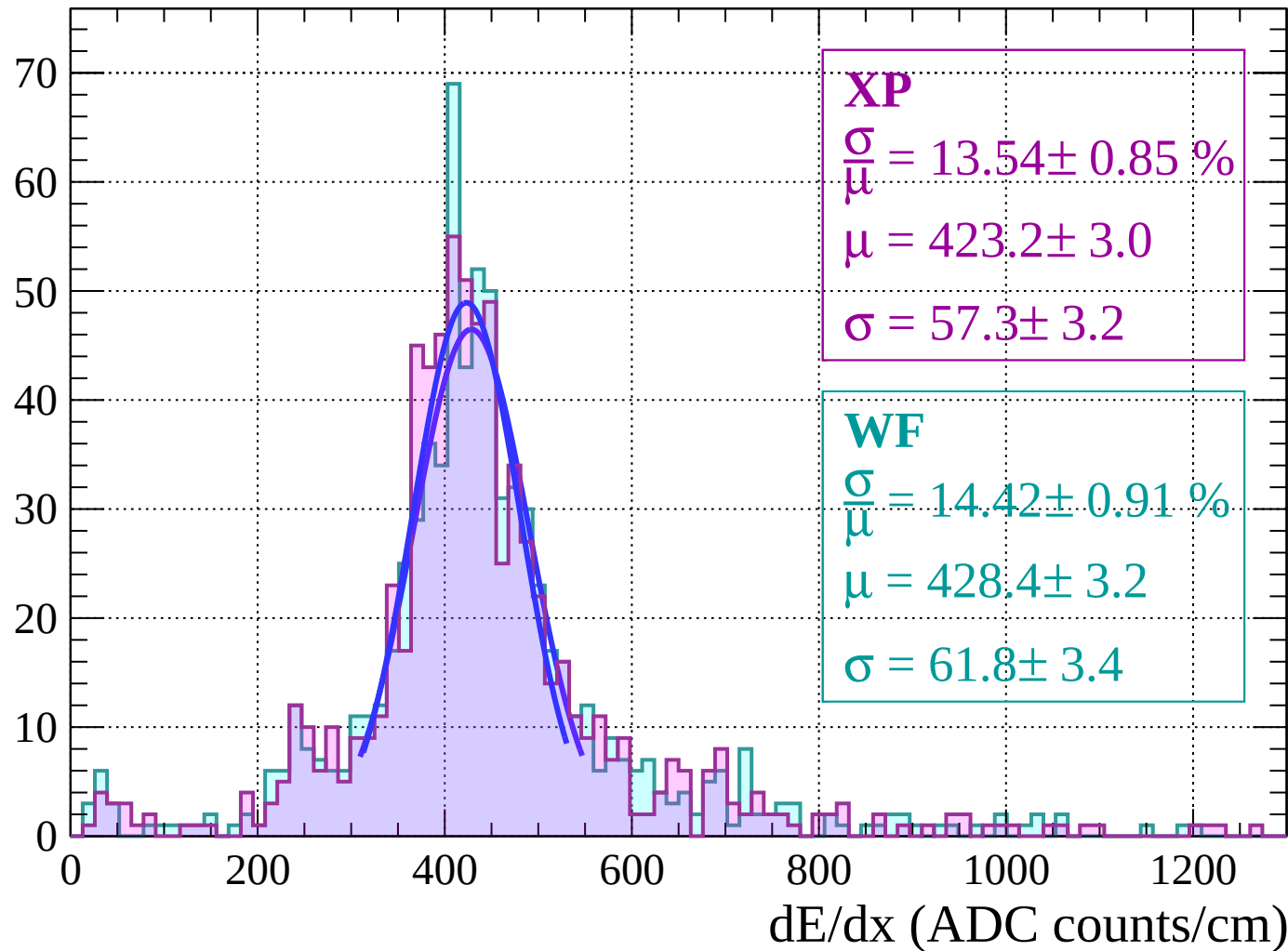
Energy loss |  $-60 < \theta < -54$

Count



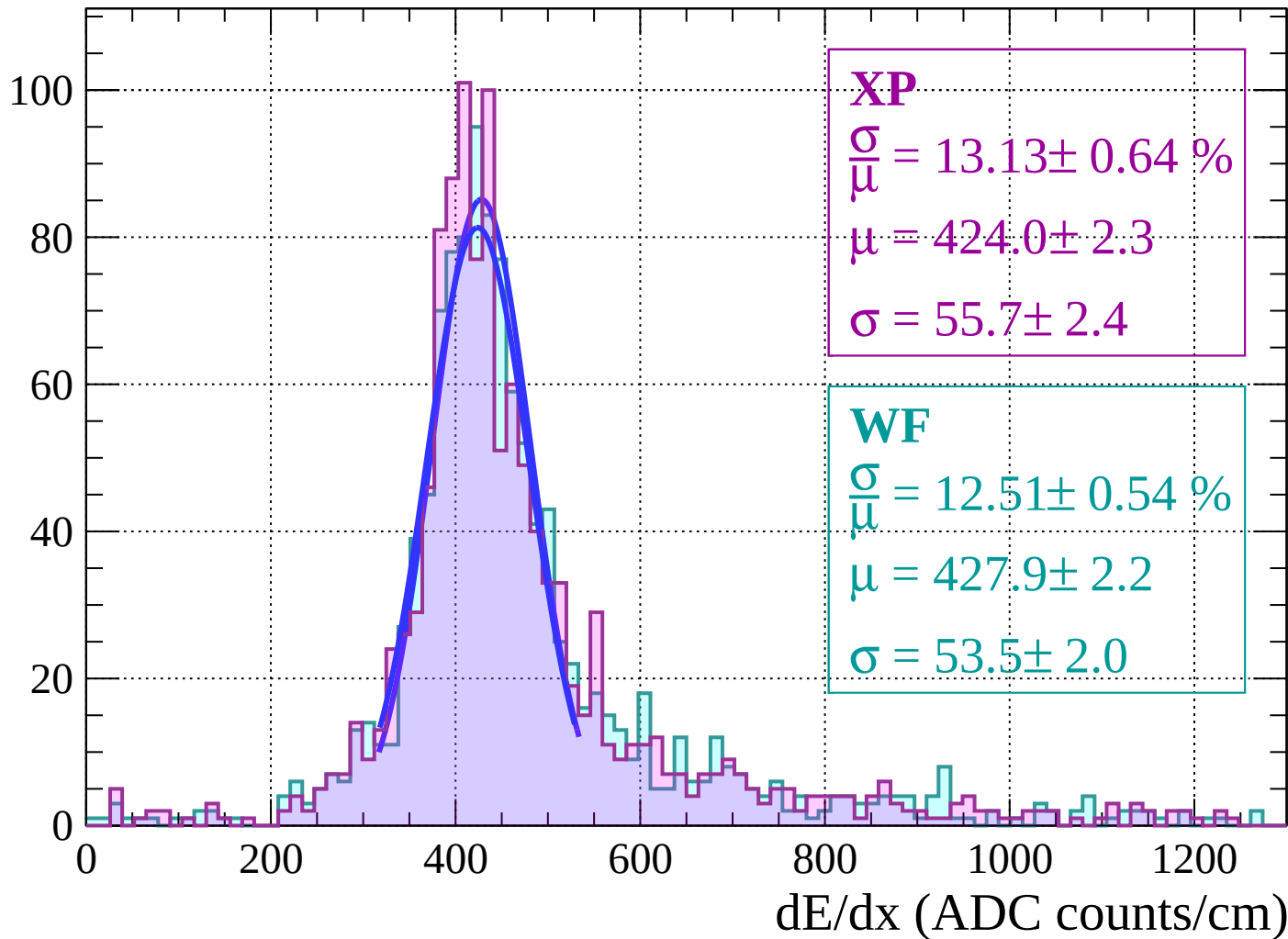
Energy loss |  $-54 < \theta < -48$

Count

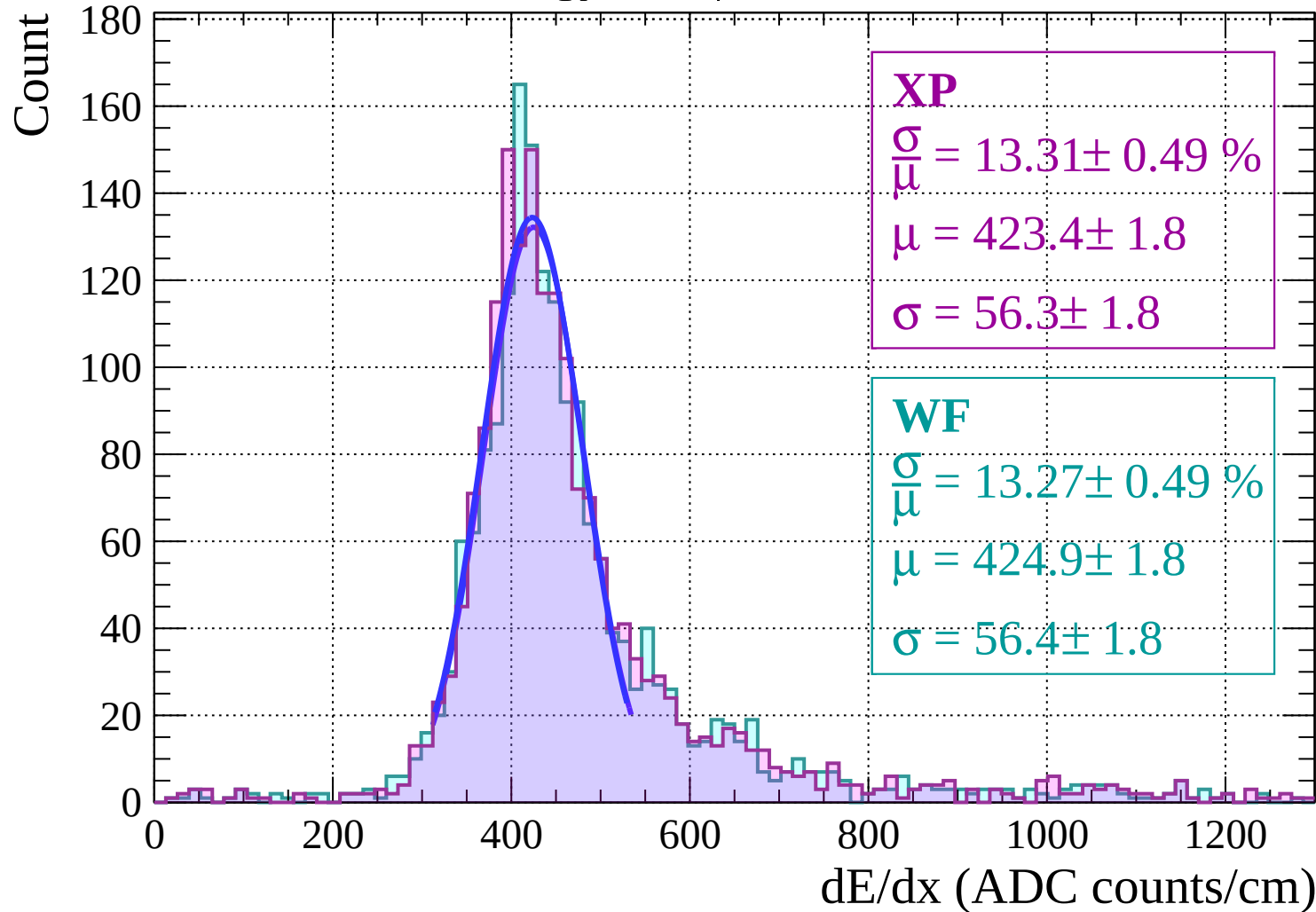


Energy loss |  $-48 < \theta < -42$

Count

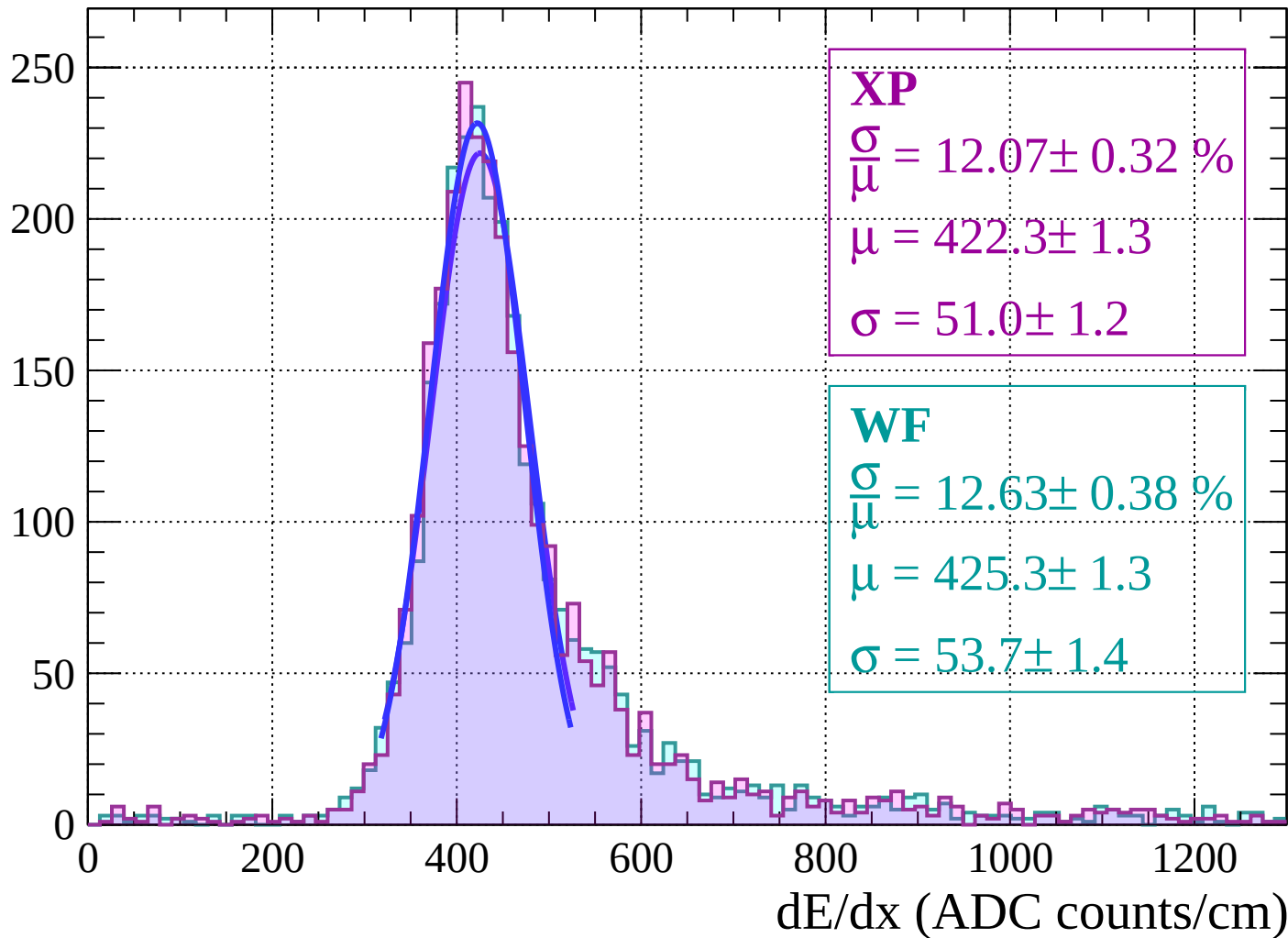


# Energy loss | $-42 < \theta < -36$

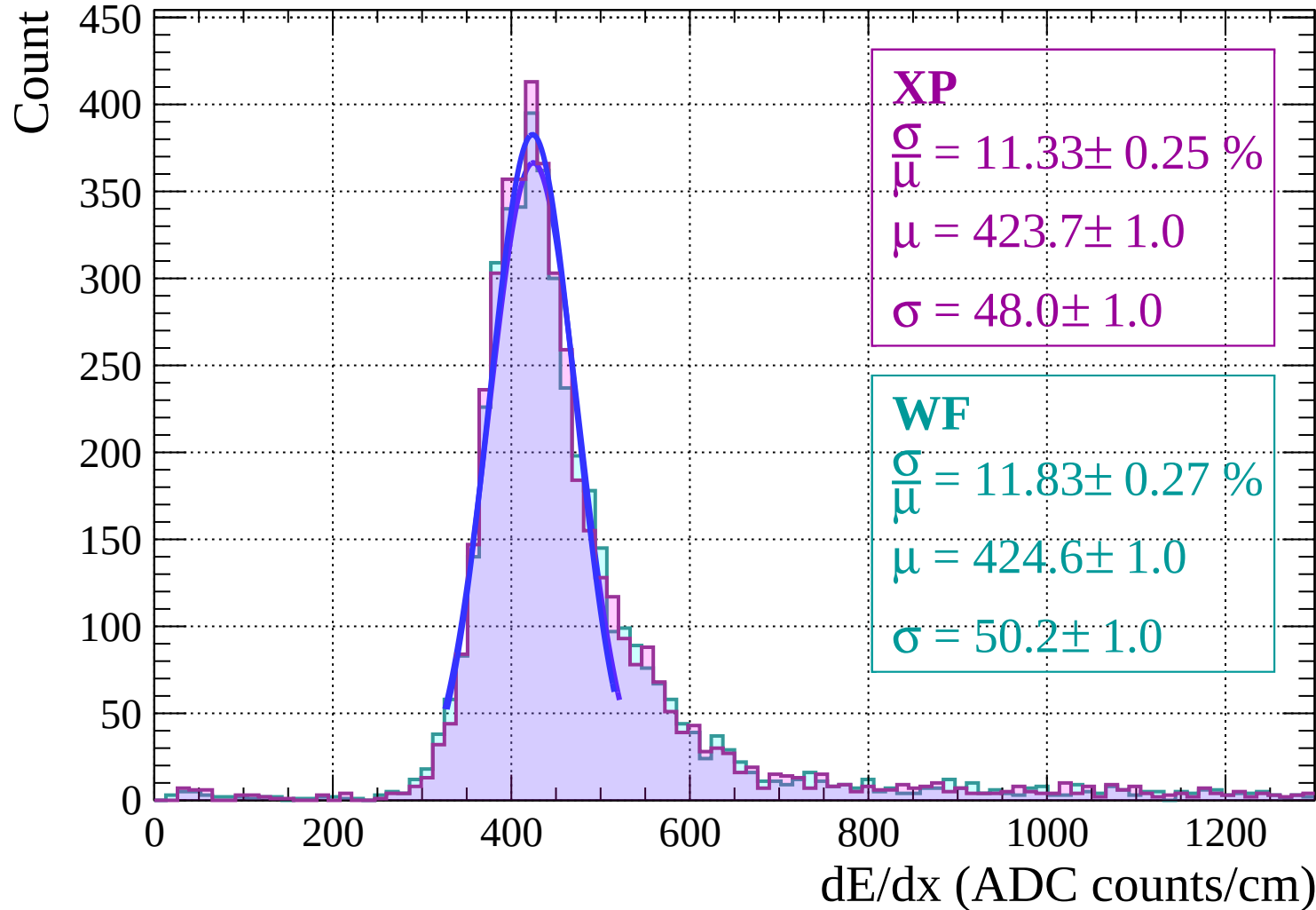


Energy loss  $|-36 < \theta < -30$

Count



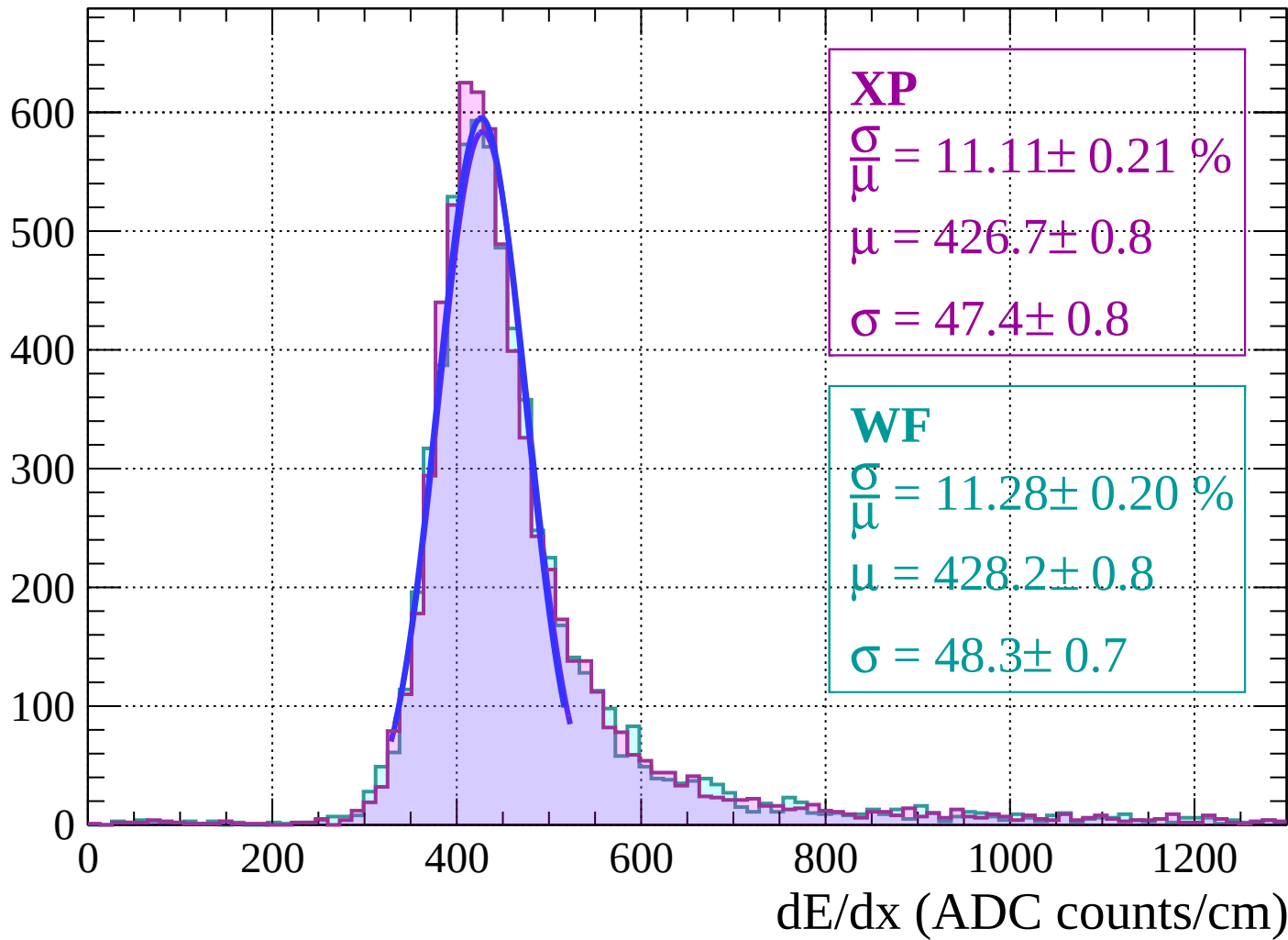
# Energy loss | $-30 < \theta < -24$





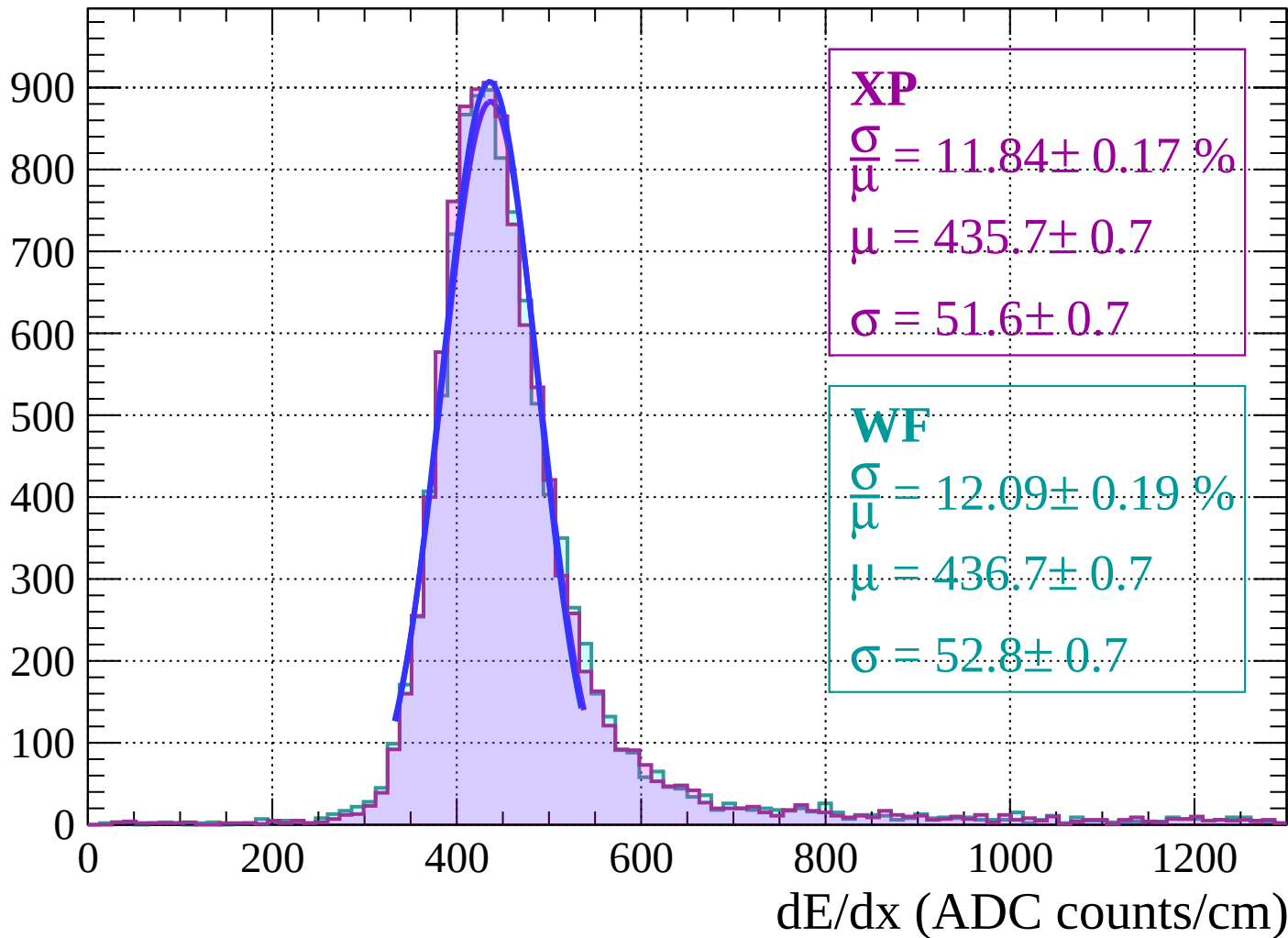
# Energy loss | $-24 < \theta < -18$

Count

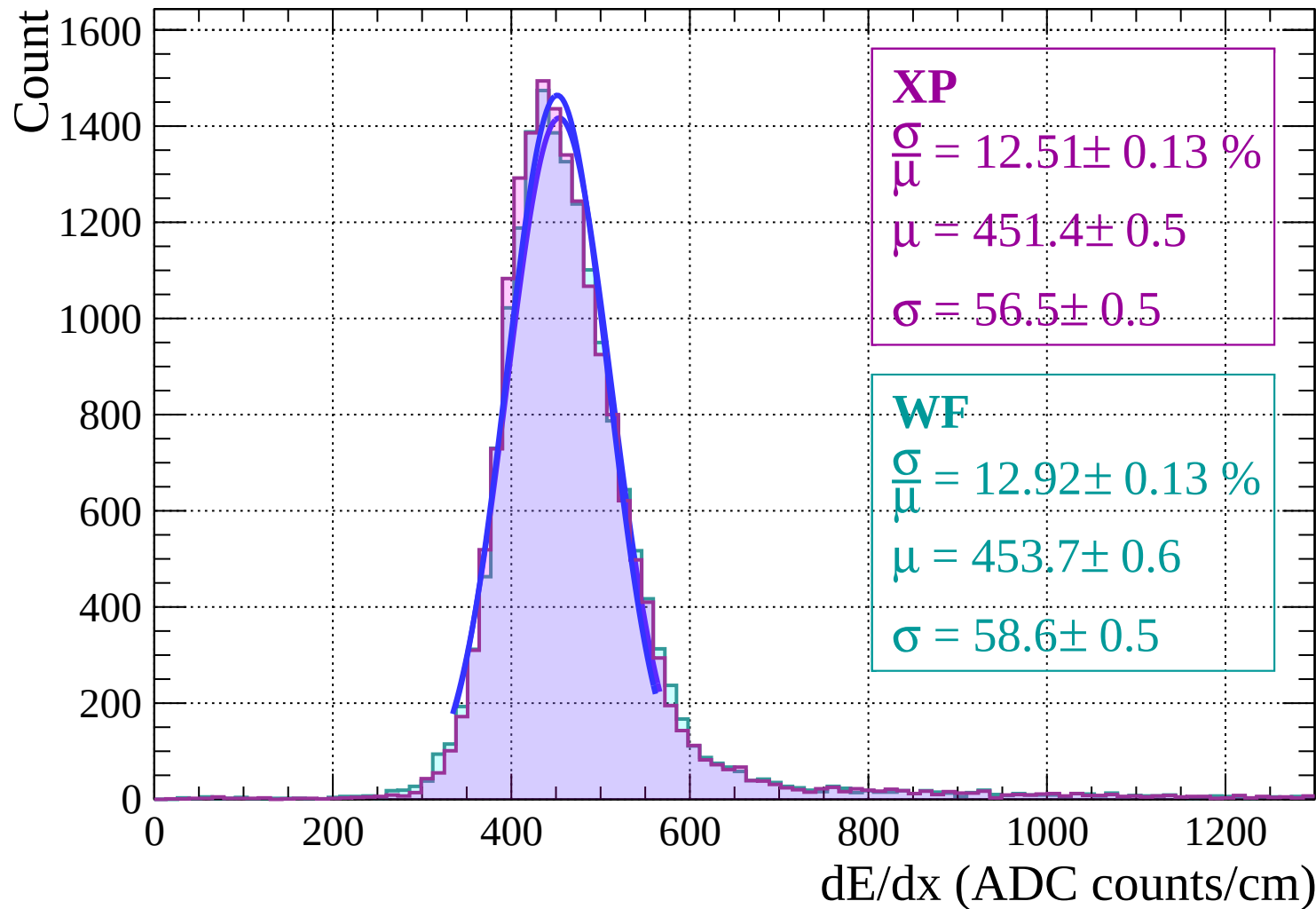


# Energy loss | $-18 < \theta < -12$

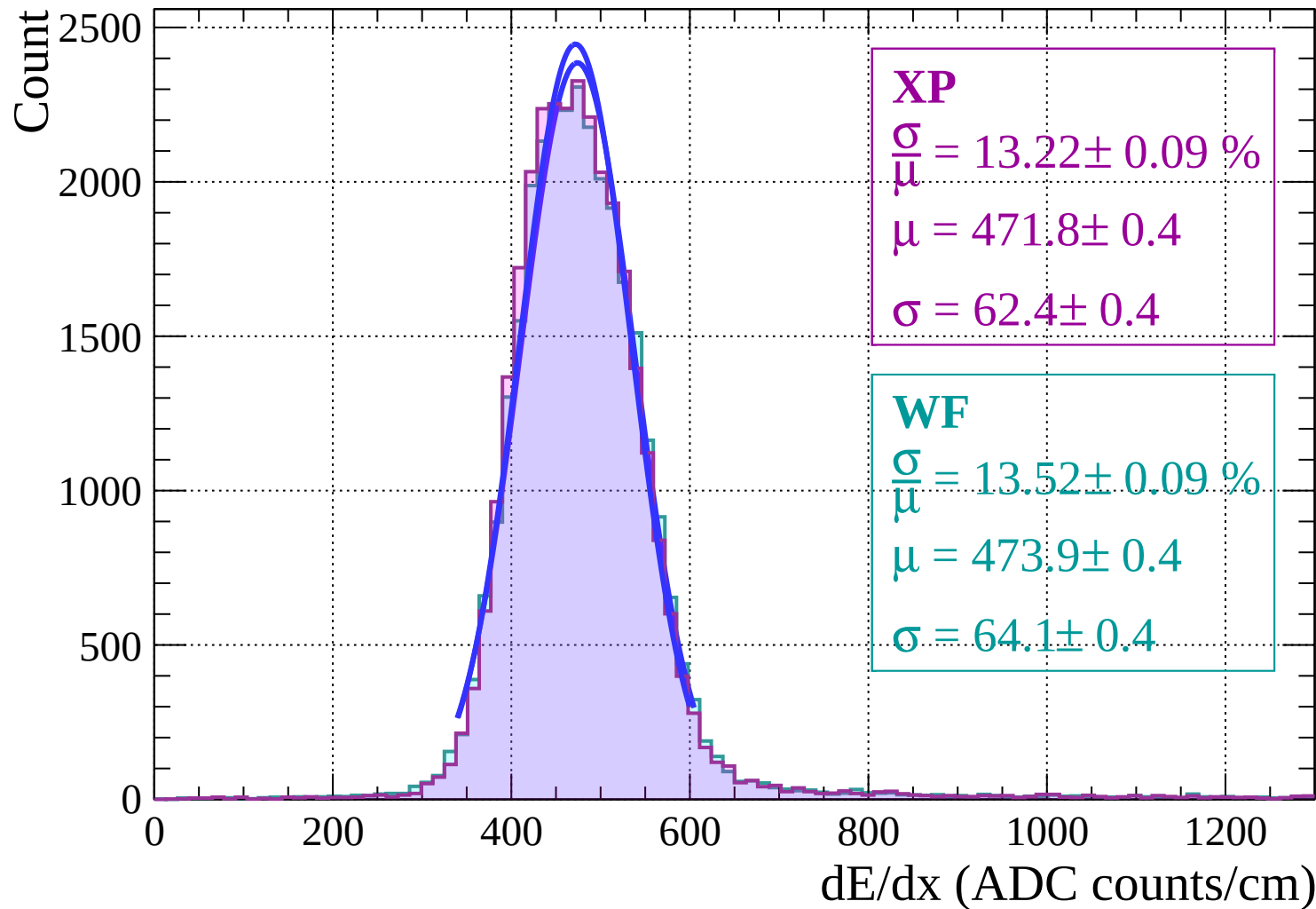
Count



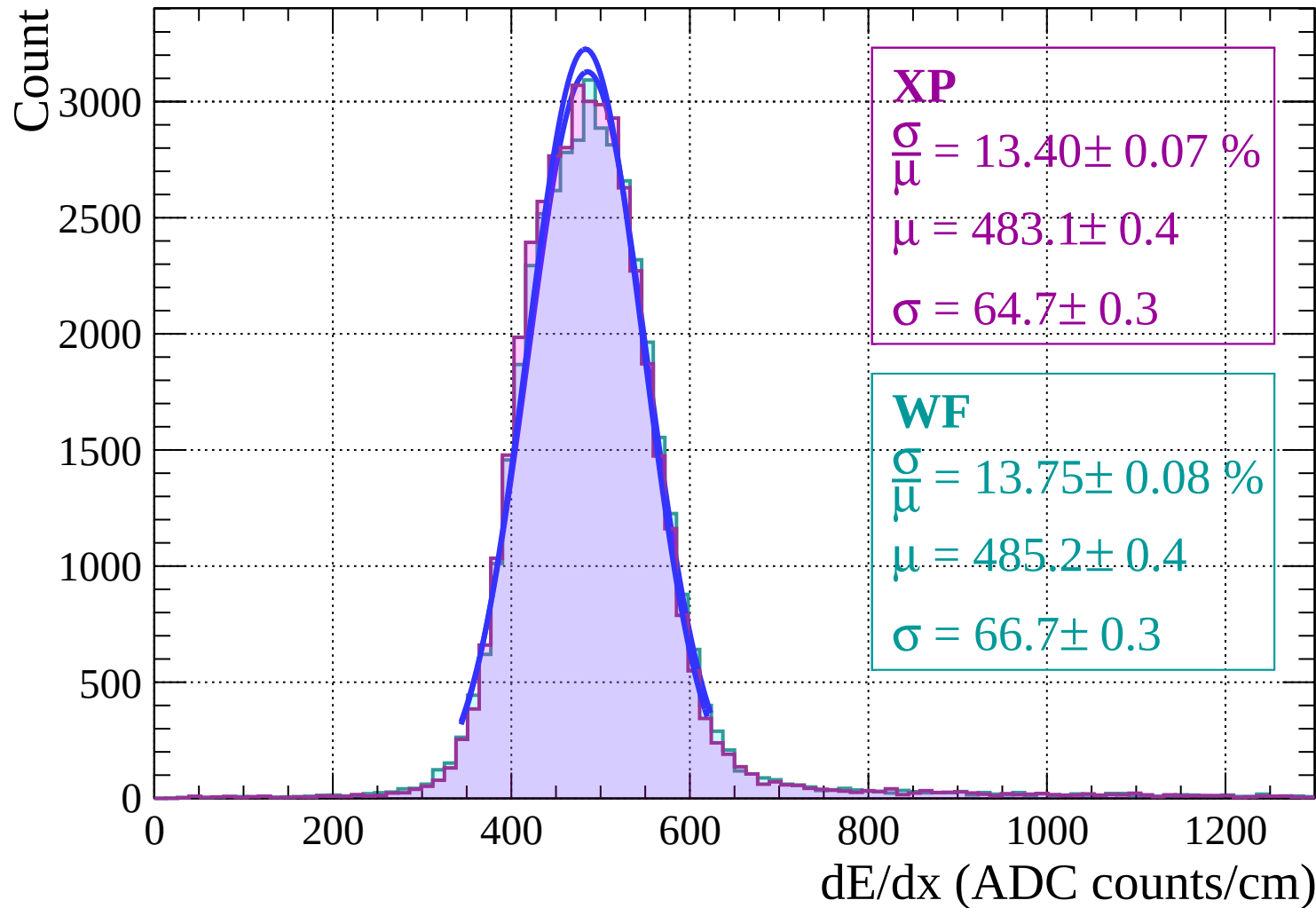
# Energy loss | $-12 < \theta < -6$



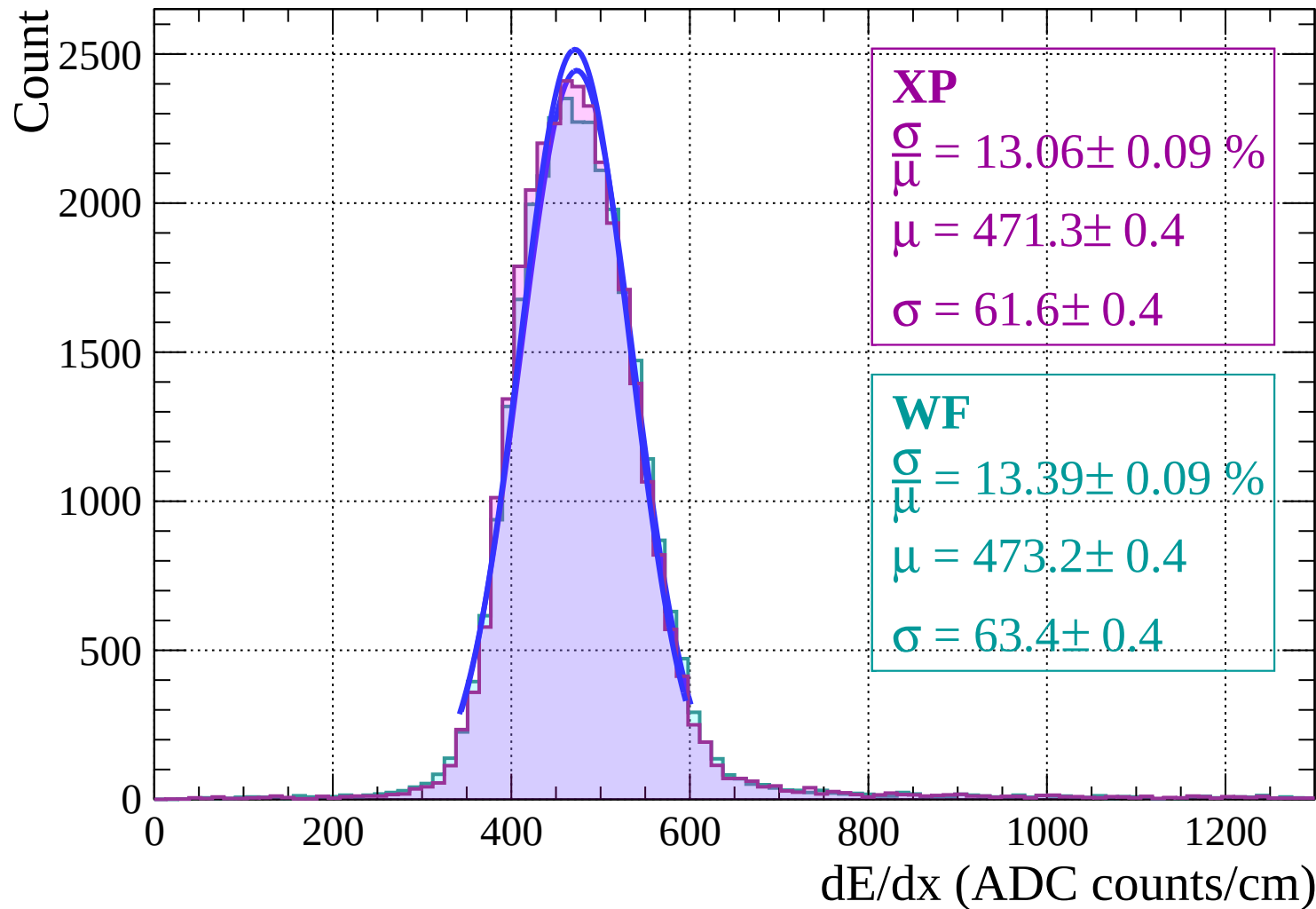
# Energy loss | $-6 < \theta < 0$



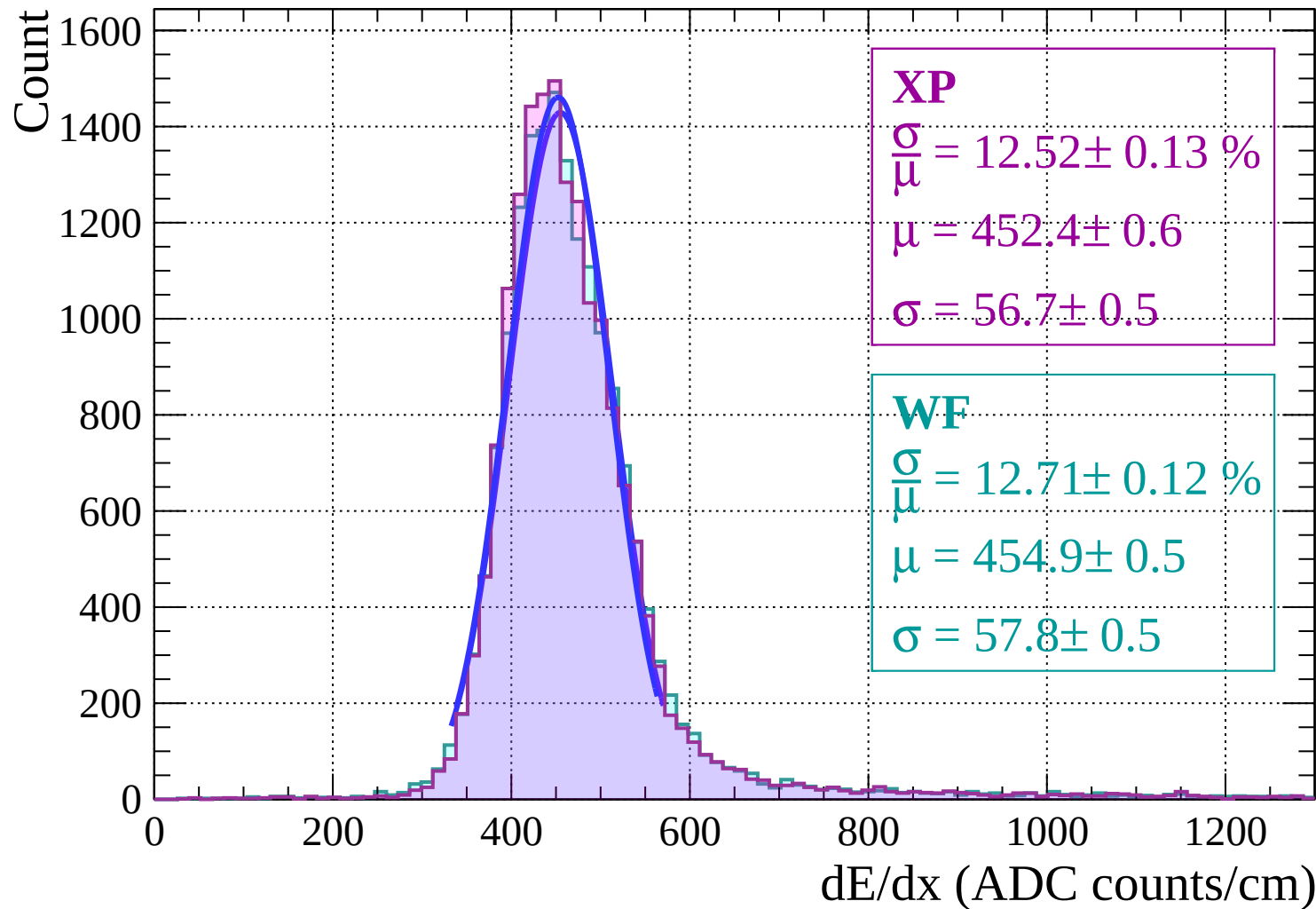
# Energy loss | $0 < \theta < 6$



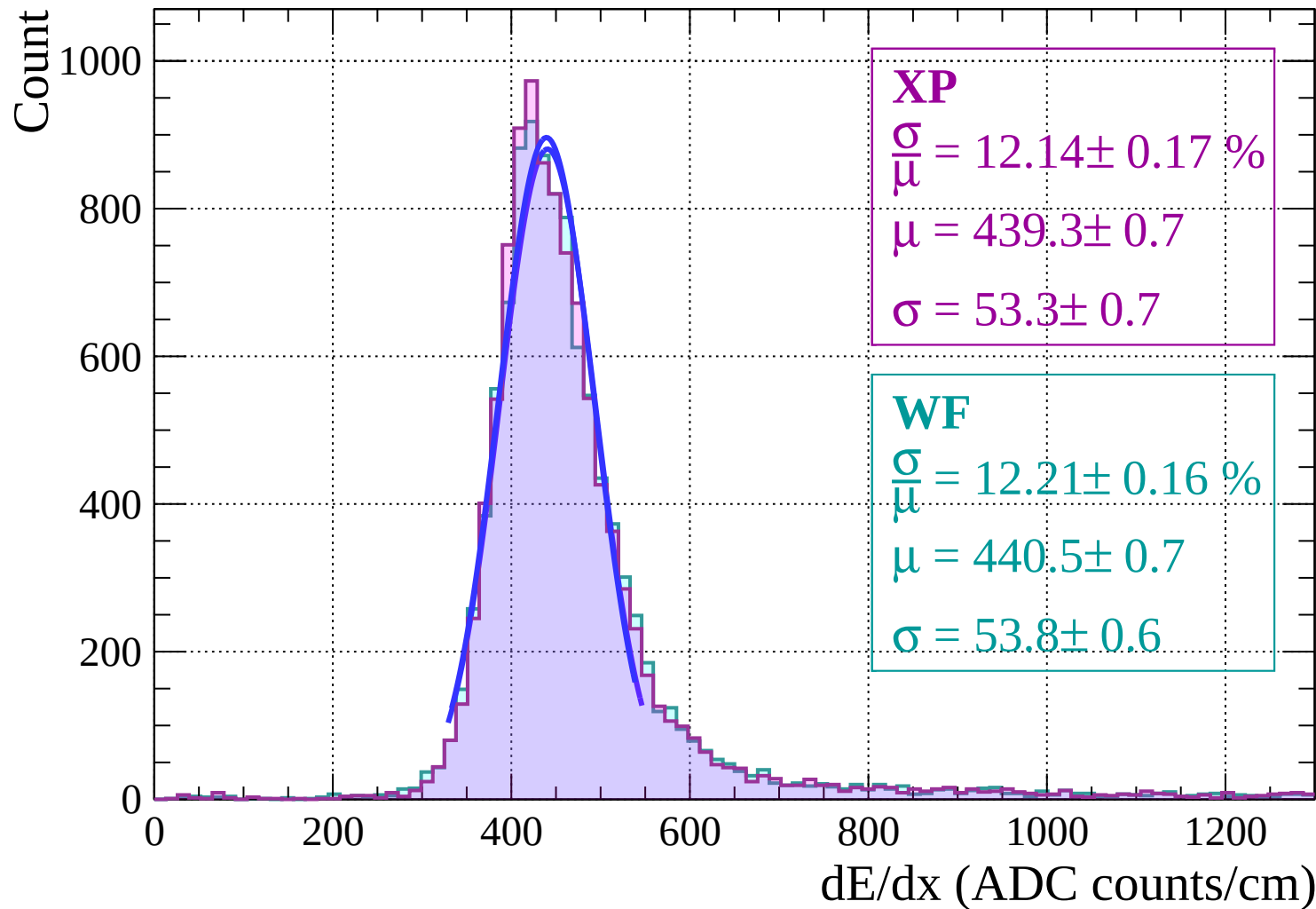
Energy loss |  $6 < \theta < 12$



# Energy loss | $12 < \theta < 18$



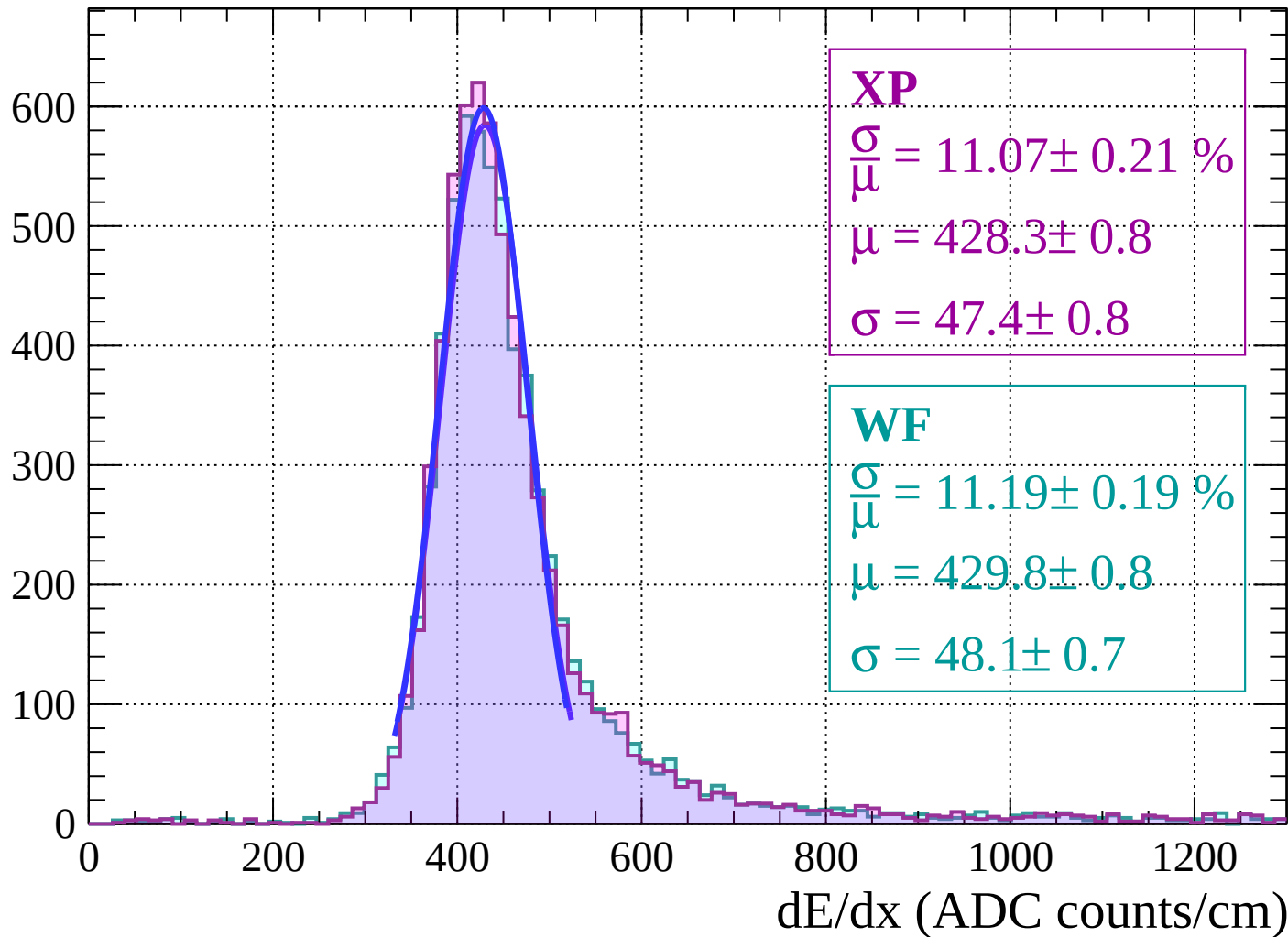
# Energy loss | $18 < \theta < 24$



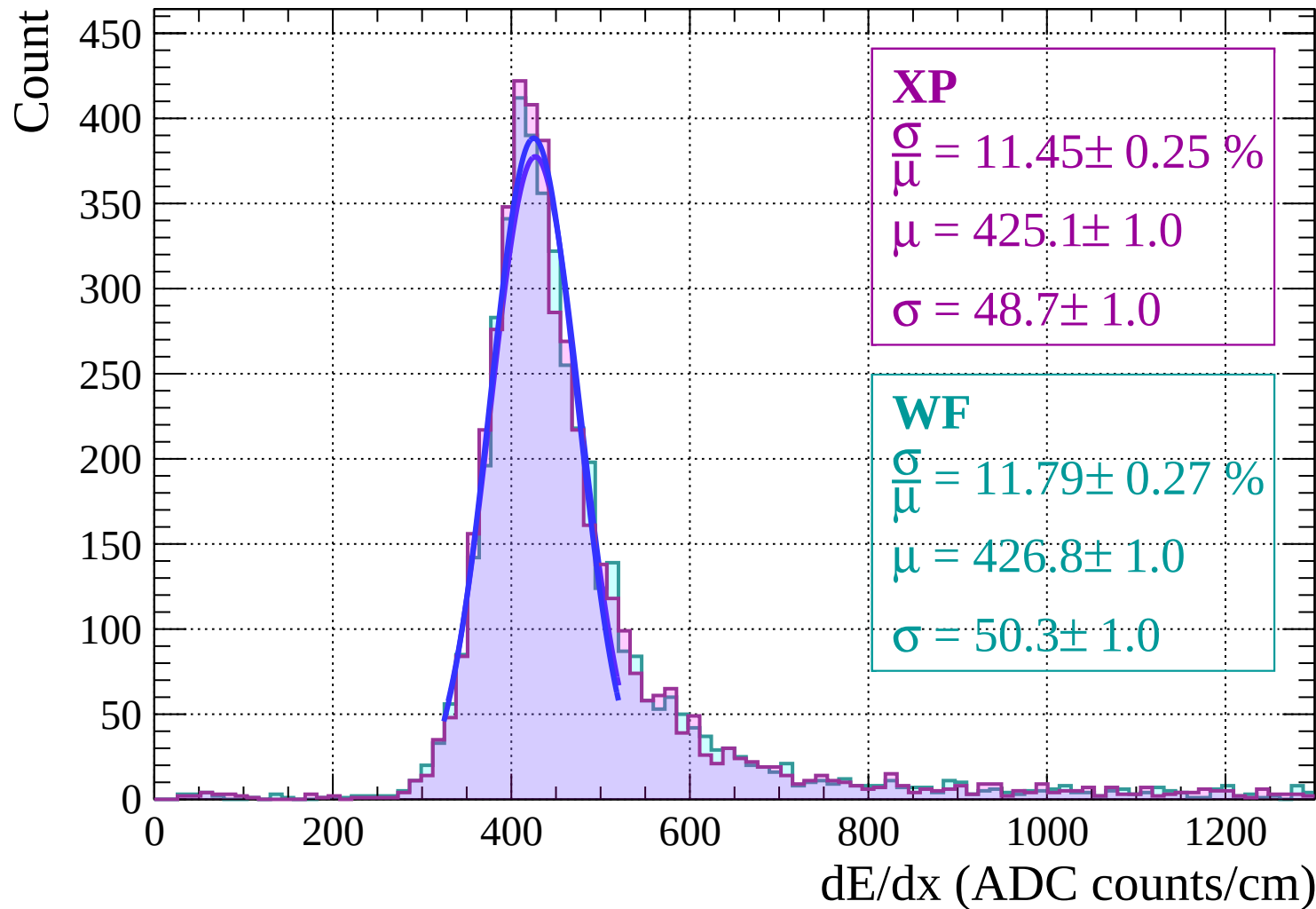


Energy loss |  $24 < \theta < 30$

Count

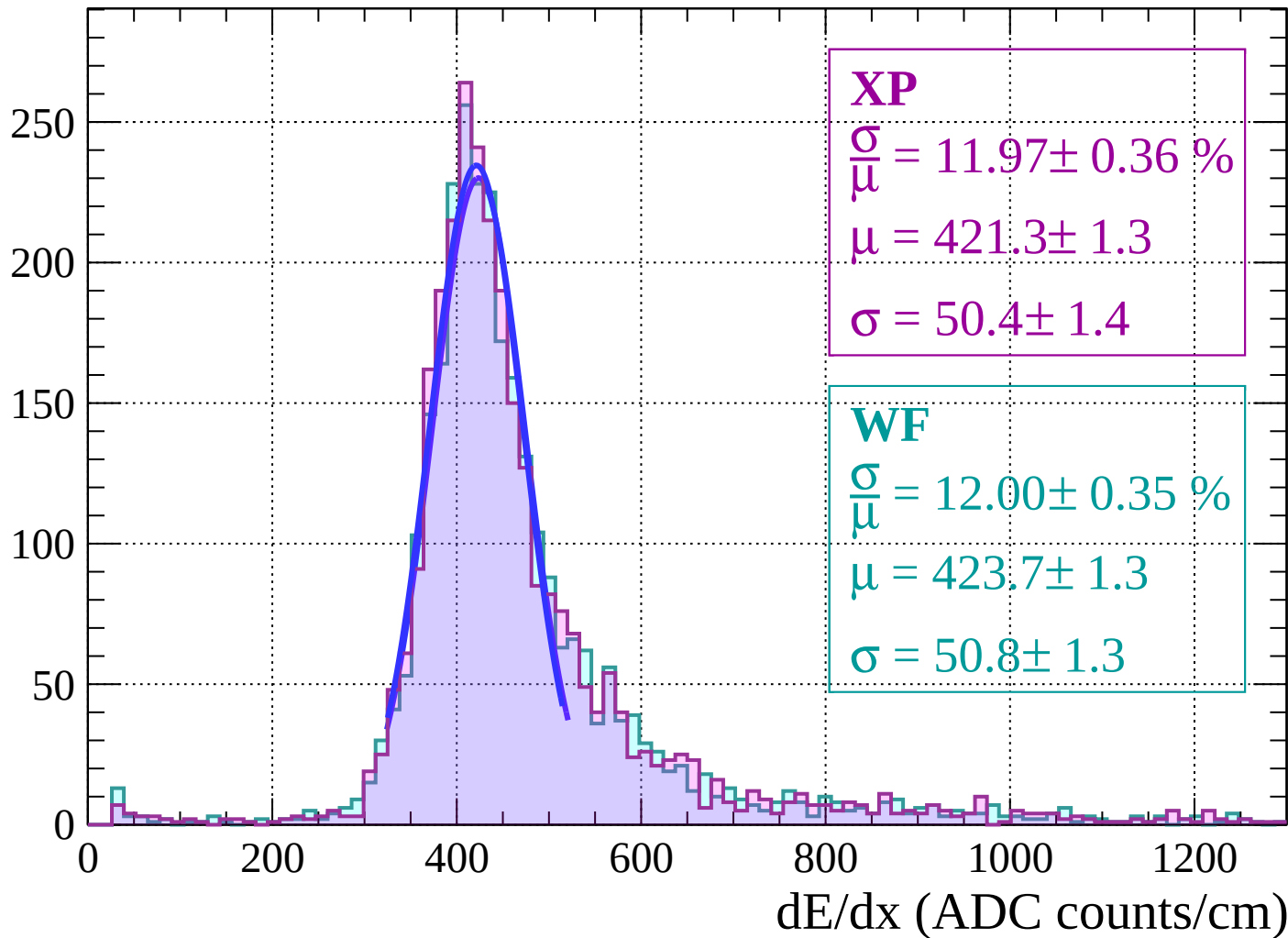


# Energy loss | $30 < \theta < 36$



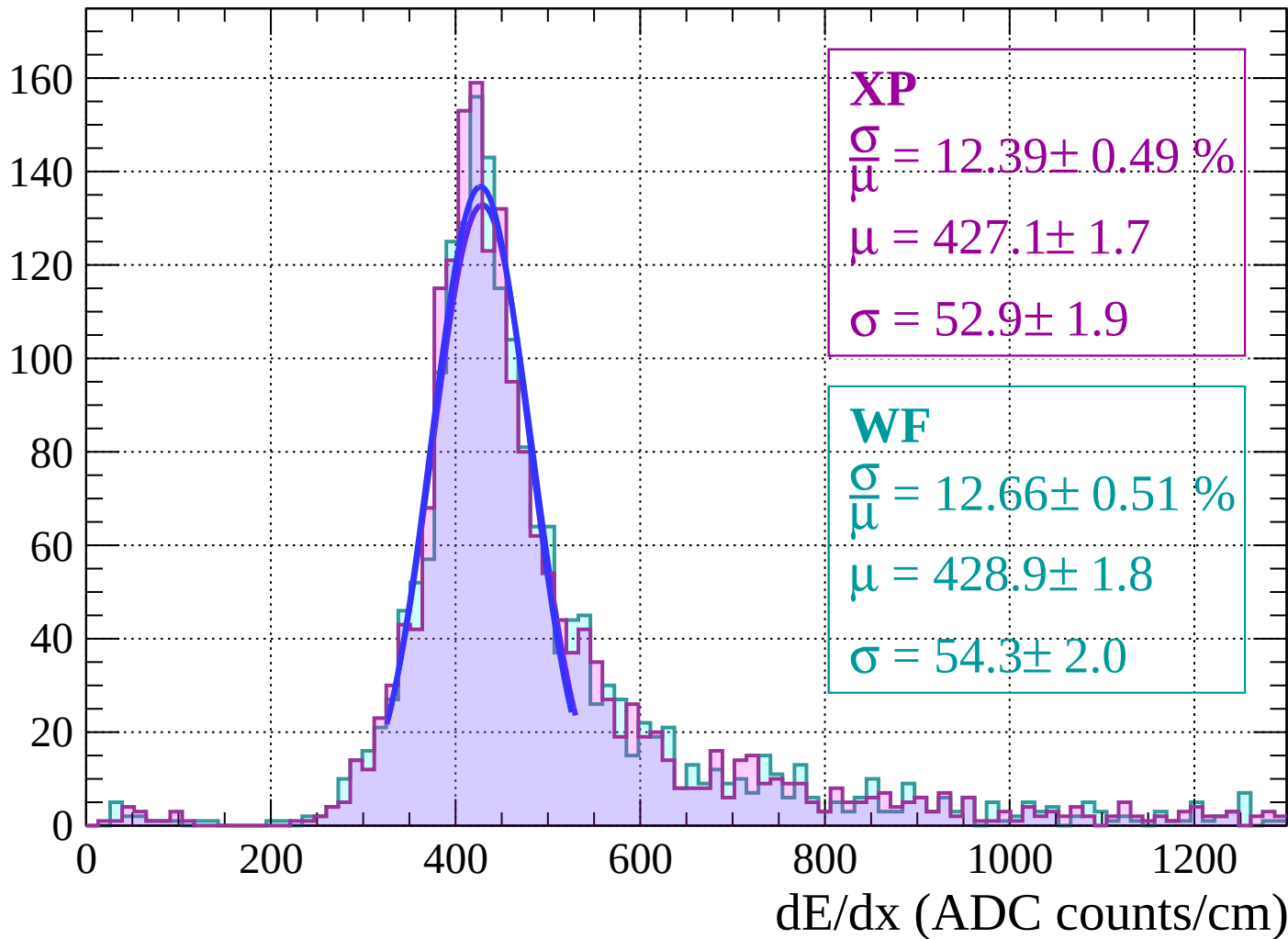
Energy loss |  $36 < \theta < 42$

Count



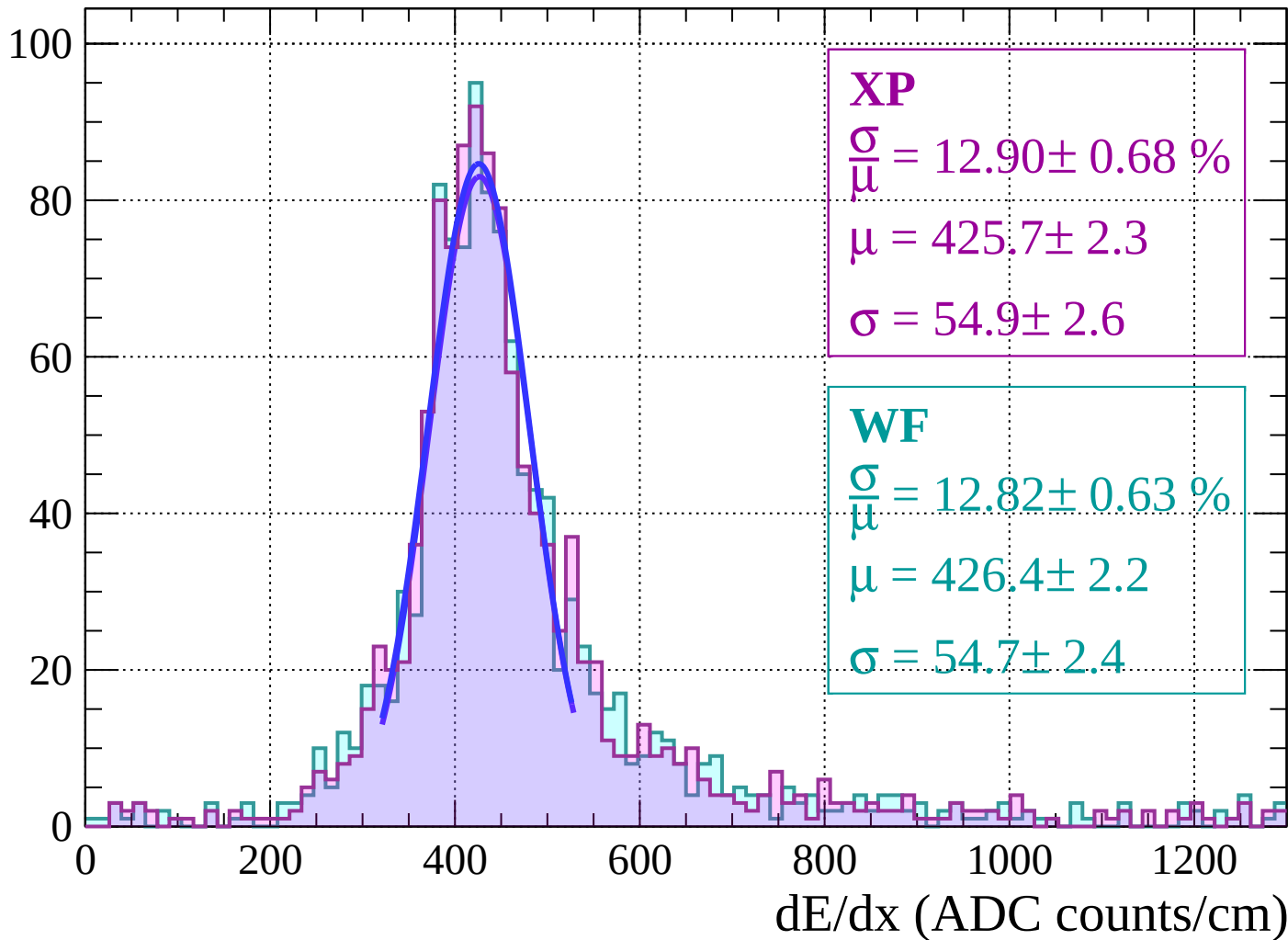
Energy loss |  $42 < \theta < 48$

Count



Energy loss |  $48 < \theta < 54$

Count



Energy loss |  $54 < \theta < 60$

Count

**XP**

$$\frac{\sigma}{\mu} = 13.85 \pm 0.80 \%$$

$$\mu = 429.9 \pm 3.1$$

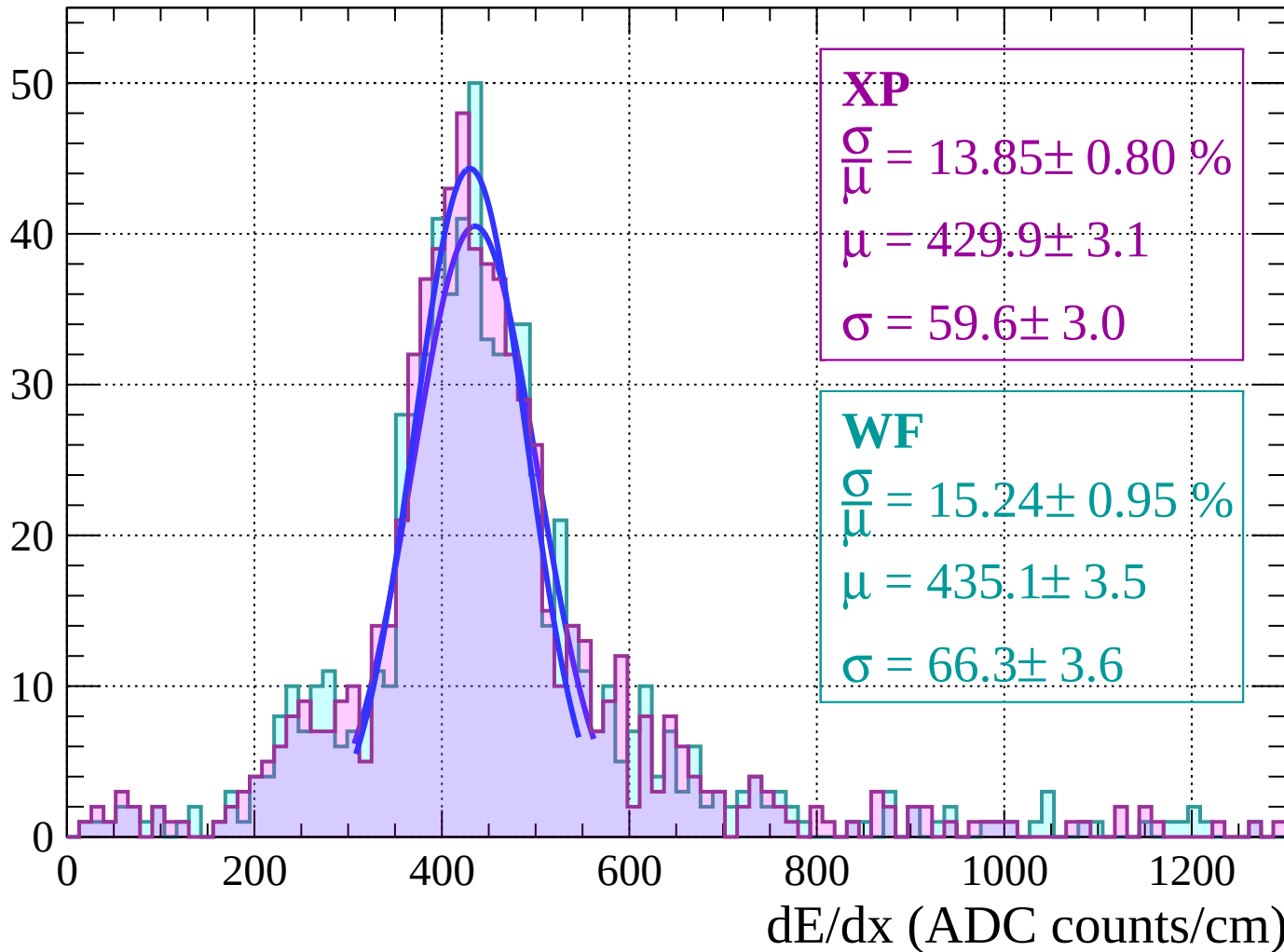
$$\sigma = 59.6 \pm 3.0$$

**WF**

$$\frac{\sigma}{\mu} = 15.24 \pm 0.95 \%$$

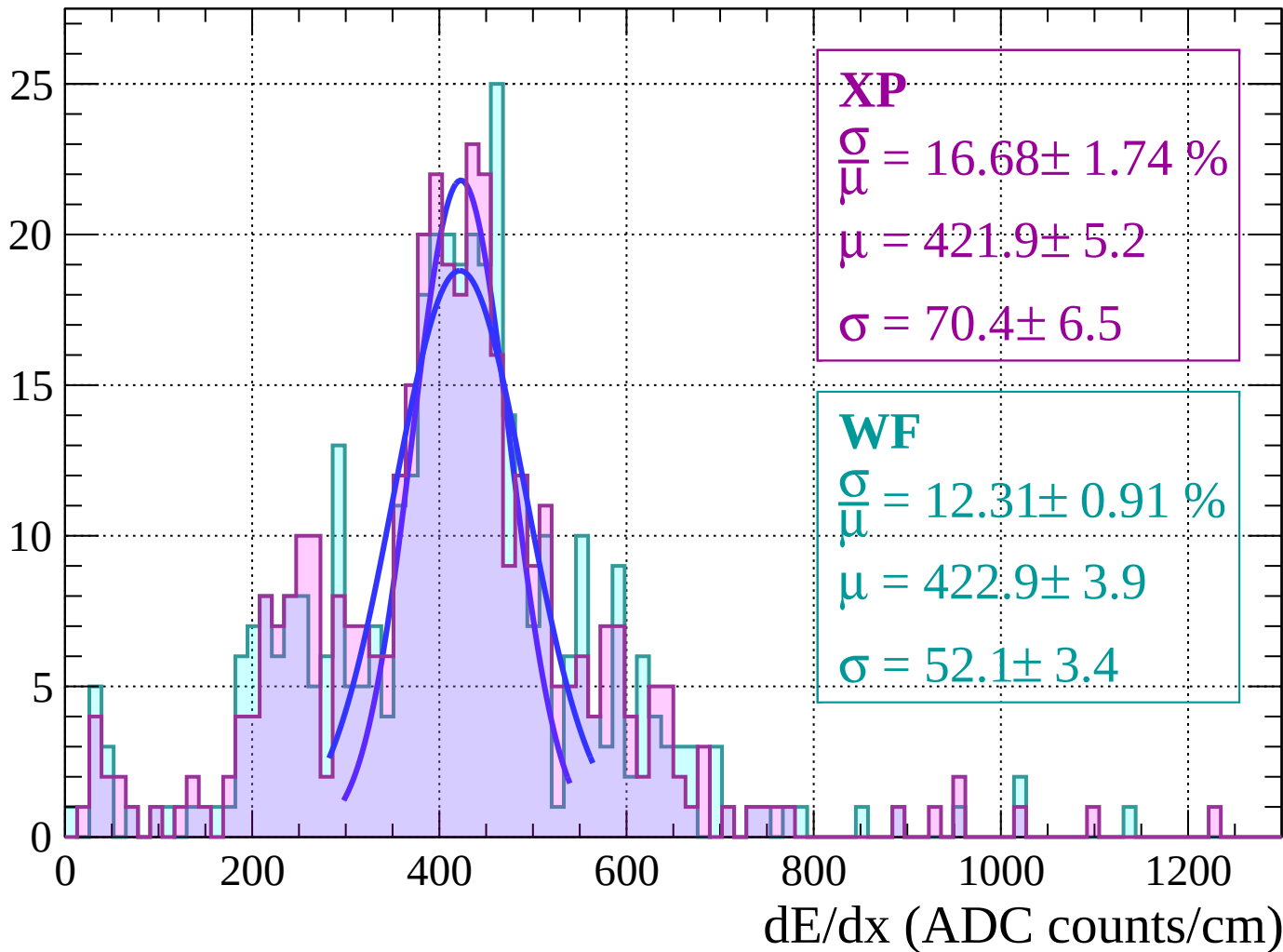
$$\mu = 435.1 \pm 3.5$$

$$\sigma = 66.3 \pm 3.6$$



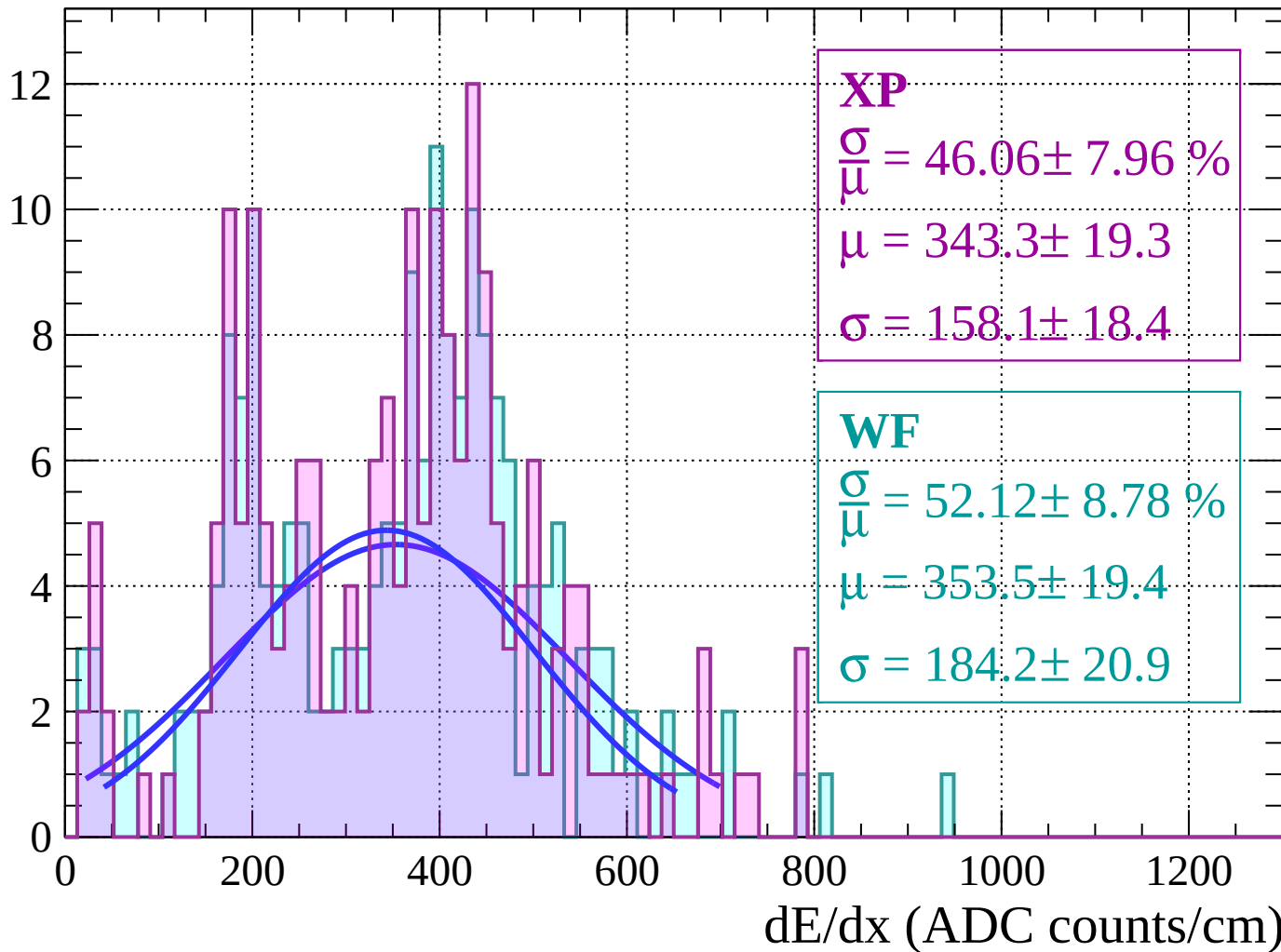
Energy loss |  $60 < \theta < 66$

Count



Energy loss |  $66 < \theta < 72$

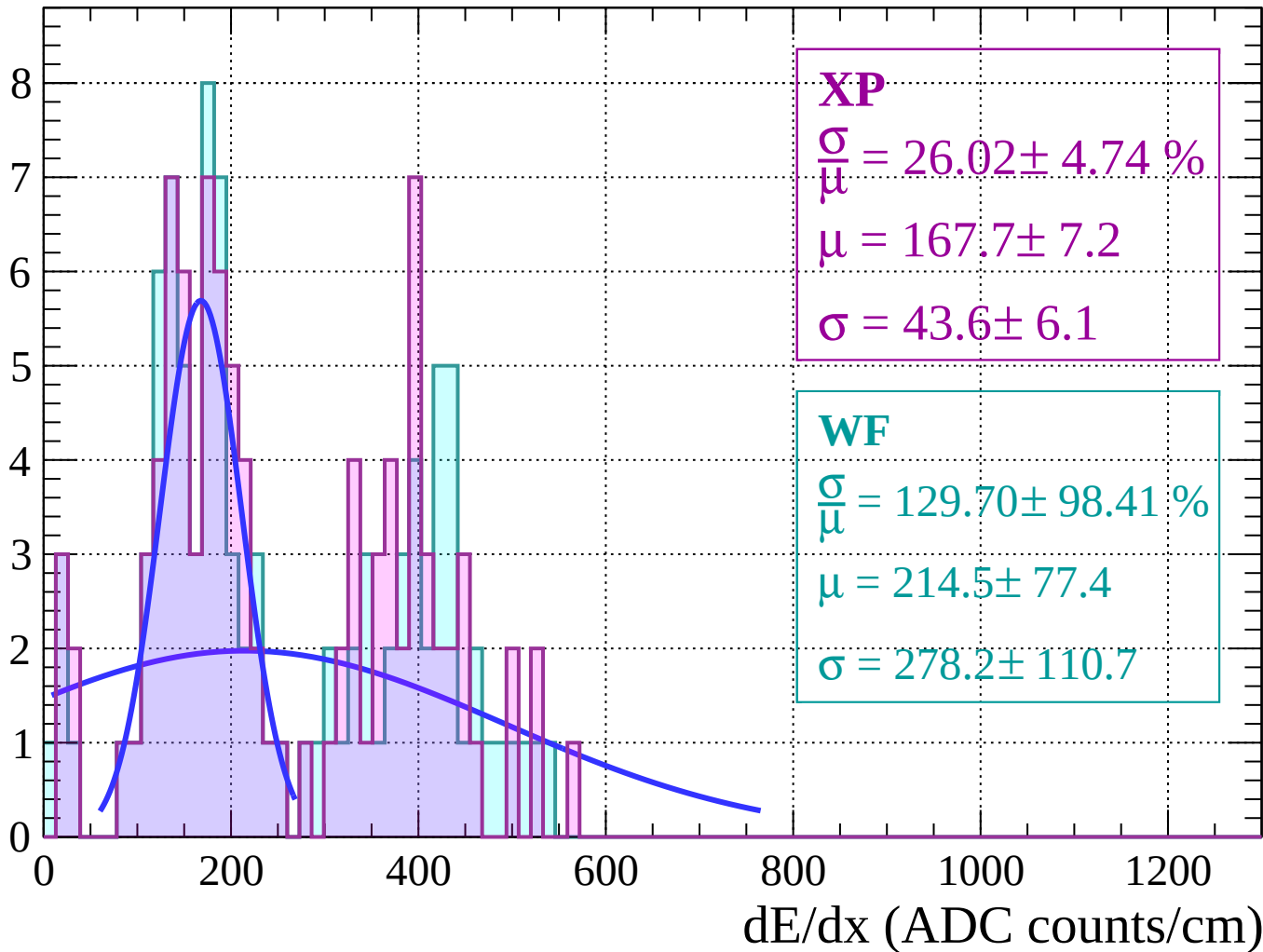
Count





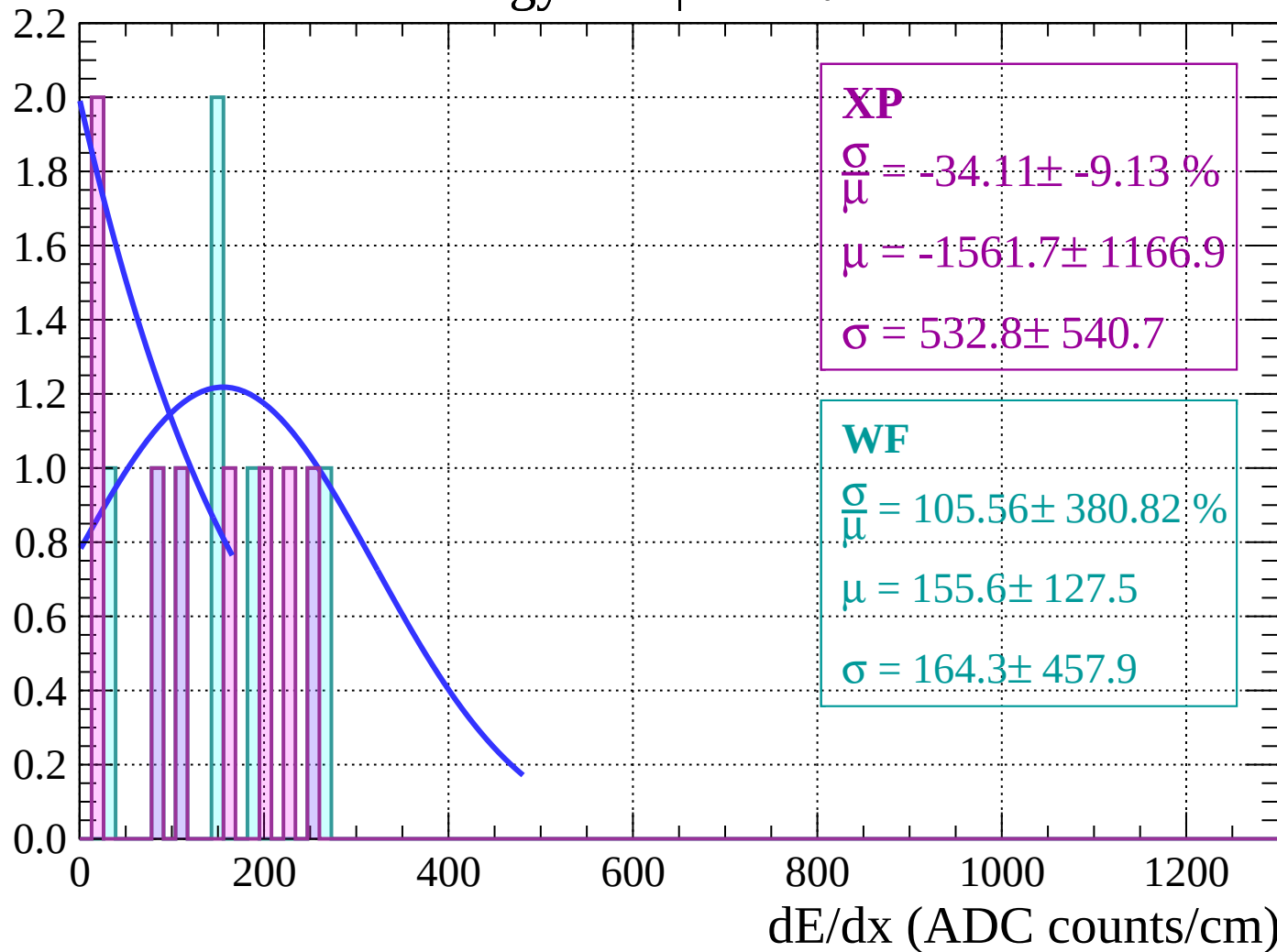
Energy loss |  $72 < \theta < 78$

Count



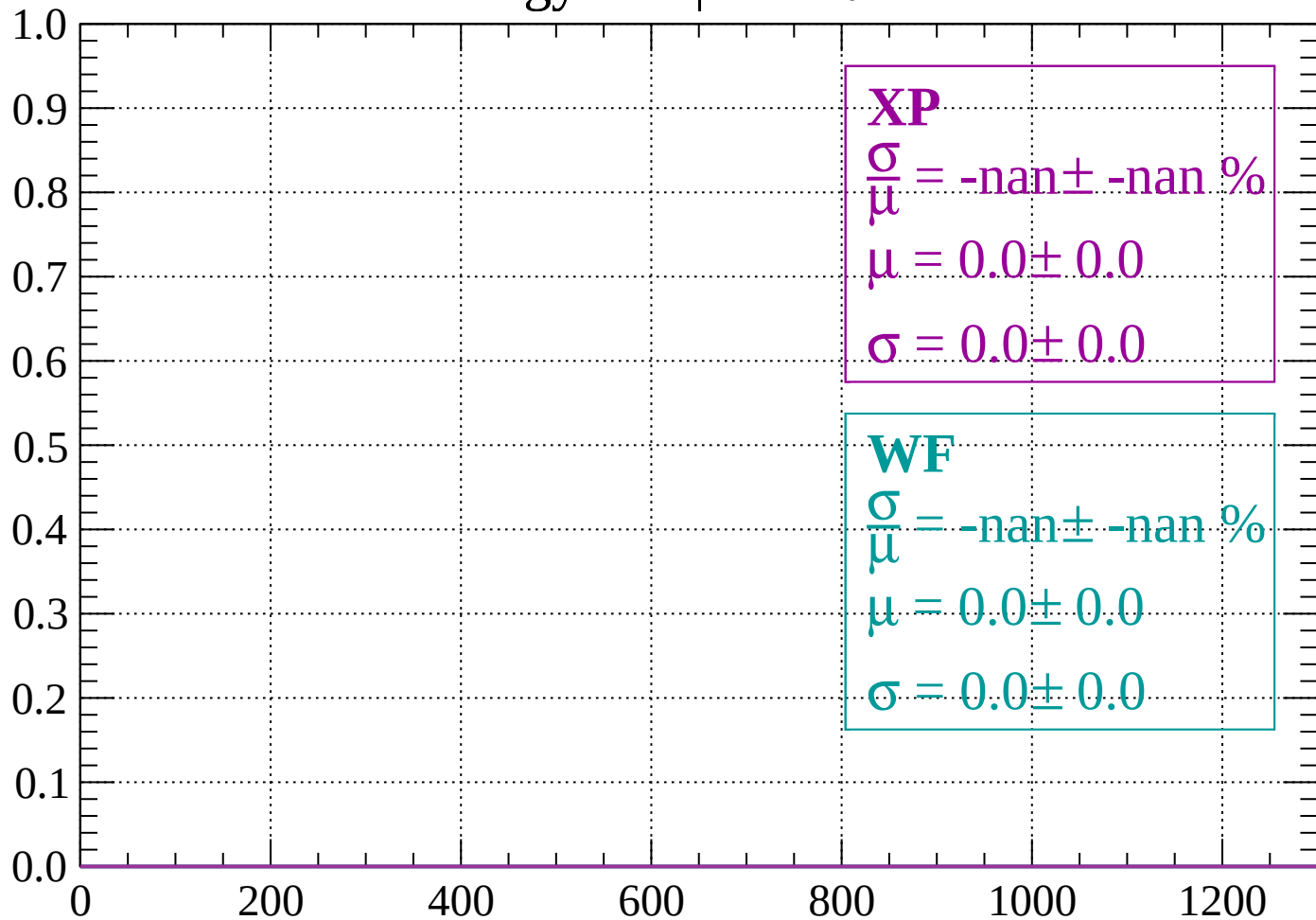
Energy loss |  $78 < \theta < 84$

Count



Energy loss |  $84 < \theta < 90$

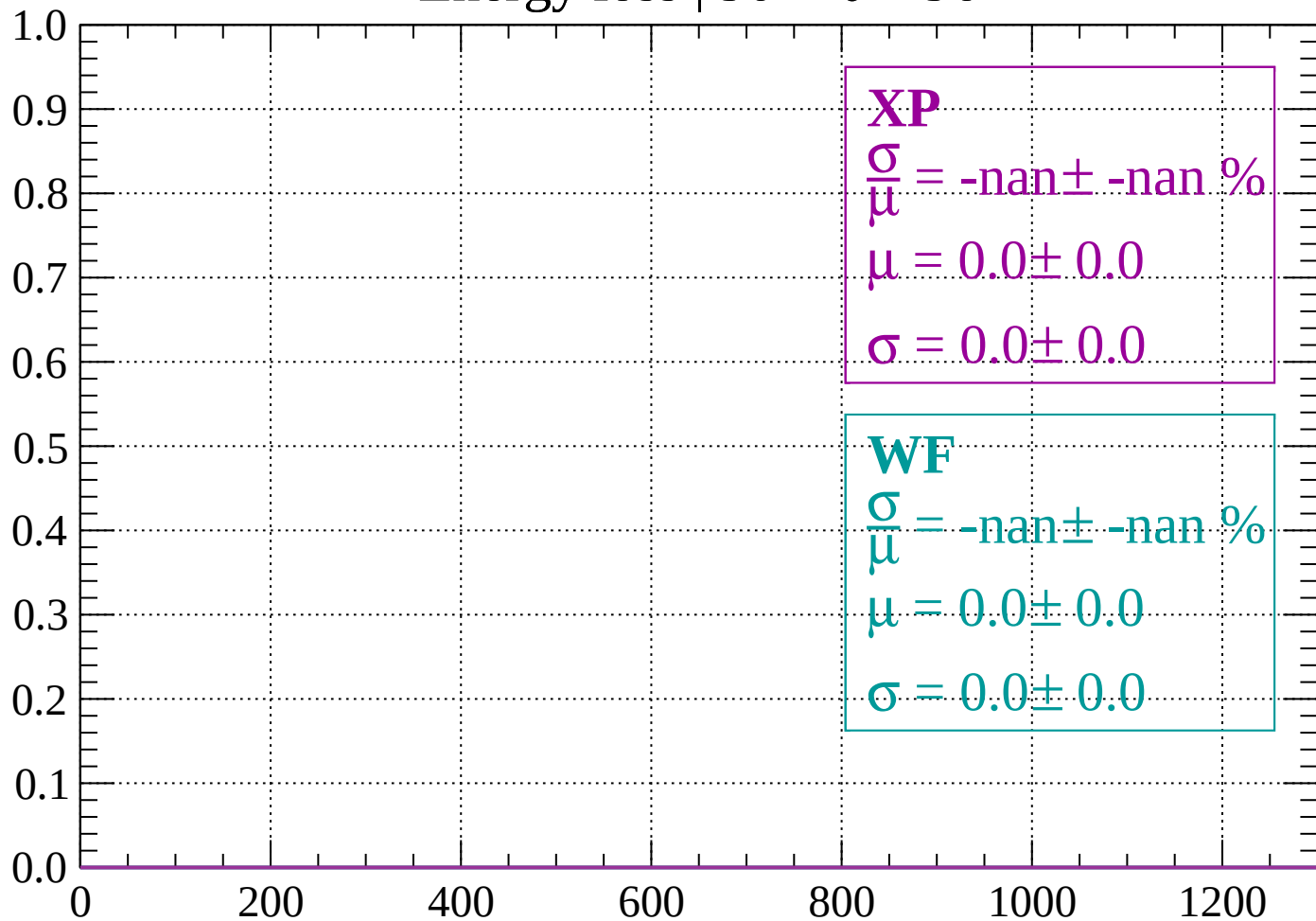
Count



dE/dx (ADC counts/cm)

Energy loss |  $90 < \theta < 96$

Count



dE/dx (ADC counts/cm)

